

EXHALE: Update Log (stage 3, from 2026-09-23)

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This log continues md/Update_EXHALE_stage2.md from 2026-09-23. The earlier logs are kept under their own names:

- docs/Update_EXHALE_stage0.{tex,pdf}: the early phase (the ATES fork up to 2026-08-15).
- md/Update_EXHALE_stage1.md (typeset docs/Update_EXHALE_stage1.{tex,pdf}): sections 1-171, 2026-06 to 2026-09-05; "Update_EXHALE section N" in the source comments and the older memos points into that file.
- md/Update_EXHALE_stage2.md (typeset docs/Update_EXHALE_stage2.{tex,pdf}): sections 1-17, 2026-09-05 to 2026-09-22.

The entries of 2026-09-23 were first written into the stage 2 log as its sections 18 to 24 and moved here the same day as sections 1 to 7; a record of that day that cites "md/Update_EXHALE_stage2.md section 18" to "section 24" means section 1 to section 7 of this file (the references known at the move were rewritten).

Each entry below is one dated section: what changed, why, which files, the check that was run and its result. Section numbers restart at 1. The plan the entries execute is md/PLAN_20260923_rev2.md.

1. The Ritz residuals of the preconditioned spectrum (2026-09-23)

The spectrum hook EXHALE_PRECOND_SPECTRUM=1 printed Ritz values with no accuracy of their own (md/ISSUES_20260909.md item (D8), plan section 16). It now prints the residual $\|A y - \theta y\|$ of each.

The residual. For a Ritz pair $(\theta, y = V_k s)$ with $\|s\| = 1$ the Arnoldi relation $A V_k = V_k H_k + h(k+1, k) v(k+1) e_k^T$ gives $A y - \theta y = h(k+1, k) s_k v(k+1)$, so the residual is $|h(k+1, k)| |s_k|$. `ritz_values_of_a_matrix_free_operator` now hands back $h(k+1, k)$, and the routine prints that residual and the residual over the value's magnitude for the six smallest values of every compression. That form assumes a linear operator and an orthonormal basis, and the finite-difference action is linear only to its truncation, so for the three smallest values of the whole operator the residual is also measured by applying the operator once more to y , once to each part of a complex pair. The operator action is now one routine, `action_of_the_preconditioned_operator`, called by the recursion and by the measured residual alike; the complex-pair packing of `dgeev` is undone in `coefficients_of_the_ritz_vector`. Every existing [diag 15] line is printed as before, and the residuals are new lines.

Verification. On the stored state of `atomic_scalar_gj1132x0.01_kzz1e9/HeH9.7 (g0002_20260919T004843Z_f7485b14)` with EXHALE_TRUST_REGION=1, EXHALE_ELEM_DIAG=1, EXHALE_PRECOND_SPECTRUM=1 and EXHALE_JFNK_MAXIT=1, a 17-second run: the 30 [diag 15] lines of the binary before the change are byte-identical after it, so extracting the action moved nothing in the recursion; and the Arnoldi estimate and the measured residual agree to five or six digits for all three measured pairs, the complex one included. MEASURED. The hook runs only under EXHALE_PRECOND_SPECTRUM=1 inside the trust-region step, which no case of the regression matrix enables, so the matrix was not rerun. Two unused local declarations of the same file (`islot`; `tv`, `net`) were removed on the way; they carried no value.

What it shows. At 200 products that spectrum is the one phase 9 of the campaign quoted (largest 9.99385E-01, smallest 2.52128E-04, ratio 3.96381E+03, reproduced to every digit), and its small end is not resolved: $h(k+1, k) = 0.168501$, and the residual over the magnitude is 12.8, 3.58, 1.86, 1.86, 1.09 and 1.09 for the six smallest values, every one larger than one.

The whole space. `n_ritz_step` became EXHALE_RITZ_STEPS=<k> (default 200, capped at the number of unknowns, below 2 refused). At $k = 1500$, the number of unknowns of this operator, the Hessenberg is similar to the operator, and the same fixture took 136 s. MEASURED: smallest 5.60717E-06, ratio 1.78560E+05, 43 values below 1E-4 and 319 below 1E-2, against 0 and 6 at 200 products. So "nothing below 1E-4" is wrong on that checkpoint, and "there is no near-null mode of the preconditioned operator" is refuted. What the recursion finds are small eigenvalues of the operator the Krylov cycle sees: the hydrodynamic block alone (three unknowns a cell; `nvar_jac = 3 + nspec_row`, with no species rows on this route), scaled and preconditioned. [Corrected later on 2026-09-23 by `md/lhs1140b_spectral_checks_20260923.md`: this read "and shifted by the pseudo-time diagonal"; the shift is in the band preconditioner only, the hook being handed `idtau_leg = 0`.] They are not established as time scales of the atmosphere, and they say nothing about the alternated composition-hydrodynamics map. [Wording corrected 2026-09-23 after the review of `md/PLAN_20260923.md`: this paragraph read "it has a cluster of slow modes".] Below about

1E-5 even the full recursion does not resolve them: the residual measured on the operator is 9.72E-06, 9.88E-06 and 9.56E-06 for the three smallest values (the first two one conjugate pair), larger than the pair's magnitude and 0.96 of the third's, so none is resolved from zero at the production probe length; the floor was later measured to be the rounding of the forward difference (it falls as $\text{arc}^{-0.97}$ with the probe length, spectral checks memo). What the measurement bounds rigorously is the smallest singular value of the action as applied: on the normalized Ritz vector its norm is at most $\|\text{thetal} + \|\text{rll}\| = 1.5\text{E-}05$. Whether the 1E-5 floor of the residual is truncation, rounding or nonadditivity of the finite-difference action is not decided by these runs; `measure_the_additivity_of_the_jacobian_action` is the narrower test. The vectors (the real and imaginary parts of the pair, and the third) carry 53 to 61 per cent in the mass rows and peak in the wind around cells 93 to 169, not at the base, but since they are not resolved their location is not established either. The documents that quoted the 200-product numbers carry a dated correction beside them. What stands is the direct measurement they were offered beside: on the alternated route the Krylov cycle reaches its tolerance in one to four products.

2. Two records the campaign left open, closed (2026-09-23)

EXHALE_RESIDUAL now writes the configuration it measured under. The route stopped at `src/EXHALE_main.f90:1335`, before `write_setup_report` and `write_resolved_config` further down, so `EXHALE_setup.out` was left empty and `EXHALE_resolved.out` absent. It now writes both before its stop, as the direct steady route already did before its own. **MEASURED** on the stored state of `atomic_scalar_gj1132x0.01_kzz1e9/HeH9.7`: 62 and 48 lines written where there were none, the printed residual byte-identical (9.806145E-01, 8.723527E-06, 1.071147E+00), and the resolved file identical to a normal run's but for two added lines, the base ghost count and its electrons as solved, present because this route evaluates the residual, and so solves the ghost, before it writes.

The conversion now asserts the charge. `molecular_seed_from_atomic_state` computed the positive charge density before and after the 2 H → H₂ transfer and never compared the two. It now does, cell by cell, against `inventory_feasible_slack` (eight units in the last place) times the charge density, the bound `element_inventory` already uses for two expressions of the same nuclei; a neutral cell must come back exactly. A breach stops the run, and the largest change is printed in units of `eps * n_chg`. A relative change was not used because a neutral cell carries no charge and the ratio is not defined there. The `molecular_seed` suite asserts it in each of its three modes: 29 of 29 pass, and the change is 0.00 units in every mode, so in memory the transfer conserves the charge bitwise, and the 1.5 units the M2 conversion audit measured were its file round trip, as that audit said. The failure path was read, not exercised.

3. Two statements of the source that were not true, corrected (2026-09-23)

Found while checking the review of `md/PLAN_20260923.md` against the code; the plan's revision is `md/PLAN_20260923_rev1.md`. Neither change moves a number: one is a comment and one is the text of a printed line.

The certificate printed a withdrawn reason. Its report says why the species rows below `cert_regime_wind_r = 1.20` are reported and do not gate, and for the band 1.10 to 1.20 it printed that the row's discretization, 4.6e-4 at the binding

radius, "stands above any tolerance one could set there". The comment above `cert_regime_wind_r` said the same, that a $1e-5$ gate would ask the residual to fall below the error of its own discrete equation. That inference was withdrawn on 2026-09-11 (`md/code_status_20260910.md` section 3.3): the algebraic residual of the discrete equations is not bounded below by their truncation error, and the same routine took a wind H2 row to $2.4e-13$ on a held background. The gating decision is unchanged; its stated reason is now the one that document gives, that no state of this code has come near $1e-5$ there and that the terms of the row are conserved only to $2e-3$ (`certification.f90`, the comment of the two regimes and the report block).

The absent-carrier threshold claimed a round-off argument. The comment on `carrier_absent_fraction` (`diffusive_photochemistry.f90`) said that $1e-20$ of the free reservoir is "four decades below the round-off of a double against the reservoir, so nothing at or under it is a resolved density at all", and that such a carrier carries nothing any reported quantity depends on. A small term vanishing from a rounded sum does not make it unresolved: stored as its own double, $4e-17$ stays $4e-17$ with a relative spacing of $1.5e-16$, and its production, loss and transport stay resolved with it. The comment now says the threshold is a choice and not a round-off limit, and that the test reads the abundance alone, not the production or the incoming flux that could raise it. The threshold itself is unchanged.

Verified: the build compiles (`EXHALE.x` md5 69233669123c368b2d58aebc132eda85; the printed string moved it from 67646328ad476a976a7953aba65bdfc), and one stationary evaluate of a stored molecular state (the pass-6 state of `kzz1e9/HeH0.55` from the 2026-09-21 alternation audit) prints the corrected block and reproduces to four digits the verdict that audit measured on it under its own binary (carrier balance H2 $9.722E-04$ at cell 500, elemental transport He/H $7.353E-04$ at cell 336). The line counts of both files are unchanged. The regression matrix was not rerun: the printed text enters no output file the matrix compares, and a comment enters no object code.

4. Four more source comments that were not true, corrected (2026-09-23)

Found while checking the second review of the LHS 1140 b plan (`md/PLAN_20260923_review1.md`) against the code; the plan's revision is `md/PLAN_20260923_rev2.md`. All four are comments in `diffusive_photochemistry.f90`; the line count of the file is unchanged and the rebuilt `EXHALE.x` has the same md5, 69233669123c368b2d58aebc132eda85, so no object code moved.

- **The boundary section of the module header said "OUTER: zero gradient, as the element operator does"**, and that a run whose carriers are not negligible there is flagged in `output/Oxygen_chemistry.txt`. The carrier operator imposes zero diffusive, eddy and settling flux on the outer face ($J_f(N) = 0$), which with a settling drift is not a zero gradient, and `Oxygen_chemistry.txt` is written only by oxygen-chemistry runs and flags no carrier. The header now states the flux condition and points to `carrier_face_coefficients`.
- **carrier_face_coefficients justified that closure by the outer cells being collisionless**, quoting $Kn_{bulk} = 1.4-2.4$ at the domain top and an exobase at $18.8-25.4 R_p$. Those numbers come from runs made before the 2026-09-13 well-balanced re-solve. MEASURED with `src/utls/collisional_validity.py` on the current certified states and on the `kzz1e9/HeH0.55` audit state: Kn_{bulk} at the top cell is $0.055-0.142$, the exobase lies above the domain, and the largest species value in the critical region is $0.41-0.59$. The comment now says the outer cells are transitional and that the zero face flux rests on the other reason it gives, the measured inward

settling flux of a ghost-built face. `md/collisional_validity.md` carries the same measurement as a dated note.

- **The row-scale floor was said to be converged by construction** ("a density that small cannot change any observable, so a row below it IS converged", and "every term of such a row is round-off"). The floor is the rate that would move a carrier by $1e-20$ of its element in one signal-crossing time, a scale that a row is judged against; it proves nothing about the row.
- **The absence test was said to separate round-off from a real row.** It reads the abundance alone, and a carrier far below the threshold can carry a real, resolved production; the comment now says that the test removes such a cell from the carrier gate without asking whether production or incoming flux could raise it, and that this is a choice.

Outside the source, the comment on the 2.26 km/s overlay shift in `LHS1140b/make_memo_figures.py` said the equivalent width is unaffected. The equivalent widths are computed from unshifted arrays, so the shift enters none of them; applied to a model it would move the fixed-window EW by $1.2e-5$ relative (MEASURED). The comment now says both.

Verified: make rebuilt EXHALE.x to the same md5; the Python file parses. Nothing was run beyond that, since no executable statement changed.

5. One equivalent-width operator for the observation and every model (2026-09-23)

The user's decision on `md/PLAN_20260923_rev2.md` section 4: the observed statistic stays as it is ($1.1084293096 \pm 0.0294774727$ %A, working target 1.108 ± 0.030), and every model is put through the same operator. A model spectrum, already convolved to the instrument resolution, is interpolated linearly onto the observation's own vacuum samples inside 10832.60 to 10834.20 Å (44 samples, 10832.622 to 10834.175 Å) and integrated with their trapezoid weights. Before, each model was integrated with a masked trapezoid on its own grid inside the nominal window, so the two integrals started and stopped at different wavelengths.

One module, six readers. `LHS1140b/he10830_equivalent_width.py` holds the window, the vacuum/air ratio, the observation, and the operator (`model_equivalent_width`, `transit_file_equivalent_width`, `observed_equivalent_width`). The six places that computed an EW each on its own now call it: `LHS1140b/models/status.py` (the results table and, since this change, the EW column of the provenance table, which it used to copy from each run's `REPRODUCE.md`), `LHS1140b/make_memo_figures.py` (the unbroadened and the broadened curves, and the observed value, whose variable and printed label no longer call the released spectrum a digitization), `LHS1140b/models/compare_archive.py`, `LHS1140b/models/write_reproduce.py` (whose line also said the window was in air; it is in vacuum), and the inline EW of `LHS1140b/models/run_case.sh`. The unused regular expression that read the EW out of `REPRODUCE.md` was removed.

What moved, MEASURED. Over the 177 EXHALE spectra of the catalog the change is at most $1.45e-4$ %A, relative $-1.1e-3$ to $+0.9e-3$; 31 spectra move in the fourth decimal, 15 of the 83 certified rows of the catalog table among them (for example the certified molecular $K_{zz} = 1e9$ He/H 2.13 and 9.7 read 1.5604 and 2.4060 where they read 1.5603 and 2.4061), and one crossing moves ($K_{zz} = 0$: 3.1011 to 3.1010). Broadened curves and p-winds curves move more, up to about 0.6 per cent, because their wings reach the window edges that the old model in-

tegral kept and the observation's samples do not (for example the matched-kernel EW of the well-mixed He/H 0.4 solution, 1.067 to 1.065). LHS1140b/MODELS.md sections 7 and 8 were regenerated by `models/status.py --write`; docs/lhs1140b_catalog_state.tex carries the new EWs, a paragraph on the operator and the recomputed comparison of its section 4 (1.374869 to 1.356745 %A, 0.60 standard deviations), rebuilt to seven pages and checked; the memo docs/lhs1140b_exhale_vs_pwinds.tex and its figures are brought to the same operator in a separate pass (md/lhs1140b_ew_operator_memo_changes_20260923.md).

6. The stage export of the stationary alternation (2026-09-23)

EXHALE_STAGE_EXPORT=1 (default off) writes, for every pass of the alternation, the state the driver holds at four named stages: `hydro_entry` (before the hydrodynamic solve; from the second pass on, the state the previous composition update handed over), `hydro_return` (the accepted solve after the ionization sweep that follows it), `certification` (the joint test, with the three residual rows), and `composition_return` (after the element and carrier updates and the rebuild of `p` and `T` at the fixed conserved state). Each goes to `output/stage_state_p<pass>_<stage>.txt` with every row, ghosts included: `r`, `rho`, `v`, `p`, `T`, `heat`, `cool`, the particle and electron densities, `H I`, `H II`, `He I`, `He II`, `He III`, `He(23S)`, `H2`, `H2+`, `H3+`, `HeH+`, the conserved variables, and at the certification the residual rows. At `hydro_entry` and at the certification the driver also asks `conservation_budget` for the next assembly under the stage's name (`conservation_budget_request_next`, `new`), so the energy row's terms of that state are exported without counting assemblies, which probes and trials also consume. It is the instrument `md/PLAN_20260923_rev2.md` P1.5 and P1.6 (b) call for (the four named-stage captures of review R4).

The writer calls nothing that sets a module variable. MEASURED on one pass from the certified `molecular_scalar_gj1132_kzz1e9/HeH2.13_g0008` state: the data rows of `Hydro_ioniz.txt` and `Ion_species.txt` are bitwise the same with the export off (twice), with it on, and with the previous binary `69233669123c368b2d58aebc132eda85`; the log differs only by the export lines. The two exported budget files carry their stage names. Binary EXHALE.x md5 `b017fe42407c980068e112a8ca3d71d3`, copied to `LHS1140b/models/EXHALE_b017fe42.x` with its manifest. The regression matrix was not rerun: every new call in the driver is guarded by the key, and the budget module gained a subroutine and a header line that are reached only on a request, so with the key off no executed statement of any route changed. The bitwise check above covers the stationary alternation; the matrix's cases, which march, were not run.

7. Three defects the family C audit found, fixed, and two small ones (2026-09-23)

The audit of the 13-decade `H2` hole in `molecular_photochem_gj1132_kzzprofile/HeH9` (md/lhs1140b_family_c_h2_slab_20260923.md, item P1.7 of `md/PLAN_20260923_rev2.md`) traced it to the molecular seed and found why no solve removed it. Each finding was checked in the source before it was changed.

The element relaxation threw away its own kept steps (`binary_element_diffusion.f90`, `relax_element_composition`). Its retry budget counts every rejected step of the relaxation, and when it ran out the routine returned `element_relaxation_failed` and restored the entry composition, even after 24 to 31 steps had been kept. The module's outcome table, and the driver that reads it, reserve that outcome for a relax-

ation that kept no step; the driver then refuses the pass and skips the carrier relaxation (EXHALE_main.f90, the outer_element_update_refused branch). This is how the carrier operator never ran on the family C case in four solves, and how the 2026-09-18 rung at He/H 1.411 ended at its second pass ("took 28 steps and found no admissible 29th", READ, L34c section 4). A relaxation that has kept steps now ends as a partial advance (element_relaxation_step_budget) with the kept steps handed back, and says so once in the log; one that kept none is still refused with its entry composition. The retry budget still counts every rejection. Test: two new rows of src/tests/element_operator/ on a test knob (element_refuse_after_accepted_for_test, -1 in every production run) that lets three steps through and refuses the rest: the relaxation reports the partial advance, three steps, and a moved composition; with no step let through it reports the refusal and returns the entry composition bitwise. Every row of the suite passes.

A sweep could rewrite a transported H2 (ionization_equilibrium.f90, the extraction of the molecular partition). With the carriers transported the sweep is handed H2 (x_h2_fixed) and solves the rest against it; a root meets the imposed value through its own row, but a non-root acceptance (class 4) or a clamp onto the element budget (class 3) returned another x(4), and the extraction wrote that back as the cell's H2. MEASURED on the seed's intermediate state of the family C case: the returned H2 differed from the handed one at exactly the 21 non-root cells, by up to 82 per cent; with the fix it differs nowhere by more than $7.8\text{e-}16$. The imposed value is now written and the atomic hydrogen takes the remainder of the budget.

The molecular seed read a root off cells the sweep had not solved (molecular_seed_from_atomic_state.f90, the local mode). The seed takes the root of each cell's own H2 row at the ions of one sweep; at cells where that sweep adopted a non-root with every ion at zero, the root was $4\text{e-}17$ of the hydrogen, thirteen decades below the neighbouring cells, and that was the hole. The sweep now records the acceptance class of each cell (ieq_cell_acceptance_class, written and never read inside the sweep), and the seed reads no root off a class-4 or class-6 cell: such a cell takes the root of its nearest solved neighbours, interpolated in log x_2 over r , and the smaller of that and the fit. MEASURED on the same reproducer: the 21 cells go from $4\text{e-}17$ (and from up to $6.8\text{e-}4$ where the rising slab was) to a smooth $1.8\text{e-}4$ to $2.1\text{e-}3$ between the solved cells 428 and 450; outside them the seed moves by at most $8.2\text{e-}4$ relative, through the H2 fix above. Two further changes the audit proposed were tried and not kept: iterating sweep and root to a common fixed point did not converge in 30 iterations and drove the base layer to its unshielded branch ($x_2 = 6.9\text{e-}4$ at 974 K), because a cell's shielding column is the H2 above it and the iteration is not a local one; and adopting the root in place of the smaller of root and fit follows from the same iteration and was withdrawn with it. The molecular_seed suite passes 29 of 29.

Smaller. The comment on nH_free in diffusive_photochemistry.f90 said it excludes the hydrogen of the ion stages and molecular ions; it is the cell's hydrogen less the HeH⁺ nucleus (carrier_element_totals). And carrier_row_terms.txt was written with ES14.7, which drops the E of a three-digit exponent ($1.0332735\text{-}100$); it is now ES15.7E3.

Not changed, reported. Two provenance attributions in the state index of the family C case are wrong (the audit memo, D4): the certificate of g0002 belongs to a later solve, and g0001's manifest names another binary than its own file header. The index is written by the catalog tools and was not edited by hand.

Verified: the element-operator and molecular-seed suites pass; the H2 fix and the seed on the fam-

ily C reproducers as above; the stationary route with EXHALE_STAGE_EXPORT unchanged. EXHALE.x md5 45190de9ba3752eb720a728d54d3a1e3. The one regression case that runs the H2 carrier operator, mol_carrier, was run with the binary before and after the element and H2 fixes (section 8).

8. The regression check of section 7, a cell writer, and corrections from the spectral and On stall tests (2026-09-24)

The one regression case that runs the H2 carrier operator. mol_carrier was run from its input with OMP_NUM_THREADS=1 EXHALE_MAXSTEPS=12000 twice, once with the binary before the fixes of section 7 (EXHALE_b017fe42.x) and once after (EXHALE_af853fe7.x), in copies (LHS1140b/models/.P1/d2d3_mol_carrier/). MEASURED: the two Hydro_ioniz.txt differ by at most $1.3\text{e-}11$ relative and the two Ion_species.txt by at most $9.4\text{e-}13$; neither run met a non-root cell or the retry budget of the element relaxation, so the difference is the H2 now written exactly at the imposed value where a root had met it to the solver's tolerance. Below 0.1 per cent, it needs no golden refresh.

A cell writer on the EXHALE_NEWTON_TEST route (EXHALE_main.f90). The route printed only the row maxima of the solver's vector F(Y), so where it differed from the EXHALE_RESIDUAL route the cell was not known (ISSUES D5). It now writes output/newton_residual_profile.txt with the columns of output/residual_profile.txt (its temperature column is 0: the route forms no temperature array). MEASURED on molecular_scalar_gj1132_kzz1e9/HeH2.13 g0008 with EXHALE_RELOAD_EQ=0: the two routes differ in cells 1 and 2 alone, where the base face mass flux differs by 9 per cent, and nowhere else in 500 cells; the mass-row maxima ($5.10\text{e-}09$ against $1.37\text{e-}09$) are that base difference.

From the checks of the preconditioned spectrum (md/lhs1140b_spectral_checks_20260923.md):

- the cost line of the additivity measurement printed the run's cumulative count of inner cell solves where it meant this measurement's own (nt_calls, now nt_calls_here, steady_newton.f90);
- the comment that the Newton model's products reuse the frozen WENO3 weights is true on the line-search route and not on the trust-region route, where the merit evaluation sets weno_mode = 0 and nothing restores it before the leg and the spectral hook; the comment now says so, and which of the two the leg should use is left open;
- the base boundary's comment that the states "sit decades from" the thresholds of the wind window now records a state that sits one decade away and a probe that crosses the switch (base_boundary.f90);
- the records that called the spectrum's operator "shifted by the pseudo-time diagonal" (ISSUES D8, stage 3 section 1, the p2_base note, the plan rev2 section 3.6) were corrected: the shift is in the band preconditioner only.

From the On stall test (md/lhs1140b_on_stall_end_to_end_20260923.md):

- Ionization transport: True was refused with Coupled carrier solve: True but not with On stall, which hands the same registry the same unknowns once the alternation stalls; it is now refused with either (input_read.f90);
- the comment on EXHALE_JFNK_MAXIT said a named cap makes the run take one stationary solve; it forbids a second entry into the stationary route and caps the solve of every pass inside the one entry, which the test

measured (EXHALE_main.f90); the plan rev2 carried the same error and is corrected. The handover itself fails: the conditions are met and the transition begins, and the re-evaluation of the snapshot's residual stops the run on the face mass flux guard, the carrier balance also reading a background the snapshot does not hold. That is a change of the solver's control flow and is reported for a decision, not fixed.

The update log. The entries of 2026-09-23 were moved from the stage 2 log (its sections 18 to 24) into this file on the user's instruction, and `src/utls/update_log_to_tex.py` now takes the stage as its argument (2 or 3; the code-size appendix stays with stage 2). `docs/Update_EXHALE_stage2.tex` was regenerated for sections 1 to 17 (310 pages, no error or undefined reference) and this log's twin built (7 pages at the time).

Verified: the element-operator and molecular-seed suites pass; the builds compile. EXHALE.x md5 46d5b9f4c0a62c1ab0cdf9bb165839ef.

9. Three defects the transient measurements found, repaired (2026-09-24)

The P4 measurements of the plan (`md/lhs1140b_p4_transients_20260923.md`, whose worker reported them and left them in place) found three defects; each is repaired and its path checked.

The default CFL number was a single-precision literal. `real*8 :: CFL = 0.6` stored 0.6000000238 (`parameters.f90`); every run that took the default marched at a step $3.97\text{e-}8$ longer than the one it declared (MEASURED: the first step on a certified molecular state is 0.54736376538957865 s with the default and 0.54736374363929763 s with `CFL: 0.6`). Now `0.6d0`. The same kind of literal set the Voronov (1997) collisional-ionization coefficients of all seventeen rows of `Cool_coeff.f90` (11.26 stored as 11.2600002), and the physics probe `voronov_carbon_ionization.f90` mirrored it on purpose to compare exactly; both are now double-precision literals, and the probe passes at its $1\text{e-}12$ tolerance (MEASURED). The rounding was below $1\text{e-}7$ relative, four decades under the precision of the fits; a declared constant is still the constant it declares.

EXHALE_UPDATE_MAP could stop before its last requested step. The diagnostic switched off the flux-spread stop, the residual gate and the JFNK hand-off, but not the step-change stop or the stall stop, and on a stationary state a step of a quarter of the CFL step already moves the state by less than `dtu_th`: a six-step map ended after three blocks. While the map runs, every stop flag is now cleared after the stop tests of each step (EXHALE_main.f90), so the map ends on its own step count and the staged secondary ionization cannot change the physics between two of its steps; the stop line says "the update map wrote its N requested step(s)". MEASURED on a copy of the failing run without `Force_start`: six blocks, `dt` from $6.6296\text{e-}05$ to $6.6296\text{e-}11$ of the code unit.

EXHALE_DUMP_IC=1 wrote placeholders as the lower ghost rows. `load_IC` fills the primitive ghost rows with cell 1's state until the first `Apply_BC`, which writes the ghosts of the conserved array only; the dump wrote the primitive array. It now takes the ghost rows of the primitive array the boundary derivation held (`bc_W_hold`). MEASURED: the two lower ghost rows read $\rho = 1.302\text{e}14$ and $1.242\text{e}14$, $T = 233.0998$ and 226.0275 K, against the placeholder $4.857\text{e}13$ and 546.06 K before, and the file's 233.1004 and 226.0281 K (the $2.5\text{e-}6$ is the ghost composition the boundary re-derives from the loaded column).

The P4 memo now carries its command record (section 7, which its worker had not written) and the repairs beside each defect.

10. The elemental flux records ride on the Riemann face mass flux; a row writer for the helium element (2026-09-24)

What was wrong. The element transport operator is handed the face mass flux of the Riemann solve (face_mass_flux_of_state), the one the mass row differences. Two records of it were not: the base budget entry (element_base_flux) and output/element_flux_profile.txt built their face mass flux as the face mean of the cell-centred density and velocity. Near the base the cell-centred velocity carries a collocated two-cell odd-even mode (project record P44), so on the certified molecular_scalar_gj1132_kzz1e9/HeH0.55 state the file read $4\pi r^2 F = -1.07e9, +1.16e8, -3.8e7$ and $+5.9e7$ g/s at faces 1 to 4, while the Riemann flux is $3.934813e7$ g/s through every face (MEASURED). The "standing base sound wave" the comments named was that mode, not a wave of the solution.

What the records are now (binary_element_diffusion.f90, certification.f90, EXHALE_main.f90):

- write_element_flux_profile takes the face mass flux of the state it describes and is called by the certification of that state, with the flux the element row was measured on, whenever EXHALE_DIFFUSION_CHECK=1 or a lower-atmosphere profile is in use; the element step no longer writes it. Before, the file held the profile of the last relaxation, taken before the last hydrodynamic solve, and a stationary evaluate never wrote it.
- The base budget entry uses the flux of its own evaluation: the relaxation's Frho_in, or on the marching path the step flux $F_0/6 + F_1/6 + 2 F_2/3$ of the three SSP-RK3 stages (the flux whose divergence is the step's density change), read once per step; with neither, it says "not measured".
- face_mass_flux_cgs converts a code face flux with $n_0 \mu v_0$ and the face mean of msum.
- The velocity is no longer an argument of element_diffusion_step or relax_element_composition (nothing read it).
- composition_residual and trace_composition_residual reduced their norm with max, whose result with a NaN is compiler dependent; a non-finite row now sets the norm to huge.
- element_transport_residual, documented as side-effect free, added its face reconstructions to the run's face counters; it now restores them.

The helium row, term by term. EXHALE_ELEMENT_ROW_TERMS=1 (default off) writes output/element_row_terms.txt from the certification's own evaluation: the diffusive and advective terms of each face, the residual, the floor, the scale and the measure, with the face F_{ρ} , X_{face} and J . Its measure agrees with the certification's full-precision anchor to the sixteenth digit.

What it measured (md/lhs1140b_element_row_terms_20260924.md): on the certified He/H 0.55 and 1.8 states the helium flux through $r < 1.10$ varies by $2.2e-3$ and $7.8e-5$ of its median, nearly all of it the step across cell 2 (the first solved cell, row $6.1e-5$ and $2.3e-5$, ungated); every other cell below $r = 1.2$ stands at or below $1.5e-7$ and $2.7e-7$. At cell 2 the residual would change the helium density in $9e9$ and $3e10$ s, against an advective cell time of $2e6$ s and a diffusive one of 47 s. The comment in certification.f90 that gave "element fluxes not conserved, spread 9.8" as the reason for not gating $r < 1.10$ had measured the face-mean artifact; it and the printed line now say the band is reported by

decision, and the gating itself is unchanged. The Phase-E statement that the LHS 1140 b overlap window is structurally empty rested on the same quantity and is marked stale where it stands (md/phase_e_flux_closure_design.md, src/utls/element_flux_closure.py, md/certification_tolerance_anchoring_20260910.md); it was not re-measured. src/utls/collisional_validity.py and md/collisional_validity.md used the same wording for the base cells; corrected (the computation, whose velocity scale binds four decades below its threshold there, is unchanged).

Checks. The solved state does not move: an evaluate of the certified He/H 1.45 state by the new binary writes Hydro_ioniz.txt, Ion_species.txt and both _adv files identical to the 46d5b9f4 evaluate apart from the provenance line, CERTIFIED, and its element_flux_profile.txt carries 7.0353671e7 g/s from face 1 outward (MEASURED; the worker measured the same on the He/H 0.55 and 1.8 states). Suites: element operator 65 of 65 (its base record rows assert the new definition and fail against the old objects by 1.54 and 1.31), diffusion_tests.x 35 of 35, certification 123 PASS, steady species rows 202, species face flux 18 (the last two by the worker). The dead velocity arrays of diffusion_tests.f90 and certification_contexts.f90 are removed, and the test T6, which claimed a uniform mixture "under an arbitrary velocity field" that the operator never read, now says what it tests.

11. The molecular $K_{zz} = 1e9$ ladder certified across the observed equivalent width (2026-09-24)

Items P1.1 to P1.3 of the plan, run under the binary EXHALE_46d5b9f4.x (the tree of section 8), every rung a stationary alternation from a certified parent mapped onto the new reservoir with the pressure held (map_state_to_grid.py --ic --reservoir He/H, the base mixing ratio from the same nucleus fraction f), 24 passes and a 3 h ceiling allowed. Run directories under LHS1140b/models/.P1/; driver .P1/make_rung.sh.

He/H	parent	passes	wall of the passes	outcome	EW [%A]	log10 Mdot
0.55	the 2026-09-21 audit state (pilot)	12, then 4	4.30 h, then 0.07 h	certified in the continuation	0.1015	7.59
1.8	certified 2.13 (descent pilot)	12, then 3	1.04 h, then 0.04 h	certified in the continuation	1.3584	7.88
1.45	certified 1.8	17	1.15 h	certified	1.0964	7.85
1.6	certified 1.8	21	1.22 h	certified	1.2155	7.86
1.52	certified 1.45	14	0.88 h	certified	1.1532	7.86
0.689	certified 0.55	24 (budget)	2.85 h	not certified: elemental row 9.9e-4, still falling	0.274 (uncertified state)	7.67

MEASURED. Every certified state was re-entered four times in a chain of Restart intent: stationary evaluate, each entry reading the state the one before wrote: CERTIFIED at every entry, the largest relative change of any field

after four entries $1.2\text{e-}12$ (the 1.8 chain; $3.5\text{e-}13$ or less for the others). The EW is the catalog's measurement operator (section 5) on the spectrum EXHALE_transit.py synthesized from the published state. The two adjacent certified rungs 1.45 and 1.52 straddle the observed $1.108 \pm 0.030\%$ A, and models/status.py puts the crossing of the log-log chord at $\text{He/H} = 1.4643$. The interval is 0.0205 dex, below the plan's 0.03 dex floor, so the refinement stopped there.

Connectivity (.P1/connectivity/compare_rungs.py, figure docs/figures/lhs1140b_kzz1e9_molecular_rungs.pdf): from 1.45 to 2.13 the temperature, velocity, density, H II and He 2³S profiles, the mass-loss rate (7.847 to 7.902), the outer Mach number (0.515 to 0.508) and the H2 front ($x(\text{H}_2) = 1\text{e-}2$ at 1.200 to 1.161 R_p) move monotonically, and neighbor differences scale with the step (about 3 per cent in T for 0.02 dex). The 0.55 state is a different wind: $x(\text{H}_2)$ above 0.5 to 1.13 R_p and near $1\text{e-}2$ in the far wind, a peak temperature of 3400 K against 5200 K and more, He 2³S thirty times lower. The interval between 0.55 and 1.45 is not sampled (the 0.689 rung is its first point), so whether the two are one branch, and whether a second crossing lies between them, is not established. The 0.689 rung's last state (uncertified) is intermediate: $x(\text{H}_2)$ above 0.5 to 1.10 R_p as at 0.55, but $1.5\text{e-}4$ at 10 R_p , near the 1.45 value.

The line shape (docs/figures/lhs1140b_kzz1e9_molecular_ew.pdf): the crossing is of the integrated EW alone. At $\text{He/H} = 1.45$ the model red line is 4.07 per cent deep, 3.3 times the depth Cherubim et al. (2026) fit ($1.24 \pm 0.22/-0.23$ per cent), and correspondingly narrower; the molecular solutions share the line-shape disagreement of the unbroadened atomic spectra.

Published (models/run_case.sh with SEED= the certified run, which solves nothing new from a certified entry, evaluates, synthesizes the spectrum and writes REPRODUCE.md): molecular_scalar_gj1132_kzz1e9/HeH0.55 g0007 and the new cases HeH1.45, HeH1.52, HeH1.6, HeH1.8 (g0002 each), added to models/make_models.py. The new cases' input.inp is the certified HeH2.13 input with the He/H changed, the input the states were solved under; make_models.py itself would write Restart intent: stationary equilibrate for a molecular case, which the older HeH0.083 carries and the certified cases do not (a difference of route, reported for a decision). The catalog inventory reads 157 state indexes, 100 certified. LHS1140b/MODELS.md sections 7 and 8 and models/CASE_INVENTORY.{md,json} were rewritten by status.py --write; docs/lhs1140b_catalog_state.tex carries the new rows (section 13) and docs/lhs1140b_exhale_vs_pwinds.tex its corrected molecular paragraph and campaign table (both rebuilt, no error, pages checked). A verification binary copied into models/ had become the "newest EXHALE_x" that status.py takes as the binary of record; it was moved to the run directory that used it.

The physical record of the straddling states (plan section 3.1, with the writer of section 10, .P1/element_row_terms/record_straddle.txt): the H2 carrier row is worst inside the gated wind ($5.16\text{e-}6$ and $7.90\text{e-}6$), and the helium element row is worst at cell 2, ungated ($2.04\text{e-}5$ and $1.27\text{e-}4$), the next worst below $r = 1.2$ being $1.07\text{e-}6$ (cell 6) and $5.4\text{e-}6$ (cell 3), both in the first cells above the base; at cell 2 the residual would change the helium density in $2.9\text{e}10$ and $4.6\text{e}9$ s, against an advective cell time of $1.5\text{e}6$ s, and the region below 1.2 passes the helium flux on to within $9.3\text{e-}5$ of its throughput.

P3 (md/lhs1140b_p3_low_xuv_20260923.md, run 2026-09-23 under EXHALE_69233669.x): the atomic He/H 9.7 branch certifies at 0.0675, 0.0662 and 0.065 of the fiducial XUV (EW 0.0239, 0.0172, 0.0132 %A) and refuses at 0.06 and 0.0624 on the elemental row; each certified rung re-entered four times, CERTIFIED throughout.

12. The regression of sections 9 and 10, and the validation of the crossing (2026-09-24)

The matrix (make check, binary ac232b34, the tree with section 9; sixteen cases run, against the goldens of 2026-09-20). Verdict REGRESSION FAIL, for three reasons, MEASURED and attributed with the binary of the day before (EXHALE_46d5b9f4.x, the tree without section 9) run on copies of the cases:

1. **The CFL literal of section 9.** The control reproduces the goldens (mol_lyman_werner identical, mol_carrier within $1.3\text{e-}11$), so every movement is today's, and in these marching cases the only change that acts is the step $3.97\text{e-}8$ shorter. The state files move by at most $9.1\text{e-}5$ (wasp_he23off; $1.6\text{e-}5$ or less in the rest), below the $1\text{e-}3$ tolerance. What exceeds it is the adv_mass_row column of Hydro_ioniz_adv.txt, the mass-row measure the post-processing writes per row, in the four molecular 12000-step snapshots that are far from stationary (du near 0.8): $1.2\text{e-}3$ to $4.3\text{e-}3$ relative (mol_lyman_werner, mol_diffusion, mol_sec_ion, mol_carrier), and 0 against $5.6\text{e-}15$ in one row of wasp_full_newton. hydrostatic_column moves one ghost-row ion density from $4.1\text{e-}185$ to $1.2\text{e-}170$ (the control is identical to its golden). None of these is a change of a physical quantity above the tolerance.
2. **oxygen_chemistry stops before its first step**, with the control as well: "ghost composition is not a fixed point", the map applied 20 times and still moving by $5.7\text{e-}7$ against an accuracy of $1\text{e-}11$, after the constrained molecular roots of the cold start were clamped onto the simplex. The golden (2026-09-20 13:53) predates the ghost fixed point (commit of 2026-09-20 16:18); the case has not run since. The refusal is the boundary's designed behavior, so what to do with it is reported for a decision, not changed.
3. Two files of iontrans_metals have no reference (MISS, not a failure).

Section 10 does not move them. The five cases that exercise the changed paths (element operator and its records, the certification's element block) were re-run with the tree of section 10 (76c3ce17) against the outputs of the matrix: mol_diffusion, lower_profile, iontrans_metals, mol_carrier, hydrostatic_column, REGRESSION PASS, all byte-identical (tolerance 0). The other cases run without element diffusion and do not reach those paths.

The goldens were not refreshed. The refresh of the sixteen cases that section 9 moves was refused by the session's permission check; the outputs of the matrix stand in the case directories and are what a refresh would snapshot.

The validation of the crossing of section 11 (plan order 12):

- A reverse continuation: He/H 1.52 reached downward from the certified 1.6 certifies (pass 13) and agrees with the 1.52 reached upward from 1.45 to $5.3\text{e-}6$ in every hydrodynamic field and $1.4\text{e-}4$ in every species, EW 1.15317 against 1.15316 %A (LHS1140b/models/.P1/reverse_HeH1.52_from1.6/).
- A refined grid (LHS1140b/models/.P1/refine_grid/): the two straddling states on 1000 cells (the base block and every other spacing halved, the outer radius unchanged) certify (passes 16, 15) and give EW 1.0640 and 1.1196 %A (against 1.0964 and 1.1532 on 500 cells), log10 Mdot 7.84 and 7.85. They still straddle 1.108; the chord moves from He/H 1.4643 to 1.505. The discretization shift, 3 per cent in EW, is of the size of the observation's working error; a 2000-cell pair is running to read the order.
- The physical record of section 11 and the operator of section 5.
- The domain test waits for a grid option (a decision).

13. The width of the line on the molecular crossing, and the catalog summary brought up to date (2026-09-24)

The broadening the line needs (user instruction: after the EW, match the width with an assumed turbulent Gaussian as docs/lhs1140b_exhale_vs_pwinds.tex does). LHS1140b/models/.P1/connectivity/broaden_molecular.py restates the memo's operation (the excess absorption of the instrument-convolved spectrum convolved with a Gaussian of FWHM f in velocity, the three-Gaussian extractor of he_line_metrics.py, the EW operator of section 5, f matched where the red-pair FWHM reaches the measured 0.841 Å) and first reproduces the memo on its atomic well-mixed He/H 0.40 state: $f = 22.22$ km/s, depth 1.229, blue 0.161, ratio 7.65, EW 1.0645 (the memo prints 1.065). MEASURED on the certified molecular $K_{zz} = 1e9$ states:

He/H	cells	EW, $f = 0$	matched f [km/s]	red [%]	blue [%]	red/blue	EW at f
0.55	500	0.1015	22.32	0.113	0.014	7.89	0.0981
1.45	500	1.0964	22.21	1.224	0.160	7.64	1.0600
1.52	500	1.1532	22.20	1.287	0.169	7.63	1.1149
1.6	500	1.2155	22.19	1.357	0.178	7.61	1.1752
1.8	500	1.3584	22.16	1.516	0.200	7.58	1.3135
2.13	500	1.5604	22.12	1.741	0.231	7.55	1.5087
9.7	500	2.4060	21.80	2.684	0.359	7.47	2.3264
1.45	1000	1.0640	22.21	1.188	0.155	7.65	1.0288
1.52	1000	1.1196	22.19	1.250	0.164	7.64	1.0825
observed		1.108 +/-		1.24	0.25	6.7	1.108 +/-
		0.030		+0.22/-0.23	+0.14/-0.12	+12.7/-3.1	0.030

At He/H 1.45 the width is reached at 22.21 km/s (sigma 9.43 km/s) and the depth, the blue component and the ratio then fall inside the observed 1 sigma intervals without being tuned. The kernel does not depend on composition or chemistry (22.12 to 22.32 over 0.55 to 2.13; the memo's atomic families 22.20 to 22.40). The Gaussian wings carry 3.3 per cent of the EW out of the window, so the broadened EW crosses the working target 1.108 at He/H 1.5112 between the certified 1.45 and 1.52 on the catalog grid; on 1000 cells the broadened EW of 1.52 is 1.0825, and the chord extended past 1.52 reaches the target at 1.55 (DERIVED, an extrapolation). On the advection-corrected profiles weighted by $n(2\ 3S)\ r$ over r , $c_s = 4.33$ km/s in the line-forming region of both straddling states, so the kernel is Mach 2.2 (sigma), 3.1 (sqrt 2 sigma, the v_{turb} of Lampon et al. and p-winds) and 3.8 (sqrt 3 sigma). Figure docs/figures/lhs1140b_kzz1e9_molecular_broadened.pdf.

A correction to the memo found on the way. The memo's c_s of the line-forming region of the well-mixed He/H 0.55 state, 4.1 km/s, did not follow from its own $T = 2499$ K and $\mu = 1.94$; recomputed with the memo's weighting ($n(2\ 3S)\ r$ integrated over r on the $_{adv}$ profiles, which reproduces $T = 2498$ K and $\mu = 1.93$) it is 4.21 km/s, and the Mach numbers of the matched kernel move from 2.3, 3.3, 4.0 to 2.2, 3.2, 3.9 (summary and section of the memo corrected, marked in place).

The catalog summary. Renamed on the user's instruction to docs/lhs1140b_catalog_state.tex

(no date in the name; it was lhs1140b_catalog_state_20260922.tex, and the references in md/du_stop_vs_stationary_20260920.md, md/session_handoff_20260920.md and the stage 2 and 3 logs now point to the new name; the stage 2 twin Update_EXHALE_stage2.{tex,pdf} still carries the old name and was not regenerated). Brought up to date rather than annotated: the counts (157 indexes, 100 certified; 103 models, 91 certified), the table by group, the table of certified models (eight new rows) and of uncertified ones (the molecular 0.55 row out, the He/H 9.7 rungs at 0.06 and 0.0624 in), the mechanisms (and the count of refusing rows, which the old text stated wrongly: the energy row, not the mass row, appeared most often), the operator-difference statement re-measured over the 91 certified rows (still at most 1.5×10^{-4} %A, 15 rows in the fourth decimal), and a new section 5 on the molecular crossing, its validation and the broadening. Rebuilt (11 pages, no error), every changed page checked.

14. The user's decisions of 2026-09-24 on the open items, and what each gave

The list of section 6 of md/lhs1140b_plan_rev2_execution_20260924.md, reordered by what it changes, was put to the user; every item was decided "as recommended". Items 2 and 3 (the rungs below He/H 1.45, and the grid of the catalog) are still running and are recorded in the next section.

Item 1, the width of the line. The added Gaussian of section 13 is an empirical parameter of the comparison, fixed by the measured width alone; every composition read from the full line is stated as conditional on it (docs/lhs1140b_catalog_state.tex section 5, the execution record).

Item 4, which cells gate the species rows: compared, not changed. A default-off diagnostic, EXHALE_CERT_GATE_DIAGNOSTIC=1 (certification.f90), prints for every carrier and element row what the verdict would be if every cell from cell 3 outward were gated against the wind tolerance (1×10^{-5}), cells 1 and 2 reported. MEASURED on all 99 ladder cases re-evaluated with it (md/lhs1140b_gating_diagnostic_20260924.md): every stored verdict reproduces; under the candidate rule 38 of the 91 certified cases stay certified and 53 would be refused, 52 of them by the helium element row in the first cells above the base (cell 3 in 45 cases, up to 9.1×10^{-2}) and one by the H2 row (molecular kzz1e9/HeH9.7, 4.76×10^{-5} at $r = 1.127$); no uncertified case would become certified. In 27 atomic K_{zz} cases the element row exceeds 1×10^{-5} in 76 to 205 cells up to $r = 1.09$ to 1.19 (on the atomic kzz1e9 He/H 1.50 and 1.60 states 1.5×10^{-3} and 1.8×10^{-3} at cell 3, 184 and 182 cells out to $r = 1.16$). The premise of the item, that below $r = 1.2$ only the first solved cell stands above 1×10^{-5} , holds on the six certified molecular K_{zz} = 1e9 states and not on the atomic ladders. The gate was not changed. The same worker's compiler warnings showed two truncations in the certification report (System_HeH_mol_metals printed as System_HeH_mol_metal, the carrier row's scale text cut at 76 characters); both fields were widened (log text only). The diagnostic shares the gate's reduction with the verdict (species_row_distance_over_gated_cells); the certification suite passes 143 of 143, and with the variable unset every output is byte-identical to the build before it (MEASURED by the worker on four states).

Item 6, a grid option that keeps the interior. Outer shells [r_face,cells]: <r_face> <cells> (K33d of md/input_schema.md, default off): the grid is built as before, and <cells> cells are appended beyond its outer face, their widths in geometric progression with the ratio that puts the last face exactly on r_face; every constructed center up to the first outer ghost, every face, width and volume and the window indices are bitwise those without the key (new test grid_and_gates/outer_shells_base_identity, 40 of 40), N and everything sized on it follow the extended grid, and a state written with shells carries them in its # grid line, so it loads only into a run with the same shells.

With the key absent every output is byte-identical to the control build (MEASURED on an LHS 1140 b evaluate and on hydrostatic_column).

Item 5, the domain test (P1.4 (c)) and the outer closure (md/lhs1140b_outer_shells_domain_test_20260924.md).

The straddling molecular states on the 500-cell grid plus 43 shells to 60 R_p (spacing ratio continuous to 4e-4) both certify (13 and 10 passes). MEASURED: the wind turns transonic, with a sonic point at 40.3 R_p (Mach 0.51 at the old 29 R_p edge becomes 0.77); the EW falls by 1.74 per cent at both compositions (1.0964 to 1.0773, 1.1532 to 1.1330), log₁₀ Mdot moves by 2e-4, and the profiles move by at most 0.13 per cent inside 2 R_p and by up to 20 to 30 per cent at the old edge. The EW crossing moves from He/H 1.4643 to 1.4886; with the matched kernel (unchanged, 22.2 km/s) both extended states lie below 1.108. The H₂ carrier's last-cell defect stays two decades below the gate on both domains (5.6e-7 at 60 R_p): the zero diffusive outer face does not decide these states; the subsonic outflow boundary at 30 R_p holds the outer wind on a subsonic branch and acts through the hydrodynamics (tentative).

Item 7, the composition update holding the pressure. Composition update holds: energy|pressure (K15j, default energy): with pressure, the composition update of the alternation keeps rho, v and p of every cell, re-closes the eliminated species at the temperature of the new composition and rebuilds the energy from the caloric equation of state (rebuild_energy_at_held_pressure, EXHALE_main.f90), printing the energy it adds. MEASURED (md/lhs1140b_composition_update_pressure_20260924.md): it removes what it was built for, the base pressure step (cells 1 to 3 held exactly; the base face mass flux handed to the next solve 4.46e-8 against the wind's 4.44e-8, where the default hands -2.84e-5), and the next hydrodynamic solve still fails in all three tests: on the well-mixed He/H 0.083 and 0.55 cases on the energy row at the H₂ front (the carrier update raises the cooling there by 31 to 44 per cent at the fixed wind, under either value), and on the K_{zz} = 1e9 rung 1.8 to 1.6, where the default recovers and certifies at pass 21, it fails from pass 2 on and never recovers. The obstacle behind the base step is the front's energy balance under a composition change at fixed wind. The default is unchanged; the option stays, default off, with its negative result recorded. With the key absent every output is byte-identical to the control build (MEASURED on the P1.6 entry and on hydrostatic_column).

Item 8, one route for every molecular catalog case. make_models.py wrote Restart intent: stationary equilibrate for a molecular case; the certified molecular cases carry stationary, and were solved under it from an atomic-to-molecular conversion as well as from molecular continuations (He/H 2.13 and 9.7, runs of 2026-09-19, READ). So instead of a rule by the kind of seed (the recommendation), every molecular case now takes stationary: make_models.py writes it, with Coupled carrier solve: False and the base-grid comment line the certified cases carry, and its output now equals the input.inp of every molecular K_{zz} = 1e9 and well-mixed case byte for byte (the two photochemical molecular cases differ only in the order of one line); molecular_scalar_gj1132_kzz1e9/HeH0.083/input.inp, the one case with equilibrate, was changed to match.

Item 9, the On stall handover: deferred.

Item 10, the goldens. Refreshed for the sixteen cases the matrix of section 12 ran (all but oxygen_chemistry), from the outputs of that matrix; the previous goldens are kept in backup/regression/golden_pre_cfl_20260924/.

Item 11, oxygen_chemistry. The case now starts from its own golden of 2026-09-20 (the 1000-step snapshot of the cold start, kept in its IC/) and marches 1000 steps from there (Load IC? True, Restart intent: relaxation; its README.md records why); the load passes the ghost fixed point and the run ends at du = 20.9, log₁₀ Mdot 9.39

(MEASURED with the control build). Its golden was snapshoted from that run.

The smaller items. The Phase-E overlap window is entry (AH) of `md/TO_BE_DONE.md`, deferred to the next time the LHS 1140 b profile runs are handled. `docs/Update_EXHALE_stage2.{tex,pdf}` was regenerated for the edit of its section 17 (310 pages, no error). `src/utils/map_state_to_grid.py` wrote nan as the center shift in its # mapped: line between grids of different cell counts; it now says the shift is not defined there.

15. The grid of the crossing (decision item 3) and the regression of sections 13 and 14 (2026-09-24)

Three grids, three compositions. The certified molecular $K_{zz} = 1e9$ states at He/H 1.45, 1.52 and 1.6 were mapped onto grids of 1000 and 2000 cells (the base block and every other spacing halved at each step, the outer radius unchanged; target grids from cold-start dumps) and solved again with the rung environment; all six certify (1000 cells: 16, 15, 15 passes; 2000 cells: 13, 13, 19). MEASURED through the catalog's evaluation, synthesis and EW operator (`LHS1140b/models/.P1/refine_grid/, .P1/connectivity/grid_convergence.txt`):

He/H	EW, 500 cells	1000 cells	2000 cells	continuum (DERIVED)	log10 Mdot 500 / 1000 / 2000
1.45	1.0964	1.0640	1.0475	1.030	7.85 / 7.84 / 7.84
1.52	1.1532	1.1196	1.1025	1.085	7.86 / 7.85 / 7.85
1.6	1.2155	1.1804	1.1625	1.144	7.86 / 7.86 / 7.86

The EW converges at first order (successive-difference ratio 1.96, $p = 0.97$, at every composition), and the continuum column is the Richardson extrapolation at that order. The crossing of the working target 1.108 moves from He/H 1.4643 (500 cells) to 1.5054 (1000), 1.5273 (2000) and 1.552 (continuum); from the broadened line of section 13, from 1.5112 to 1.5547, 1.5780 and about 1.60 (the last just above the 1.6 rung); every other value is interpolated between certified rungs. The matched kernel does not move with the resolution (22.18 to 22.21 km/s on all nine states). The 500-cell catalog overestimates the EW by about 6 per cent, twice the observation's working error; a formally second-order scheme converging at first order in the EW points to a first-order error in the line-forming region, whose source is not established. Combined with the domain factor of section 14 (DERIVED, the two measured separately), the crossing stands near 1.58 from the EW and near 1.63 from the broadened line. Which grid the catalog should use is left to the user with these numbers.

The regression of sections 13 and 14. After all the code of this series had landed (the diagnostic of item 4, the grid option of item 6, the composition option of item 7, the widened report fields, and the mapper record), the tree was rebuilt (`EXHALE.x md5 2df491cb020e8200be247e2469964a63`, no warning) and the whole matrix run against the goldens refreshed in section 14: REGRESSION PASS, all seventeen cases byte-identical (66 files; the two `_adv` files of `iontrans_metals` have no reference, as that route writes none), `oxygen_chemistry` included on its warm start.

16. The user's decisions of 2026-09-24 on the four remaining items

Put to the user after sections 14 and 15; decided the same day.

The catalog's grid: 500 cells, the compositions quoted from the grid continuum. The catalog keeps its 500-cell grid (every certified state, the chains and the ladders stay as they are); a composition read from it is quoted as the continuum value of section 15, with the difference between the 500-cell and the continuum value as the grid uncertainty of every catalog number: from the EW, He/H 1.55 (the 500-cell crossing is 1.46, 0.025 dex lower, because the 500-cell EW is about 6 per cent high), from the broadened line about 1.60 (1.51 on 500 cells); with the domain factor of section 14, 1.58 and 1.63 (DERIVED, the two effects measured separately). docs/lhs1140b_catalog_state.tex section 5 states the rule.

The gating candidate: not adopted. The species rows keep their current gate; the diagnostic EXHALE_CERT_GATE_DIAGNOSTIC=1 stays as a default-off report.

The pressure-holding composition update: kept, default off. Composition update holds: pressure stays in the code with its negative result (section 14, item 7); the default remains energy.

The energy imbalance at the H2 front: explored. The obstacle item 7 exposed, the energy row at the H2 front failing after a composition change at fixed wind, is being diagnosed (LHS1140b/models/.P1/h2_front_energy/); its result is recorded in a later section.

17. The atomic-data audit of 2026-09-24, applied

md/EXHALE_ATOMIC_DATA_AUDIT_2026-09-24.md (an audit of the atomic data against the sibling code MoCHII) was checked item by item in the code and, where it could be obtained, in the original source, and applied; the four memos carry every verdict, source, size and command: md/atomic_data_audit_applied_core_20260924.md (the H/He cooling core), md/atomic_data_audit_metal_tables_20260924.md (the CHIANTI metal tables), md/charge_exchange_detailed_balance_20260924.md (the charge-exchange table) and md/atomic_data_audit_applied_docs_20260924.md (comments, unreachable code, the detailed-balance test). Every item the audit raised was found to be real; the corrections are physics corrections and are on by default.

The H/He cooling core.

- Recombination cooling of every ion is $kT\alpha(3/2 + \ln\alpha/\ln T)$ on the coefficient the balance itself uses (B1, B7, B12). Hui & Gnedin (1997) print the He III fit with a factor 8 where the hydrogenic scaling gives 2: He III removed 3.05 kT per capture at $1e4$ K and now 0.76 kT. The relation reproduces Hummer (1994) Table 1 to 0.1 per cent.
- The post-process temperature root and eval_cool now call one cell assembly, radiative_cooling_of_cell (B2); the two coolings differed by a factor 1.04-1.60 on the wasp_full post-process and now agree to $1.3e-14$.
- **The $2^3S \rightarrow 1^1S$ electron-impact de-excitation (B3)**, the detailed-balance reverse of q13, is in every place the triplet balance is written, with the 19.82 eV returned to the gas. Oklopčić & Hirata (2018) leave it out as small against $2^3S \rightarrow 2^1S$, 2^1P ; that holds at $1e4$ K (ratio 0.055) and not below: 0.15 at $5e3$ K, 0.57 at $3e3$ K, 2.8 at $2e3$ K (MEASURED), the temperatures of the LHS 1140 b line-forming region.
- Voronov (1997) fits take Voronov's fitted dE again (B4: Mg II $\times 1/1.47$, C I $\times 1/1.10$ at $5e3$ K); $n = 2$ recombination is the balance's case-B coefficient split with Draine's 2s fraction (B6, +2.6 per cent at $1e4$ K); He I, He II (A3) and H I (B8) collisional-excitation cooling are CHIANTI 11.0.2 sums with the observed level energies ($\times 1.11$,

x1.29 and x1.11 at 1e4, 1e4 and 5e3 K), H I on one rate set shared with the $n = 2$ model; single-precision literals are gone from the four files (B10); use_2lev_cool, lambda_2level, melem_Z, cool_FeII_func and the other unreachable wrappers are removed (B11).

The metal line cooling (A2). Mg II took the CHIANTI .scups scaling energy (4.27 eV) as its transition energy where the observed g-weighted 4.4300 eV belongs, 1.15 times too high at 1e4 K and 1.39 at 5e3 K; it is refitted with observed energies and every bound channel (0.2 per cent). Ca II and Mg I carried the resonance line alone; they now take (T, n_e) statistical-equilibrium tables with every channel (at $n_e = 1e9 \text{ cm}^{-3}$ the old coefficient was 1.7-4 times low for Ca II and 5.8-480 times low for Mg I over 1e4-3e3 K). Fe I stays on its Boltzmann (LTE-limit) table, now stated as an upper limit at most 2.0 times the low-density value; the data for a density-resolved Fe I solve are not available here.

Charge exchange. The reverse rows of Huang et al. (2023) Table 4 depart from detailed balance with internal partition functions by factors 0.18 to 2.9. Each measured or computed direction is kept as the original paper gives it, and the other is computed from it by detailed balance with NIST partition functions; four reverse rows are not carried (A2, A10, C4, C6), their forward channel ending in an excited or radiative product; three transcriptions are corrected to the originals (A12 Si²⁺ + H: 1.23e-9 and the sign of its exponent; C5: Zhao et al. 2004 eq. 19 in full; A15: Stancil et al. 1998 Table 1 directly). Ten pairs whose originals could not be obtained (mostly Physical Review A) stay as printed and are marked KNOWN_BLOCKED in the new test src/tests/charge_exchange_detailed_balance; among them N + H⁺ (its endothermic direction falls with T) and Na + H⁺ (both directions below $1e-60 \text{ cm}^3 \text{ s}^{-1}$, so no Na-H charge exchange acts).

Other corrections of the same series. Comments that contradicted the code or the source (C1, C4 to C9, C11) are corrected; rk_R21_H_Hep_cx, rk_R22_Hp_He_cx and the three unreachable Lyman-Werner fits are removed after a call-graph search; the Lyman-Werner warning of ionization_equilibrium.f90 and write_output.f90 no longer cites a DB96 fit the band share has not used since 2026-09-06 (NH2_db96_max removed); src/tests/a2_m1 compiles again against a built tree and its saved output was regenerated (the 0 K photolysis thresholds; md/a2_reaction_audit.md carries a correction block); the Makefile flag stamp now hashes the first line of \$(FC) --version, whose conda-forge package build number -dumpfullversion drops: the previous build/ held 35 objects of gcc 16.2.0-4 and 92 of 16.2.0-5 linked into one binary.

Checks. The tree was rebuilt from make distclean (EXHALE.x md5 551dae520985d63c9ea8b15c21106589, no warning; copied to LHS1140b/models/EXHALE_551dae52.x with BINARY_MANIFEST_551dae520985.txt); physics_probe, charge_exchange_detailed_balance (65 pass, 10 known blocked) and charge_exchange_rows pass on it. The regression matrix and the remaining suites are recorded in the next sections.

Effect (MEASURED by the workers, on builds of the same physics). wasp_full: log10 Mdot -2e-4 (cooling core), +4e-4 (charge exchange), -2.7e-3 (metal tables, a du-stopped case, so below its path spread); Fe III x0.48 at 1.2 R_p. LHS 1140 b: the He 10830 EW of the certified molecular states falls by 14.9 to 16.7 per cent on evaluation, almost all from B3, and by 20 per cent after one stationary re-solve (He/H 1.45: 1.0964 -> 0.8775 %A; 1.52: 1.1532 -> 0.9180), because $n(2^3S)/n(\text{He})$ falls by 41-58 per cent at 5-10 R_p, where n_e q31g is 3-100 times A31. No certificate of the catalog holds under the corrected physics. **Every LHS 1140 b number of this log before this section, and of docs/lhs1140b_catalog_state.tex, is of the physics before the audit.**

18. The H2-front energy imbalance, the H3+ non-LTE interpolant, and the descent below He/H 1.45 (2026-09-24)

The H2 front (decision of section 16). Diagnosed with the tree binary of section 15 and no change to the source (md/lhs1140b_h2_front_energy_imbalance_20260924.md, figure docs/figures/lhs1140b_h2_front_energy_imbalance.pdf). MEASURED: across the composition update one term of the front's energy row moves, the H3+ infrared cooling (99.6 per cent of the cooling there): on the well-mixed He/H 0.083 update H2 at the front rises 3.2 per cent, H3+ answers with +8.6 per cent (+17.5 with the pressure held), and the cooling of the spike cell rises by 31 (44) per cent against a heating that rises by 1 per cent. The code path is consistent (the next solve assembles the relaxation's cooling to 1e-7 and 3e-5). About two thirds of the jump came from the interpolant of the H3+ non-LTE factor (below). With the composition frozen a hydrodynamic root exists and the trust-region solve reaches it ($\|R\|$ 1.5e-7 and 1.6e-8), where the line search cannot take its first step; the well-mixed He/H 0.083 alternation still does not converge with the pressure held and the trust region (6 passes, the front walking outward, $\dot{M}$ -26 per cent). The trust region inside the alternation is a solver-control change and was not made.

The H3+ non-LTE factor is now a power law between the columns of Miller et al. (2013) Table 6 (h3p_cooling.f90, s_table). The emission per H3+ ion can grow at most as fast as the collision rate that populates the emitting levels, so the local index $d \ln s / d \ln n(\text{H2})$ lies between 0 and 1, and the table's node-to-node indices do (0.00 to 0.89). The interpolant linear in s over $\log n$ gave 12.6 just above the $1\text{e}8\text{ cm}^{-3}$ column and 0.17 just below it at 2000 K (DERIVED), and over $r = 1.25\text{--}1.40 R_p$ of the well-mixed He/H 0.083 state an H3+ cooling 1.8 to 5.6 times the power law's. In T the interpolant stays linear; its form there is not fixed by any bound and is written at the code with the row ratios. The correction is on by default and is in the binary of section 17.

The descent below He/H 1.45 (decision item 2 of section 14) ran under the P1.2 rules on EXHALE_46d5b9f4.x (the physics before section 17; LHS1140b/models/.P1/descend_from_1.45_record.txt): 1.152 from 1.45 not certified (18 passes, 3 h); 1.2924 from 1.45 certified (16 passes); 1.152 from 1.2924 not certified (24 passes, the H2 carrier row $3.1\text{e-}4$ at cell 247); 1.2202 from 1.2924 not certified (24 passes, the He/H element row $8.6\text{e-}5$ at cell 234 and falling by about 10 per cent a pass); the driver stopped on two consecutive failed refinements. The interval 0.55 to 1.29 stays unsampled, and the one new certified state is of the physics before section 17.

The molecular family under the corrected physics (user decision of 2026-09-24: this family first). The certified He/H 1.45, 1.52, 1.6, 1.8 and 2.13 states are being re-solved from themselves with EXHALE_551dae52.x (LHS1140b/models/.P1/newphys/, the rung environment, 24 passes, 4 h); their result, the new crossing and what follows for the rest of the catalog are recorded in a later section.

19. The second round of the atomic-data audit (2026-09-24/25)

The audit's author rechecked the applied tree from MoCHII (~MoCHII/MoCHII_v1.00/md/EXHALE_AUDIT_RECHECK_2026-09-24.md and ..._SECOND_RECHECK_2026-09-24.md): every item of section 17 was confirmed as applied correctly, and the open ones were listed. Each was verified in the code and the original source before it was applied; memos md/atomic_data_audit_round2_core_20260924.md, md/charge_exchange_detailed_balance_20260924.md

section 10 and md/chianti_descaling_spline_20260924.md.

Recombination and the helium network.

- The ground-state capture coefficient α_1 is the Milne relation on the photoionization cross section the transfer uses, for H I, He I and He II (0.998 of Hummer 1994 for H I, 1.006-1.011 of Hummer & Storey 1998 for He I); case B = Badnell α_A minus that α_1 (the Mao & Kaastra fit, 2.7 per cent low for He I at $1e4$ K, is removed).
- The Oklopčić & Hirata (2018) α_1 is the capture to 1^1S alone; with the singlet/triplet coupling off the excited-singlet captures were missing. Case B is now split into triplet and singlet captures with Hummer & Storey (1998) Table 5 in both configurations.
- **H ground captures escape where the gas is thin** (new key `H_rec_escape`, default True, K14d2 of md/input_schema.md): $\alpha_{HII} = \alpha_B + y_{HI} \alpha_1$ with the local escape weight of the He II construction, case A in the thin outer wind and case B in the thick base; False is case B everywhere. A correction of the case-B assumption where it does not hold, hence on by default.
- The He III recombination cascade (ground capture, direct capture to $n = 2$, He II Ly-alpha, the two-photon continuum of Nussbaumer & Schmutz 1984, the $2s-2p$ mixing of He II) is absorbed on the spot under `He_rec_coupling`.
- The collisional feed of 2^3S through the higher triplets ($feed/q_{13} = 0.018, 0.101, 0.328$ at $5e3, 1e4, 2e4$ K, CHIANTI 11.0.2) is added to q_{13} wherever the triplet balance reads it, and so are the $2^3S \rightarrow n^1L$ ($n \geq 3$) excitations.
- $A(2s \rightarrow 1s) = 8.2249 \text{ s}^{-1}$ (Nussbaumer & Schmutz 1984; was 8.26); the $2s-2p$ l-mixing of H by protons, He+ and He2+ from Pengelly & Seaton (1964) with their own cut-off rule (the Lamb shift or fine structure for 2s, not the Debye radius); the 2^3S electron-impact ionization as Black (1981) eq. 12 prints it; `coll_rate_prefactor` = $8.629132e-6$ from CODATA, one parameter; the 2^3S triplet cooling is a CHIANTI sum (Black's 13179 K was his fitted threshold, not an energy); the (T, n_e) cooling tables above their top density continue with the last slope clipped to $[-1, 0]$; `q31g` is recorded in the diagnostic states (state 28 of `stage_channels.txt`).

Energy terms that were missing. The metal recombination energy ($k T \alpha (3/2 + d \ln \alpha / d \ln T)$) and the metal collisional-ionization energy (potential times the Voronov rate) are in the cooling; the charge-exchange energy defects heat the gas (new heat channel 20, `heat_charge_exchange`: per reaction the defect less the energy radiated by an excited product, product states from the sources read; the ledger closes to $1.4e-8$ eV an event). On `wasp_full` the charge-exchange heat is 18 per cent of the heating at the base, 3.9 at $1.1 R_p$, 0.4 at $1.3 R_p$; the base share is mostly $Fe + H^+$, whose product state could not be read (Rutherford & Vroom 1972) and is taken as ground, an upper bound.

Charge exchange. $N + H^+ / N^+ + H$ are now Kingdon & Ferland (1996) Table 1 (Butler & Dalgarno 1979) with the reverse by detailed balance (the printed Huang rows were 69 and $1.5e7$ times low at $1e4$ K); $Na + H^+$, $Na^+ + H$ and $K + H^+$ are not carried (no reading of the printed Table 4 form gives a physical rate; the originals, Dutta et al. 2001 and Watanabe et al. 2002, could not be obtained); D26 and D30 are derived by detailed balance from the source directions; the gate now also requires every carried rate to obey the threshold law (205 pass, 6 known blocked, 2 known defects).

CHIANTI descaling. CHIANTI prescribes the natural cubic spline through the .scups points (`descale_scups.pro`,

CHIANTI Technical Reports 4 and 13); the scripts of `cooling_data/` used the not-a-knot spline (as ChiantiPy). Every CHIANTI-based coefficient was regenerated with the natural spline: He I cooling moves by -2 to +0.5 per cent over $3e3$ - $3e4$ K (Upsilon(1^1S - 2^1P) by -16 per cent at $1e4$ K), H I by +0.7 per cent, O I fine-structure Upsilon by +1.3 to +6.7 per cent, most others below 1 per cent.

Other repairs of the series. The Lyman-Werner band share the H₂O/OH continua see is now summed cell by cell along the path at each cell's own (T, n_H) (it was one uniform slab at the local T : the band's photon ledger was 18 per cent short on the warm-started oxygen_chemistry state; only Oxygen chemistry: True runs reach it); six test suites were repaired (`md/test_suite_repairs_20260924.md`).

Build and checks. Tree rebuilt from `make distclean`: EXHALE.x md5 b059a4be0459d65fdc3dff9894ba5d2f, no warning, copied to `LHS1140b/models/EXHALE_b059a4be.x` with `BINARY_MANIFEST_b059a4be0459.txt`. All 42 suites run on it (MEASURED): 38 pass; `grid_and_gates` fails its three `stationary_evaluate_*` rows on the pinned fixture `backup/regression/wasp_full_newton/IC/` (solved under the old physics; a re-solve under this physics certifies, `info = 0`, `||R||` $2.3e-9$, and waits in `LHS1140b/models/.P1/suite_repairs/wfn_IC_r2/` for the golden refresh); `residual_determinism` fails its known closure-spread row ($1.17e4$, the solver item open since 2026-09-18); the two `steady_*` suites need the log of a finished steady solve as input.

Effect (MEASURED by the workers). Of the core items alone: `wasp_full log10 Mdot` 13.35 -> 13.32, the H ionization front outward ($x(H^+)$ 0.36 lower at $1.27 R_p$), T up to +11 per cent; `mol_metals` -0.004. Of the charge-exchange items: `wasp_full` +4.2e-3, N II down by up to 1.8 dex. On the LHS 1140 b He/H 1.45 state the core items lower the He 10830 EW by 7 per cent on evaluation, mostly through the triplet/singlet split of case B (0.041 %A), `H_rec_escape` (0.020) and the Milne `alpha_1` (0.011).

The LHS 1140 b molecular family between the two rounds (binary EXHALE_551dae52.x, `LHS1140b/models/.P1/newphys/`): re-solved from the certified states, no rung certified in 24 passes (the H₂ carrier row at the front falling by 0.72-0.86 a pass), then all five certified in 7-11 more passes. Their EWs (MEASURED, evaluate + synthesis + the EW operator): He/H 1.45 0.9813, 1.52 1.0185, 1.6 1.0587, 1.8 1.1500, 2.13 1.2765 %A; the 1.108 crossing near He/H 1.70. These are of the physics of section 17 only; the states under this section's physics are being solved from them.

20. The molecular $K_{zz} = 1e9$ family under the audited physics (2026-09-25)

The user's decision of 2026-09-24 was to re-solve this family first. The five states He/H 1.45, 1.52, 1.6, 1.8 and 2.13 were re-solved from the states certified under the first round (section 19) with EXHALE_b059a4be.x (`LHS1140b/models/.P1/newphys/HeH<x>_r2/`, the rung environment, `resolve.sh`): all five certified in 13 passes (the H₂ carrier row at the front falling from 0.5 to $6e-6$ - $9e-6$). Each was then evaluated, synthesized and measured with the EW operator (`evaluate_ew.sh`, `eval_r2/`; every evaluation certified). MEASURED:

He/H	EW, audit round 2	round 1	before the audit	log10 Mdot, round 2
1.45	0.9387	0.9813	1.0964	7.899
1.52	0.9776	1.0185	1.1532	7.906
1.6	1.0198	1.0587	1.2155	7.913

He/H	EW, audit round 2	round 1	before the audit	log10 Mdot, round 2
1.8	1.1159	1.1500	1.3584	7.927
2.13	1.2497	1.2765	1.5604	7.945

The 1.108 crossing of the log-log chord moves from He/H 1.4643 (before the audit) to **1.7833, between the certified 1.6 and 1.8**, on the 500-cell grid; the EW falls by 14 to 20 per cent at fixed He/H and log10 Mdot rises by about 0.05. The grid continuum factor and the domain factor of sections 15 and 14 were measured under the old physics and are not re-measured here. Figure docs/figures/lhs1140b_kzz1e9_audit_stages.pdf (..P1/newphys/compare_physics_stages.py): the EW against He/H at the three stages, the line at He/H 1.8, and T, v, $n(2^3S)$, $x(H^+)$ and $x(H_2)$ at He/H 1.8. At He/H 1.8 (MEASURED on the _adv profiles, before the audit / round 1 / round 2): the line keeps its shape and loses depth; the peak temperature is 5439 / 5315 / 5600 K; the velocity at the outer edge 1.21 / 1.40 / 1.36 km/s; $x(H^+)$ at 20 R_p 0.059 / 0.056 / 0.050 (the ground captures escape in the thin far wind); the metastable density at its peak (1.35-1.38 R_p) 33.3 / 33.3 / 31.8 cm⁻³, and its radial integral over 1-1.5, 1.5-3 and 3-30 R_p 11.1 / 10.9 / 10.2, 28.8 / 28.3 / 27.6 and 28.4 / 24.2 / 23.4 cm⁻³ R_p. The loss is in the cool outer wind beyond 3 R_p (-18 per cent), where the superelastic $2^3S \rightarrow 1^1S$ collisions outrun the radiative decay, and the EW falls by 18 per cent with it.

21. The code-size appendix moves to this log (2026-09-25)

On the user's instruction the hand-written code-size appendix docs/Update_EXHALE_appendix.tex is now input by this log's typeset twin instead of the stage 2 twin (src/utis/update_log_to_tex.py, STAGES); docs/Update_EXHALE_stage2.{tex,pdf} were regenerated without it. Its references to "this log" now name the stage 2 log where they meant it, and it states that its numbers (measured 2026-09-22) do not yet include the code of stage 3; they are re-measured with src/utis/codesize.py when the atomic-data series is closed.

22. The originals supplied on 2026-09-25, read and applied

The user supplied the papers the two audit rounds could not obtain; each was read and the code checked against it (memos md/charge_exchange_detailed_balance_20260924.md sections 11 and 11.8, md/atomic_sources_hhe_20260925.md).

Charge exchange.

- Na + H⁺ (Dutta et al. 2001, non-radiative, product H(n=2), endothermic by 1.682 eV) and Na, K + H⁺ (Watanabe et al. 2002, radiative): neither paper gives a fit and Huang et al.'s Table 4 forms are in neither; the rate is now the Maxwellian average of Dutta's cross sections (their own rate figure violates the threshold law below about 5000 K) plus Watanabe's radiative values: 8.4e-12 at 1e4 K against 4.4e-171 as coded before. The reverse is not carried (it starts from H(n=2)). K + H⁺ from Watanabe.
- Fe + H⁺, O⁺, N⁺ (Rutherford & Vroom 1972 Table I): the 300-1200 K values in log-log, held above 1200 K; the measured cross sections start at 8 eV, so every wind rate is an extrapolation, bracketed by 1.4e-9 and 4.0e-9 at

1e4 K (DERIVED; the held value is the source's own extrapolation to 6 per cent; Pequignot & Aldrovandi 1986 infer 2.7e-9). The product state is not determined in the paper; the near-resonance reading (Fe+ z4F/z4D, 1.18 eV of heat an event instead of the full 5.70 eV defect) is the one Pequignot & Aldrovandi's Table 3 also takes; the range is stated at the code.

- S + H+ (Zhao et al. 2005 Table III), Si + He+ (Satta et al. 2013 eq. 8, Table 1), S + C+ (Chenel et al. 2010 Table 2), Na + Ca+ (Kwlek et al. 2019, measured at 0.015-0.09 eV, the Langevin value kept as an upper bound) replace the Table 4 transcriptions; the reverses of channels whose forward ends in an excited product (Fe rows, Si + He+, Na + Ca+) are not carried. C + Si+ (D1/D2) stays blocked: the paper Table 4 cites for it does not contain it. The detailed-balance gate: 194 pass, 0 fail, 1 known blocked.
- Effect: wasp_full log10 Mdot -3e-4; the charge-exchange heat at the base falls from 18 to 6.7 per cent of the heating; Na I D2 transit depth 1.169 -> 1.156 per cent.

Hydrogen and helium.

- The He I collision strengths of the metastable network (q13, q31g, q31a, q31b) were exponential fits the ATES code carried, not coefficients of Lampon et al. (2020), who take the values from Bray et al. (2000). Bray tabulates 5.6e3-5.6e5 K; the fits passed through the first four table values and were an unsupported extrapolation below 5.6e3 K (1.10-1.16 times the new strengths at 1.5e3 K). Now: Bray's table inside its range (PCHIP in log-log; one misprint of Table 2 corrected), the CHIANTI he_1 temperature dependence joined continuously outside it; the feed through the higher triplets, the 2³S cooling, the 2³S -> n¹L excitations and the He I cooling sums were rebuilt on Bray. On the LHS 1140 b He/H 1.8 state the He 10830 EW rises by 2.3 per cent on evaluation.
- H 2s-2p mixing by electrons: the coded rate (Janev et al. 2003, from Chibisov 1969) was ten times below Seaton's (1955) first-order dipole constant; now Seaton (1955) eq. (55), approximation V (5.78e-5 cm³ s⁻¹ at 1e4 K against his Table 1 5.7e-5; Osterbrock & Ferland Table 4.10 agree).
- A(2s -> 1s) = 8.22461 s⁻¹ (H) and 526.823 s⁻¹ (He II), Drake (1986) eq. (27) (section 19 wrote 8.2249, the Nussbaumer & Schmutz 1984 value it replaced).
- The case-B share of recombinations into H 2s is now Pengelly's (1964, Table I) hydrogenic fraction, the one the He II branch already used; the quadratic in T that section 17 (item B6) called "Draine's 2s fraction" is Christie et al.'s (2013) fit, and Draine (2011) gives only Tables 14.2 and 14.3.
- The atomic-mode (no 2³S) He I case-B photons are now sent at their own exits (584 Å, 19.8 eV, two-photon) split by the network's rates, instead of one averaged energy (3.10 instead of 4.12 eV of heat a capture at high density; z = 0.966 and 0.660 at low and high density against Draine's 0.96 and 0.67).
- Voronov (1997): all 20 rows equal the published Table I; its stated range (1 eV-20 keV) is written at the code with what the fit does below it. Anderson et al. (2000) confirm the CHIANTI H I strengths to 2.5 per cent at 0.5-5 eV; Yan & Babb (2023) published values equal the arXiv ones; Dere et al. (1997) and Del Zanna et al. (2015) say nothing about the descaling spline, which rests on the CHIANTI IDL code.

Build. Tree rebuilt from make distclean: EXHALE.x md5 50aa18d79bcd25f219b5b59baa31b8b8, no warning, copied to LHS1140b/models/EXHALE_50aa18d7.x with its manifest. The manifests of 2026-09-24/25 listed files as changed that were not (the header lines of the previous manifest matched the file-name search); the three lists were rebuilt from

the md5 lines of consecutive manifests. The refined-grid runs of the round-2 physics that were still running (He/H 1.6 on 2000 cells, 2.13 on 1000 and 2000) were stopped as superseded; the molecular family, the regression fixture, the regression matrix and the suites are being re-run on this binary.

The texts the corrections made stale (md/stale_texts_20260925.md): the physics document (docs/physics_overview/*.tex, the PDF rebuilt, 70 pages), docs/transmission_spectrum.{tex,md,pdf} and md/atomic_data_EXHALE_vs_MoCHII.md were brought to the code. The transit post-processor built the $n = 2$ populations of the Balmer lines with Christie et al.'s rates while claiming to mirror the solver; it now uses the solver's rates, one transcribed function each (agreement with the Fortran 3.1e-13, MEASURED), and keeps only its own Ly-alpha and Balmer fields (which the profiles do not carry). H-alpha peak depth of HD 209458 b -1.73 per cent, of wasp_full +0.14 per cent (MEASURED), almost all from the proton l-mixing, which lowers $n(2s)/n(2)$ at 1.3-3 R_p from 0.53-0.69 to 0.26-0.34. The $n = 2$ threshold cross section $1.4e-17 \text{ cm}^2$ is Seaton (1959) at the $n = 2$ edge ($1.378e-17$ DERIVED), not Osterbrock & Ferland.

23. The Markdown documents move to md/ (2026-09-25)

On the user's instruction the 350 Markdown files directly under docs/ moved to the new top-level md/; docs/ keeps the LaTeX and PDF documents, the figures and the dated audit directories, whose 13 Markdown files stayed with the files they describe. Done with mv, not git mv: git sees the 35 tracked files as deleted under docs/ and new under md/. Paths that followed:

- .gitignore: the rule that keeps the development record off the public repository (docs/*.md with 35 exceptions) is now md/*.md with the same 35 exceptions under md/.
- 73 relative links of 24 moved files into docs/ subdirectories now read ../docs/...
- 3079 references docs/<name>.md in 416 maintained files (sources and their comments, tests, scripts, README.md, README_HOWTO.md, the md/ files themselves, the docs/ LaTeX, the regression README and scripts) now read md/<name>.md; four of them are strings the binary prints (the setup report, one lower-atmosphere message, one test message), so the tree binary changed (EXHALE.x md5 b30e56f63335431dcb22b9fe3f80f948, no warning); the state files the regression compares carry none of them.
- src/utils/update_log_to_tex.py reads the logs from md/.
- Left as they were: run records and products (inputs, outputs, state files, REPRODUCE.md, logs, the .P1 run areas, archived regression snapshots), whose text records the paths of their day and whose bytes other records hash; and the one string LHS1140b/models/make_models.py writes into the 94 stored catalog inputs, so that it still reproduces them byte for byte.
- The workspace CLAUDE.md and the session memory point to md/.
- The 17 docs/ documents whose text names these paths were rebuilt on the user's instruction (the two log twins regenerated first): every build without error or undefined reference, the new paths in the PDFs.

24. The last items of the MoCHII comparison, and the molecular crossing under the corrected physics (2026-09-25/26)

md/EXHALE_MOCHII_ATOMIC_COMPARISON_2026-09-25.md (a side-by-side of the atomic processes of EXHALE and MoCHII, written from the MoCHII side) found the recombination construction, the recombination cooling, the He 2³S network and the H 2s-2p mixing now the same in both codes to four digits at 1e4 K, and listed five open items for EXHALE; each was verified and applied (memos md/charge_exchange_detailed_balance_20260924.md section 12 and md/photoionization_584_auger_20260925.md).

- **N2+ + H -> N+ + H+** (Barragan et al. 2006 Table 2, $9.70\text{e-}10\text{ cm}^3\text{ s}^{-1}$ at 1e4 K, the product N+ 2s2p³ 3D0, so no reverse; heat 4.57 eV an event; metals.inp: cx_N2p_H <scale>, default 1). On wasp_full N III falls by up to 5.4 decades.
- **He2+ + H -> He+ + H+ + photon** (Kingdon & Ferland 1996 Table 1, $1.0\text{e-}14$, radiative, no reverse and no heat), in every system that carries He III; He III falls by 0.69 dex at the wasp_full base. The original of the rate (West, Lane & Cohen 1982, Phys. Rev. A 26, 3164) could not be obtained.
- **Density-dependent detailed balance.** Where the source names the product level, the reverse is the thermal reverse times the population-weighted factor of the metastable levels at the local (T, n_e) (five-level statistical equilibrium of N I, S II, C I from CHIANTI 11.0.2), and the heat of the forward channel removes the forbidden-line energy a metastable product radiates at low density; elsewhere the thermal reverse is exact at every density. wasp_full (n_e 8e8-3.5e9) moves by less than 0.2 per cent. A 0/0 below 60 K in the metastable Boltzmann weights (one LHS 1140 b base cell sits at 40.8 K) was guarded.
- **He I 584 A.** The line is resonantly scattered by He0 inside the cell, and each scattering ends in absorption by H I (or H2, or a metal ion), in conversion to 2¹S through the 2.06 um branch (1/912) and the existing two-photon exit, or in escape (Wood, Mathis & Ercolano 2004 sect. 4.1, eqs. 15-17, generalized to every continuum absorber of the cell and to escape); their eq. 17 carries He/H = 0.1 in its coefficient.
- **Inner-shell photoionization.** Each subshell absorption heats with the photoelectron plus the Auger energy, the fluorescence and the further ionization energy are not heat, and the neutrals of C, N, O, Mg, Si, Ca and Fe are carried straight to X++ where the vacancy autoionizes (Kaastra & Mewe 1993 Tables 2 and 3; Ca II 3s is treated as radiative, since the measured energies forbid its autoionization). At 98 eV the electrons of a Mg II absorption receive 0.333 of the photon energy instead of 0.847.

Build. EXHALE.x md5 f4da21f29579d87370efec2a4220f4c4, no warning, copied to LHS1140b/models/EXHALE_f4da21f2.x with its manifest. On LHS 1140 b the two He items move the He 10830 EW by +0.05 per cent (He/H 1.8), but a state of the previous physics does not certify under them, so the molecular states are re-solved on this binary.

The molecular K_{zz} = 1e9 family under the physics of section 22 (EXHALE_50aa18d7.x; .P1/newphys/, .P1/refine_grid_r3/). He/H 1.8 and 2.13 re-solved from the round-2 states; 1.6 and 1.52 as rungs from 1.8 (1.45 failed its rung; the H2 carrier row held at 1e-1 through four passes). MEASURED, every evaluation certified except the one marked:

He/H	EW, 500 cells	1000 cells	2000 cells	continuum (DERIVED)
1.52	1.0038			
1.6	1.0475	1.0187	1.0039	0.988
1.8	1.1468	1.1162	1.1003 (a)	1.083
2.13	1.2853	1.2516	1.2343	1.216
crossing of 1.108	1.721	1.783	1.819	1.860

(a) the re-evaluation of the certified state refuses the base mass row by a factor 1.9 (1.4e-8 against 7.4e-9); its EW is quoted for reference.

The EW converges at first order with the grid (successive-difference ratios 1.93-1.95, $p = 0.95$ -0.97), as under the physics before the audit, and the crossing moves from He/H 1.46 (500 cells) and 1.55 (continuum) before the audit to 1.72 and 1.86. The 60 R_p domain factor of section 14 was measured before the audit and is not applied here.

The documents brought to the code (md/user_manual_update_20260926.md): docs/EXHALE_user_manual.tex (rebuilt, 103 pages, no error or undefined reference, every changed page looked at), README.md and README_HOWTO.md now describe the input keys (H_rec_escape, He_rec_coupling, the metals.inp keys cx_N2p_H, cx_O2p_H, cx_full), the 24 columns of Heating_breakdown.txt (charge-exchange heat in column 24), the stage_channels.txt layout and the atomic data of sections 17-24. Found in the code and fixed: H_rec_escape was parsed but missing from the list of recognized keys (a spurious "unrecognized input line" warning, and a second statement not refused); the missing-metals.inp message promised built-in abundances that are zero; comments said cx_full toggles the He-H pair and that O2+/N2+ + H are dropped; the stage_channels.txt header omitted the metal charge-exchange channels 27-32. md/input_schema.md K14e (still "Koskinen 2013") and its metals.inp table (no group-E keys) were brought up to date. Not changed: H_rec_escape and He_rec_coupling, like the other physics switches of their group, are neither echoed in the setup report nor restart option tokens.

Checks of the binary of this section (EXHALE_f4da21f2.x). The suites: physics_probe did not compile one new probe (inner_shell_photoabsorption_ledger called cx_set_cell without the electron density the density-dependent reverses added; the two items were written at the same time); fixed, and the suite passes. grid_and_gates fails its fixture rows plus outer_iteration_ending_is_the_stagnation_one and restart_lower_ghost_is_the_derived_boundary (7.9e-14 against 1e-14), all on the pinned fixture of the previous physics; residual_determinism keeps its known row; the two steady_* suites need a solve log. The regression matrix ran all 17 cases; every one differs from the goldens by the physics (wasp_full log10 Mdot and the base inflow: 214 inflow cells against 257). **The user held the golden refresh (2026-09-26).** The stationary re-solve of the fixture from the previous one did not converge (info = 2: the energy row 3.1e-2 at cell 406, the closure of System_HeH_TR_metals 4.6e-3 at cell 340), while the matrix case wasp_full_newton (marching, then Newton) certified (info = 0, ||R|| 4.4e-9); that output is the candidate fixture. The tree was rebuilt after the comment and message corrections of this section (EXHALE.x md5 0a7532d5087f14df41142fb499b68240; it differs from EXHALE_f4da21f2.x in printed strings and comments only).

The molecular states on EXHALE_f4da21f2.x (.P1/newphys/HeH<x>_r4, re-solved from the certified states of the

previous physics; certified in 9, 9, 8 and 7 passes; every evaluation certified). He 10830 EW, 500 cells: He/H 1.52 1.0047, 1.6 1.0483, 1.8 1.1477, 2.13 1.2862 (+0.06-0.09 per cent against the table above); the 1.108 crossing 1.7194 (1.7213 before). The refined grids were not re-solved on this binary; the crossing's grid factor above stands within the size of this change.

25. The helium-hydrogen charge exchange from its originals (2026-09-26)

The He²⁺ + H original the user supplied (West, Lane & Cohen 1982) and an analysis made on the MoCHII side of the He⁺ + H exchange (~MoCHII/MoCHII_v1.00/md/heii_h_charge_exchange/) were checked against the published papers and applied (md/charge_exchange_detailed_balance_20260924.md section 12.9). Every rate below is DERIVED from the cited cross sections unless marked READ.

- **He²⁺ + H -> He+(1s) + H⁺ + photon.** The 1.0e-14 cm³ s⁻¹ of Kingdon & Ferland (1996) is not a result of any calculation: it is the value they set for every reaction without a significant channel (READ, their sect. 2). Now the Maxwellian average of the West et al. (1982) radiative cross sections (Figs. 4-6, with the resolved resonances): 2.0e-13 at 1e3 K, 1.65e-13 at 1e4 K, 1.57e-13 at 3e4 K, 16-20 times the old value; a calculation above 1e3 K, an extrapolation below.
- **He⁺ + H -> He + H⁺, non-radiative.** The Kingdon & Ferland fit reproduces the singlet-conditional cross sections of Zygelman et al. (1989) without the 1/4 of the spin statistics, and those cross sections are themselves hundreds of times those of Kimura et al. (1993) and Loreau et al. (2014). Now the Loreau et al. (2014) H⁺ + He cross section by microscopic reversibility (weights 4:1, threshold 13.65 eV/u, zero below 15 eV/u): 4.4e-17 at 1e4 K instead of 9.0e-15, 1.5 per cent of the radiative channel; the band at the code (up to 2.9x at 1e4 K from the threshold treatment; 0.4-2.5x above 3e4 K). The reverse He + H⁺ (B1) follows by detailed balance (5.1e-22 at 1e4 K instead of 1.0e-19); the heat of an event is unchanged.
- **He⁺ + H, radiative.** Stancil, Lepp & Dalgarno (1998) Table 1 row 19 prints 1.20e-15 (READ); the code carried the 1.25e-15 of Glover & Jappsen (2007). Now 1.20e-15.

Effect (MEASURED): wasp_full log10 Mdot -0.002 (inside the marching spread), He III down to 0.94 dex at the base, He II and He 2³S +48 and +52 per cent at 1.1 R_p, the charge-exchange share of the heating 5.6 -> 0.9 per cent at 1.05 R_p. **LHS 1140 b He/H 1.8 (evaluation of the certified state): the old He⁺ + H exchange supplied up to 10.7 per cent of the heating; without it the peak temperature falls from about 5600 to 5150 K, He II and He 2³S rise by 14 and 15 per cent at 1.5 R_p, and the He 10830 EW rises from 1.1477 to 1.1818 %A (+3.0 per cent).** The certificate does not hold under the corrected rates (energy row 8.4e-2), so the molecular states are re-solved on the binary of this section (EXHALE.x md5 59c6be36820b8ba5c958c294ad9e6f02, copied to LHS1140b/models/EXHALE_59c6be36.x with its manifest).

The molecular states re-solved on EXHALE_59c6be36.x (.P1/newphys/HeH<x>_r5, from the certified states of section 24): He/H 1.6, 1.8 and 2.13 certified in 25, 23 and 21 passes, every evaluation certified. He 10830 EW, 500 cells (MEASURED): 1.6 1.0577, 1.8 1.1550, 2.13 1.2902 (+0.9, +0.6 and +0.3 per cent against section 24; the +3.0 per cent of the evaluation at fixed wind is mostly given back by the solved wind); the 1.108 crossing moves from 1.7194 to 1.7026 (DERIVED). He/H 1.52 stopped in its second pass: a JFNK trial iterate with ||R|| near 2 left a lower

ghost whose base H2 handoff partition did not close in 30 passes (last move $2.0\text{e-}6$ against $1\text{e-}10$), and the guard of `ionization_equilibrium.f90` (a boundary nothing stands behind is refused) ends the run with error `stop` even on a trial state. Whether a trial should instead be refused and the step shortened is a solver-control question put to the user.

26. A trial state no longer stops the run at an unclosed lower ghost (2026-09-26)

User decision: as recommended (`md/ghost_closure_trial_refusal_20260926.md`). The guard of `ionization_equilibrium.f90` that stops the run when the lower ghost's base H2 handoff partition does not close against its own ionization balance (tolerance $1\text{e-}10$, 30 passes) now distinguishes a state a caller can refuse from one it cannot: the stationary JFNK's trial evaluations (`eval_residual` of `steady_newton.f90` when it screens admissibility: line search, trust region, Levenberg-Marquardt and pseudo-time trials, the finite-difference probes) and the relaxation's trial chemistry (`equilibrate_chemistry_at_fixed_conserved_state`) declare it before the sweep, and there an unclosed ghost refuses the trial (logged as `CANDIDATE REFUSED`, counted in the ledger, refusal code 7, closure reason 5), the solver then shortening the step as for its other refusals; an accepted state, the state written, the marching sweep and every other path keep the stop. The same rule now covers the ghost-composition fixed point one step later. MEASURED: the He/H 1.52 re-solve of section 25 passes the trial that stopped it (the $\lambda = 1$ trial of JFNK solve 3 iteration 7; the halved trials close) and certifies at pass 24; an evaluate of He/H 1.6 and a copy of `mol_carrier` are byte-identical to the build before (`EXHALE_59c6be36.x`) in every data line; the suites pass, the two failing ones (`residual_determinism`, `grid_and_gates` fixture rows) failing identically on the build before. `EXHALE.x md5 7db5afe9b1d80ce757fcc07d1b499330`, copied to `LHS1140b/models/EXHALE_7db5afe9.x` with its manifest.

The He/H 1.52 state on this physics: EW 1.0149 %A (certified evaluation); with 1.6, 1.8 and 2.13 of section 25 the 500-cell ladder is 1.0149, 1.0577, 1.1550, 1.2902 and the crossing of 1.108 stays at 1.7026.

The grid of the crossing on this physics (`.P1/refine_grid_r5/`, binary `EXHALE_59c6be36.x`, the 1000-cell He/H 1.8 and 2.13 continued on `EXHALE_7db5afe9.x`, identical physics; every state and evaluation certified). He 10830 EW, MEASURED; continuum DERIVED (Richardson at the measured order):

He/H	500 cells	1000 cells	2000 cells	continuum
1.6	1.0577	1.0297	1.0150	0.999
1.8	1.1550	1.1249	1.1094	1.093
2.13	1.2902	1.2571	1.2403	1.223
crossing of 1.108	1.703	1.764	1.797	1.838

First order again (successive-difference ratios 1.91-1.96, $p = 0.93\text{-}0.97$). Against the physics before the audit (1.464 on 500 cells, 1.552 in the continuum) the crossing has moved to 1.703 and 1.838; the continuum value is interpolated between the certified continuum points of 1.8 and 2.13.

27. The molecular family under the corrected physics: catalog publication and the width of the line (2026-09-26)

Catalog. The certified He/H 1.52, 1.6, 1.8 and 2.13 states of sections 25 and 26 were published into LHS1140b/models/molecular_scalar_gj1132_kzz1e9/ (run_case.sh, SEED= the certified run, EXHALE_7db5afe9.x): every re-measurement "original claim REPRODUCED", every evaluation certified; EW 1.0149, 1.0577, 1.1550, 1.2902 %A, log10 Mdot 7.89, 7.90, 7.91, 7.93. The catalog's He/H 0.55, 1.45 and 9.7 molecular states are of the physics before the audit and were not re-solved.

The added Gaussian (.P1/connectivity/broaden_molecular_final.py, the operations of section 13; output broaden_molecular_final.txt, figure docs/figures/lhs1140b_kzz1e9_molecular_broadened_final.pdf). The memo check (atomic well-mixed He/H 0.40) reproduces $f = 22.22$ km/s. MEASURED: the matched kernel is 22.10-22.19 km/s FWHM on every state and grid (sigma 9.39-9.42 km/s), and at it the red depth, blue depth and ratio of the He/H 1.8 state lie inside the observed 1 sigma (red 1.289, blue 0.170, ratio 7.60). The broadened EW (MEASURED on each grid, continuum DERIVED):

He/H	500 cells	1000 cells	2000 cells	continuum
1.6	1.0227	0.9955	0.9813	0.966
1.8	1.1167	1.0876	1.0726	1.057
2.13	1.2474	1.2155	1.1992	1.182
crossing of 1.108	1.781	1.851	1.890	1.933

As turbulence the kernel is supersonic: on the He/H 1.6 and 1.8 states the line-forming region ($n(2^3S)$ r weighting) has $T = 2650$ and 2756 K, $\mu = 1.94$ and 2.03 , $c_s = 4.33$ km/s, so $\sigma = 9.4$ km/s is Mach 2.2 as a line-of-sight dispersion, 3.1 as $\sqrt{2}$ sigma and 3.8 as an isotropic speed, as before the audit. From the EW alone the crossing is 1.703 (500 cells) and 1.838 (continuum); from the broadened line 1.781 and 1.933.

28. The catalog summary brought to the tree's state, and why the molecular winds do not certify (2026-09-26)

docs/lhs1140b_catalog_state.tex (13 pages, rebuilt, pages 5 to 8 checked by eye):

- The certified table carries the molecular $K_{zz} = 1e9$ rows the catalog now publishes, READ from each case's run.log and models/status.py: 1.52, 1.6, 1.8 and 2.13 under EXHALE_7db5afe9.x (note a); 1.45 under EXHALE_b059a4be.x, the second round of corrections, EW 0.9387 (note b; under EXHALE_50aa18d7.x neither its re-solve nor its rung from 1.52 certified within the time ceiling, and it was not attempted under the binary of record); 9.7 (note c, below). The rows of the table before this entry were the pre-audit values (1.6: EW 1.2155, now 1.0577).
- A new subsection of section 3, "Why the molecular winds do not certify", assembled from md/lhs1140b_stationary_L34c_20260918.md, md/lhs1140b_alternation_audit_20260921.md, md/lhs1140b_coupled_trial_20260921.md and md/TO_BE_DONE.md L22: the composition half of the

partitioned route refuses (the H2 carrier row in all six cases), by two measured mechanisms (the composition update takes the wind out of solution in the well-mixed cases; a drift under a saturated movement bound in kzz1e9/HeH0.083); the coupled route fails in its Krylov solve; the open item is a preconditioner that holds the species rows. Every one of those measurements predates the atomic-data corrections and none was repeated under EXHALE_7db5afe9.x.

- The two molecular cases that published no state (molecular_scalar_gj1132_wellmixed/HeH2.13, molecular_photochem_gj1132_kzzprofile/HeH2.09) are now named.

Found and fixed: LHS1140b/models/status.py printed the not_solved.md reason beside molecular_scalar_gj1132_kzz1e9/HeH9.7, which is certified. The certified generation g0008_20260920T052941Z_c3c466a4 is the 2026-09-20 evaluation of the state written on 2026-09-19; the not_solved.md is that of a re-solve made later the same day (g0006_20260920T000506Z_5b2061aa, composition_refusal, the H2 carrier row 5.13×10^{-4} at cell 262 after 40 passes), which did not replace it. A certified row with a not_solved.md beside it now reads "a later re-solve, not the state shown: ..." (MEASURED on the table output). MODELS.md was not rewritten.

29. Errors fixed and models solved first: the plan of 2026-09-26

User instruction (2026-09-26): re-checking solutions that move little against the observational error is not the priority; fixing errors and solving the models that do not solve is. Plan md/PLAN_20260926.md; the user approved its recommendations (goldens and fixture refreshed; the composition-side solver work reopened if continuation does not close the cases; diffusion_check taken out of the model count). New physics only where it can change results seriously; the rest is recorded, not applied.

29.1 An OpenMP data race in the metal sweep (fixed)

meg_g02 (the inner-shell Auger share of neutral photoionization to X++, added with binary f4da21f2 on 2026-09-25) was missing from the private list of the He + metals cell sweep (ionization_equilibrium.f90, then ~2137, a default(shared) loop), so threads overwrote each other's value between the write and the two reads. MEASURED: route-B stationary restart of wasp_full_newton failed at 8 threads (composition elimination never contracts, info = 2) and certified at 1 thread; two 8-thread runs of one input differed before the solve started (md/metal_sweep_openmp_race_20260926.md). Every metals-on run at more than one thread since f4da21f2 was affected; the regression matrix (1 thread) and every metals-on catalog certificate (last published 2026-09-20) were not. Fix: meg_g02 private, and all four cell loops of ioniz_eq converted to default(none) with explicit shared lists; an audit of every parallel region of radiation/ and lower_atmosphere/ found one more, formal, race (two table-ready flags read without synchronization; now omp atomic read/write seq_cst). New suite src/tests/thread_count_identity/ (200 steps of wasp_full at 1 and 8 threads, data lines bitwise): 7db5afe9 FAIL (3.2×10^{-3} Hydro, 5.4×10^{-2} Ion), fixed binary PASS. The r0 case (route B, 8 threads) now certifies, info = 0, ||R|| 6.50×10^{-9} . No 1-thread run moved.

29.2 Fix batch (errors confirmed in the source)

F1 free-free Gaunt factor: van Hoof et al. (2014) eq. (21) uses the infinite-mass Rydberg (READ, their sect. 2), so $Ry/k = 157887.51$ K, not 157807 (the Hui & Gnedin 1997 constant, kept in that fit); F2 the carrier diffusivity read by outputs and diagnostics was zero on a stationary path (MEASURED $D_{H2} = 0$ at every cell; now $7.9e6 - 8.2e9$ cm²/s); F3 the determinism replay now passes u to `cell_row_scales`; F4 pressure formed before the composition refresh at 3 of 9 sites (coupled-block entry, after `solve_steady_*`, trial end; the other 6 verified not stale); F5 the CO column, dissociation rate and shielding factor joined the step checkpoint; F6 an inert Coupled carrier solve is reported; F7 five wrong comments; F8 the `_adv` headers carry every provenance field; F10 seven callerless `Cool_coeff` wrappers removed and the Voronov H/He fit defined once (rates bitwise unchanged). Effect on the regression inputs (single thread): F1 alone, at most $1.4e-4$ relative (wasp_full velocity), $\log_{10}$ $\dot{M}$ unchanged to 4 digits.

29.3 The interdiffusion enthalpy flux (new physics, serious)

The energy equation of a diffusing mixture carries $q_d = \sum_i h_i J_i$ (Cook 2009, Phys. Fluids 21, 055109, eqs. 11-13; Schunk 1977, Rev. Geophys. Space Phys. 15, 429, eqs. 15b, 16, 20c; READ). EXHALE moved He relative to H and carried no such term in either energy equation. MEASURED size on LHS 1140 b `He_diffusion` states: 13-26 % of the local energy budget at 1.1-5 R_p (md/interdiffusion_enthalpy_flux_20260926.md). Implemented in both routes (binary model: $q_d = J_{He} (h_{He} - h_H)$, sensible enthalpies with the electrons each ion carries; formation energy verified to be carried already); key `Interdiffusion enthalpy flux`: default True with `He_diffusion` (the term belongs to the equation), False only to match published models that omit it (Koskinen 2013, 2022; Yelle 2004); restart token `interdiff_enth` (older states restart under `Restart option change: interdiff_enth`). Suite `src/tests/interdiffusion_enthalpy_flux/` 18/18. Effect on the molecular He/H 1.8 state (24 + 24 passes, carrier row $1.6e-5$ at the end, not yet certified): T -166 K median over 1.3-2 R_p (largest -236 K, -4.4 %), He/H -3.1 %, the $n(2^3S)$ column -1.3 %, $\log_{10}$ $\dot{M}$ +0.002.

29.4 He+ + H non-radiative charge exchange near threshold

Section 25 took the Loreau et al. (2014) $H^+ + He$ cross section forward by reciprocity with $\sigma_r = 0$ below 15 eV/u, where their calculation starts. That closes an open channel: the forward (exothermic) rate then fell as $\exp(-E/kT)$ at low T ($5e-34$ at 300 K). Adopted, as in MoCHII (the user's decision relayed by the MoCHII session), $\sigma_r(x) = \sigma_r(15) \sqrt{(x - x_{th})/(15 - x_{th})}$ between $x_{th} = 13.655$ eV/u and 15 eV/u, i.e. σ_f v constant at low energy (ion-induced-dipole capture with a constant transfer probability; an assumption no calculation tests). Rate: $9.64e-17$ at $1e4$ K (was $4.41e-17$), $9.56e-17$ at $1e3$ K (was $1.5e-21$); 3.3 % of the radiative channel at $1e4$ K. Generator `cooling_data/heii_h_charge_exchange_rate.py` (reproduces the old table to 0.4 % with the old cut).

The radiative channel, the user asked, is one coefficient in both codes: Stancil, Lepp & Dalgarno (1998) Table 1 row (19), $1.20e-15 (T/300)^{0.25}$, the Zygelman et al. (1989) Table I DIRECT column times the 1/4 of the singlet approach (READ; $1.17e-15$ at 300 K from the original). MoCHII moved from the Glover & Jappsen (2007) R26 $1.25e-15$, which cites the same data and cannot be reproduced from it (2026-09-26); EXHALE unchanged. A digitization of Courtney et al. (2021) Fig. 3 agrees with this fit within 2 % over $1e3 - 1e4$ K and lies 1.9x above it at 100 K; a fit to their tabulated

rate needs the data file (University of Georgia Molecular Opacity Project, not reachable from here).

29.5 Goldens, fixture, suites

Goldens re-snapshotted from the tree binary 4e275f1d (the physics of 7db5afe9) after a full matrix (all 17 cases differed from the round-3 goldens, as expected); backup golden_pre_round45_20260926. Fixture wasp_full_newton/IC replaced by the 1-thread route-B certified state of 7db5afe9 (backup IC_pre_round45_20260926). Every suite on the integrated binary: all pass except grid_and_gates (3 rows: the fixture is not a root of the fixed physics, to be re-made with the final binary), residual_determinism (known row), the two suites that need run logs.

Closing the series (2026-09-27): the matrix on the final binary 6844f188 (sections 29.1-29.4, 29.7, 29.9) differs from the goldens of 7db5afe9 in every case beyond 1e-3, largest in the helium columns of the lower cells (3.7 % at row 1 of wasp_full, the He+ + H rate at low T) and in the molecular cases (the ghost start, the progress control); goldens refreshed from it (backup golden_pre_final_20260927). Fixture re-solved on 6844f188 at 8 threads (info = 0, ||R|| 4.11e-9; backup IC_pre_final_20260927); grid_and_gates 345/345. Every suite passes except the known residual_determinism row and the two suites that need run logs.

29.6 Models solved

MEASURED, .P1/s1_remeasure/, .P1/s2_continuation/: every failure of the catalog's unsolved molecular and atomic cases stands under 7db5afe9 (six passes each); continuation then solved:

- molecular_scalar_gj1132_kzz1e9/HeH1.45: rung from the certified 1.52, certified in 13 passes.
- molecular_scalar_gj1132_wellmixed/HeH2.13, 1.8, 1.4: the well-mixed molecular cases differ from the $K_{zz} = 1e9$ ones ONLY by He_diffusion off (their base.inp carries Kzz_base 1.0e9, which the carrier transport reads). From the certified kzz1e9/2.13 state, He/H set uniform (new map_state_to_grid.py --uniform) and diffusion off: certified in 10 passes; then rungs 1.8 (10 passes) and 1.4 (14 passes) with make_rung.sh (UNIFORM=1). The direct re-solve of the catalog's well-mixed 2.13 state is refused by the progress control at pass 5, so the continuation is what solves it.
- molecular_photochem_gj1132_kzzprofile/HeH9: on the final binary of the series (2dd9a865, face at the level), mapped from its 40-pass state on 6844f188 (.P1/s2_continuation/photo9_f, row 7.4e-5 falling 0.77 a pass) and continued 40 passes: certified (H2 carrier row 9.2e-6 at cell 217), .P1/s2_continuation/photo9_v4 (2026-09-27). Its earlier stops were the lower-ghost closure defect of section 29.9.
- Ionization-stage transport on the molecular 1.8 state stalls on the He++ row under the progress control; the cause is the control (29.7).

29.7 A progress-control defect of the composition relaxation

Diagnosed (.P1/iontr_diag/): each pass of the carrier relaxation ends on its particle-count cap, the state moves a fixed amount a pass, the relative gated measure stays flat because residual and scale fall together, and the outer control reads that as no progress, halves the cap and ends the loop. With the cap held (diagnostic override) the same state certified in 14 passes.

Fixed (EXHALE_main.f90 progress control, binary d4b571e8): a pass whose consumed carrier update ended on its movement bound, whose solve converged and whose joint distance did not rise by more than 1.1x is not judged by the relative measures and does not halve the bound; the bound doubles after such a pass up to 0.5 and halves when a bound-limited update raises the distance of a converged wind or precedes the first failed solve; omega follows the same rule and no longer drops on hydro entry failures; EXHALE_CARRIER_TRUST_HOLD still freezes the bound. Also fixed: the scale x_{ref} of the carrier fixed-point test was the H2 maximum alone, so with transported stages and no molecules it sat at its $1e-30$ floor and the test could never pass (logged displacement $\sim 1.7e28$, MEASURED on iontrans_metals). MEASURED: the hot-Uranus stagnation fixture certifies at pass 42 (it stagnated; the suite row that needs that ending now holds the bound); iontrans_metals accepts at pass 17 instead of 48, the two certified states within $7.6e-5$; the molecular He/H 1.8 state with transported stages certifies in 14 passes from a starting bound of 0.5 (EXHALE_CARRIER_TRUST=0.5) but not from the default $1e-2$, where the hydro solve fails at pass 2 under the old control and the new one alike; with the interdiffusion flux on, that state's hydro solve fails on entry and a slow mode on the He front (0.86-0.93 a pass) remains. REGRESSION on kzz1e9/HeH0.083: a growing bound makes the H2 row at the outer cells 498-500 overshoot (up to $1.5e-1$) and the bound saws; neither control certifies that case (the old one stalls on the elemental row at cell 338). mol_carrier bitwise unchanged (the outer loop is not entered). Open: the starting bound for transported stages, the H2 outer-cell overshoot, the hydro entry failures with the interdiffusion flux.

29.8 Incidents of this series

- A sed insertion with an empty line address put a three-line comment before every line of src/tests/boundary_state/molecular_seed_of_an_atomic_state.sh; the 153 blocks were removed and the file checked by diff against the afternoon copy (identical except the intended line).
- The tree binary was rebuilt while a regression matrix was running on it; the matrix was stopped and is re-run after the series.

29.9 The lower-ghost H2 closure stalled with a zero move (fixed)

The molecular photochemical-column cases (He/H 9 and 2.09) stopped at passes 1-3 with "the base handoff partition of a lower ghost did not close": the closure's own move exactly zero and its residual 9-41 times the $1e-10$ tolerance. Diagnosed (.P1/ghost_stop/): each closure pass after the first started the molecular solve from the previous pass's accepted state, whose $x(4)$ held the OLD imposed H2 partition; the pinned row $x(4) - x_{\text{h2_fix}}$ then entered hybrd1 carrying the whole secant move, and hybrd1 ($\text{xtol} = \sqrt{\text{eps}}$, $\text{diag} = 1$) can return a move below $2 \text{xtol} \|\mathbf{x}\| \sim 3e-8$ untouched with $\text{info} = 1$ (hybrd.f90: first $\text{delta} = \text{pnorm}$, halved on the rejected step, then $\text{delta} \leq \text{xtol} \text{xnorm}$). The write-back then imposed the value, the next pass's map saw no move, and 28 identical solves followed (27 of 27 traced events; 41 of 41 refusals and stops in the S1 logs, residuals $8.2e-10 - 2.8e-8$, all below that floor). Fix: impose_transport_fractions sets every unknown a transport solve owns (H2 partition, oxygen carriers, ionization stages) to its imposed value before both the molecular and the atomic hybrd1 solves; the stop report prints the pin error. MEASURED with the fix (worker build): 0 of 127092 ghost closures unclosed, pin error exactly 0, no refusal and no stop; the He/H 9 case then fails on its H2 carrier row (0.88), a separate problem. Binary 6844f188. Every molecular base-handoff regression case is reached (ghost composition moves by $1e-12 - 1e-9$).

29.10 The _adv post-process made second order

User decision (2026-09-27): the catalog quotes the He 10830 EW of the _adv post-process. That post-process was a first-order upwind recursion in r (backward Euler of $dx/dr = R/v$ with the rates at r_j and the residence time at r_{j-1}), the main source of the first-order grid convergence of the EW (md/ew_grid_order_20260926.md). Now the variable-step BDF2, $y_j - a_1 y_{j-1} + a_2 y_{j-2} = g h_j F(y_j, r_j)$ with $w = h_j/h_{j-1}$, $a_1 = (1+w)^2/(1+2w)$, $a_2 = w^2/(1+2w)$, $g = (1+w)/(1+2w)$ (derived at post_process_adv.f90, coefficients as eq. (1.4) of Li & Liao, arXiv:2201.00527; L-stable, so a population that equilibrates within a cell does not ring as under the trapezoid rule), for every composition system and the energy equation, rates and $1/v$ at the same point; backward Euler on the first step, after a refused cell, for $w > 1 + \sqrt{2}$, and as a retake of a step with a negative population or no temperature root (0 retakes on the nine states). MEASURED on the nine certified grid states (He/H 1.6, 1.8, 2.13 on 500, 1000, 2000 cells), with the state held fixed: the post-process share of the EW error falls from +3.0 % / +1.0 % ($p = 1.00$) to below $3e-4$, and the _adv 2³S column converges at $p = 1.98 - 2.14$ (1.00 - 1.01 before); a closed-form column test gives $p = 1.989$. The EW itself changes by -3.9 % (500 cells) to -0.97 % (2000 cells) at He/H 1.6 (1.05692 → 1.01571 on 500 cells) and still converges at first order: what remains is the solved state's first-order base closure (being fixed) and the transit integration ($p \sim 1.4$, opposite sign).

Also fixed: the $h \operatorname{div}(\rho v)$ term of the _adv energy equation was built from differences of the cell-centred $\rho v r^2$, which carries the odd-even pattern of the cell-centred velocity at the base ($1.8e12$ to $3.4e12$ between neighbours while the face mass flux changed by $1e-10$), giving 805 / 4008 / 579 K at the first corrected cell on 500 / 1000 / 2000 cells; it is now the state's own mass row, the face-flux divergence. Below about 1.13 R_p the molecule-free post-process lacks the molecular heating and H3⁺ cooling that set the run's energy balance there; by the user's decision (2026-09-27) the run T is kept there with status 3 "unsupported" wherever the omitted heating and cooling exceed the carried ones (196 / 389 / 776 cells on 500 / 1000 / 2000 cells, up to 1.134 - 1.137 R_p ; EW change at most $1.2e-4$ relative); the user manual's status table says so. Binary ca651768. Suites: adv_static_limit 53, transit_state 12, transit_census 21, physics_probe 2199, thread_count_identity 9, all pass. Every _adv golden moves (5 short cases: T up to 5.9 %, n(2³S) up to 9.4 %). The catalog EWs predate this change by -1 % to -4 %. Found, not changed: thermal_damkohler_number is the net rate $|H - C|/u$, near zero at a balanced cell, not the relaxation rate its comment describes. Docs: the transit tool now says a stationary SOLVE writes no _adv products and its evaluation does; the project CLAUDE.md says the matrix compares the _adv pair as well.

29.11 The lower face at the reservoir level

The reservoir (the base.inp handoff or the Photochem profile's matching level) is stated at $r = 1$, but the domain's lower face sat at $r_{\text{edg}}(0) = 1 + dr/2$, and base_boundary.f90 carried the reservoir pressure up to that face along the reservoir's own isentrope: a half-cell slab of 226 K gas whose thickness scales with dr lay between the level and the domain (MEASURED: the lower layer offset by -2.2 % / -1.15 % in density on 500 / 1000 cells, md/ew_grid_order_20260926.md section 3). Now the face is the level in every grid mode (Uniform, Stretched, Mixed: define_grid.f90, $r(0) = 2 - r(1)$, refusals if a ghost reaches $r \leq 0$ or the face is not 1), the face state is the stated reservoir with no transport along any isentrope, the ghosts hold the reservoir continued below the level, and the Pho-

tochem matching level is the face on every grid; boundary model id ..._face_at_level_v4. MEASURED on the molecular He/H 1.8 state, 500 / 1000 / 2000 cells, all certified: the pressure gap between the interior at the level and the reservoir falls from 2.93 / 1.44 / 0.70 % to 0.114 / 0.067 / 0.038 %; the lower-layer density offset falls 20-40x; the wind quantities stay first order with errors 1.6-1.9x smaller (log10 Mdot 7.9147 / 7.9113 / 7.9094 -> 7.9116 / 7.9098 / 7.9086; _adv EW 1.1548 / 1.1248 / 1.1091 -> 1.1482 / 1.1214 / 1.1074 %A, measured before the _adv change of 29.10). The remaining first-order error is likely the thermal layer at the contact, which the grid does not resolve (cell-1 T 800 / 695 / 604 K against the 226 K reservoir, 13 % lower at every halving; not shown to drive the wind). Two other first-order terms were found and left: the element-diffusion Dirichlet He/H is imposed at the cell-1 center, $dr/2$ above the level (binary_element_diffusion.f90 946-959; the He/H change near the base is $1.5e-4$), and the conduction bath at the ghost center $r(0)$ (viscous_conduction.f90 485; conduction off in these runs). Every cell center moves by $dr/2$, so the grid guard refuses every state written before: such a state loads only as a seed made with map_state_to_grid.py --ic onto the new grid (a 1-step cold run gives the target grid). The regression fixtures and goldens are migrated in the same series (MEASURED, 2026-09-27: every fixture of the default matrix and every fixture a suite reads mapped with map_state_to_grid.py --ic onto a 1-step cold-run grid; wasp_full_newton/IC re-solved and certified, info = 0, ||R|| 1.43e-9, and iontrans_metals/IC a copy of it; the hp_* seeds, whose recipe from mol_base_handoff is now refused by the carrier restart token, mapped from their previous pairs with the seeds kept (x(H II) 0 and $1e-12$); jfnk_hd189, a named case outside the matrix whose legacy pair has no # columns header, not migrated; the two suites that read catalog states now map them into their own scratch; every suite passes, and the standing residual_determinism row now passes ($5.9e-2$ against < 1); goldens refreshed from the matrix run on 2dd9a865, backup golden_pre_facelevel_20260927); the LHS 1140 b catalog states stay records of the previous boundary (their EWs move by -0.6 % at 500 cells, below the observational error). Binary 2dd9a865.

29.12 A restartable state after every outer pass

The stationary route wrote its state only when the run ended, so a run stopped from outside (a wall-clock timeout) lost every pass it had made: MEASURED on 2026-09-28, three LHS 1140 b continuations lost 35, 13 and 9 passes. Now each outer pass that does not end the iteration writes the state the next pass starts from, output/Hydro_ioniz_last_pass.txt and output/Ion_species_last_pass.txt, overwritten each pass, by the same writer and headers as the final state files, rebuilt from (u, f_sp) in the order of rebuild_state_from_checkpoint with every side effect on the caloric mixture undone; both files carry certified=F cert_reason=pass_snapshot_p<n> and a # pass_snapshot line (the movement bound and omega of that pass, which a restart does not inherit: pass them by EXHALE_CARRIER_TRUST / EXHALE_DIFF_OMEGA), so a pair mixed from two passes is refused by load_IC and no tool reads a snapshot as certified. Each file is written to <name>.part and renamed. MEASURED: 0.03 - 0.05 s a pass on 500 cells (passes of 90 - 135 s); a run killed during pass 3 left the complete pass-2 pair, which restarted and continued as the uninterrupted run did; the final state files of a normal run are bitwise those of the previous binary. LHS1140b/models/.P1/s2_continuation/config_rung.sh continues from the pair when the parent has no final state. Binary a7255a03. The regression matrix on it, strict (REGRESSION_REL_TOL=0): all 17 cases byte-identical to the goldens of 2dd9a865 (MEASURED).

29.13 The unsolved molecular cases by continuation (state of 2026-09-28 16:30)

All runs under LHS1140b/models/.P1/s2_continuation/ (scripts config_rung.sh, make_rung.sh in .P1/, chain.sh, facelevel_continue.sh; env as the catalog: EXHALE_PTC_DTAU0=1.0, 8 threads). Certified states (MEASURED, certified=T, the JFNK ||R|| of the last solve 6.5e-9 - 2.9e-8):

family	He/H	passes	binary	note
kzz1e9	1.45	14	7db5afe9	rung from 1.52
kzz1e9	1.30	5 (after 24)	7db5afe9	
kzz1e9	1.223	9 (after 40 + 40)	2dd9a865	moved to the face-at-level grid
kzz1e9	1.186	24 (after 30)	2dd9a865	
kzz1e9	1.168	20	2dd9a865	
kzz1e9	1.15	46	a7255a03	carrier bound held at 1e-2
wellmixed	2.13, 1.8, 1.4, 1.1, 0.85, 0.7603, 0.68, 0.6116	10 - 21 each	7db5afe9	path finding, previous grid
wellmixed	0.6116, 0.55, 0.4	16, 18, 17	2dd9a865	on the face-at-level grid
photochem kzzprofile	9	41	2dd9a865	after 40 passes on 6844f188

Not solved yet: kzz1e9 below 1.15 (1.12 running, gated row falling to 1e-4 then flat at 1.3e-4 over passes 14-16); wellmixed 0.3347 and below; photochem kzzprofile 2.09 (gated row oscillating 0.08 - 0.54, hydro info = 2 on most passes, about 40 min a pass); kzz1e9 0.083 and the wellmixed 0.083 are the far ends of the two chains.

What was learned on the way (MEASURED):

- The well-mixed molecular cases are the kzz1e9 ones with He_diffusion off (their base.inp carries Kzz_base 1.0e9, which the carrier transport reads); a kzz1e9 state with He/H set uniform (map_state_to_grid.py --uniform) certifies with diffusion off in 10 passes, which is how the family was entered.
- Runs stopped by a wall-clock timeout lost every pass (35, 13 and 9 passes), which section 29.12 fixed for every later run; and config_rung.sh ran under set -e, so a run that timed out never wrote its EXIT.txt and a waiting continuation never started (fixed).
- The growing carrier bound of section 29.7 drives a sawtooth in the H2 cases below He/H ~1.2 (kzz1e9 1.15: the gated row between 5e-3 and 2e-1 over 35 passes with hydro info alternating 0/2) and in the well-mixed 0.3347 rung (between 2e-4 and 1e-2, the refusing cell moving between the front near cell 357 and the outer cell 500). Holding the bound at 1e-2 (EXHALE_CARRIER_TRUST_HOLD=1e-2) certified kzz1e9 1.15 in 46 passes, and 1.12 then fell 2.6x a pass to 1e-4; it did not stop the well-mixed 0.3347 oscillation (1e-3 - 2.7e-3 over 19 passes). The rule that grows the bound is therefore not right for the H2 carrier; an open item.

Documents brought to the code of this series (2026-09-28): the user manual (input keys Interdiffusion enthalpy flux, Outer shells, Composition update holds, H2 double ionization, H2 neutral dissociation and the retired Oxygen transport added from known_keys; the _last_pass output files; the face at the level; the BDF2 post-process; the progress control; the thread-count note; stale statements corrected, among them the tokens a Restart option change: line may not name, four not six), the physics and algorithms document (the interdiffusion enthalpy flux in the energy equations, the lower face, the grid constructions, the He+ + H pair, Ry/k, the post-process step, the progress control and the snapshot; stale statements corrected, among them the default of H2 double ionization, chung80, and the claim that every extension defaults to off) and README.md. Two source comments corrected (base_boundary.f90 model id, steady_residual.f90 energy row scale; comments only), and src/tests/interdiffusion_enthalpy_flux/run.sh made executable: make test had skipped it silently.

30. The base contact held through each steady solve, and the Intel build (2026-09-28/29)

30.1 OpenMP thread count: measured, and the default team

MEASURED on lart2 (2 x Xeon Gold 6154, 36 cores, 72 hardware threads), EXHALE_a7255a03.x (gfortran 16.2, -O3), one run at a time; A is wasp_full marching for 300 steps, B one stationary outer pass of the well-mixed molecular He/H 0.3347 state (500 cells, not converged). Records: LHS1140b/models/.P1/omp_scaling_lart2/.

threads	A [s]	A speed-up	B [s]	B speed-up
1	201.7	1.00	1535.6	1.00
8	47.4	4.26	234.8	6.54
18	34.7	5.81	126.2	12.17
36	32.2	6.26	87.8	17.49
72	41.8	4.83	89.7	17.12

The output is bitwise the same at every thread count. The second hardware thread of a core adds nothing (72 against 36), so the default team when OMP_NUM_THREADS is unset is now the number of physical cores the process may run on (init.f90, physical_core_count: the processors of Cpus_allowed_list grouped by their thread_siblings_list; where Linux does not provide the lists, every processor counts), in place of min(cores, 16), a 2026-07 marching measurement. Checked on lart2: 36 with nothing set, 4 under taskset -c 0-3, 1 under 0, 36, 5 with OMP_NUM_THREADS=5. README_HOWTO.md updated. The rung scripts take the team from NTHREADS (default 8).

30.2 The Intel compiler

lart2 has no classic ifort: in oneAPI 2026.1.1 and 2025.3.1 ifort is a link to ifx. The ifx build refused diffusive_photochemistry.f90: the derived types carrier_module_state, carrier_checkpoint and thermochemical_state were given the PUBLIC attribute twice (the module's public :: list and the type, public :: statement), which the standard forbids and gfortran accepted; the , public of the type statements was removed. The regression case mol_carrier stayed bitwise identical (REGRESSION_REL_TOL=0). The Intel build now records the oneAPI compiler and MKL libraries in its rpath (Makefile), so it runs under the env -i of the run scripts.

MEASURED on lart2 against the gfortran build of the same source (cases as in 30.1; B normalized by the closure evaluations, because the three builds take 27, 67 and 38 JFNK iterations from the same state):

threads	A gfortran / ifx [s]	B gfortran / ifx [ms per closure evaluation]
1	201.4 / 104.6	1.81 / 0.99
8	47.3 / 27.1	0.275 / 0.147
18	34.6 / 20.5	0.151 / 0.078
36	34.5 / 19.2	0.107 / 0.054

ifx is 1.7 to 2.0 times faster; -march=native gains gfortran at most 10 per cent. The hydrodynamic columns agree within $3.4\text{e-}9$ and the species within $1.3\text{e-}4$ (trace neutral metals) of the gfortran output. By the user's instruction every calculation started on lart3 and lart4 from 2026-09-28 uses the ifx build (EXHALE_b037d586.x, manifest BINARY_MANIFEST_b037d5869f76.txt); the goldens stay gfortran snapshots.

30.3 Why the molecular continuations alternated for 60 to 80 passes

The well-mixed He/H 0.3169 rung (.P1/s2_continuation/wm8_0.3169) ran 80 passes with the hydrodynamic solve converging on every other pass and the carrier update reversing itself each pass (cosine of consecutive $x(\text{H}_2)$ increments -0.990 to -1.000 , amplitude $6.3\text{e-}3$ at the H_2 front). The stage exports show that the wind itself alternated between two base states: the converged one with the reservoir owning the level (lower ghosts at 232 K) and the failed one with the interior owning it (ghosts at 1200-1212 K). The upwind side of the base contact is decided at every evaluation from the spread d_{window} of the wind-window mass flux (reservoir where $d_{\text{window}} \leq 2\text{e-}3$) and otherwise from the sign of the matched face velocity (base_boundary.f90). MEASURED over three stalled runs:

run	solves that converged returned	solves that failed returned
wm8 0.3169, passes 60-79	$d_{\text{window}} 1.355\text{e-}4$, ghost 232 K	$d_{\text{window}} 2.000\text{e-}3$ to $2.25\text{e-}3$, ghost 1200-1212 K
wm6 0.3347, passes 47-60	$1.33\text{e-}4$, 232 K	$2.00\text{e-}3$ to $2.35\text{e-}3$, 1222-1236 K
kzz7 1.09, passes 30-43	$8.4\text{e-}5$, 234 K	$2.00\text{e-}3$ to $4.07\text{e-}3$, 1136-1413 K

Several failed solves stopped at $d_{\text{window}} = 2.000\text{e-}3$ to four digits: the iterate was held at the threshold, where the residual steps. The composition update of the default contract moves the base pressure, the first steps of the next solve send a mass-flux pulse through the window, d_{window} crosses $2\text{e-}3$ and the base goes over to the interior side; the root is on the reservoir side, across the step. The code comments stated that the threshold stands far from every operating point; that holds for written states and not for the iterates of a solve, and the comments now say so. The rule that grows the carrier bound (TO_BE_DONE (g)) harmed the H_2 cases for the same reason, as far as these records show: a larger composition step makes a larger pulse.

Tests from the pass-80 state of wm8_0.3169 (ifx, carrier bound held at $3\text{e-}3$, MEASURED):

- T1, the reservoir side held for every solve by a measurement setting (EXHALE_BASE_WIND_SPREAD=1e3): certified in 6 passes, every solve $\text{info} = 0$, ghosts at 232 K throughout; continued under the unmodified rule it certified

again in 2 passes (wm9_0.3169_spread_check6), the two states within $1e-3$ of each other.

- T2, Composition update holds: pressure alone: the solves of passes 2-4 failed on the reservoir side (the line-search stagnation recorded in md/lhs1140b_h2_front_energy_imbalance_20260924.md) and the run stopped on its no-progress rule. Not a remedy here.

30.4 The side of the contact is held through each steady solve

base_boundary.f90 / EXHALE_main.f90. Before each steady solve of the stationary route the side of the contact is fixed by the first evaluation of the solve (its entry state) and held for every residual and Jacobian evaluation inside it (hold_base_contact_direction_from_entry, release_base_contact_direction). After the solve the rule is applied to the returned state (base_contact_reverses_in_state); where a converged solve returns a state whose rule selects the other side, one more solve from it holds that side. A root is therefore accepted only on the side its own rule selects, as before; only the path to it changes. The marching route and every evaluation outside a steady solve read the rule at every evaluation. EXHALE_BASE_CONTACT_PER_EVALUATION=1 restores the old reading inside the solves (a control). Each solve logs the side it held, the number of evaluations, and at how many of them the rule read the other side. Documented in docs/physics_overview/boundaries_initial.tex.

MEASURED (gfortran build d9047b32, the pass-80 state of wm8_0.3169):

- H1, the code as changed: certified in 6 passes with the same worst rows as T1 ($3.41e-4$, $7.37e-4$, $6.44e-4$, $7.71e-5$, $2.37e-5$, $1.65e-6$); in passes 2 to 5 the rule read the interior side at 822 of 2702, 850 of 3014, 259 of 2741 and 109 of 2341 evaluations, in pass 6 at none.
- H0, the same build with EXHALE_BASE_CONTACT_PER_EVALUATION=1: passes 1-6 alternated info = 0, 2, 0, 2, 0, 2, the old cycle.

Regression (the full default matrix of 17 cases, gfortran d9047b32 against the goldens of 2dd9a865, MEASURED 2026-09-29): 16 cases bitwise identical in all four files; wasp_full_newton, the one case whose steady solve met the contact threshold, agrees within $1.7e-9$ of every column's own maximum. Its first solve held the interior side (the rule read the reservoir side at 5365 of 5968 evaluations inside it), the returned state's rule read the reservoir side, and the second solve, held there, returned the old root. The harness flagged one entry of Hydro_ioniz_adv.txt, the rounding-level residual column adv_mass_row (0 against $3.1e-15$ in a column whose largest value is $7e-3$), as a relative difference of 1; it is not a difference of the state. Goldens are not refreshed (the user's decision: later).

30.5 An output argument that was never set

hydrodynamic_rows_body.inc: hydrostatic_equilibrium_of_each_cell_fv declared pL_asm and pR_asm INTENT(OUT) and never set them (ifx warning 6843); the caller forms both itself before any use, so no value was wrong. The two arguments were removed from the routine and its one call.

30.6 Runs

- chain_verified.sh (new, .P1/s2_continuation/): chain.sh with every certified rung continued for up to 6 passes under a second environment (CHECK_ENV), the rung counting only if that continuation certifies too.

- Well-mixed chain `wm10` from the re-certified 0.3169 state (ifx, T1's setting, every rung verified under the unmodified rule): 0.3 (9 passes), 0.26 (21), 0.22 (24), 0.19 (22) certified and verified by 09:00 on 2026-09-29; continuing to 0.083.
- `K_zz = 1e9` chain `kzz8` (ifx, carrier bound held at $3e-3$): 1.09 (32 passes), 1.075 and 1.06 (the first 1.06 failed), 1.03, 1.0, 0.97, 0.94, 0.91, 0.88, 0.85, 0.8, 0.75, 0.7, 0.65 certified by 08:50 on 2026-09-29.
- photochem `kzzprofile` He/H 2.09: `photo209_v5` (default control, binary 2dd9a865) certified at pass 72; its launcher failed after the run because `config_rung.sh` was edited under it from another host (its `EXIT.txt` was written by hand, with the reason).

31. The code audit of 2026-09-29: findings checked and applied

`md/CODE_AUDIT_20260929.md` (Codex, with its probes in `docs/audit_20260905/code_audit_20260929/`). Each finding was checked against the source before anything was changed. The audit document was made ASCII (curly quotes and dashes); its disposition section records the same.

- **F2, the admissibility predicate accepted infinities: FIXED.** `UW_conversions.f90`, `positive_density_and_internal_energy`: `+Inf` density or energy compares true against zero and passed. Every conserved component and the internal energy must now be finite. The audit's probe (`guard_probe.f90`, linked against the fixed objects) now reads finite accepted, `+Inf` energy refused, `+Inf` density refused, NaN refused. A finite state takes the same branch as before, so no output moves.
- **F3, a steady route wrote a state without certifying it: FIXED.** The direct `EXHALE_PTC=1` route wrote whatever certificate the solve had left; it now calls the same final certification as the stationary restart (`certify_the_state_to_be_written`, one routine for both: the claim, the boundary rebuilt from the written composition, the residual assembled from that state, the certification evaluated on it). And a solve whose root stays on the other side of its own contact rule after the second solve (section 30.4) now returns `info = 2` instead of 0. The restart route calls the same statements in the same order as before.
- **F4, the transit tool replaced non-finite transmission by 1 or 0: FIXED.** `EXHALE_transit.py`, `_checked_transmission`: a non-finite value or a departure from `[0, 1]` beyond $1e-12$ stops the synthesis and names the first wavelength; the rounding of the Ly-alpha disk average is still clipped (the only line that was clipped before), and He 10830, H-alpha and H-beta are checked without clipping. The comment that `exp(-tau)` overflows in a saturated wing was wrong (it underflows) and is gone. MEASURED on the molecular `K_zz = 1e9` He/H 2.13 catalog output: every number of the five `tpm_*` files is identical before and after.
- **F5, the equivalent-width operator: FIXED.** `LHS1140b/he10830_equivalent_width.py`: the model curve is refused when it is not one-dimensional, not of one length, not finite (a NaN wavelength used to pass) or not strictly ordered, and a transit file with a non-finite column or a non-positive continuum; the observational error now uses the trapezoidal weights of the estimator itself. MEASURED: EW 1.1084293096 %A unchanged, error 0.0294775 -> 0.0288956 %A (0.030 -> 0.029 to two figures). The error is ours, computed from the released spectrum under independent samples, not a number of Cherubim et al. (2026).
- **F7, a changed link library did not relink: FIXED.** `Makefile`: a second stamp whose name hashes the link

command (compiler, flags, LDLIBS); the executables depend on it. MEASURED: make -q with an unchanged command returns 0, with LDLIBS=-l_exhale_missing_library it returns 1, and make -n then shows the link alone, no compilation.

- **F8, the formation-energy diagnostic proposed a double count: CORRECTED (text).** src/utls/formation_energy_flux_diagnostic.py said the sensible-energy equation lacks -div(F_form) and a chemical source; in the sensible-energy form the chemical conversion is already the reaction heating and cooling ($h_{\nu} - I$ for a photoionization), and adding either term counts it twice. The header now states the correct balance and what the script measures (the chemical share of the energy ledger); no computation changed.
- **F1, the sensible enthalpy carried by the diffusing carriers is not in the energy equation: CONFIRMED, MEASURED, OPEN.** The carrier transport (diffusive_photochemistry.f90) moves H2 relative to the gas and returns no energy flux; interdiffusion_enthalpy_active requires He diffusion and covers the elements only, so an H2 counterflow at zero hydrogen flux carries $q = (h_{H2} - 2 h_H) \Phi_{H2}$ that nothing represents. DERIVED on two certified states (well-mixed He/H 0.19 and $K_{zz} = 1e9$ He/H 0.65), from the face diffusive fluxes of the carrier row terms and $h_{H2} - 2 h_H = k(u_{rv} - 5T/2)$ with a rigid rotor and a harmonic vibrator (0.4 per cent from the code's own H2 caloric at 1000 K): $|\text{div } q|$ is 1 to 3 per cent of the heating over the molecular layer (median 1.8 per cent over $1.001 < r < 1.03$), 8 to 11 per cent at the base cell, and as large as the net heating or larger where heating and cooling cancel; the integral of $|\text{div } q|$ is 1.4 to 1.5 per cent of the integrated heating, while the SIGNED integral vanishes ($-1e-18$ of it), as the closed diffusive boundary of the carrier operator requires (script, inputs with their md5 and the numbers: docs/audit_20260905/code_audit_20260929_F1/). It can move the temperature of the H2 layer; the implementation (the carrier face fluxes into the energy rows of the steady residual and the marching update, the compensating atomic flux from the element constraints, tests) is the next item.
- **F6, the pass snapshot is two renames, not one publication: OPEN, a decision.** A reader-interface and data-layout change (a state generation with a manifest and one published pointer); left for the user's decision.
- Sections 11 to 13 of the audit (the wind-only certification scope, the abundance mask of absent carriers, trace caloric approximations, general EOS Riemann tests, the outer diffusive closure, parallel tests, optimization order) are statements of scope and proposals, recorded in md/TO_BE_DONE.md; none is a defect to fix in place.

32. The follow-up review of the audit corrections (2026-09-29)

md/CODE_AUDIT_20260929_review1.md (Codex; probes in docs/audit_20260905/code_audit_20260929_recheck/) confirmed F2, F4, F5 and F7 and the substance of F3 and F8, and named four remaining points. All four were checked in the source and applied.

- **R1, the direct steady route stopped with status 0 whatever the final certificate said.** It now ends with certification_stop_uncertified, the exit contract of the other steady routes: a stationary claim the final certificate refuses exits 2 after the state and the resolved configuration are written; an unfinished snapshot exits 0.
- **R2, the formation-energy diagnostic described more than it computes.** It computes the advective part u_{form}

v only; the header now says so, gives the stationary identity with the relative transport term, lists what a difference contains (omitted relative transport, the species residual of the saved state, missing species, the flux reconstruction), and states the total energy balance of an open, gravitating wind with its work and boundary terms instead of "only radiation".

- **R3, the admissibility test could overflow before rejecting.** The sign of ρe is now taken as the sign of $2 \rho E - (\rho v)^2$ on the components divided by the largest of them, so nothing can overflow (admissible_conservated_cell, one definition used by the stage test and by the positivity repair, whose own cell test had accepted infinities as well). Under -ffpe-trap=overflow,invalid the 1e160 cell of the review is admitted ($\rho e = 5e159$) with no trap, and its negative-energy variant, a tiny-density large-momentum cell and a negative-energy ordinary cell are refused. The three marching stages now test the interior before Apply_BC builds the boundary on it: Apply_BC reads the physical cells and writes only the ghosts, and neither the test nor the repair reads the ghosts of the trial, so the decisions are unchanged and an inadmissible interior no longer reaches the boundary and EOS routines.
- **R4, the transmission messages quoted metres as Angstrom.** The wavelength arrays of EXHALE_transit.py are in metres; the messages now quote 1e10 times them, and the function states its unit.
- **The F1 estimate** is now kept as a script with its inputs and the signed and absolute integrals (section 31 corrected: the quoted 1.5 per cent was the integral of $|\text{div } q|$; the signed integral vanishes).

Checks of this section (MEASURED): the overflow probe above; the audit's admissibility probe (finite accepted, +Inf and NaN refused); the six regression cases that march or take the stationary restart route (hp_zero_seed, hp_trace_seed, hp_front, hydrostatic_column, oxygen_chemistry, iontrans_metals) bitwise identical to their goldens with REGRESSION_REL_TOL=0 (gfortran EXHALE.x 67c12cd1); the transmission function on a NaN and on a rounding excursion. The production ifx build of this source is EXHALE_64954795.x (manifest BINARY_MANIFEST_64954795cb2a.txt), built with make -B on lart2 because /nfs/mocafe is mounted with a 30-minute attribute cache there and a plain make did not see the sources edited on lart4 (it relinked the old objects; the md5 stayed that of the previous build).

Two findings of the runs of this morning, recorded here and in md/TO_BE_DONE.md (m): the well-mixed chain (wm10, the contact held) certified 0.3 to 0.1 and stopped at 0.09, where every hydrodynamic solve converged and the H2 carrier row of the OUTERMOST cell (500, $r = 30 R_p$) grew slowly pass after pass (0.09: 1.2e-3 to 2.5e-3; 0.09487: 3.5e-3 to 4.0e-3) with every carrier relaxation ending where the carriers stopped moving, the symptom the 2026-09-24 memo recorded at He/H 0.083 and a candidate for the zero diffusive carrier flux at the outer face (audit section 12.3); and the four reduced-XUV atomic He/H 9.7 cases fail their first solves with the contact held, so their obstacle is not the contact threshold.

33. The state of every pass published as one generation (audit F6)

Audit finding F6 (md/CODE_AUDIT_20260929.md section 8, md/CODE_AUDIT_20260929_review1.md section 9): the stationary route wrote the state of every outer pass as output/Hydro_ioniz_last_pass.txt and output/Ion_species_last_pass.txt, each half through its own .part file and its own rename. An interruption

between the two renames left halves of two passes, and the pass number of `cert_reason=pass_snapshot_p<n>` did not tell two runs apart (two runs both write pass 1), so a mixed pair from two runs with compatible metadata was not excluded. Changed with the user's approval.

What changed (src/modules/files_IO/write_output.f90, src/modules/files_IO/load_IC.f90, src/modules/functions/utilities.f90, src/EXHALE_main.f90):

- **A run identity.** `establish_run_identity`, called once after `init`, forms `run_identity = yyyymmddThhmmss.sss_c<count>` from `date_and_time` (to the millisecond) and the 64-bit `system_clock` count, both standard Fortran; characters outside `[A-Za-z0-9._-]` are replaced.
- **One generation for each pass.** `publish_pass_state_generation` (it replaces `write_pass_state_pair`) writes, with the writers and header lines the pair had, `Hydro_ioniz.txt`, `Ion_species.txt` and `manifest.txt` into `output/pass_state/.<gen>.tmp/`, with `<gen> = r<run_identity>_p<pass>`. The manifest states, one key=value a line, `state_id`, `run_identity`, `pass`, `previous_state_id`, the two file names with their sizes in bytes (`INQUIRE size=`), `certified=F` and `cert_reason=pass_snapshot_p<pass>`.
- **The order of publication.** (1) the three files are written and closed in the temporary directory; (2) `rename`(2) of that directory onto `output/pass_state/<gen>`; (3) `output/pass_state/current.part` is written with `<gen>`, closed and renamed onto `output/pass_state/current`, the one `rename` that publishes; (4) the generation current named before is kept, and the one it had replaced (its manifest's `previous_state_id`) is removed file by file under its explicit name and then `remove`(3)d as an empty directory. Directories are made with a `C mkdir` binding (mode 0755) and removed with `C remove`, beside the existing `rename` binding; no shell command and no pattern is used. A name read from a file is removed only if it starts with `r` and has only `[A-Za-z0-9._-]` characters.
- **An identity in both halves.** The coupling line of both halves of a generation ends in `state_id=<gen>` (`write_coupling_state_header` takes it as an optional argument; no other file carries it). `load_IC` reads it with the certification pair (`parse_certification_claim`) and `adopt_certification_claim` refuses a pair whose halves state two different ids, and a pair in which one half states an id and the other none. A pair in which neither states one (every final state, every older file) loads exactly as before; a pair that does prints the generation.
- **A measurement key.** `EXHALE_PASS_STATE_STOP_AT=<step>[:<pass>]` stops the run with `stop 3` right after step 1, 2 or 3 of the publication of that pass (of every pass if none is named). Unset: no effect.

What changes for the user: the files moved from `output/{Hydro_ioniz,Ion_species}_last_pass.txt` to `output/pass_state/<gen>/{Hydro_ioniz,Ion_species}.txt`, and the generation to take is the one `output/pass_state/current` names; the old names are no longer written. The final `Hydro_ioniz.txt` / `Ion_species.txt` of a run are unchanged (no `state_id`). An interruption before step 2 leaves a `.<gen>.tmp` directory, one between steps 2 and 3 a complete generation no pointer names; neither is read and neither is removed by a later run. The continuation script of the LHS 1140 b rungs is being executed by running jobs, so it was not edited in place: the new version is `LHS1140b/models/.P1/s2_continuation/config_rung.sh.new`, to be moved onto `config_rung.sh` when no process uses it. Its entry precedence is (a) the parent's final pair, (b) the generation current names, read once and taken only if its three files exist and the pointer, the manifest's `state_id` and both headers' `state_id` agree,

otherwise refused, (c) a legacy `*_last_pass.txt` pair with the checks it had. `run_check.sh` clears `output/*.txt` of a case before a run and leaves `output/pass_state/`, which holds at most two generations and is not compared. `src/tests/steady_selfconsistent_residual/run.sh` never read the pair. Documents updated: `README.md`, `README_HOWTO.md`, `md/input_schema.md` appendix D, `docs/EXHALE_user_manual.tex` (section "pass_state/", label `sec:passstate`, and the key in the table of environment variables), `docs/physics_overview/steady_state.tex`, `boundaries_initial.tex`, `postprocessing.tex`; both PDFs rebuilt without errors or undefined references. Checks (MEASURED 2026-09-29 on `lart4`, gfortran 16.2, binary built in its own object directory `build_f6`, md5 `b8cf7a21f2fa1149b1d9131cafb44a0`; the tree also held other uncommitted edits in progress at that moment):

- **Publication and retention.** `iontrans_metals` inputs and IC in a scratch directory, `EXHALE_OUTER_PASSES=4`, 4 threads, exit 0: passes 1, 2 and 3 published; afterwards current named `..._p3`, `..._p2` was kept, `..._p1` was removed, no `.tmp` or `current.part` was left; each manifest's byte sizes (94446, 448089) equal the files, and the manifest and both headers state the same `state_id`. The final pair carries no `state_id`. The regression run of `iontrans_metals` (below) published 14 passes and left exactly `..._p13` and `..._p14`, current naming `..._p14`.
- **Interruption.** Three runs started within 30 ms of one another, each on a directory holding the generations of the first run, with `EXHALE_PASS_STATE_STOP_AT=k:2`, all exiting with status 3, got three different identities (`...228.498_c...163735`, `...228.527_c...351728`, `...228.522_c...722562`). `k = 1`: current named the run's complete `p1`; `.<p2>.tmp/` held its three files. `k = 2`: current named `p1`; `p2` was complete and unnamed. `k = 3`: current named the complete `p2`; `p1` and the previous run's `p3` were still present (step 4 not reached). At every point current named a generation whose three files existed and whose manifest and headers stated its name. The previous run's `p2` had been removed at the publication of the new `p1`, across the two runs.
- **load_IC.** Hydro of `p2` with Ion of `p3`: refused, exit 1, both ids printed. Hydro of `p3` with a final `Ion_species.txt`, and the reverse: refused, exit 1. Both halves of `p3`: loaded, exit 0, "the state is the pass state generation ...". The legacy certified IC pair of the case (no `state_id`) loaded as before (the runs above).
- **config_rung.sh.new**, with `BIN=/bin/true` on dummy parents: a final pair is taken first even with generations beside it; a parent with only generations gets the pair current names (byte-identical copies, `LAUNCH.txt` seed "LAST PUBLISHED PASS (pass_snapshot_p3, output/pass_state//)"); a manifest naming another id, a generation missing `Ion_species.txt`, and a pointer to an absent generation are refused (exit 1); a legacy `*_last_pass.txt` pair is taken; a legacy pair of two passes and an empty parent are refused.
- **Final outputs.** `REGRESSION_EXE=<that binary> REGRESSION_REL_TOL=0 run_check.sh check iontrans_metals hp_front`: all six compared files data identical (bitwise) to their goldens, `REGRESSION PASS`.

34. The enthalpy the transported carriers carry (audit F1), and the regression comparison of residual columns (2026-09-29)

The physics. A molecular carrier (H_2 ; OH , H_2O , CO with the oxygen chemistry) is moved by the carrier operator relative to the rest of its own elements: its diffusive, eddy and settling number flux $\Phi_{i,c}$ moves nuclei that the element fluxes do not, and the non-carrier species of the same elements move the other way, nucleus for nucleus. Cook (2009,

eqs. 11-13) then puts $q_{\text{car}} = \sum_c \Phi_{\text{c}} [h_{\text{c}} - \sum_{\text{el}} n_{\text{u}\{\text{c}, \text{el}\}} \bar{h}_{\text{el}}]$ into the energy equation, h_{c} the enthalpy of one carrier particle and $\bar{h}_{\text{el}}$ the enthalpy of the non-carrier species of element el per nucleus, every particle with the electrons its charge gave up, as in `component_specific_enthalpies`; an H_2 molecule rising through atomic hydrogen at zero hydrogen flux carries $k(u_{\text{rv}} - 5T/2)$, about -1.5 kT at 1000 K . Before this section the energy equation carried the element part only (section 29.3), which does not see a carrier moving within its element.

The code.

- `diffusive_photochemistry.f90`: `carrier_enthalpy_face_flux` (the flux at every face from the carrier operator's own coefficients and face rule, `carrier_face_coefficients` and `carrier_face_flux`, with the temperature and the particle count of the state passed, so it follows the iterate of a steady solve), `carrier_enthalpy_divergence_of_state` (cells $1 \dots N$ on the carrier rows' geometry; both ends of the carrier column are closed, so the column sum is zero) and `carrier_enthalpy_active`. Writes no module state.
- The stationary energy row (`steady_residual.f90`, `assemble_residual`) adds the divergence into `Sidf`, the one record of the enthalpy of relative transport. The residual module cannot use the carrier module (which uses it), so the carrier module registers the procedure in a pointer of the residual module when the carrier set is decided (`carrier_set_init`); a run without the term leaves it null.
- The marching step (`EXHALE_main.f90`) subtracts $dt \text{ div } q_{\text{car}}$ after an accepted carrier step, from the fluxes of the composition that step returned (the step's own fluxes to first order in dt , exactly at a fixed point, so the two routes solve one energy equation).
- One key for both terms, `Interdiffusion enthalpy flux`: (default `True`): it now also acts with `Molecular carrier transport`: `True`. A new restart token `carrier_enth` (absent in older files, read as `F`), so a restart from a state written without the term names it on a `Restart option change`: line.
- Not covered: the ionization stages of `Ionization transport` (their face flux contains the element's advection; the relative part is not formed), recorded in `md/TO_BE_DONE.md` (j).

Checks (MEASURED).

- New suite `src/tests/carrier_enthalpy_flux/` (10 rows, all `PASS`) on a synthetic HI/H_2 column at zero hydrogen flux: both end faces carry nothing; the column sum of $V \text{ div } q$ is $4\text{e-}17$ of the sum of $|V \text{ div } q|$; the face flux equals $\Phi_{\text{H}_2} k(u_{\text{rv}} - 5T/2)$, Φ_{H_2} recomputed from the operator's coefficients, to $3\text{e-}15$; the key off and a run without carrier transport give exactly zero.
- The certified well-mixed He/H 0.19 state (`wm10_0.19_check`) reloaded with the term (`LHS1140b/models/.P1/audit_F1/wm019_F1`, `gfortran b029a1e8`): the code's divergence, from the conservation budget of pass 1, is a median 1.84 per cent of the heating over $1.001 < r/R_{\text{p}} < 1.03$ (the independent estimate of section 31: 1.83), 1.6 per cent in absolute integral (1.5), zero in signed integral, largest 9.9 per cent at $r = 7.2 R_{\text{p}}$. That state is not stationary with the term: the gated carrier row stood at $7.1\text{e-}3$ on the first pass and fell by about 18 per cent a pass ($1.7\text{e-}4$ at pass 20), every hydrodynamic solve converging. So every molecular state certified before this section is a state of the previous energy equation.

The regression comparison of residual columns (user decision of 2026-09-29). `backup/regression/compare_within_tolerance.py` measured every entry against itself, so a residual column that

is rounding at a solution read 0 against $3.1\text{e-}15$ as a relative difference of 1 (adv_mass_row of wasp_full_newton, section 30.4). A column whose header name ends in _row is now measured against the largest value of its golden column; every other column keeps the value-by-value measure (a state column spans decades, and a column-wide scale would hide a change of its small values). The flagged file now reads within $2.5\text{e-}9$.

What the term changes (MEASURED). The He/H 0.19 reload above certified again after 35 passes (certified_in_wind). Against the state certified without the term: T higher by 8 to 19 K in the H2 layer ($r < 1.01 R_p$), lower by a median 4.8 K over $1.3\text{--}3 R_p$ and by at most 24 K ($r = 4.0 R_p$, 1774 $\rightarrow$ 1749 K); the wind-window mass flux -0.55 per cent; the He 2^3S density higher by a median 1.9 per cent over $1.05\text{--}5 R_p$, at most 6.3 per cent. The equivalent width was not measured (the stationary route writes no _adv pair); a change of the order of the He 2^3S change, about 2 per cent, is DERIVED, against an observational error of 2.6 per cent. Figure: LHS1140b/models/.P1/audit_F1/wm019_before_after_F1.pdf.

The regression matrix and the goldens (MEASURED 2026-09-29). The full matrix on the F1 + F6 source: twelve cases bitwise or within $1\text{e-}3$; the four cases with Molecular carrier transport: True differ as F1 makes them (mol_carrier, hp_zero_seed, hp_trace_seed, hp_front); and oxygen_chemistry wrote nothing, because its IC predates the carrier_enth token: its input now names Restart option change: carrier_enth, and it is then bitwise its golden, since in that 1000-step cold transient the carrier interval is never covered (1049 of 1049 step attempts) and the carriers never move, so F1 adds nothing by construction. The goldens of the four changed cases were re-snapshotted (the previous ones kept in backup/regression/golden_pre_F1_20260929/), with iontrans_metals so that its reference directory is current, and the four re-checked bitwise PASS.

35. Why the mapped atomic He/H 9.7 seeds at reduced XUV do not solve (2026-09-29, open)

The four atomic He/H 9.7 cases at 0.01, 0.06 and 0.0624 of the fiducial XUV (LHS1140b/models/.P1/atomic97/) were moved from the catalog states onto the face-at-level grid (facelevel_continue.sh, section 29.11) and restarted stationary. The hydrodynamic solve ended info=2 on every pass. Measured on the 0.06 case, LHS1140b/models/.P1/atomic97/diag_R/ (one directory for each measurement, Restart intent: stationary evaluate unless named otherwise):

run	state and binary	residual norm	largest row
D1	catalog state, previous grid, EXHALE_69233669.x (the binary that wrote it)	$5.6\text{e-}7$	the catalog value
D2	the same state, EXHALE_7db5afe9.x (2026-09-26, previous grid)	$5.7\text{e-}2$	energy, $r = 1.072$
D4 / D5	mapped seed, EXHALE_6dd54955.x, element enthalpy flux off / on	1.88 / 1.88	mass, cell 13 (all of it below $r = 1.03$)

run	state and binary	residual norm	largest row
D6	mapped seed, EXHALE_a7255a03.x (the first face-at-level binary)	1.88	as D4
D9	D4 with v below $r = 1.003$ set to the wind flux / (ρr^2)	1.86	mass, cell 13
D10	D4 with $\ln \rho$, $\ln p$ and T below $r = 1.03$ interpolated as cubics	1.82	mass, cell 110 (where the two interpolations meet)
D11	D4 with 50, 110 or 200 cells rebuilt in the discrete well-balanced equilibrium	1.54	mass, cell 6
D13 / D14	D11 (110 cells) with cell 1 anchored to the reservoir pressure, without / with the smooth velocity of D9	1.54 / 1.18	mass, cell 6 / cell 1

What this shows, and what it does not:

- The seed is a wind: its mass-flux spread above $r = 1.03$ is 0.32 per cent, the same as the catalog state's. The spread of 0.42 to 0.48 in the logs of the stationary restarts is what the first JFNK solve left (27 to 42 non-monotone accepts), not the seed.
- CORRECTED after the review (35.1): the two bullets written here first read D9 to D14 from the maximum over all cells and excluded the interpolation of the lower layer and the reconstruction. That was wrong. The element enthalpy flux is not the direct cause of the fixed-state mass maximum (D4 = D5 in the mass row), and nothing more is excluded.
- The heating of the lower layer at fixed T is higher by about 8 per cent under the face-at-level binary than under EXHALE_7db5afe9.x at the same radius ($1.139\text{e-}9$ against $1.053\text{e-}9 \text{ erg cm}^{-3} \text{ s}^{-1}$ at $r = 1.010$); the increase is the charge-exchange channel (35.1), while cooling is below 2 per cent of heating from the base to $1.1 R_p$ in both.
- A marching relaxation does not settle the seed: 5000 steps take the mass-flux spread from 0.003 to 3.8 and back to 1.9, and 41000 steps (D8) leave it at 1.09 to 1.10 from step 15000 on. That relaxation ran its first stage with PLM (Restart intent: relaxation enters the two-stage PLM -> WENO3 marching), while the state and every stationary solve above are WENO3.
- Restart intent: relaxation does not relax a seed whose spread is below the Newton switch ($1\text{e-}2$): a restart arms the hand-off at load ($\text{du_newton_armed} = \text{do_load_IC}$), and the 0.06 seed was handed to JFNK at marching step 2.

The two 0.01 cases were already refused in the catalog with the same signature (flat mass flux, mass and energy rows of order 1 in the first cells); the 0.06 and 0.0624 cases acquired it with the grid and code change. The whole account,

with the next steps considered, is `md/atomic_heh97_reduced_xuv_20260929.md`. The four runs are stopped (their `EXIT.txt` say why). The same failure, a first JFNK stalled at $\|R\|$ 1.32 on the mass row of cell 2, stopped the F1 resolve of the mapped `m7_HeH1.45` state; that He/H is instead being reached from `kzz_1.223_v4b`, a state solved on the face-at-level grid (`LHS1140b/models/.P1/audit_F1/launch_F1_kzz_1.223_to_1.45.sh`).

35.1 The review of the account (2026-09-29)

`md/atomic_heh97_reduced_xuv_20260929_review.md` (its retained evidence is the directory `atomic_heh97_review_20260929/` of `docs/audit_20260905/`: a diagnostic linked to the production objects, built with bounds checks, reproducing the production face fluxes exactly) was checked here and taken over:

- The continuity defect is a numerical face flux: on the reconstructed traces of the mapped seed the Roe pressure-jump term ($-\Delta p_{WB} / 2c$, Δp_{WB} about $1e-7$ of p at Mach $4e-8$) reverses the mass flux at face 12 and removes most of it at face 13, and at face 0, whose two traces differ by $4.9e-4$ of the pressure, carries about $1e4$ times the wind flux inward. The low-Mach velocity-jump correction scales Δv and leaves this (1.876).
- A lower layer projected into the reconstruction's discrete equilibrium with the momentum set to the wind flux takes cell 13 from 1.876 to $1.3e-4$; cell 1 stays at 1.18 and the join at cell 111 rises to 1.50 (READ from the retained results). The interior is repaired; the base boundary and the join are not.
- The base boundary continues cell 1 along a hydrostatic isentrope, the reconstruction balances each cell with a constant-density equilibrium, and `Rec_BC` keeps the interior trace of face 0: three constructions, not one discrete equilibrium. The comment in `base_boundary.f90` (`characteristic_base_face_state`) that claimed the boundary exactly balanced in any hydrostatic atmosphere at any resolution was corrected (comment only; no binary change beyond line numbers).
- The heating difference is the nonradiative $\text{He}^+ + \text{H} \rightarrow \text{He} + \text{H}^+$ channel, $8.785e-11$ against zero at $r = 1.010$ (`Heating_breakdown.txt` of D2 and D4, MEASURED here as well): the rate of section 25, $9.6e-17 \text{ cm}^3 \text{ s}^{-1}$ at 1057 K against $2.8e-21$ before, whose threshold continuation carries a band of $1.5e-5$ to 3.6 times at $1e3 \text{ K}$. A deliberate physics change, not a defect; a sensitivity test moves the forward rate, its reverse and the heat together. Heating cannot make the fixed-state continuity defect.
- The first JFNK solve cut the certification distance from $7.96e10$ to $1.19e9$ and the mass row from 1.88 to 0.099 while spoiling the outer wind; a failed multi-scale solve, not by itself evidence against the non-monotone acceptance.

The next steps in the review's order (a face-flux budget diagnostic, discrete equilibrium tests of the boundary with the reconstruction, one discrete equilibrium at the base, a seed that satisfies the lower-layer equations, continuation, and only then the nonlinear formulation) are listed in `md/atomic_heh97_reduced_xuv_20260929.md` section 5; none is started, and the base boundary change needs its design agreed first.

36. The face-flux budget export and the discrete equilibrium tests of the base (2026-09-29)

Steps 1 and 2 of `md/atomic_heh97_reduced_xuv_20260929_review.md` section 8. No equation, flux, source or boundary changed.

The face-flux budget export (schema 3). EXHALE_CONSERVATION_BUDGET=<n> (src/modules/time_step/conservation_budget.f90, called from assemble_residual in steady_residual.f90) now appends to the 42 columns of schema 2, for the upper face of each row: the reconstructed left and right traces (rho, v, p) the flux was formed from, wb_dp_eq and the two departures, the pressure jump Delta p_WB the Roe flux uses, the central mass transport of the physical flux, the pressure-jump part of the numerical mass flux (production Num_flux with the jump minus the same call with the jump set to zero: an attribution, not a flux), and the recomputed numerical mass flux; for each cell, the absolute and normalized continuity residual (the scale the stationary solve divides by) and the signed cumulative mass budget. A header block states the lower boundary of the assembly: the reservoir, the interior state continued to the face along cell 1's isentrope (recomputed from the inputs the boundary used and checked bitwise against the values it stored), the characteristic face velocity, the two traces of face 0 and the numerical base flux. Columns that a branch does not define are NaN, with the reason in the header. The flux branch counters the extra Num_flux calls would move are saved and restored. MEASURED: on the review's 0.06 input the new columns reproduce the review's table (face 0: +3.28224e-4, 9995.84 times the wind flux; faces 12 and 13: pressure-jump parts -4.05157e-8 and -2.59041e-8 against central transport 3.28e-8), the continuity rows telescope from the exported face fluxes with a difference of 0, and the state files are the same with the export on and off (data identical; only the provenance time differs); hydrostatic_column with the export off is bitwise the binary before the change. The base block of that input shows the two interior pressures of face 0 that do not agree: the isentropic continuation the boundary compares with the reservoir, 0.999877 p0, and the reconstructed interior trace the flux sees, 0.999509 p0.

The discrete equilibrium tests. Two measurement programs in src/tests/grid_and_gates/ (base_reservoir_discrete_equilibrium.f90, low_mach_face_mass_flux.f90, a shared module base_equilibrium_columns.f90, input base_reservoir_equilibrium.inp; run.sh names base_reservoir_equilibrium and low_mach_face_flux), each on a ladder of three lower cell widths (2e-4 x 50, 1e-4 x 100, 5e-5 x 200 R_p), production Apply_BC, Reconstruct, Num_flux and assemble_residual with no sources. Columns: a_iso, a_warm built exactly in the reconstruction's discrete constant-density equilibrium (wb_dp_eq = 0) and anchored to the reservoir pressure at face 0; b_iso an analytic isothermal column; c_isen an analytic isentrope. MEASURED (16 PASS, 37 FAIL, the failures being the findings; the README marks both tests RED by design):

- The well-balanced interior keeps its discrete equilibrium to rounding (interior momentum rows 1.2e-14 and 4.6e-14 of their scale on a_iso, a_warm). The base does not: on a_iso the face-0 flux is -1.46e-4 of (rho c)_res, v_b = -2.6e-4 c_res, the ghost-0 / cell-1 mismatch is wb_dp_eq(0) = 2.9e-4 p, and the cell-1 momentum row is 3.1e-3 of its scale.
- These are truncation errors: the face fluxes converge at order 2.0, the cell-1 momentum row at order 1.0, on every column.
- Against a manufactured wind at Mach M they grow as 1/M: on a_iso the face-0 flux misses the wind by -0.136 at M = 1e-3, -14.6 at 1e-5 and -1.47e4 at 1e-8 (1.47e4, 3.61e3, 8.93e2 on the three grids), while the interior faces 2-10 stay within 5.3e-4 at every M. On b_iso (continuum but not discrete balance) the interior faces are off by 1.03 at M = 1e-8 as well.

Interpretation (not measured as such): two mechanisms act at face 0. The outgoing relation takes p_i from the isentropic continuation of cell 1, which differs from cell 1's own constant-density equilibrium at the face by about $(1/2 \gamma)(dr/2H)^2$ ($4.3e-4$ here) and drives v_b ; and the reconstructed right trace departs from p_{res} because WENO3 forms its departure from ghost 0, which lies on the reservoir isentrope and not on cell 1's equilibrium. Refinement cannot bring the base flux to 1 per cent of a Mach $1e-8$ wind (a base cell about 300 times narrower than the finest tested would be needed); the boundary has to share the reconstruction's discrete equilibrium (review step 3), and a seed has to be projected onto it (step 4).

Also corrected: the header of `hydrostatic_residual.f90` named a `SAVEd tol = 1.0` in `define_grid` as the reason one grid is built in one process; `define_grid` now sets `tol` before each search, and the reason is the operator's module arrays allocated at the first N. The grid-and-gates README count ("Twenty-one tests") and its list of names were stale and were corrected.

37. One discrete equilibrium at the base (2026-09-29)

Step 3 of `md/atomic_heh97_reduced_xuv_20260929_review.md` section 8. Under `Well balanced: True` the base condition reads the interior pressure at the level face as cell 1's own constant-density equilibrium, $p_i = P_{dn}(1) = p_1 + \rho_1 (\phi_c(1) - \phi_i(0))$, the pressure the reconstruction extrapolates cell 1 to (`base_boundary.f90`, `interior_state_at_level_face`), in place of cell 1's hydrostatic isentrope. The two ghosts continue the same equilibrium below the level, $P_{up}(0) = p_b$ and $P_{up}(-1) = P_{dn}(0)$, at the reservoir-side temperature (`base_ghost_averages`). Once the contact rule has chosen the face density ρ_b , the face velocity carries the interior's mass flux at that density ($v_b = v_i \rho_i / \rho_b + (p_b - p_i) / (\rho_i c_i)$): a stationary face with flow through it has no contact. The reservoir still states the pressure and entropy of inflowing gas; the equilibrium shared is mechanical, not thermal (Kaeppli & Mishra 2016, A&A 587, A94, section 2.1). Without the option the previous construction is unchanged. The model identity is `..._face_at_level_discrete_equilibrium_v5` with the option and the unchanged `..._face_at_level_v4` without it (`base_boundary_model_id`).

MEASURED (the tests of section 36): on columns built in the discrete equilibrium, v_b , the face-0 flux and the base momentum rows went from $2.6e-4$ c, $1.46e-4$ ρ c and $3.1e-3$ of the weight to below $1e-14$; the base face carries a manufactured wind of Mach $1e-3$ to $1e-9$ to $2.1e-3$ (isothermal column) and $2.5e-3$ (warming column), first order in the cell width, where the previous boundary missed it by $1.47e4$ at Mach $1e-8$. Carrying the flux at ρ_1 instead of ρ_b would have left 2.8 and 6.7 per cent; a temperature contact at the level with flux through it is not a stationary flow and was moved to a static-only test column. Analytic columns (continuum, not discrete, balance) now show their $O((dr/H)^2)$ offset at the base face (order 2.0). `grid_and_gates`: 361/37 -> 393/11 PASS/FAIL, the eleven all the analytic isothermal column. On the mapped atomic 0.06 seed the cell-1 row went $1.378 \rightarrow 1.206$ and the face-0 flux $1.0e4 \rightarrow 2.2e4$ times the wind: the seed's cell 1 lies $5.5e-4$ of p off the discrete equilibrium, which the isentrope had partly offset; the seed must be projected (section 38). Two continuation states certified under v4 ($K_{zz} = 1e9$ and well-mixed He/H 0.55, with the carrier enthalpy flux) re-certified under v5 in 2 and 4 passes, starting at $\|R\|$ 1.14 and 1.13. Also corrected: the v2 identity in `md/input_schema.md`, the v3 literal in `src/tests/boundary_state/ghost_composition_fixed_point.sh`, the missing `carrier_enth` token in `molecular_seed_of_an_atomic_state.sh`.

38. The atomic He/H 9.7 cases: the seed projection, and the element enthalpy flux as the remaining obstacle (2026-09-29)

The seed projection. Measurement key EXHALE_SEED_HYDROSTATIC_CELLS=K (src/modules/init/init.f90, seed_layer_in_discrete_equilibrium, off when unset, requires Well balanced: True): after load_IC, cell 1 is set so that its constant-density equilibrium reaches the reservoir pressure at the level face, cells 2 to K so that wb_dp_eq of faces 1 to K-1 vanishes, every cell above K (upper ghosts included) is scaled by the one factor that balances face K, all with T and the mass fractions of every cell held; the momentum of cells 1 to K carries the mass flux of cell K+1 and above K the velocity is divided by the density factor. MEASURED on the mapped 0.06 seed under v5 (LHS1140b/models/.P1/atomic97/diag_R/P<K>_v5_projected): mass and momentum rows 1.88 and $5.0e-3 \rightarrow 0.133$ and $3.2e-5$ (K = 50), $3.6e-2$ and $8.5e-6$ (110), $3.7e-3$ and $7.6e-7$ (200 and 300); deeper than local Mach $1e-3$ the neglected momentum flux shows ($1.5e-5$ at K = 400). The energy row stays at 1.05: it is not projected.

What the energy row is. From the budget export of the projected seed (cell, dF_energy, heat - cool, Sidf): cell 50, $8.6e-6$, $9.4e-6$, $+1.44e-5$; cell 200, $4.3e-6$, $4.6e-6$, $+5.4e-6$; cell 2, $5.0e-6$, $3.0e-5$, $-5.4e-4$. Without Sidf the energy row would be about 8 per cent of the heating from cell 3 up; the element interdiffusion enthalpy flux (section 29.3), absent when the seeds were solved, is 1.1 to 1.5 times the heating over $1.0 - 1.15 R_p$. It is physical: helium stays while hydrogen escapes, and the escaping hydrogen carries $(5/2) kT$ per particle relative to the barycentric velocity, an enthalpy flux of the order of the wind's own. Cell 2's spike is the helium diffusive flux through face 1, 2.8 times that of the faces above, because cell 2's helium fraction is about $1e-6$ off the diffusive steady state against the reservoir composition of cell 1; the seed's composition is not in its element steady state (the catalog refused these states on the elemental He/H row). With the term off, cell 1 still reads 0.94: the heat of the base cell, which the previous boundary's spurious face fluxes had been carrying away.

Solves. From the projected seeds with the term on (*_v5proj, EXHALE_710ab9ca.x), the hydrodynamic solve returned info = 2 on every pass with the mass and momentum rows unchanged, the energy row grew $1.09 \rightarrow 1.40$ over the element steps, and 0.06, 0.0624 and the photochemical 0.01 were refused by the progress control at pass 4 (the scalar 0.01 was stopped in its second pass). With the term off (*_v5proj_enthoff), the same seeds solve: hydro info = 0 on pass 1 in 83 s, ||R|| $5.1e-8$ (0.06) and $4.8e-8$ (0.0624); the elemental He/H row, $2.2e-2$ and $1.1e-2$ at pass 1, moves between $5.7e-3$ and $1.4e-2$ over passes 2 to 5, then falls by about a factor 2 a pass, and both states CERTIFY at pass 24 (He/H row $8.5e-6$ and $9.9e-6$ against $1e-5$; run logs of *_v5proj_enthoff). A marching test of the projected 0.06 seed (20000 steps, term on) settles at a mass-flux spread of 1.09 to 1.10, as before the boundary change.

The continuation factor. Measurement key EXHALE_INTERDIFF_ENTH_SCALE=s, $0 \leq s \leq 1$, 1 when unset (binary_element_diffusion.f90, interdiffusion_enthalpy_divergence, the one place both the stationary row and the marching update take the term from): the divergence is multiplied by s, so a state of the full equation ($s = 1$) can be reached from one solved without the term in steps. A state at $0 < s < 1$ is a step of that continuation, not a prediction.

39. The review of the 2026-09-29 handoff: state identity of the continuation factor, the projected seed, the output precision; the continuation runs (2026-09-29/30)

The review is `md/session_handoff_code_review_20260929.md` (evidence in `md/session_handoff_code_review_20260929_evidence/`). Every finding was checked against the code before anything was changed; the verdicts:

item	verdict
R1 continuation factor absent from the state identity	CONFIRMED: EXHALE_INTERDIFF_ENTH_SCALE changes the energy equation, the interdiff
R2 certificate carried onto a transformed state	CONFIRMED (the projection kept <code>ic_certified</code> ; so did a restart that changes a named option
R3 list-directed output loses precision	CONFIRMED with ifx 2026.1: 15 significant digits, up to 12 ulp on reload (gfortran: 0 ulp)
R4 projection without admissibility check	CONFIRMED on the review's synthetic grid (negative density and pressure returned)
R5 terminal success of the continuation scripts	CONFIRMED in <code>adaptive_enthalpy_scale_chain.sh</code> and in <code>xuv_descent_chain.sh</code> (th
Q1 element enthalpy budget over cells 2..N	not a defect: cell 1's composition is steady, so the reservoir supplies through face 0 the element
Q2 expected low-Mach limits reported as failures	improvement adopted

Checks (`ifx` and `gfortran` builds of the edited tree; only the paths the edits touch). `src/tests/interdiffusion_enthalpy_flux/run.sh`: 18 PASS. Gate tests `low_mach_face_flux` `output_state` `restart_round_trip` `restart_option_change` `restart_intent` `base_reservoir_equilibrium`: 120 PASS, 11 XFAIL, 0 FAIL. IC dumps (`ifx`) of scratch copies of the certified 0.06 term-off state: `plain` `certified=T` `certified_in_wind` `interdiff_enth=F`; projected (`K = 244`) `certified=F` `projected_seed`; `s = 0.125` with the term named `certified=F` `option_change_seed` `interdiff_enth=0.12500`. Dump, reload, dump: physical rows differ by at most 1 ulp in ρ , v , p , 6 ulp in T and 5 ulp in the species (the unit conversions and T from p , not the format); ghost velocity $1e-8$ (the boundary rebuilds the ghosts, as the reader does). The review's own probe, linked against the new objects: the projection is refused at cell 2 and nothing is adopted. The regression matrix was not run (user instruction of 2026-09-29).

What the correction leaves as it was. The binaries in use by the running chains (`LHS1140b/models/.P1/atomic97/EXHALE_acb07e93.x`) predate R1: every intermediate rung they write at $0 < s < 1$ states `interdiff_enth=T (*_adNN_s<value>/`, the value in the directory name and in `LAUNCH.txt`); those states are continuation steps and are not published. State files written before 2026-09-29 by `ifx` builds carry 15-digit values; they are left as they are.

The continuation runs (binary `EXHALE_acb07e93.x`, `ifx`; `lart3` and `lart4`):

- Element enthalpy flux factor from the certified term-off states, `adaptive_enthalpy_scale_chain.sh`: `s = 0.125` CERTIFIES at both 0.06 and 0.0624 (19 and 20 passes); from there a step of 0.1125 fails (hydro info = 2, energy row 0.27 to 1.8), and the chain at 0.06 advances in steps of about 0.02 to 0.03 (`s = 0.153`, 0.174 converged; 0.195, 0.206 failed).
- XUV descent at `s = 0` from the certified 0.06 term-off state, `xuv_descent_chain.sh` (new): 0.06 -> 0.05 fails (hydro info = 2 all 24 passes; the JFNK starts at `||R|| 0.18` with the energy row of the outer wind, `r = 15`, and stagnates near $1e-2$), the chain retries at `q = 0.913`.

- The photochemical case: the catalog's certified 0.10 state (previous grid, HLLC) mapped onto the face-at-level grid and projected ($K = 230$; the density moved by $1.5\text{e-}3$, against $5.2\text{e-}2$ on the unsolved 0.01 seed), solved with ROE and the term off: hydro info = 2 on all 9 passes, refused; loaded energy row 0.51 at cell 1 and 0.45 at $r = 13$. The same run with HLLC loads to four digits the same residual (energy 0.5075 at cell 1, first JFNK $\|R\|$ 0.4455 against 0.4457), so the flux choice does not carry the mismatch; it was stopped.
- The scalar 0.01 case from its projected seed with the term off (the run of section 38, lart3): hydro info = 2 on every pass; its mass and energy rows fell from $6.1\text{e-}2 / 1.1\text{e-}1$ at pass 1 to $5.35\text{e-}4 / 6.97\text{e-}4$ by pass 17 and stay there (passes 17 to 19: the hydrodynamic solve is not accepted), while the element row falls ($1.9\text{e-}4$ at pass 19).

40. Why the atomic He/H 9.7 continuation fails: two mechanisms, and the heat conduction the models omit (2026-09-30, open)

Diagnosis of the failed rungs of section 39, read from their run logs (LHS1140b/models/.P1/atomic97/*_adNN_*, *_xd01_*); every number MEASURED.

A. The alternation between the hydrodynamic solve and the element step diverges (0.06, $s = 0.174$ to 0.2375).

Each hydrodynamic solve converges (info = 0, quadratic in its last iterations, $\text{lam} = 1$, the linear solve at its tolerance in 1 to 6 products), but the elemental He/H row grows from pass to pass and the under-relaxation of the element step does not stop it; the hydrodynamic solve fails only once the composition has moved far:

```
| s = 0.195, pass | hydro start ||R|| | hydro | He/H row | omega | |---|---|---|---| | 1 | 0.092 | info = 0 | 9.0e-3 | 0.5 | | 2 |
0.17 | info = 0 | 2.3e-2 | 0.5 | | 3 | 0.32 | info = 0 | 1.4e-1 | 0.25 | | 4 | 0.72 | info = 0 | 3.8e-1 | 0.125 | | 5 | 0.39 | info = 2 |
3.5e-1 | 0.125 |
```

The rung at $s = 0.174$ that the chain accepted ended at pass 14 with the He/H row at 0.51 (hydro info = 0): the acceptance on hydro info=0 alone (section 39, R5) carried an unconverged composition into every later rung. The last states of the 0.06 chain that are converged are $s = 0.125$ (certified) and $s = 0.153$ (He/H row $1.05\text{e-}4$, not certified); the chain stopped at 01:07 with its step below 0.005.

B. The stationary solve does not move from its start (0.0624 from $s = 0.125$ to 0.2375 ; the XUV step $0.06 \rightarrow 0.05$ at $s = 0$). The composition converges at the held hydrodynamics (He/H row $9.9\text{e-}6 \rightarrow 2.7\text{e-}8$ over 12 passes), but every pass starts the solve at the same $\|R\|$ (0.2671; 0.183) and ends there. Within a solve the pseudo-time steps are accepted at $\text{lam} = 1$ while dtau doubles; near $\text{dtau} \sim 2\text{e}3$ a step overshoots ($\|R\|$ $0.06 \rightarrow 0.24 \rightarrow 0.28$, the ionization root of 1 to 2 cells outside the simplex in the trial evaluations), dtau is cut, $\|R\|$ returns to 0.053 while the fixed-scale merit rises, and after 40 iterations without improvement of either functional the solve stops and returns its best judged iterate, which is the entry state (every move raises the continuity row against its $3\text{e-}12$ tolerance). The linear model is not the obstacle (the cycle reaches its tolerance in one product); the path the pseudo-transient has to take is longer than the solve's stagnation window.

The energy equation of these models omits heat conduction ($\text{cond}=\text{F}$, the default). Evaluated on the stored states with the conductivity the code carries (viscous_conduction.f90: Watson, Donahue & Walker 1981 neutral atomic hydrogen, $4.45\text{e}4 (T/1000 \text{ K})^{0.7} \text{ erg cm}^{-1} \text{ s}^{-1} \text{ K}^{-1}$), $|(1/r^2) d/dr(r^2 \kappaappa dT/dr)|$ against the local heating, and the conductive flux F_c against the advected enthalpy flux of the wind

(LHS1140b/models/.P1/atomic97/conduction_vs_heating.py <Hydro_ioniz.txt>, from the columns of the state file and R0 of its grid line):

state	r = 1.001	1.01	1.05	1.1	1.2	1.5	2.0 R _p
atomic scalar 0.06 He/H 9.7 (section 38, certified)	99	55	5.2	51	42	22	17
atomic photochem 0.10 He/H 9.7 (catalog)	67	31	3.6	30	23	9.0	7.1
atomic scalar 0.33 He/H 9.7 (catalog)	15	8.0	2.5	7.9	6.8	2.9	1.2
atomic scalar 1.0 He/H 9.7 (catalog)	4.1	2.1	0.8	2.3	1.9	1.0	0.34
atomic scalar 1.0 He/H 0.55 (catalog)	0.8	1.1	0.5	0.5	0.6	0.4	0.06
molecular photochem He/H 9 (catalog)	73	4.0	1.6	1.5	0.02	1.1	0.45

($\text{conduction} / \text{heating}$.) On the 0.06 state F_c is 1.5 to 3.5 $\text{erg cm}^{-2} \text{s}^{-1}$ from the base to 1.3 R_p against an advected enthalpy flux of 0.014 to 0.15; on the fiducial atomic He/H 9.7 state 2 to 3.4 against 0.3 to 2.8. The lower layers of the LHS 1140 b catalog are therefore solved without a term of the energy equation that is of order one and, at reduced XUV, one to two orders above the heating. *Likely* this is also why the reduced-XUV models are so stiff: without conduction the temperature of a layer whose wind runs at Mach $1e-8$ is set by the heating divided by the mass flux, and the element enthalpy flux, of the order of the wind's own, moves it by a large factor (section 38). Not established: whether the temperature structure with conduction differs by as much as the ratio suggests; the published models that include conduction (Koskinen et al. 2013a, 2022; Yelle 2004) state it as small for their hot-Jupiter cases, whose base wind is 2.5 km/s (Koskinen 2022) and not Mach $1e-8$.

Conductivity of a helium-dominated gas. The code applies the atomic hydrogen value whatever the composi-

tion. Yelle (2004), eqs. 10-11, forms the neutral conductivity of a mixture from the constituents' values (Banks & Kockarts 1973) with a Wilke-type rule; at He/H = 9.7 the helium value decides it. Banks & Kockarts (1973) is not in references/; the value is not adopted until it is read in the original.

Test. From the certified 0.06 term-off state, Conduction: True switched on at once (binary EXHALE_a1c14b9c.x, tree source of section 39; *_cond_enthoff, *_cond_enthon with the element enthalpy flux on at s = 1): the loaded ||R|| is 1.00 and 1.84 (the conduction term alone), and the first stationary solve of each STAGNATED after 90 iterations and returned the entry state. A state with conduction is not reached from one without it in one solve either; both runs were stopped after their first pass.

Open, for the user: whether conduction becomes part of the LHS 1140 b model (it changes every model of the catalog), with which conductivity for the helium-rich mixtures, and by which route the states with it are reached (a continuation factor on kappa as on the element enthalpy flux, or time marching with the implicit conduction update).

41. Heat conduction with the conductivity of the mixture, and its continuation factor (2026-09-30)

User decision of 2026-09-30: heat conduction enters the LHS 1140 b model, reached by continuation in the conductivity; the no-conduction runs were stopped (the 0.0624 s chain had ended at s = 0.181, the XUV descent was stopped at x = 0.0573, rung 4; the scalar 0.01 run had been refused at pass 20).

The conductivity (viscous_conduction.f90, section 2, Eq. 4'), read in the published versions:

$$\kappa = (1/n) [n_e 1.2e-6 T^{2.5} + n_{\text{HI}} 379 T^{0.69} + n_{\text{HeI}} 299 T^{0.69}] \quad \text{erg cm}^{-1} \text{ s}^{-1} \text{ K}^{-1}$$

Salz et al. (2015, A&A 576, A21), section 3.4.2, Eqs. 24-26 (electron-ion, Spitzer; neutral hydrogen; the number-weighted sum), and Sutton et al. (2015, JGR 120, 6884), Eq. 5 (the helium coefficient $299 T^{0.69}$ in the same weighted form); both quote Banks & Kockarts (1973), which was not available. n counts every particle (electrons, atoms, ions, metals); the ions add nothing of their own (Yelle 2004, Eq. 9, gives $7.4e-8 T^{2.5}/\sqrt{\mu}$, below $1e-2$ of κ_{H} at $1e4$ K). The previous coefficient, Watson et al. (1981) $4.45e4 (T/1000 \text{ K})^{0.7}$ (verified in Erkaev et al. 2016, Eq. A6), was the neutral hydrogen value applied to any composition; the hydrogen term now is Salz Eq. 25, within 2 per cent of it from 300 to 7000 K (1.013 at 300 K, 1.0006 at 1000 K, 0.981 at 7000 K). On the 0.06 state the mixture coefficient is 0.81 of κ_{H} where helium is 90 per cent of the particles. The viscosity keeps the hydrogen value (5); it is not used by any LHS 1140 b case. None of the three sources gives an H₂ coefficient; the H₂ term is $n_{\text{H}_2} k_{\text{H}_2}(T)$ with k_{H_2} the tabulated conductivity of Incropera et al. (2007, Fundamentals of Heat and Mass Transfer, 6th ed., Table A.4), 100 to 2000 K, linear in (ln T, ln k) and continued outside with the exponent of the end interval (an extrapolation), h2_conductivity (the user supplied the book, 2026-09-30; Holman, Heat Transfer, 10th ed., Fig. 1-4, shows H₂ and He only up to 500 C). The same table gives helium 0.152 and 0.354 W m⁻¹ K⁻¹ at 300 and 1000 K, against $299 T^{0.69} = 1.53e4$ and $3.51e4$ erg cm⁻¹ s⁻¹ K⁻¹; the helium coefficient of Sutton et al. agrees to 1 per cent. (A first version of this entry stopped molecular runs with conduction; that stop is removed.) The table was first interpolated in (ln T, ln k); at the user's request (2026-09-30) it is replaced by a fitted function, $\kappa_{\text{H}_2} = A T^s [1 + c E(\theta_v/T)]$, $E(x) = x^2 e^x/(e^x - 1)^2$, $A = 272.523$, $s = 0.735652$, $c = 0.370446$, $\theta_v = 5987$ K (the $v = 0 \rightarrow 1$ spacing of the Roueff et al. 2019 ladder the code carries); the Eucken form, fitted over 200-2000 K, largest deviation

2.5 per cent (200 K), rms 1.05 per cent; valid 200-2000 K, an extrapolation outside (+7.6 per cent at 150 K, +20 per cent at 100 K where rotation freezes out). Other forms measured on the same table: a power law 6 per cent, Sutherland 2 to 12 per cent, a cubic in $\ln T$ 1 per cent (untrustworthy outside). The fit has a continuous derivative (the interpolation kinked at 25 nodes); above 2000 K it lies 3 and 6 per cent below the end-slope continuation at 3000 and 4000 K, neither being a measurement. Linked probe: 200/300/1000/2000 K give 1.025, 0.989, 1.013, 0.986 of the table. No run with molecules and conduction has been made with either form. Yelle (2004) Eq. 10, the Wilke-type mixture rule, was not used: as printed it is not dimensionally the Wilke form (a factor $1/r$ in the sum), *likely* a typesetting error.

Code. `thermal_conductivity(Tcell, f_sp, kap)` forms the number fractions from the composition of the same cells (`calc_ne`, `calc_ntot`: the equation of state's own counts), and the composition is passed down `thermal_conduction_coeffs`, `thermal_conduction_source`, `viscous_conduction_sources` (steady residual) and `viscous_conduction_step` (marching); the test driver `src/tests/energy_update/conduction_floor_tests.f90` passes a neutral-hydrogen composition. Continuation factor `EXHALE_CONDUCTION_SCALE = s` (`conduction_scale()`), 1 when unset; the `cond` token of the options field carries `s` wherever `s /= 1`, so a restart across a change of `s` names `cond` (the rule of section 39, R1); the input echo and the setup report state the coefficient and the factor.

Checks. `gfortran` and `ifx` builds; `src/tests/energy_update/run.sh`: 54 PASS, 0 FAIL. Restart intent: stationary evaluate of the certified 0.06 term-off state with Conduction: True and the budget export: the conduction term of the energy row (Sene, viscosity off) over the heating agrees with an independent evaluation of Eq. 4' from the written columns to five digits at every radius tested (-46.04 at 1.003, +41.40 at 1.10, -34.07 at 1.20, +12.42 at 3.0 R_p).

Continuation, first attempt (`.P1/atomic97/factor_continuation_chain.sh`, new: a rung is accepted only when certified, the chain ends successfully only at `s = 1` certified with exit 0, every rung names its token; binary `EXHALE_0c5924c7.x`): from the certified term-off 0.06 and 0.0624 states, the first rung `s = 0.005` returns its entry state from every hydrodynamic solve (hydro info = 2, $\|R\|$ 1.000 at the energy row of cell 1). The row of cell 1 is dominated by the conduction through the base face, where the operator holds the reservoir's temperature (Dirichlet, `conduction_base_level_T`) and the no-conduction states carry a large temperature jump across half a $2e-4 R_p$ cell; since the row is measured against its own largest term, its relative residual stays at 1 for any `s`. Both chains were stopped at pass 6 of rung 1. A smaller factor cannot help; the obstacle is mechanism B of section 40.

Mechanism B measured (`.P1/atomic97/diag_R/B1_jump_scan_cond0.005`, `EXHALE_RESID_JUMP_SCAN=1` at the entry state of that rung): along the first Krylov direction the residual is smooth from 2 arcs down to $3e-5$ arcs of the probe step; below that the only steps are rounding (the largest moves its row by $4e-7$ of it, 0.6 to 1.7 times the arithmetic bound $\epsilon \times \text{face state} / \text{width}$). A discontinuous residual is therefore not what rejects the steps of $\text{lam } 1e-6$ at $\text{dtau } 8e-5$. At such dtau the step is $dY \sim -\text{dtau } F$, the explicit pseudo-time step, and its rejection means the merit (the scaled 2-norm of F) rises along the physical relaxation direction. *Likely* this is the obstacle: the pseudo-transient continuation is globalized by a merit line search, which can refuse a relaxation that is stable in time but not monotone in that norm. Kelley & Keyes (1998, SIAM J. Numer. Anal. 35, 508), abstract: "Standard globalization strategies such as line search or trust region methods often stagnate at local minima. Ψ_{tc} succeeds in many of these cases by taking advantage of the underlying PDE structure of the problem"; their pseudo-time step is set by the residual reduction (SER, their Eq. 1.5),

and they report a nonmonotonic steady-state residual norm at the outset of the iterations for iterates far from the steady state. Changing the acceptance of the solve is a solver change and is held for the user (the stage 2 solver program is closed, 2026-09-10).

42. Pseudo-transient steps accepted by the pseudo-time control (2026-09-30)

User decision of 2026-09-30: the acceptance of the pseudo-transient solve may be changed (the stage 2 solver program is reopened for this item).

What was changed (steady_newton.f90, solve_steady_jfnc; measurement key, off by default, so no solve without it moves). EXHALE_PTC_SER=1 | 2: a trial of the line search is accepted when it is admissible (the existing admissible test of eval_residual: positive, finite, a chemical root in every cell) and no cell's density or total energy moves by more than EXHALE_PTC_SER_MAX_CHANGE (default 0.3) of itself; otherwise lam is halved. No merit descent is asked. The returns to the best iterate after 20 iterations without improvement, the stagnation stop after 40, the return after a run of damped steps, and the damped Gauss-Newton escape replacing an accepted step are not taken; the solve ends at its gates, at its iteration cap or after 12 consecutive iterations without an admissible trial, and hands back its LAST iterate unless the best one met the gates (its iterates are a relaxation path, not a search). The pseudo-time: =1, SER (Kelley & Keyes 1998, Eq. 1.5), $\text{dtau} \leftarrow \text{dtau} \min(2, \max(1/2, f2/f2_try))$; =2, doubled on a full step; both times lam on a shortened step. The outer alternation adopts the returned state whatever info is (EXHALE_main, after solve_steady_jfnc), so progress carries over.

Measured (.P1/atomic97/diag_R/SER/, one outer pass, cap 400 iterations, binaries EXHALE_d0d5bf27.x (=1) and EXHALE_9d0b8d4c.x (=2); every fixture returned its entry state under the merit line search):

fixture	merit line search	SER=1	SER=2
T1: 0.06, conduction factor 0.005 (section 41 rung 1)	entry state,		R
T3: 0.0624, no conduction, element enthalpy factor 0.125 -> 0.2375 (section 40, B)	entry state,		R
T2: 0.06, conduction factor 1 at once	entry state		

Under =2 every step of T1 is taken at lam = 1 with dtau doubling from 2 to 4.2e6; T3's ||R|| rises from 0.06 to 1.2 at iterations 10 to 16 (a nonmonotone transient the merit line search refused) and then converges quadratically. =1 keeps dtau near its entry value because the merit falls by about a per cent per step. =2 is the one used from here. Pass 1 of T3 leaves the elemental He/H row at 0.23; mechanism A (the diverging hydro / element alternation, section 40) is a separate question, which the continuation chains meet with their certified-only acceptance.

Continuation restarted (.P1/atomic97/launch_conduction_then_enthalpy.sh, lart3, CHAIN_ENV=EXHALE_PTC_SER=2): from the certified term-off 0.06 and 0.0624 states, conduction by

EXHALE_CONDUCTION_SCALE ($S_0 = 0.005$), then, from the certified full-conduction state, the element enthalpy flux by EXHALE_INTERDIFF_ENTH_SCALE ($S_0 = 0.05$); logs <case>_ser2_cond_chain.log, <case>_ser2_cond_interdiff_enth_chain.log.

43. The continuation chains, first three hours, and their step control (2026-09-30)

With EXHALE_PTC_SER=2 (section 42) every conduction rung below $s \sim 0.14$ certified in 7 to 29 passes (5 to 15 minutes): 0.06 up to $s = 0.0673$, 0.0624 up to 0.137, the photochemical 0.10 up to 0.0673 (its term-off state had certified first, in 9 passes, where the merit line search had refused it). Two defects of the chain script were measured and fixed (.P1/atomic97/factor_continuation_chain.sh):

- the step ratio grew as $q^{1.5}$ on every success, so 0.06 jumped from 0.0673 to 0.698 ($\times 10.4$) and the photochemical case to 0.32; a rung that fails under EXHALE_PTC_SER=2 runs to the solve's iteration cap (3000, the stagnation stop being off there), 2 h 20 min for the 0.06 rung (hydro info = 2, elemental row 0.89). Now the ratio is capped ($Q_CAP, 2$) and every solve of a chain carries EXHALE_JFNK_MAXIT=400 (the certified solves take 22 to 28 iterations);
- 0.0624 was refused twice at 30 passes with the elemental row at $1.35e-5$ and $1.07e-5$ against $1e-5$, i.e. still converging; a refused rung whose hydrodynamic solve converged and whose elemental row is below NEAR ($1e-4$) is now continued once at the same factor from its own state before the step is shortened. The factor at which the alternation of 0.0624 diverges (mechanism A) is between 0.137 (certified) and 0.225 (refused, elemental row $7.8e-2$ at pass 5). The chains were stopped at 06:35 and restarted from their last certified rungs (S_START) by one wrapper for all three cases, .P1/atomic97/launch_full_equation_from.sh (conduction to 1, then the element enthalpy flux, then for 0.06 and the photochemical case the XUV descent to 0.01).

The cond and interdiff_enth tokens carry the factor in F7.5, five decimals: two factors that differ below $1e-5$ carry one token.

44. The outer under-relaxation sets the pace of the chains: omega = 1 (2026-09-30)

In every certified conduction rung the elemental He/H row falls, after a transient of 5 to 8 passes, by 0.505 per pass (three rungs tabulated: photochemical 0.32 $\rightarrow$ 0.453, scalar 0.0624 0.194 $\rightarrow$ 0.306, scalar 0.06 0.190 $\rightarrow$ 0.247), which is $1 - \omega$ at the default $\omega = 0.5$ of the element step (EXHALE_main, comp_omega): the undamped alternation contracts by about 0.01 per pass there, and the damping alone makes the 21 to 22 passes of a rung. The existing key EXHALE_DIFF_OMEGA starts omega at the value given (the halving on a pass without progress stays). Three known rungs rerun at $\omega = 1$, everything else as the chains (.P1/atomic97/diag_R/omega1/):

rung	$\omega = 0.5$ (chain)	$\omega = 1$
photochemical 0.32 $\rightarrow$ 0.453	certified, 22 passes, 1 h 53 min	certified, 6 passes, 40 min
scalar 0.06 0.190 $\rightarrow$ 0.247	certified, 21 passes	certified, 6 passes, 51 min
photochemical 0.453 $\rightarrow$ 0.761	refused at pass 5 (He/H row 0.099, mechanism A)	certified, 7 passes, 48 min (one pass at hydro info = 2 halved c

At $\omega = 1$ the He/H row falls by about ten a pass ($6.5\text{e-}2$, $3.2\text{e-}2$, $3.8\text{e-}3$, $3.4\text{e-}4$, $2.8\text{e-}5$, $2.3\text{e-}6$). *Likely* part of mechanism A (section 40) was the damping itself: an element step held back by half while the hydrodynamic solve answers in full. The chains were stopped at 13:09 and restarted with EXHALE_DIFF_OMEGA=1.0 added to their CHAIN_ENV (launch_full_equation_from.sh) from the highest certified states: photochemical 0.761 (the $\omega = 1$ rung), scalar 0.0624 0.410, scalar 0.06 0.300. The default of comp_omega is not changed.

45. The first state with heat conduction: a conductive thermosphere below an exobase at 1.5 R_p (2026-09-30, open)

The photochemical K_{zz}(p) He/H 9.7 case at 0.10 of the fiducial XUV reached conduction factor 1 (the equation: Conduction: True, mixture kappa of section 41, element interdiffusion enthalpy flux still off) certified at 13:27 (.P1/atomic97/atomic_photochem_gj1132x0.10_kzzprofile_HeH9.7_ser2_cond01_s1, 14 passes, 17 min, from the $\omega = 1$ rung at 0.761). Against the certified state of the same case without conduction (..._v5proj_enthoff_ser2), MEASURED (.P1/atomic97/compare_conduction_on_off.py, figure .P1/atomic97/diag_R/cond_on_off_photochem_x010.png):

r [R _p]	T [K] off -> on	n [m _H cm ⁻³] off -> on	v [cm/s] off -> on
1.01	1041 -> 494	$6.4\text{e}12$ -> $4.4\text{e}12$	$4.0\text{e-}2$ -> $2.1\text{e-}4$
1.10	3463 -> 940	$7.2\text{e}10$ -> $1.2\text{e}9$	3.0 -> 0.64
1.50	7246 -> 933	$1.2\text{e}9$ -> $2.0\text{e}6$	94 -> 210
3.0	3182 -> 900	$2.8\text{e}7$ -> $1.3\text{e}3$	$1.0\text{e}3$ -> $7.9\text{e}4$
10	740 -> 719	$2.9\text{e}5$ -> 19	$8.7\text{e}3$ -> $4.9\text{e}5$

The thermosphere is nearly isothermal at 900-940 K from 1.05 to 5 R_p (the XUV heat conducted down to the 226 K reservoir), in place of a 7290 K peak at 1.43 R_p; the mass-loss rate falls from $7.3\text{e}6$ to $2.7\text{e}4$ g/s (median over $r \geq 1.1$ R_p, $\times 0.004$); the He(2³S) density falls by decades above 1.1 R_p. src/utls/collisional_validity.py on the two states: without conduction the domain is subsonic (max Mach 0.07) with Kn_{bulk} 0.40 at the outer boundary; WITH conduction the exobase (Kn_{bulk} = 1) is at 1.52 R_p, Kn_{bulk} = 0.1 at 1.30 R_p, the critical point at 8.8 R_p, and Kn reaches 859 in the acceleration region: the hydrodynamic Mdot (log 4.41) is UNVALIDATED, and Jeans escape through the exobase (T = 932 K, $\lambda_J = 17$) gives 9.8 g/s. *Likely* the low-XUV LHS 1140 b atmosphere with conduction is a conduction-limited thermosphere in the Jeans regime, the structure of the Earth's thermosphere, and not a hydrodynamic wind; the continuum solution above ~ 1.3 R_p is outside its validity. Not established: the fiducial-XUV cases with conduction (not yet solved), and the element enthalpy flux, whose continuation from this state is running.

46. Heat conduction in the fiducial-XUV cases (2026-09-30, running)

User decision of 2026-09-30: beside the low-XUV chains, switch conduction on in the fiducial-XUV cases the observations are compared with, to decide the direction of the catalog. Started 14:16 on lart4

(.P1/fiducial_conduction/launch_fiducial_conduction.sh, binary EXHALE_e37b8e27.x with the H2 conductivity fit, EXHALE_PTC_SER=2, cap 400, omega = 1):

- molecular_scalar_gj1132_kzz1e9/HeH1.45 (the He/H at which the model EW crosses the observed 1.108, section 14 of this log) and molecular_photochem_gj1132_kzzprofile/HeH9: certified 2026-09-29 on the face-at-level grid with the element and carrier enthalpy fluxes on (boundary model v4; loaded by the v5 binary as a seed); conduction by its factor from 0.005 to 1 (factor_continuation_chain.sh; rungs and logs in .P1/atomic97/).
- atomic_scalar_gj1132_kzz1e9/HeH9.7 (previous grid, 2026-09-20): mapped, 230-cell projection, term-off solve, then conduction and the element enthalpy flux by launch_full_equation_from.sh (whose fixed binary EXHALE_9d0b8d4c.x differs from e37b8e27 only in the H2 conductivity, absent in an atomic run). Entry inputs carry the spectrum file by absolute path (the catalog's .././../sed/ does not resolve from .P1/).

47. The photochemical 0.10 case on the full equation (2026-09-30)

From the certified conduction state of section 45, the element interdiffusion enthalpy flux by its factor (0.05 -> 1, interdiff_enth chain, omega = 1): rungs 0.2, 0.4, 0.8 and 1 certified in 10 to 15 passes each; the full equation (conduction and the element enthalpy flux, factor unset) certified at 14:22 (.P1/atomic97/atomic_photochem_gj1132x0.10_kzzprofile_HeH9.7_ser2_cond_interdiff_enth06_s1).

Against the conduction-only state, MEASURED (compare_conduction_on_off.py): Mdot 2.711e4 -> 2.725e4 g/s (+0.5 per cent), T within 0.5 K at every radius tested (943.4 K at 1.2 R_p), n and v within 0.5 per cent; collisional_validity.py: exobase 1.518 R_p, critical point 8.81 R_p, Kn up to 855 in the acceleration region, Jeans escape 9.88 g/s; HYDRODYNAMIC RESULT UNVALIDATED as in section 45. With conduction on, the element enthalpy flux, which without it was 1.1 to 2 times the heating of the lower layer (section 38) and the hardest term to switch on, moves the solution by less than a per cent: conduction carries the energy of that layer. The XUV descent 0.10 -> 0.01 on the full equation started from this state (atomic_photochem_gj1132_kzzprofile_HeH9.7_full_xuv_descent.log).

48. The fiducial-XUV atomic case with conduction (2026-09-30)

atomic_scalar_gj1132_kzz1e9/HeH9.7 (fiducial XUV), mapped from the previous grid and projected (section 46): the term-off solve certified in 2 passes; conduction by its factor 0.005 -> 1 certified in 9 rungs of 1 to 2 passes each, 14 minutes, reaching the equation at 14:34 (.P1/atomic97/atomic_scalar_gj1132_kzz1e9_HeH9.7_fid_ser2_cond09_s1; the element enthalpy flux continuation follows). Against the term-off state, MEASURED (compare_conduction_on_off.py, figure .P1/atomic97/diag_R/cond_on_off_atomic_scalar_fiducial.png; collisional_validity.py):

	no conduction	conduction
T max	7826 K at 1.434 R_p	6342 K at 1.419 R_p
T at 1.05 / 1.5 / 3.0 R_p	2433 / 7784 / 4365 K	2798 / 6298 / 3741 K

	no conduction	conduction
Mdot (median $r \geq 1.1$)	7.57e7 g/s	5.25e7 g/s (x0.69)
critical point	none in domain (max Mach 0.508)	none (max Mach 0.51)
Kn_bulk at the outer boundary	0.060	0.066

At the fiducial XUV conduction redistributes the heat (the lower layer warms, the peak cools by 19 per cent) and lowers the mass-loss rate by 31 per cent; the solution stays a subsonic continuum wind with the exobase above the domain, UNVALIDATED on the same grounds as before (Kn 0.7-1.1 in the acceleration region, no critical point in the domain). At 0.10 of that XUV (section 45) the same term turns the wind into a conductive thermosphere with an exobase at 1.5 R_p . The effect on the He 10830 equivalent width is not yet computed.

49. The advected temperature correction keeps the transport terms (2026-09-30)

Defect. The advection-corrected profiles (post_process_adv, the *_adv.txt files the He 10830 synthesis reads) re-derive the temperature from the steady internal-energy equation (E), $\text{div}(\mathbf{u} \cdot \mathbf{v}) + p \text{div} \mathbf{v} = \text{heating} - \text{cooling}$, which holds neither heat conduction nor the enthalpy flux of the element and carrier fluxes. On a run that solved them the corrected temperature is then the answer to another equation: MEASURED on the fiducial atomic case on the full equation (section 48), 7939 K against the run's 5489 K at 1.2 R_p and 8479 K against 6259 K at 1.5 R_p .

Fix (user decision 2026-09-30: the consistent form, conduction of the corrected temperature). T_equation takes $\text{heating} = \text{photoheating} + \text{heat_extra} + \text{cond_diag} \ T$ (new teq_state fields, zero for a run without the terms: the residual is unchanged); post_process_adv fills them from the conduction triplets of the run's operator (thermal_conduction_coeffs, now public) and the enthalpy flux divergence of the element (interdiffusion_enthalpy_divergence_of_state) and carrier (carrier_enthalpy_divergence) fluxes. Conduction couples each cell to both neighbors, so after the marching sweep (cell below at its corrected value, cell above at the pass's profile) the corrected cells are solved together by a Newton iteration on the column (energy_column_newton: banded Jacobian, two sub- and one super-diagonal, by one-sided differences; the conductivity and the enthalpy flux divergence re-evaluated at each iterate; steps capped at 30 per cent; converged at a relative step of 1e-10; a solve that fails returns the marching profile and is reported). Cells that keep the run's temperature are fixed values of the solve.

Checks. gfortran and ifx builds; src/tests/adv_static_limit: 53 PASS, 0 FAIL; src/tests/energy_update: 54 PASS (section 41). A run without the terms (the fiducial term-off state): the column solve does not run, the solved state is bitwise unchanged, and the *_adv rows move by at most 1.4e-11 relative (the ifx build re-associates the heating sum; below the standing 0.1 per cent rule). The full-equation fiducial state: the column solve converges in 4 to 7 iterations each pass (relative step 5e-11, 502 cells); the corrected temperature follows the run's where conduction carries the energy (1129/5507/6346 K against 1127/5489/6259 K at 1.01/1.2/1.5 R_p) and departs from it beyond 3 R_p , where the non-equilibrium composition changes the balance (figure .P1/atomic97/diag_R/adv_transport_terms_fiducial.png).

He 10830 EW (WINERED kernel, the catalog's operator, .P1/fiducial_conduction/ew_of_state.sh; observed 1.108 +/- 0.029):

state	adv	solution (no advection correction)
term-off (no conduction, no element enthalpy flux)	1.573	12.99
full equation, adv without the transport terms	0.946	7.52
full equation, adv with them	0.649	7.52

On this case heat conduction and the element enthalpy flux lower the model EW from 1.57 to 0.65 (x0.41); the He/H at which the model EW crosses the observed value (1.5 in the catalog, section 14) will move. The previous EW of every conduction run is superseded by this operator.

50. A floor of the energy row near convergence (2026-09-30; located in section 53, explained and removed in section 58)

Two chains slow down on rungs whose hydrodynamic solve ends at the 400-iteration cap (info = 1): the 0.0624 conduction rung at factor 0.751 (He/H row $1e-7$, energy row $1e-5$ to $5e-5$ against its tolerance $1e-6$ for twelve passes of 140 s each; the rung then CERTIFIED at pass 15, 15:44, when one pass's state fell inside every tolerance) and the first XUV descent rung of the photochemical case ($0.10 \rightarrow 0.0833$, passes of 900-1000 s, running). The rungs are not stuck but their certification rests on a fluctuation. In both the solve reaches $\|R\| \sim 5e-5$ within 25 iterations and then stays between $5e-6$ and $5e-5$ for the rest of the cap, at $d\tau = 1e14$, with every step accepted.

Tried: EXHALE_PTC_SER_RNORM (default $1e-3$), the pseudo-time acceptance of section 42 only while $\|R\|$ is above the switch and the merit line search below it (binary EXHALE_1543a8f2.x). The two fixtures of section 42 still certify in 22 and 28 iterations (T1 $\|R\| 3.5e-7$, T3 $1.5e-7$). On the 0.0624 rung $0.579 \rightarrow 0.751$ (.P1/atomic97/diag_R/ser_phase/P1_*) the transient is crossed as before ($\|R\| 1.9 \rightarrow 5.7e-5$ in iterations 19 to 24) and the merit phase then wanders at the same $1e-5$ to $5e-5$ for 376 iterations, the Krylov forcing term pinned at its ceiling 0.1 (the residual not falling, the Eisenstat-Walker rule does not tighten it). The stall is therefore not the acceptance rule. *Likely* the energy row carries a floor of that size on this state (candidates: the tolerance of the composition elimination inside the residual, an ionization root switching branch); not measured. The test was stopped after pass 1. The switch is kept (it changes nothing on the fixtures and is the standard terminal phase), with this record that it did not cure the stall.

51. The molecular photochemical fiducial case with conduction (2026-09-30; EW corrected by section 54)

molecular_photochem_gj1132_kzzprofile/HeH9 (fiducial XUV; the molecular network with the element and carrier enthalpy fluxes, section 46) reached conduction factor 1 certified at 16:22 in 9 rungs of 10 to 15 passes (.P1/atomic97/molecular_photochem_gj1132_kzzprofile_HeH9_cond09_s1; binary EXHALE_e37b8e27.x, the first molecular run with the H2 conductivity). Against its catalog state (no conduction; boundary model v4, re-measured by the current binary, whose evaluate refuses the old certificate, exit 2, as expected across the boundary change), with the advected profiles of section 49 (binary EXHALE_1543a8f2.x) and the WINERED operator, MEASURED:

	no conduction (catalog)	conduction
T max	7202 K at 1.525 R _p	6194 K at 1.441 R _p
T at 1.001 / 1.1 / 1.5 / 3 R _p	985 / 2878 / 7197 / 4350 K	444 / 3788 / 6168 / 3732 K
Mdot (median r >= 1.1)	9.78e7 g/s	5.95e7 g/s (x0.61)
He 10830 EW, adv	1.836	1.103
He 10830 EW, solution	14.81	8.56
critical point / Kn at the outer boundary	none (Mach 0.57) / 0.059	none (Mach 0.53) / 0.065

(Figure .P1/atomic97/diag_R/cond_on_off_molecular_photochem_fiducial.png.) The observed EW is 1.108 +- 0.029 (he10830_equivalent_width.py). **Correction (section 54):** in this evaluation all ten column energy solves of the advected profile were rejected (marching profile kept: T, .P1/fiducial_conduction/ew_M2_s1/eval.log). The 1.103 is therefore the EW of the retained marching profile, not of a converged corrected-energy profile, and its agreement with the observation is not established. With the atomic fiducial case (section 49, 1.57 -> 0.65) conduction lowers the model EW by a factor 0.4 to 0.6 at the fiducial XUV, so the catalog's inference of the He/H that reproduces the observation (the crossing near 1.5 of section 14, molecular scalar family) no longer holds. Where the crossing now lies is undetermined: this case and the molecular scalar He/H 1.45 case of section 52 belong to different families. Both hydrodynamic results stay UNVALIDATED on the grounds of section 48 (no critical point in the domain, Kn up to 1 in the acceleration region).

52. The molecular scalar He/H 1.45 case with conduction (2026-09-30)

molecular_scalar_gj1132_kzz1e9/HeH1.45 (the He/H at which the catalog's model EW crossed the observed value, section 14) reached conduction factor 1 certified at 16:39 (9 rungs of 12 to 20 passes; .P1/atomic97/molecular_scalar_gj1132_kzz1e9_HeH1.45_cond09_s1). MEASURED as in section 51 (advected profiles of section 49, binary EXHALE_1543a8f2.x; figure .P1/atomic97/diag_R/cond_on_off_molecular_scalar_HeH1.45.png):

	no conduction (catalog state, re-measured)	conduction
He 10830 EW, adv	0.864	0.153
He 10830 EW, solution	9.17	2.15
T max	5036 K at 1.465 R _p	3711 K at 1.316 R _p
T at 1.001 / 1.01 R _p	1432 / 1453 K	481 / 993 K
Mdot (median r >= 1.1)	7.75e7 g/s	2.92e7 g/s (x0.38)
critical point / Kn at the outer boundary	none (Mach 0.57) / 0.090	none (Mach 0.65) / 0.172

The catalog carries EW 1.0964 for this case (section 14); re-measured here by the current binary with the advected profiles of section 49 it gives 0.864. Which part of that difference is the operator (the transport terms of the advected energy equation) and which is the re-evaluation by the current boundary (v5) was not separated. With conduction the EW falls by a factor 0.18, to 0.14 of the observed 1.108 +- 0.029, while the molecular photochemical He/H 9 case falls

onto it (section 51): with conduction the He/H that reproduces the observation is higher than the catalog's 1.5, between 1.45 and 9 on these two families (not the same family; not yet bracketed on one).

53. Where the energy-row floor of section 50 comes from (2026-09-30; the cause is in section 58)

The floor of section 50 (hydrodynamic solves ending at the iteration cap with $\|R\|$ $1e-5$ to $1e-4$ against the energy tolerance $1e-6$) was measured on the 0.06 conduction rung at factor 0.773561 (IC: pass state r20260930T172935.418_c1790756975418473_p22 of atomic_scalar_gj1132x0.06_kzz1e9_HeH9.7_ser2_cond08_s0.773561; runs in .P1/atomic97/diag_R/floor_*, binaries EXHALE_9d0b8d4c.x and, for the new diagnostics below, EXHALE_dfb51387.x). Every run starts at the same $\|R\| = 3.882e-5$, energy row of cell 296.

The row. At cells 290 to 312 ($r = 1.75$ to $2.06 R_p$) the energy row is the difference of the advective divergence and the conduction source, each $\sim 1e-10$ in the row's units, with heating $\sim 4e-12$; the residual is $\pm 1e-15$ to $4e-15$, alternating in sign from cell to cell, i.e. $1e-5$ to $4e-5$ of the row's scale. The temperature there is 748 K and flat: the relative first difference of T is $4.3e-5$ to $5.3e-5$ per cell and the relative second difference $1.4e-6$ to $2.1e-6$ per cell. The H ionization crosses $x(H^+) = 0.5$ between cells 306 and 307.

Excluded, by measurement.

candidate	test	result
acceptance rule	EXHALE_PTC_SER_RNORM (section 50)	same floor
scaling of the linear model	EXHALE_JUDGED_ROWS, EXHALE_JUDGED_MERIT	cap at $1.5e-5$ / $5.3e-5$
probe length of the Jacobian action	EXHALE_JV_PROBE_ARC = 10, 100; EXHALE_JV_COLUMN_SCALE=1	40 iterations, best $8.2e-6$
frozen WENO3 weights in the model	new EXHALE_JV_FRESH_WEIGHTS=1 (products with the weights recomputed)	first 9 iterations identical
tolerance of the per-cell ionization solve	EXHALE_IEQ_TOL = $1e-12$, $1e-14$	start
sonic point, Roe entropy fix	Mach number of the band	0.005 to 0.013: subsonic,

The additivity test (EXHALE_JV_ADDITIVITY=1) puts its defect in cells 1 to 32 (mass and momentum rows; along directions off the Krylov basis the second difference does not scale with the arc at cell 1), not in the band. That non-smoothness at the base is recorded here and not pursued.

Located. EXHALE_LINEAR_ROWS=1 now prints, cell by cell over a band (EXHALE_LINEAR_ROWS_CELLS=lo,hi, default 290,312), the energy row F , the model's $F + J dY$, the row actually found at $Y + dY$, their ratio, and the difference quotient $(F(Y + t dY) - F)/t$ for $t = 0.1$ to $1e5$. (The "linear residual left over the row of F " printed before is the column maximum in raw units, which the base cells set; it says nothing about this band.) On the first iteration:

- cells 290 to 304 and 308 to 312: actual change / predicted = 0.90 to 1.13, the row falls by ~ 10 per step;
- cells 305, 306, 307: 1.91, 2.05, 1.41; the row changes sign at the same magnitude (306: $+1.42e-15 \rightarrow -1.49e-15$) and does so again on the next iteration (ratios 2.03, 4.62, 0.43).

The difference quotient along dY is constant within 10 to 20 per cent for $t \geq 10$ and at $t = 0.1$, and doubles at $t = 1$ in

cells 305 to 307 only (306: -1.28, **-2.91**, -1.59, -1.39, -1.43, -1.42, -1.44 e-15): the row carries a deterministic jump of 0.6 to 1.5e-15 between states 1e-12 apart, i.e. 1e-5 of its scale. Heating and cooling in cell 306 jump by 3e-11 (relative) at the same t, and the H I / H II fractions of cells 306 and 307 by 1e-11 to 2e-11, ten times their neighbors', with a relative step of ~1e-12 in every cell.

The source. With EXHALE_EQ_MEASURE_TRACE=1 every residual evaluation stops the composition elimination after 1 pass, on a particle-count increment of 6e-12 at cell 307 (the closure target is 1e-8). Forcing more passes (EXHALE_RESID_EQ_TOL=1e-15, EXHALE_RESID_SC_MAX=12), the increment at cell 307 stays at 3.0e-11 for all 12 passes: the elimination does not converge there. EXHALE_IEQ_REPORT_CELL=307 shows x(H+) of that cell alternating between two values on successive passes,

5.030758860692102E-01 5.030758860599341E-01 (difference 9.3e-12)

each returned with a reaction residual of 1e-16 and solver info 1; with the per-cell tolerance at 1e-15 the two values are the same to the last digit and the solver returns info 4 (no further progress). *Likely* the elimination alternates between two nearby states. It is not established that the per-cell residual is flat (section 54): the printed residual is the normalized one, whose hydrogen turnover bound includes the Penning term with the total He density, and successive passes change the coefficients of the cell problem (T, n_e, attenuation), so the two x values can be roots of two slightly different problems.

The amplification. An indeterminacy of 1e-11 in the composition is an indeterminacy of 1e-11 in the particle count and in $T = p / ((n_{\text{tot}} + n_e) k)$. Where the row is the balance of advection and conduction on a profile whose relative curvature is 1.7e-6 per cell, that moves the conduction divergence by $1e-11 / 1.7e-6 = 6e-6$ of itself, which is the row's scale. MEASURED jumps 0.6 to 1.5e-15 against a scale of 1e-10 agree with this. Newton cannot bring the row below the jump; the solve therefore wanders at 1e-5 to 1e-4 and a rung certifies only when one pass lands inside the tolerance (section 50).

The same place in the other chains. The worst energy cell of the last 40 iterations against the x(H+) = 0.5 crossing:

rung	worst energy cells (count of 40)	crossing between cells
0.0624, factor 0.913	297 (30), 290, 296, 292	296 / 297
photochemical descent, x = 0.0634	287 (20), 284, 282, 299	288 / 289
0.06, factor 0.774	301 (10), 303, 308, 307	306 / 307

Not decided, for the user. (i) Condition the per-cell ionization residual at $x(\text{H}^+) \sim 0.5$, so that the composition is determined to rounding there. (ii) The energy row's certification scale in a conduction-dominated layer: it is the largest of the row's terms, and the conduction term is itself the difference of two face fluxes that are, by the first and second differences above, ~30 times larger; measured against the face fluxes the same jump would be ~2e-7 of scale (an estimate from those differences, not measured). Neither was changed. Remedy (ii) is withdrawn (section 54): a larger denominator admits a 30 times larger absolute imbalance and does not make the physical residual smaller.

Also seen. EXHALE_RESID_SC_TRACE=8 from a second seed (the column's composition shifted by five cells) converges at fixed Y to a different composition (max relative difference 4.8 at cell 266, ||R|| 1.78 against 3.9e-5). The shifted

seed changes the elemental totals too (hydrogen at cell 144 by 37 per cent, section 54), so the two runs solve different elemental problems. They do not show multiple fixed points of one problem.

Code. EXHALE_JV_FRESH_WEIGHTS=1 (steady_newton.f90, measurement only) and the band and ladder printout of EXHALE_LINEAR_ROWS (one residual evaluation per ladder length) were added; with neither key set the run is unchanged (iteration 1 of the diagnostic run reproduces $\|R\| = 1.945e-5$). The comment of residual_norms (steady_residual.f90) said the two regions' rows are made dimensionless by different physics; residual_row_scale has no region switch, and the comment now says so. The chains were not touched.

54. The review of the 2026-09-30 handoff: findings checked (2026-09-30)

The review is md/session_handoff_20260930_review.md (source tree 60a90fb). Each finding was checked against the code or the saved products before being accepted. The plan that carries them out is md/PLAN_20260930.md, revised as md/PLAN_20260930_rev1.md after its review md/PLAN_20260930_review.md, and as md/PLAN_20260930_rev2.md after md/PLAN_20260930_rev1_review.md; md/PLAN_20260930_rev3.md is revision 2 re-checked against the code and the stored runs (the ionization-transport route's failed 09-23 attempt on this planet, the chains untouched by the derived-state exit code, He 2³S transport expected). No code was changed at this step; the statements of sections 51 and 53 and of the handoff that the review showed to be wrong were corrected in place.

review item	checked
4.3 column rejection not propagated	ew_M2_s1/eval.log: marching profile kept: T on 10 of 10 passes; ew2_fid_atomic_full
4.1 transport on the input composition	transport_terms_of_profile passes f_sp_in to thermal_conduction_coeffs, the element
4.2 step-only convergence	exit on dmax < 1e-10; no final residual evaluation
4.4 fixed ten outer passes	do k = 1,10 with no convergence test
10.1 metal re-solve leaves an outdated n_e	calc_ne before the metal re-solve; calc_mmw and p_out after it use that n_e (only pp_metals=
5 factor serialization	write(opt_value, '(F7.5)'): 0.773561 and 0.773564 give one token
7 EW launcher	ew_of_state.sh defaults to EXHALE_e37b8e27.x, continues after a failed evaluation or transit,
10.3 wrapper exit	launch_full_equation_from.sh ends exit 0 after the XUV descent
8 control bytes	factor_continuation_chain.sh line 56 has literal 0x01 and 0x02 in place of \1, \2
10.2 H2 conductivity overflow	x*x*exp(x)/(exp(x)-1)**2 guarded only at x > 700; exp(x)**2 overflows above x ~ 355, i.e.
2.1 label	report_accepted_cell_state prints x(min(4,nx)) as x(H2 nuclei); in the atomic He 2 ³ S b
1 shifted seed	comp_b(k,:) = f_sp(min(N,k+5),:) moves elemental totals
2 rounding-floor interpretation	the printed residual is normalized_reaction_residual; passes change T, n_e and attenuation
2.2 composition stopping measure omits transport	increment_of_what_the_residual_reads reads particle count, H2 share, net radiative source
3 face-flux certification scale	-
6 crossing inference across families	-

55. One failure contract for the advection-derived product (2026-10-01)

Plan md/PLAN_20260930_rev3.md A1, A2 and A6; the defect is finding 4.3 of section 54 (a rejected column energy solve left its rows labeled corrected and the evaluation exited 0). Implemented by the worker and checked on the diff and the runs.

The column solve (energy_column_newton, post_process_adv.f90) returns an outcome: *rejected* with its reason (linear solve, nonfinite, non-positive T, metal temperature band, iteration cap) and the triggering cell, or *step_passed_unverified* (the step test passed; no residual is formed after the last step, so it is not called converged). A rejected column keeps the marching profile and writes its WHOLE unknown set adv_failed (decision D2(i): no schema change; the ghost rows are written failed but not counted). For a metal-band rejection the log prints the triggering cell's energy row term by term in $\text{erg cm}^{-3} \text{ s}^{-1}$, at the column's T and at the run's T ($R = q_0 \text{ fv}/(\mu_{\text{pp}} \mu_{\text{mum}} \text{ dr_step})$, the scaling of T_equation).

The derived-state record. post_process_adv returns adv_derived_state (chemistry, column outcome and reason, physical-cell counts, outer = unverified: the outer loop is still ten fixed passes, n_pp_passes); both callers print it and write_output writes it as # adv_derived_state: ... into both *_adv headers (schema and columns unchanged; the regression comparison excludes # lines). The evaluate route exits 7 when the record is rejected (2 stays the certification refusal of the input state, written first; 7 is used by nothing else in src/ or the shell consumers). read_adv_validity (exhale_transit_lib.py) refuses a row when adv_T_status $\neq 0$ OR adv_comp_status = 2, returns refused_reason and derived_state (None = unknown); EXHALE_transit.py prints the record and refuses a rejected adv spectrum unless EXHALE_TRANSIT_DIAGNOSTIC=1, which writes the refusal into the spectrum metadata. ew_of_state.sh requires the binary, a fresh work directory, checks every step, records the md5 and state_id of both state files, the binary md5 and the EXHALE_* overrides, judges solution and adv separately, and states the EW in percent-angstrom (the former mA? was wrong). run_case.sh accepts 7 as "solved state certified, adv product refused": the state is published with its certification, the refusal and the verdict line go into run.json (adv_product, adv_derived_state), REPRODUCE.md and the last log line, the transit pass is skipped (the transit tool itself exits 7 on a rejected record) and earlier tpm_* products of the case are moved to superseded_case_products/ so that none can be read as this run's line measure; an unknown exit still fails the pass. Launcher tests (LHS1140b/models/tests/): evaluate_entry.sh 25/25 (the five exit-7 checks fail on the former tools), runner_records.sh 18/18 after a fixture defect of its own was fixed (its synthetic tree lacked he10830_equivalent_width.py and the observation CSV that status.py imports; 14/18 on the former fixture), and the other five launcher suites unchanged.

A6. After every composition update (the H/He sweep, the metal re-solve and its failure branch) n_tot, n_e, mmw and the caloric quantities are refreshed, then xion, then heating, the energy solve and p_out. wasp_full with pp_metals 2: $\max |p - (n_{\text{tot}} + n_e) k T|/p$ of the written composition $4.4\text{e-}8 \rightarrow 7.8\text{e-}16$; the _adv T column moves by $1.0\text{e-}6$. The xion refresh acts in every mode: on the fiducial atomic full-equation state the outer passes now reach their fixed point (column iterations $20.4 \rightarrow 20.1$ over the ten passes) and the _adv T moves by $2.9\text{e-}10$. **Separate defect found and fixed:** the pp_metals = 2 metal re-solve never set ieq_cell%T_K, so its charge exchange ran at the temperature of the last equilibrium cell and the cx_require_cell guard aborted every pp_metals = 2 run (reproduced on wasp_full; one line).

The M2 rejection explained. ew_M2_s1 (molecular photochemical fiducial, section 51): reason metal_T_band at cell 437 ($r = 10.35 R_p$) on 10 of 10 passes: the column's T 2445 K is 2.007 times the run's 1218 K. The row at 2445 K closes to $9.5e-12$ of its largest term (compression work $5.6e-13 =$ photoheating $3.7e-13 +$ conduction $1.6e-13$; radiative cooling $1e-16$), and with that cell alone put back at 1218 K the conduction from its neighbors ($7.9e-10$) leaves a residual equal to the largest term. So 2445 K is a root of the corrected, conduction-coupled equation and only the band refuses it; the band is a guard of the cell-by-cell Brent solve against spurious hot metal-cooling roots and has no stated validity for the column solve. Whether the column is to be accepted on its residual instead (plan B2) is held for the user; the band is unchanged and the case stays rejected (EW 1.1032 diagnostic; the solution spectrum 8.563 is a result).

Checks. src/tests/adv_static_limit: 53 -> 63 PASS, 0 FAIL, with the gfortran and the ifx binary (rows K: a stored certified state evaluated plain and with each of the five injected rejections, EXHALE_ADV_TEST_REJECT=<reason>, measurement-only; rows L: the reader on composition-only failure, retained/unsupported/not-evaluated kept apart, an absent record read as unknown). Evaluate-only reruns in .P1/fiducial_conduction/verify_w1/: M2 exit 7 with the record above; fiducial atomic exit 0, step_passed_unverified, EW 0.6491 unchanged. Known and left as it is: a gfortran runtime error exits 2, the code of the certification refusal (TO_BE_DONE (o)).

56. Continuation factors written exactly; legacy F7.5 tokens read as intervals (2026-10-01)

Finding 5 of section 54 (plan A3). load_IC.f90 writes a factor $s \neq 1$ of cond and interdiff_enth as ES23.16E3 (factor_token_text, adjustl; $s = 1$ stays T), and the writer, reader, comparison and absent-token values share one length, 32 (previously 8/16/16/8); with both factors the options field is 296 of its 1024 characters, and a field or token that would not fit is refused, never cut. A legacy F7.5 token (factor_token_kind, these two tokens only) states the interval $[v - 5e-6, v + 5e-6]$: a requested factor inside it loads the state as a seed with the claim dropped (legacy_factor_token, one printed line, a # legacy_factor_token history line keeping the original text); a factor outside it is a physics change under the existing rule (named on Restart option change:, or refused). Malformed values are refused even when named. The run log (conduction_scale, interdiffusion_enthalpy_scale) and the setup report print the factor ES13.6 and the report names a migration. gfortran and ifx write 0.773561, $1e-6$, $1 - 1e-12$ and 0 as the same token text and each reloads the other's state at the same factor. Test restart_factor (grid_and_gates 25): 38/38 PASS with both compilers (15 of the first 27 FAIL on 60a90fb); the gate tests of section 39 unchanged (gfortran 119 PASS, 11 XFAIL, 0 FAIL; the 16 ifx FAILs are the fixtures' 1-ulp compiler difference, present before). grid_and_gates/run.sh uses -module and -qmk1 for ifx (before, ifx could not link the Fortran drivers). Round trip: rho, v, p bitwise (gfortran) or 1 ulp (ifx), T and species 1-3 ulp, the loader's own arithmetic. States written on 2026-09-29/30 with F7.5 tokens (every chain rung) keep them and load only as seeds.

57. The H2 conductivity without overflow (2026-10-01)

Finding 10.2 of section 54 (plan A4). h2_conductivity evaluated $E(x) = x^2 e^x / (e^x - 1)^2$ ($x = 5987 \text{ K/T}$) as written: $(e^x - 1)^2$ overflows above $x \sim 355$ ($T < 16.9 \text{ K}$), where the form returned 0 in place of $E < 1e-148$, and $x^2 e^x$ above $x = 696.69$, so $8.553 \text{ K} \leq T < 8.593 \text{ K}$ gave NaN, and thermal_conductivity added $n_{H2} \times \text{NaN}$ even where $n_{H2} = 0$. (The review and section 54 had said "NaN above $x \sim 355$ "; the NaN interval is the narrow one.)

E is now $\text{einstein_heat_capacity}$: $x^2 e^{-x}/(1 - e^{-x})^2$ for $x \geq 0.05$, the series $1 - x^2/12 + x^4/240 - x^6/6048$ below (truncation $< 2.3e-16$; the two agree at the switch to $6.7e-16$), 0 above $x = 700$; h2_conductivity stops on a non-positive or non-finite T; the H2 term is formed only where $n_{\text{H2}} > 0$, in the same summation order, so an atomic mixture is bitwise unchanged. Fit and range (200-2000 K) unchanged. `src/tests/energy_update`: 54 -> 69 PASS (8.57 K = 1.3235866213750098e3, 200/748/2000 K unchanged to $1e-14$, the $1e6$ K series limit, continuity, $n_{\text{H2}} = 0$ mixture bitwise atomic). Stationary evaluate of the molecular scalar He/H 1.45 conduction state, HEAD against HEAD + fix (ifx): `Hydro_ioniz` and `Ion_species` bitwise; `*_adv` at rounding ($T \ 7e-14$).

58. The energy-row floor was a rounding staircase of the absorbed fraction (2026-10-01)

Plan C1-C3, on the state of section 53 (`diag_R/floor_jvadd`). Keys added, default off: `EXHALE_IEQ_DUMP_CELL=<j>` writes every He 2^3S -branch solve of cell j to `ieq_cell_dump.txt` (tags, T, densities, every rate coefficient, start and returned vectors, dimensional rows, turnover bounds, normalized residual, the forward-difference Jacobian) and the cell's photoionization rates at perturbed entry states to `ieq_cell_rate_probe.txt`; `EXHALE_RESID_SC_TRACE_SEED=shift|h2|partition` selects seed B of the two-seed trace (partition: the elemental inventory and the charge closure kept, a fraction of H I and He I moved into H II and He II). A5: `report_accepted_cell_state` names unknown 4 by the system (He 2^3S in the atomic triplet branch). With every key unset the outputs equal the HEAD build's.

Verdict: hypothesis (b) of section 53, and the cause. At cell 307 the elimination alternated between $x(\text{H}^+) = 0.50307588606921$ and 0.50307588605993 ($1.8e-11$ apart). The dumped rows are reproduced from the dumped coefficients in binary64 (146 records, to $1.8e-17$ of the turnover bound). Each returned x is the 50-digit root of ITS OWN pass's coefficients to 1 ulp and $1.4e-11$ (normalized) from the other's; the binary64 residual is monotone across 41 ulps, not flat. The coefficients differ because P_{HI} , the cell's photoionization rate, is a staircase function of the cell's own entry $n(\text{H I})$: steps of $4\text{--}5e-11$ relative every $1.7e-11$ in $n(\text{H I})$. Cause: `absorbed_fraction_per_unit_depth` (`utilities.f90`) took the closed form $(1 - e^{-d})/d$ for $d \geq 1e-8$, whose rounding is a staircase with relative steps eps/d ($1.5e-9$ at this cell's $d = 1.5e-7$). The smooth composition map contracts (-0.073 per pass, measured with the partition seed), but its fixed point lay on a step, so a period-2 orbit resulted, and that orbit was the floor: two seeds end on it in opposite phase, $\|R_{\text{B}} - R_{\text{All}}\| = 1.7e-5$. The "flat residual" of section 53 is withdrawn.

Fix. The Taylor series through d^8 (Horner) for $1e-8 \leq d < 1e-2$ (truncation $< 3e-25$), the closed form above (error $\leq 2.2e-14$); the old comment's truncation figure for the three-term series at $1e-8$ was $4e-25$ and is $4.2e-26$. Kahan's $(1 - u)/(-\ln u)$ does not work: ifx -O3 -xHost folds $\ln(\exp(-d))$ to $-d$ and the output was bitwise the closed form's. Reach: every XUV photoionization and photoheating rate (`photoionization_field_at_cell_H` and `_HHe` through `cell_mean_attenuation`, the `a_tau = 0` branch) and the H2O/OH FUV band rates. Effect on that state (max over physical cells, new against the control): $x(\text{H}^+) \ 7.8e-11$, $T \ 1.3e-11$, $\rho \ 1.6e-10$, $v \ 2.2e-6$ at the base cell only (below $1e-9$ elsewhere), the mean $\rho \ v \ r^2$ over $r \geq 1.1 \ R_{\text{p}} \ 2.7e-10$.

Result. `floor_jvadd` with the chain settings (`EXHALE_PTC_SER=2`, cap 400, omega 1, factor 0.773561): ONE outer pass, 2 JFNK steps, info = 0, CERTIFIED, energy row $1.16e-7$ (tolerance $1e-6$), 1.3 s; the HEAD build with the same settings: 400 iterations, info = 1, energy row $3.56e-5$, still $9.0e-6$ after three passes, never certified. Test `absorbed_fraction_switch` (`physics_probe 28`): 14 assertions (quad-precision error on each side, continuity at both

switches, mpmath at 1.5e-7, 1e-3, 0.1, 1, 10, no staircase over 41 points at 1.5e-7), all pass with gfortran and ifx at -O0; the former form fails 5. Runs in `diag_R/w4_*`; figures in `diag_R/w4_figures/`.

Consequence for the chains (stopped 2026-10-01, section 55 of the handoff): their slow rungs were this staircase, not the acceptance rule or the stopping measure of the elimination (plan C4, which stays a comment defect of that measure, not a cure).

59. One $(1 - e^{-x})/x$ for the code, and the other small-value forms (2026-10-01)

`one_minus_exp_over_x` (`utilities.f90`: the series through x^8 below $1e-2$, the closed form above, error $\leq 2.2e-14$) is now the only evaluation of the quotient. Evaluated through it, each with its own name, limits and meaning: `absorbed_fraction_per_unit_depth` (section 58), `absorbed_fraction_over_tau` and `absorbed_photon_fraction` (`util_ion_eq.f90`: the case-A/B ground capture weights and the on-the-spot absorption of recombination radiation; former switch $1e-4$ with errors $2.2e-12$ and $4e-14$), the thin branch of `line_escape_probability_one_face` (`Cool_coeff.f90`, the metal and helium fine-structure line escape; former closed form from $1e-8$, error up to $2e-8$), and the Ly-alpha Sobolev escape `beta_sob` (`lya_rt.f90`), which returned 0 in place of 1 for τ_S below $\sim 1e-16$ and at $\tau_S = 0$ (a limit defect; reached only with excited hydrogen on, which the LHS 1140 b runs do not have). Then the three forms that lose accuracy where their value is small: the stimulated-emission factor $1 - e^{-E_k/T}$ of `line_center_opacity_lte` and the absorbed Lyman-continuum fraction of `excited_hydrogen.f90` as $y(1 - e^{-y})/y$ (limit y), and the lower incomplete gamma functions of the I-mixing rate, `lower_gamma_2` and `lower_gamma_3`, as their power series in Horner form below 0.25 (through x^{12}) and 0.5 (through x^{17}), closed form above, errors $\leq 1.1e-14$ (the former three-term series switched at $1e-2$ with jumps of $6.7e-8$ and $8.4e-8$ there). `fine_structure_escape_probability` takes its thin reference in quadruple precision (its double reference was the cancelling form). Test `one_minus_exp_over_x_sites` (`physics_probe 46`): 87 assertions (mpmath values, continuity at every switch, no staircase over 41-point scans, the limits), all pass with gfortran and ifx at -O0; the former code fails 36. Evaluate-only: `floor_jvadd` moves $\leq 5.7e-14$ beyond section 58; the molecular photochemical s1 state $\leq 1.2e-10$ (H II, O II at a ghost cell), CERTIFIED before and after (mass $2.7e-9$, energy $2.3e-8$); the last three forms change neither run (excited hydrogen off; y of order 0.1-1 on the closed-form side).

60. The combined build and the tests of sections 55-59 (2026-10-01)

The tree with every edit of sections 55-59 built with gfortran (md5 a68670605f8d55c3a84e8c5ce4e04ca8) and ifx (fe48f7b774aac4490bd00334644fca6b); the suites the edits touch, run once on that tree: `energy_update` 69 PASS, 0 FAIL; `grid_and_gates` 423 PASS, 12 XFAIL, 0 FAIL (every program passed); `adv_static_limit` 63 PASS, 0 FAIL with the gfortran binary and 63/63 with the ifx binary; `physics_probe` 2295 PASS, 0 FAIL. After section 61 the tree is rebuilt (EXHALE.x aacba6fd93e4a547902ae732ad0b9bb2, EXHALE_ifx.x bff027c3c4ab416a429b30b5b70e9a00) and `adv_static_limit` (62 PASS, rows K with five reasons), `energy_update` and `evaluate_entry.sh` rerun on it (counts in section 61). The launcher suites of LHS1140b/models/tests/ were run by the worker of section 55 (`evaluate_entry.sh` 25/25, `runner_records.sh` 18/18, five others unchanged). The regression matrix was not run and no golden was touched (user instruction of 2026-09-29; the measured effects are stated with each change, the

largest $2.2\text{e-}6$ in v at the base cell of the diagnostic state, section 58). The workers' intermediate binaries (EXHALE_w*.x, build_w*/, diag_R/w4_bin/) are superseded by these two.

61. The column energy solve accepted on its residual; the refused rows carry the solution (2026-10-01)

User decision of 2026-10-01 ("잔차"): the conduction-coupled column of the advected profile is judged on its residual, not on the metal temperature band (TO_BE_DONE (p), plan rev3 B2). Implemented by the worker of section 55 and checked on the diff, the tests and the three evaluate-only runs.

Acceptance. After the iteration stops (step $1\text{e-}10$ or 100 iterations, a stopping rule only) the transport terms are formed again at the final profile, and every unknown's energy row $R = q_0 \text{ fv}/(\text{mup mum dr_step})$ must satisfy $|R| \leq \text{atol_E} + \text{cert_tol_energy } S_E$, with S_E the largest of the grouped terms ($\text{div}(\rho e \ v)$ as one term, $p \ \text{div } v$, heating, cooling, the conduction divergence, the element and the carrier enthalpy divergences) and $\text{atol_E} = 10 \ \text{eps}$ times the parts the row is assembled from (the grouped terms, the two advective halves, the three conduction face terms). MEASURED: the stagnated floor is 0.73 to 0.99 eps of that sum on the three conduction states, while the grouped sum alone under-states it by $1\text{e}3$ to $1\text{e}5$, the conduction divergence being a small difference of three large face terms. Today the absolute term never decides: the achieved $|R|$ is $1\text{e}4$ below $\text{rtol } S_E$. The $[0.5, 2]$ T_{in} band is removed from the column (it guards the cell-by-cell Brent solve, which keeps it, and had refused a root, section 55). Outcome converged_on_residual; reason residual_above_tolerance with the worst cell's term table; record fields column_worst_cell, column_worst_residual, column_worst_residual_rel.

Cells without a marching root. Cells whose marching Brent step finds no root (its bracket $[0.05, 4]$ T_{in}) are column unknowns by default and the residual decides (cells_marching_unrooted in the record; EXHALE_ADV_COLUMN_EXCLUDE_UNROOTED=1 restores the old set, measurement only). On the photochemical s1 state three cells at 24.5-28.5 R_p have their roots at 4.0-4.3 T_{in} ; with them in the column every pass solves the same set (the 354/351 alternation is gone).

Refused rows carry the solution. The status-3 rows of that product (1.0-1.09 R_p) were T-unsupported but composition-corrected, and their corrected $n(\text{He } 2^3\text{S})$ was 70 to 5600 times the run's: the molecule-free advection systems omit the $\text{He}(2^3\text{S}) + \text{H}_2$ destruction that the solution's balance includes. A composition class (formed with the particle-count class on the state handed in, no threshold) now refuses a cell whose H^+ , He^+ or $\text{He}(2^3\text{S})$ loses faster to H_2 ($R_{10} + R_{13}$, $R_{17} + R_{23}$, $\text{ioniz_HeI}23\text{S_H}_2$, from mol_rates) than to the carried channels; the class is applied inside the passes, so the cascade takes the run's composition as its history there; the metal re-solve of $\text{pp_metals} = 2$ skips refused cells; and every refused row is written with the run's exact numbers (species and metal stages when the composition field is refused; T when the T field is 1, 3 or 4; p, heat, cool when both are refused): the restored fields equal the solution's columns to 0 relative on the three states; a row with corrected T and refused composition is not the solution as a whole. The class is a threshold rule on the input state and does not bound the omitted H_2 contribution. The transit census still counts them as refused (173 rows to 1.089 R_p , photochemical; 246 rows to 1.322 R_p , scalar He/H 1.45).

Results (evaluate-only, ew_of_state.sh, tree ifx binary; EW in percent-angstrom, adv / solution):

| state | column | unknowns (unrooted) | worst | R|S_E | EW adv | EW solution | |---|---|---|---|---|---| | fiducial atomic full-equation | converged | 500 (0) | 2.8e-11 | 0.6491 | 7.5179 | | molecular photochemical He/H 9, conduction | converged | 329 (3) | 1.6e-11 | **0.7705** | 8.5630 | | molecular scalar He/H 1.45, conduction | converged | 256 (0) | 2.9e-12 | **0.1408** | 2.1471 |

All three exit 0 with the COLUMN accepted; the derived state as a whole stays unverified: the transport coefficients are formed on the input composition (plan B1), the outer iteration is the fixed ten passes (B3) and the momentum balance is not measured (B4). The photochemical adv EW of an ACCEPTED corrected profile is 0.7705 against 1.1032 of the refused marching profile (section 51, now superseded) and the observed 1.108 +- 0.029: 30 per cent low. Its He 10830 line-center optical depth peaks at 0.59 at $b = 1.2 R_p$ (the solution's at 1.45), 85 per cent of it from 1.1-3 R_p , where T_{adv}/T_{run} is 1.00-1.04 and $n(\text{He } 2^3\text{S})$ is 0.93 to 0.60 of the solution's (the advected, non-equilibrium population); T_{adv} reaches 2 to 4.4 T_{run} beyond 10 R_p , which carries under 1 per cent of τ . The correction of the scalar He/H 1.45 product moved from 0.1525 to 0.1408 (-7.7 per cent) because its composition class reaches 1.32 R_p . Tests: `adv_static_limit` 62 PASS (63 before: rows K have five reasons now), `evaluate_entry.sh` 25/25.

62. The Codex review of the 2026-10-01 summary, applied (2026-10-01)

`md/code_improvements_20261001.md` summarizes sections 53-61; Codex reviewed it in place (its comments R1-R8 stay in that file). Its statements were checked against the tree before anything was changed.

- R2, R5, R8 (wording): "exact root" is agreement to floating-point accuracy (0.65 and 1.64 ulp on the last two records; the two $x(\text{H}^+)$ are $9.2\text{e-}12$ apart, $1.8\text{e-}11$ relative); refusal of an `_adv` row is per field (composition, or T), so a row with corrected T and refused composition is not the solution as a whole, and the H2 class is a threshold rule that does not bound the omitted contribution; "profiles move by $\leq 1\text{e-}10$ " was false as a general statement: on the diagnostic state the maximum relative change over physical cells is $x(\text{H}^+) 7.8\text{e-}11$, $T 1.3\text{e-}11$, $\rho 1.6\text{e-}10$, $v 2.2\text{e-}6$ (the base cell only), the mean $\rho v r^2$ over $r \geq 1.1 R_p$ $2.7\text{e-}10$. The summary, sections 58, 60 and 61 and the handoff now say so.
- R3 (HIGH, correct): what section 61 accepts is the column subproblem with lagged coefficients (the transport coefficients on the input composition, the fine-structure escape probabilities of the start of the pass); the full-state energy residual on one derived composition (plan B1), the outer iteration (B3) and the momentum balance (B4) are not tested. The stale comments of `post_process_adv.f90` (the derived-state block) and of `adv_derived_state_verdict` (`exhale_transit_lib.py`), which still said the column residual is untested, now list what is and is not tested.
- R4 (MEDIUM, correct): `EXHALE_transit.py` continued on an unknown derived-state verdict (no record) with a warning; it now exits 7 unless `EXHALE_TRANSIT_DIAGNOSTIC=1`, as for a rejected record, and the metadata carries `adv_derived_state_unknown` and the override. `ew_of_state.sh` passes the key on an unknown record in its diagnostic mode; the manual and the READMEs say so. `adv_static_limit` rows M (the rows-K product with its record stripped): refused without the key, a curve with the key stating the unknown record; 64 PASS (62 + 2). Left as it is: `run_case.sh` calls the transit without the key, so a product written by a binary older than the record would fail its case; the current binary always writes the record.

- R6 (correct): the two bitwise rows of absorbed_fraction_switch (H2O and OH band rates against the XUV cell mean) fail under ifx -O3 -xHost by 1 ulp because the optimizer reassociates the test's reference product; they now assert 2 ulps (`..._to_2ulp`, 4.5e-16) with a DIAGNOSTIC line of the bitwise-unequal counts: 14/14 with the suite's -O0 flags, with gfortran -O3 (0 unequal) and with ifx -O3 -xHost (552 and 405 unequal of 1506, max 2.2e-16). A second copy of the function would still be caught (its difference is the closed form's 1e-9 to 1e-14 error below $d = 1e-2$).
- R7: the saved EWs reintegrate to 0.6490803760 / 7.5179432737, 0.7705246807 / 8.5629850578, 0.1407893262 / 2.1470782882 (adv / solution); recorded in the summary.

63. The reviewed part of the LHS 1140 b EW worktree, taken into the tree (2026-10-03)

`md/lhs1140b_ew_codex_review_20261003.md` reviews the changes an autonomous Codex run made in the detached worktree `../LHS1140b_EW_20261001/code` (same base, 3152b74). Its sections 3.1 and 3.2 (accept; fix first) and 4 (a defect of this tree) are applied here; section 3.3 (molecular-ion transport, joint species-flux modules, metastable balance diagnostic, HeH+ Newton rows) is not. Each item was re-derived against this tree's code, not copied.

Physics corrected.

- Recombination photons absorbed by H2 (`util_ion_eq.f90` `recombination_radiation_absorbed`, `ionization_equilibrium.f90`): the whole H2 share made H2+ and heated with one threshold. It is now split into the four final states of `h2_photo_channels` (M H2+, S dissociative ionization, D double ionization, N neutral dissociation) in the ratio of the channel cross sections the stellar field uses (`h2_photoabsorption_cross_sections`, shared with `set_energy_vectors`), total absorption unchanged, heat = photoelectron(s) + fragment kinetic energy of the channel. The capture kinetic energy of a continuum channel goes to every final state (energy conservation; the Codex version gave it to the ionizing ones only). Channel shares at the default switches, MEASURED: S 1.9-6.3% at 19.8-31.1 eV; S 11.4% and N 1.1% at 40.8 eV (He II Ly-alpha, inside the neutral window the old comment said it was outside of); S 17.0% and D 0.37% at 54.4 eV.
- `h2_channel_threshold` of N: 33 -> 32 eV, the lower continuity anchor where `frac_H2_neutral_dissociation` turns on (32-33 eV absorptions had a cross section and no energy recipient). No current recombination channel lies in 32-33 eV.
- The recombination-coupling condition at the two sweep sites is `use_h_rec_escape .or. (use_he_rec_coupling .and. thereis_He)`; the old form could read the `_hrc` arrays unset in a helium-free run.
- Carrier enthalpy flux (`diffusive_photochemistry.f90` `carrier_enthalpy_face_flux`): the recoil set of the energy row is now the set `carrier_write_back` rescales. Before, H+ took part of the hydrogen recoil although a carried H+ (Ionization transport: True) is written from its own row, and HeH+ took part although the write-back never moves it. Physics document `docs/physics_overview/molecules_metals.tex` updated.
- Class 6 of the ionization sweep: a constrained projection that fails `ionization_fractions_physical` keeps the entry composition and records the element-budget violation (it was accepted as a root); the class-6 residual

is that of the retained state (also stated in `steady_newton.f90`). `impose_transported_fractions` is applied before the admissibility test and at write-back. The decay-rate diagnostic releases the He II / He III fixed flags as well.

- Jeans escape flux of `collisional_validity.py`: $n \bar{v} / 4 (1 + \lambda) e^{-\lambda}$; the old prefactor was $2/\sqrt{\pi} = 1.128$ too large. The planet radius and mass of a run are read from its `EXHALE_resolved.out` by one rule in `examples/exhale_io.py` `load_run` (the lower-atmosphere profile and `base.inp` overwrite `R0`); `collisional_validity.py` and `formation_energy_flux_diagnostic.py` use it. He mass $3.9715 m_H$ (the table's), not 4.

Restart and output.

- Exact code-unit restart: `conserved_state.txt` (new `conserved_state_restart.f90`) beside every 'eq' write and every pass-state generation, read as `conserved_state_IC.txt` when it belongs to the IC pair (checksums, grid, species, boundary). A state that cannot be written exactly is not written (warning, existing file removed); a file that does not match, an option-change restart and a seed run use the dimensional pair with a notice. Exit statuses unchanged. `README`, manual (`conserved_state.txt` subsection), `README_HOWTO.md` pass-state restore, `md/input_schema.md`, GUI copy, mapper refusal and `element_flux_closure.py` (post-processing step = Restart intent: stationary evaluate) follow.
- Species densities re-formed after `Apply_BC` at the six output and evaluation sites (the Newton residual's order); ghost output rows formed from the conserved ghost cells by the tree's own routines (`boundary_gas_state_from_conserved`); `write_resolved_config` after the certification on both branches of the loaded-restart route.
- Comments corrected: subsonic outflow ghost (one incoming characteristic), ion-neutral friction (Langevin friction independent of T , rigid core as $T^{(1/2)}$; $1/D = 1/D_{\text{pol}} + 1/D_{\text{hs}}$ stated as an interpolation, up to a factor 2 in friction at the crossover), the Banks & Kockarts diameter (2.99 Å, not 2.7 Å; also `md/binary_diffusion_design.md`), `mol_rates` R14 and R19 sources (coefficients unchanged), `n_car_max` derived from the species table.

Not taken, with reason. The element eddy-flux outputs (no caller once the row dumps of 3.3 are left out); the `spherical_flux_divergence` refactor (stops on a NaN trial; the loop stays); the Codex wording that weakened two correct statements of `collisional_validity`.

Tests. Four builds from one source (3152b74; all changes; the enthalpy recoil set only; the recombination channels and chemistry only), run single-threaded on copies of `hp_front` (100 steps) and `carrier_model_a_newton` (stationary restart), MEASURED largest relative change over physical cells against the control build:

case	change	largest relative change
<code>hp_front</code>	enthalpy recoil set	v $3.0e-7$, HeH^+ $1.2e-8$
<code>hp_front</code>	recombination channels	$\text{H}2^+$ $1.1e-4$, $\text{H}3^+$ $7.0e-6$, HeH^+ $8.8e-7$, cool $7.5e-7$
<code>carrier_model_a_newton</code>	enthalpy recoil set	HeH^+ $1.6e-4$, cool $1.5e-4$, T $7.0e-5$, ρ $4.2e-5$
<code>carrier_model_a_newton</code>	recombination channels	0 (<code>He_rec_coupling</code> : False)

case	change	largest relative change
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Both are snapshots, not converged states, so these are the changes of one evaluation and not of a solution. Full profiles (T, v, rho, H2, H+, H2+, H3+, HeH+, He 2³S) overlaid and inspected (session scratch cmp/cmp_*.pdf).

make test on the merged tree (EXHALE.x 60bd2895135134813116b4b0ac63fcaa): 44 of 48 entries PASS; steady_completion_flag and steady_selfconsistent_residual fail without EXHALE_STEADY_RUNLOGS (they read the logs of a steady run and have no default; not a code result); boundary_state and physics_probe failed on the test defects below and pass after them (physics_probe 2295 PASS, 0 FAIL). carrier_enthalpy_flux gains section (e): a column with H+ (a third of the atomic hydrogen) and HeH+ (1e-3), checked face by face against Phi_H2 [h_H2 - 2 hbar_H] with H+ local and with H+ carried; both rows pass (1.8e-15) and both FAIL on the 3152b74 code (1.2e-3 from HeH+ alone; 1.11 with H+ carried, the old flux being about twice the right one there). Three test defects found and corrected: contact_law_at_the_base.f90 called thermal_conduction_source without the composition argument added in 60a90fb (compile error); helium_584_resonance_line.f90 called absorb_recombination_channel without the new channel-split argument; boundary_of_the_installed_composition.sh asserted a bitwise zero face-state gap, which the density re-formation after Apply_BC moves to rounding (MEASURED 1.6e-15; the entry-composition order it guards against leaves 7.5e-7), now a 1e-13 relative bound. The exact restart was exercised on wasp_he23off inputs: two write-reload round trips bitwise in u and f_sp; a negative ghost fraction is not written and the run exits 0; stale and mismatched files are ignored with a notice. Not run: molecular-seed, pass-state generation, the closure driver, GUI.

Open, for the user. R19 (HeH+ + H -> H2+ + He): 9.1e-10 cm³ s⁻¹ is a room-temperature measurement; De Fazio et al. (2014) Table 2 gives 1.54e-9 (1000 K) and 1.65e-9 (2000 K), Esposito (2015) via Sil et al. (2025) 1.82e-9 exp(-121/T); not changed. Esposito, Coppola & De Fazio (2015) was then read in its accepted manuscript (the published version could not be obtained): a Boltzmann average over all 178 rovibrational states of HeH+, 10-15000 K, with the authors' own polynomial fit (1.52e-9 at 1000 K, a maximum 2.00e-9 near 6000 K; 1-6% from De Fazio up to 2000 K; the Sil Arrhenius form is 9-11% below it at 4000-8000 K). The R19 comment of mol_rates.f90 now states this. The constant is kept because it is the Koskinen et al. (2022) Table 1 entry the Model A comparison uses; the replacement is the user's decision (decided the same day: section 64). python/paper_data.py still reads the input.inp planet radius.

64. R19 is the thermal rate of Esposito, Coppola & De Fazio (2015) (2026-10-03)

HeH+ + H -> H2+ + He (mol_rates.f90 rk_R19_HeHp_H) was the Koskinen et al. (2022) Table 1 constant 9.1e-10 cm³ s⁻¹, a measurement near 300 K (Karpas et al. 1979) held at every wind temperature. By the user's decision of 2026-10-03 it is now the thermal rate of Esposito, Coppola & De Fazio (2015, J. Phys. Chem. A 119, 12615), read in the accepted manuscript (the published version could not be obtained): close-coupling cross sections for v = 0, j = 0-5 and quasi-classical trajectories for the other states, Boltzmann-averaged over all 178 rovibrational states of HeH+, in the authors' fit $k = \sum_{i=0}^6 a_i (\log_{10} T)^i$ (max error 2.5%; their eq 4 prints the upper limit 7 with seven coefficients, and the seven reproduce their Fig. 6). T is held to the authors' 10-15000 K (the polynomial is negative below about 6 K and above about 3e4 K). The function now takes T; its two callers (System_HeH_mol.f90

set_mol_coeffs, molecular_reaction_heat.f90) pass the temperature they already pass to the other rates.

MEASURED values of the built function: 1.088e-9 (300 K), 1.524e-9 (1000 K), 1.753e-9 (2000 K), 1.997e-9 (5354 K), 2.004e-9 (6000 K), held 1.571e-9 above 15000 K and 5.25e-11 below 10 K; 1-6% from De Fazio (2014) Table 2 at 100-2000 K, within 3% of Garcia Munoz (2025) Table A.6 at 500-5000 K. The Sil et al. (2025) Arrhenius form of the same rate is not used (9-11% below it at 4000-8000 K).

Effect, MEASURED against the build of section 63 (same source but R19), single-threaded, largest relative change over physical cells; full profiles of T, v, rho, H2, H+, H2+, H3+, HeH+ and He+ overlaid and inspected (session scratch cmp/r19_*.pdf):

case	HeH+	H2+	H3+	H2	T, v, rho
hp_front (100 steps)	-46%	+68% (outer wind)	-18%	+1.4%	$\leq 5.8e-7$
carrier_model_a_newt35	35%	+0.07%	-0.02%	+0.02%	$\leq 3.7e-5$

docs/physics_overview/molecules_metals.tex gains the R19 paragraph and the two references; docs/koskinen2022_model_a_comparison.tex now says that its "matched reactions" row holds for the runs of that document and that the code departs from Table 1 in R12, R15, R17, R19 and R20.

make test on this tree (EXHALE.x 3d27cc49f507582c6cdc9683e6aedca7): 46 of 48 entries PASS (5176 PASS lines, 11 XFAIL); the two that fail are steady_completion_flag and steady_selfconsistent_residual, which need EXHALE_STEADY_RUNLOGS (section 63).

65. The Codex review of sections 63-64, checked (2026-10-03)

Codex reviewed sections 63-64 and the documents of the day in md/claude_code_physics_review_20261003.md (with _checks.json, _timescales.json, _profiles.pdf, _carrier_test.txt, _r19_probe.txt) and changed five files (examples/exhale_io.py, the two md documents, src/utis/collisional_validity.py and the scattering assessment). Each change was checked against the code.

- examples/exhale_io.py load_run: EXHALE_resolved.out is now read from the parent of the output directory, not from the directory of the input file. Correct: EXHALE writes the resolved file in the run directory and the outputs in its output/, so an input file from elsewhere could pair an output with another run's radius and mass (Codex reproduced a factor 2 in $\dot{M}/M_p$ with a foreign file). Exercised on lower_profile with and without an input file: both read that run's own resolved file.
- md/binary_diffusion_design.md, md/collisional_validity.md, src/utis/collisional_validity.py: HeH+ is stated not to be a friction carrier of the element operator (true: binary_element_diffusion.f90 leaves it out of both carrier lists), and the "omitted channels are conservative" statement is limited to the retained coefficients. Both correct.
- md/lhs1140b_ew_codex_scattering_assessment_20261003.md: Codex added a qualification. Its substance was right: the lifetime had been compared with bulk advection only. The diffusion and drift times are now MEASURED on the same state (HeH+ diffusion by Blanc's law from the pair coefficients of

collisional_validity.py, $K_{zz} = 1e9$, settling in gravity and the ambipolar field $eE = -kT \frac{d \ln(n_e T)}{dr}$ plus gradient drift, n_e the production charge budget): at the He 2³S column peak $t_{chem} = 11$ s against $t_{adv} 2.8e5$, $t_{diff} 4.9e7$ and $t_{drift} 1.2e7$ s, so Da stays $2.5e4$, also with every Coulomb partner dropped (D 30 times larger; $t_{drift} 3.9e5$ s). Advection is the shortest transport time except in 210 cells of the slow base layer below $1.18 R_p$ (diffusion shorter; 0.4% of the radial He 2³S column). The qualification is replaced by these numbers and the overstated sentences are corrected. Two Codex follow-ups in the same review file then showed that the ambipolar field adds to gravity in 197 of the 500 cells (a field-free drift is not an upper bound; eE/mg down to -0.92 with the metal-ion electrons, -0.49 without), that the neutral-only row is a sensitivity and not a bound on the pair law, that the opening claim was too categorical, and that the first field-inclusive table still carried field-free drift times and a wrong count ("advection shortest in all but one cell", which had missed the cells where diffusion is shortest). All were reproduced and corrected; the assessment states the field, the margin at the peak (D 43 times larger for drift to reach advection, $1.1e6$ times for it to reach chemistry), the local scope of the claim, and that the column shares are radial, not EW weights. Da at the column peak is $2.5e4$ throughout.

Points of the review that stand as open items: the recombination photoelectrons of `util_ion_eq.f90` heat locally with no secondary ionization or excitation, where the stellar path partitions the photoelectron energy (stated at the code site; at 54.4 eV an H2 channel leaves a photoelectron above the secondary-ionization threshold; effect not measured); and the thermal R19 rate assumes HeH⁺ internal states in equilibrium, which chemical relaxation does not show.

66. The LHS 1140 b EW plan started: phases 0 and 1 (2026-10-03)

The plan `md/PLAN_20261003_lhs1140b_ew.md` was approved by the user with all five decisions of its section 3. Runs are in `LHS1140b/models/.P2/ew_fit/`; the binary is `LHS1140b/models/.P2/EXHALE_f036290f.x` (ifx, clean export of `dd4a591`, manifest `.P2/BINARY_MANIFEST_f036290f65fd.txt`).

Phase 0 (`.P2/ew_fit/NOTES_phase0.md`).

- Entry `molecular_photochem_gj1132_kzzprofile_HeH2.09.entry/` from the certified generation `g0001` (2026-09-29) of that case; its input differs from the He/H 9 entry in the planet name and He/H only.
- Evaluate-only on `f036290f`: both the HeH2.09 state and the He/H 9 conduction state are NOT CERTIFIED. HeH2.09: cell-1 rows `O(1)`, from the base being rebuilt as `discrete_equilibrium_v5` from a state written under `v4` (the He/H 9 entry was in the same position and certified through its conduction chain). He/H 9 conduction (`v5` already): carrier H2 0.838, energy $2.6e-2$, from the equation changes of sections 63-64.
- The stored EWs reproduce bitwise on their own binaries: `0.7705246807` (He/H 9 conduction, `verify_w1d/EXHALE_w1d.x`) and `1.5008683274` (HeH2.09, `EXHALE_dd4a5955.x`). On `f036290f` (diagnostic, refused state) they are `0.7705418877` and `1.4601886445`.
- Conduction continuation of HeH2.09 started on `lart4 15:33 KST` (`.P2/ew_fit/factor_continuation_chain.sh`, a copy writing into `.P2/ew_fit`, with `config_rung.sh` requiring `BIN`); `rung 1`, $s = 0.005$, certified in 21 passes.

Phase 1 (`.P2/ew_fit/phase1/NOTES_phase1.md`; all on the He/H 9 conduction state, a solution of the `w1d` equations, each number with its binary).

- Where the line forms (EW with $n(2^3S)$ zeroed beyond r_{cut}): on the local-balance solution state (EW 8.563) 17% inside 5 R_p and 18% beyond 13 R_p ; on the advected product of the same state (EW 0.7705, MEASURED here, `phase1/ew_adv/truncation_adv.txt`) 17% inside 2 R_p , 55% inside 5 R_p , 77% inside 8 R_p , 95% inside 13 R_p . $Kn_{\text{bulk}} = 0.1$ at 6.97 R_p . The solution and advected EWs of one state differ by a factor 11: carried ionization is a first-order effect on the EW.
- He 2^3S local balance (solution state): first-order EW change +0.013 to +0.019 percent-angstrom (2.2-3.3 times the numerical target 0.0058).
- Ion-neutral drift: 0.3-1.1 of the element velocity at 3-30 R_p (H+/H I and He+/He I; the resonant charge-exchange cross section used in one bracket is not verified in a published source). The one-velocity stage flux is not valid there.
- Stage enthalpy (`TO_BE_DONE.md` (j)): up to $2e-2$ of the energy-row scale; required whenever ionization transport is on.
- Recombination photoelectrons above 10.2 eV deposited as heat: below $5e-5$ of the heating, but up to 43 times the energy-row tolerance at 1.0-1.5 R_p ; $< 2.3e-7$ of the row scale where the line forms.
- Energy coupling to Photochem: the wind conducts $4.25 \text{ erg cm}^{-2} \text{ s}^{-1}$ into its base (35% of the heat it absorbs; the enthalpy flux there is 120 times smaller). The Photochem column at the level is an isotherm at the climate skin temperature with no energy equation; an LTE thin estimate gives a warming of 120-240 K within 0.2-0.4 H below the level. The one-way coupling is not justified for this planet.

Corrected in the tree. The ENERGY comment of `recombination_radiation_absorbed` (`util_ion_eq.f90`) listed He II Ly-alpha on H I (27.2 eV) among the channels above the 30 eV threshold and omitted the He III ground capture on He I, H2 and the metal ions; it now lists them. The comment of `conduction_base_level_T` called the isentropic continuation the lower atmosphere's own temperature; it now states that this holds only for an adiabatic lower atmosphere and gives the measured 3.5 K offset from the profile isotherm. The catalog publisher `.P1/publish_rows/publish_solved.sh` wrote its EW (percent-angstrom) under the key `he10830_red_pair_equivalent_width_mA`; the key is now `..._percent_angstrom`. The 59 `run.json` records already published keep the old key (held for the user).

Held for the user. The bath temperature itself (profile vs isentrope), the 10.2-30 eV electron partition (Xu & McCray 1991 not in references/), the energy coupling to Photochem, and the ion-neutral drift: decisions, see the report of this date.

67. Phase 2: carried ionization stages certify on LHS 1140 b (2026-10-03)

`.P2/ew_fit/phase2/NOTES_phase2.md`. The failed 2026-09-23 attempt (`.P1/iontrans_atomic_kzz1e9_HeH2.13`: He++ carrier row $2.0e-2$ at cells 416-419, omega halved to 0.125) was re-run from its own state, mapped onto the face-at-level grid, on `EXHALE_f036290f.x`, with the 09-23 conditions (conduction off, Interdiffusion enthalpy flux: False). With `EXHALE_DIFF_OMEGA=1.0` it certifies at outer pass 11 (`phase2/C_p40`, `certified=T`; checked here): He++ row $8.6e-6$, H+ $2.3e-8$, He+ $5.2e-7$, He/H $6.8e-9$. `EXHALE_PTC_SER` does not change the outcome; omega = 0.5 refuses (the outermost element row halves each pass while the hydro solve fails on the cell-500 momentum row).

- The 09-23 failure was the state and the solver of that binary, not the He++ equation: the row's terms show the locally balanced $x(\text{He III})$, steep outward, advected with no source (measure about half of $d \ln x$ per cell over 7-29 R_p). Beyond about 3 R_p the He III recombination time exceeds the flow time, so the stage is frozen, 20-50 times below the local root.
- Physical effect of carrying the stages (no conduction): $x(\text{H II})$ at 11 R_p 0.34 $\rightarrow$ 0.042, n_e at 29 R_p about 10 times lower, outer T 460 $\rightarrow$ 1175 K, edge velocity 1.2 $\rightarrow$ 1.9 km/s, $\log \dot{M}$ +0.012 dex; unchanged inside about 1.5 R_p .
- EW of the certified state (MEASURED here, phase2/ew/C_p40): solution spectrum 1.0483 percent-angstrom (status result), advected product 1.0182 (conditional). The two agree within 3% once the stages are carried, against a factor 11 between the solution and advected EWs of a local-balance state (section 66): the bracketing-on-adv, accepting-on- solution route of the plan's decision 2 is consistent.
- Open (proposed, not coded): the element operator's outer face carries zero diffusive flux (binary_element_diffusion.f90 219-228, 3245-3250), a wall condition at an open outflow; at the top $D_{12} = 3.7e13 \text{ cm}^2/\text{s}$ and $Pe = 2.8$, so it makes a 1-2 cell layer ($X_{\text{He}} -0.30\%$ at cell 500). An outflow condition (gradient by the zero-gradient ghost, settling and thermal drift kept with the cell-N coefficients) is proposed as a default-off option, in phase 4. With the enthalpy flux on the hydro solve fails on the cell-500 momentum row from pass 7 (phase2/G_from_C_enthon, stopped): phase 4.
- Comment corrected: carrier_row_terms_write (diffusive_photochemistry.f90) described two printed scales; it prints one, rowdump_scale, as its file header says, and the helium-stage production and loss columns are the positive and negative parts of the net source (checked at carrier_source).

68. Phase 3: the He/H 2.09 reference with conduction and carried ionization stages (2026-10-03)

- Conduction continuation from the certified HeH2.09 entry (.P2/ew_fit/molecular_photochem_gj1132_kzzprofile_HeH2.09) nine rungs $s = 0.005 \rightarrow 1$, each certified (10-37 passes), 15:33-17:39 KST; final state ..._HeH2.09_cond09_s1. EW of that state (local ionization balance): advected product 0.6255, solution spectrum 7.6754 percent-angstrom.
- Ionization transport switched on (iontrans, omega 1): refused at pass 12 (..._cond_iontrans01), the H2 carrier row at cells 217-240 oscillating 0.08-0.18. Diagnosis (phase3/iontrans_diag/NOTES.md): a limit cycle of the progress control, not of the equations. The relaxation of the carried stages stopped every pass on the particle-count movement bound in the outer wind (5.7-13.8 R_p), where the carried stages differ most from the local ones ($x(\text{H II})$ 0.35 against 0.040 at 11 R_p); the wind solve then responds (dT/T $6e-4$ to $2e-3$ at cells 217-246), and the nearly cancelling H2 cycle at 1.20-1.33 R_p (production and loss $8.6 \text{ cm}^{-3} \text{ s}^{-1}$, $1.5e4$ times the transport divergence) amplifies that into the row (measured: $2.8e-4$ after the composition update, $3.7e-2$ after the next wind solve at fixed carriers); the bound rule halves on that rise and doubles on its fall, so the relaxation never reaches the carrier fixed point. Once it does, every run contracts by 0.62-0.63 per pass.
- With EXHALE_CARRIER_TRUST_HOLD=0.32 (or EXHALE_CARRIER_TRUST=0.32), omega 1, the same parent certifies in 24-25 passes; continuing the refused pass-12 state with the original controls certifies at pass 33. The three

certified states agree to $\text{ld ln nl} \leq 4\text{e-}7$ (checked here: `certified=T, iontrans=T cond=T`).

- Carried stages against the local ones (He/H 2.09, conduction on): $\log \dot{M}$ 7.819 $\rightarrow$ 7.805; $x(\text{H II})$ at 11 R_p 0.35 $\rightarrow$ 0.040; outer $x(\text{He II})$ about 0.8 $\rightarrow$ 0.2; edge T 469 $\rightarrow$ 1113 K, edge v 1.5 $\rightarrow$ 3.2 km/s; at the H2 front (1.15-1.17 R_p) $\text{He II} \times 1.8$, $\text{H2} \times 0.44$.
- EW of the certified state `phase3/iontrans_diag/B2_hold0.32` (MEASURED here, `phase3/ew_B2_iontrans`): solution spectrum 0.5830, advected product 0.5680 percent-angstrom (2.6% apart). Observed 1.1084 ± 0.0289 .
- Not yet in this state's equations: the stage enthalpy of `TO_BE_DONE.md` (j) (section 66: up to $2\text{e-}2$ of the energy-row scale with ionization transport), the ion-neutral drift (held for the user), $\text{He } 2^3\text{S}$ transport (section 66: $+0.013$ to $+0.019$ on a local-balance state), the 10-30 eV photoelectron partition (being implemented), the energy coupling to Photochem (held for the user).
- Proposed, not coded: the progress rule `bound_step_harmed` (`EXHALE_main.f90` near 7848) should count a row rise as harm only if the same row, evaluated at the composition the relaxation returned (before the next wind solve), also rose; or should not shrink the bound when the bound-limited cells and the cells that set the joint distance differ. Section 29.13's open item ("the rule that grows the bound is not right for the H2 carrier") now has its mechanism.

69. The 10.2-30 eV photoelectron partition of Furlanetto & Stoeve (2010), a default-off option (2026-10-03)

Photoelectrons of 10.2-30 eV were deposited entirely as heat (`E_sec_ion`, the bottom of the Dalgarno, Yan & Liu 1999 tables), although above 10.2 eV they excite and ionize H I and He I. New key Low-energy electron partition: `Furlanetto2010` (default none, the previous behavior); notes `LHS1140b/models/.P2/ew_fit/sec_partition/NOTES.md`.

- Source: Furlanetto & Stoeve 2010, MNRAS 404, 1869 (published PDF in references/), and the authors' electronic tables (paper footnote 9; the archive `elec_interp.tar.gz` is no longer on the authors' site, the copy used is the Wayback Machine snapshot of 2017-11-02, SHA-1 equal to the archive's CDX digest; `references/Furlanetto_2010_tables/`). Rows 10.0-32.2 eV at all 14 ionized fractions are carried as parameter arrays in the generated `src/modules/radiation/low_energy_electron_tables.f90` (generator `src/utls/furlanetto2010_tables_to_fortran.py`, which checks the archive digest and the bytes of every table file). The physics is in `low_energy_electron_degradation.f90`; one routine decides for every bin which of the three bands (above 30 eV, 10.2-30 eV, below 10.2 eV) a photoelectron falls in, for the stellar and the recombination paths alike.
- Interpolation as the authors' `elec_interp.c` (linear in E and in x), except two stated differences: everything below 10.2 eV is heat (the paper's section 7), and x is clamped to the table range $[1\text{e-}4, 0.999]$. Against the C code at 3042 (E, x) points: largest differences $2.6\text{e-}7$ (f_{heat}), $3.4\text{e-}7$ (n_{Lya}), $7.7\text{e-}8$ ($n_{\text{ion,HI}}$). Energy closure: the tables themselves close to $2.7\text{e-}5$; the code's partition to $3.6\text{e-}5$ on rows and $9.8\text{e-}5$ between rows. Xu & McCray 1991 (ApJ 375, 190, the ADS scan of the published article) Figs. 4-5 agree qualitatively.

- Accounting: heat $f_{\text{heat}} E_0$; $n_{\text{ion,HI}}$ and $n_{\text{ion,HeI}}$ added to the existing secondary-ionization rates ($n_{\text{ion,HeII}}$ is zero below 30 eV); the excitation energy leaves as line radiation, as the existing partition treats its non-heat, non-ionization share (Ly-alpha trapping in thick H I is not followed: the heat is a lower bound there). Branching between H I and He I renormalized to the cell's composition by the existing rule from the tables' He/H 0.087. Cells with $n(\text{H}_2) > 0$ keep the heat-only treatment (no source covers H₂ at 10-30 eV).
- Restart contract: a token `lowE_partition` on the # options line, so a state is not silently restarted under the other setting; old states read as F. Every state file now carries `lowE_partition=F` when off.
- Default path: control and measured builds give identical numerical outputs on `wasp_he23off` (20 steps) and `hp_front` (100 steps), with and without `Secondary_ionization: Immediate`; `physics_probe 2295 PASS`; the new suite `src/tests/low_energy_electron_degradation 14 PASS` (rerun here).
- Discontinuity at 30 eV where the source changes (atomic gas): Delta f_{heat} -0.011 ($x = 1e-4$), -0.006 ($1e-2$), -0.002 (0.1), 0 (0.5, 0.9); not smoothed (no source).
- Option on: `wasp_he23off` heating -0.67%, P_{HI} +0.63%, T -0.41% at the ionization front (1.03-1.05 R_p). On the molecular LHS 1140 b and hot-Uranus states every cell has $n(\text{H}_2) > 0$, so the main solve is unchanged, while the advected post-process, which rebuilds an H/He gas without molecules, applies it everywhere (He/H 9 conduction state: adv T -2.9%, heat -12%, He 2^3S +3.8%). This inconsistency of the gate is held for the user. `md/input_schema.md` K14c's " $E_0 > 40$ eV" corrected to 30 eV.
- Not changed, reported: photoelectrons above 30 eV on the recombination path (He III ground-capture channels) are still deposited as heat, where the stellar path partitions them.

70. The conduction bath below the level is the profile's temperature (2026-10-03)

User decision. `base_reservoir_temperature_at` (`base_boundary.f90`), the bath temperature the conduction operator holds at the base ghost (`conduction_base_level_T`), continued the reservoir's hydrostatic isentrope below the level even where a lower-atmosphere profile is handed over; on the LHS 1140 b photochem profiles the column there is an isotherm at the clima skin temperature, and the isentrope put the bath 3.5 K above it (188.89 against 185.42 K), lowering the conductive flux into the level by about 4.6% at a fixed cell-1 temperature (section 66). With a profile it now returns the profile's temperature at that radius, through a new accessor `lap_value_at_radius` (`lower_atmosphere_profile.f90`, linear in r between the bracketing levels, nearer end outside); without a profile, the isentrope as before. Comment of `conduction_base_level_T` updated.

- Checked on the certified He/H 2.09 state with conduction and carried stages (phase3/iontrans_diag/B2_hold0.32), evaluate-only, scratch ifx build of the tree (md5 3c236561) against EXHALE_f036290f.x (`.P2/ew_fit/bath_T/`): the only row that moves is the energy row of cell 1 ($1.6e-8 \rightarrow 4.4e-2$ of its scale, refused); every other row equal to the digits printed. The state is to be re-solved with the fixes now in progress, then the matching-level temperature sensitivity (plan section 8).
- Runs without a profile, and the base ghost state of the advection (its own isentrope in `base_ghost_averages`), are unchanged.

71. One H2 gate for both products; recombination photoelectrons partitioned as on the stellar path (2026-10-03)

Notes: LHS1140b/models/.P2/ew_fit/sec_partition2/NOTES.md.

- **H2 gate (user decision).** The 10.2-30 eV partition of section 69 is gated by $n(\text{H}_2) > 0$ in one place (`low_energy_electron_partition`); every path now passes the SOLVED state's H2 density (the advected post-process passes `nmol_eq(:,1)`, the same column its `_adv` files carry). On the He/H 9 conduction state (500 of 500 cells with H2) option on and off now give identical solved and advected products in every cell (before: the advected product differed in 329 cells); on the atomic `wasp_he23off` the option acts as before (advected heat -0.67%, $x(\text{H}^+)$ -4.6%).
- **Recombination photoelectrons (defect).** They were deposited whole as heat; above `E_sec_ion` the stellar path partitions the same energies. Each recombination photoelectron is now passed, at its fixed representative energy, through the stellar path's bands and partition (`photoelectron_partition_at_energy`, the same bracket and interpolation as `photoelectron_partition_on_grid`, 0-1.2e-16 apart), under the stellar path's switch (`use_sec_ion` and `sec_ion_active`, the STAGED default included): `f_heat` of it is heat, its H I, He I and H2 ionizations go to `dP_HI`, `dP_HeI`, `dP_H2` (dissociative one per 22 H2+), the rest leaves as line radiation. Channels concerned: He III ground capture on H I, He I, H2 and metal ions with thresholds below 24.4 eV, and He II Ly-alpha on metal ions below 10.8 eV (Na I, Ca I, Mg I, Fe I; previously missing from the comment). Inner-shell metal absorptions partition the photoelectron and the Auger electron separately, as the stellar path does. With the 10.2-30 eV option on, the recombination path now uses the He I ground singlet as the He I target, as the stellar path does (it used the summed He I).
- Tests: `physics_probe 2499 PASS`, 0 FAIL (new driver `recombination_photoelectron_partition.f90`, 204 PASS: energy closure 1.2e-15, partition-whole equals heat-only to 4.4e-16, gate closed bitwise with solved H2); `low_energy_electron_degradation 14 PASS`; rerun here on the merged tree.
- Effects (control vs measured, OMP 1): `wasp_full` and `hp_front` with the default staged coupling: numerically identical (the coupling is not yet switched on in these snapshots); with `Secondary_ionization: Immediate`: `wasp_full` heat -5.3e-5, `T` -2.7e-5, $x(\text{H}^+)$ +3.4e-5 at 1.036-1.041 `R_p` (the He recombination heating channel itself -0.23%); `hp_front` below 1e-7. The certified He/H 2.09 transport state: wind energy row 8.4e-16 -> 1.3e-11, gated `H+` row 1.6e-8 -> 3.3e-8, far below tolerance.
- Not traced: on the He/H 9 conduction state (option off) one cell, 118 (1.035 `R_p`, where `n_e` falls off), moves He 2^3S -1.2% and He III +1.4%; every other cell below 1.1e-4.
- Limit: each channel is partitioned at its representative energy, not averaged over the capture continuum.

72. The enthalpy of the carried ionization stages; the movement-bound rule (2026-10-03)

Notes: LHS1140b/models/.P2/ew_fit/phase3/stage_enthalpy/NOTES.md; work in a worktree, merged into the tree by a three-way merge (the two option tokens added at the same time, `lowE_partition` and `stage_enth`, are both in the vocabulary, `n_opt = 25`, every token array checked at 25 entries).

- **Stage enthalpy** (TO_BE_DONE.md (j), now fixed for the stages). The stage flux $F_k = x_k N_{el} + E_k$, $E_k = -n_{el} K dx_k/dr$, moves the stage relative to its element by E_k ; by Cook (2009, eqs. 11-13) it carries $q_{st} = \sum_{el} \sum_k E_k [h_k - \bar{h}_{el}]$, $\bar{h}_{el}$ the enthalpy per nucleus of the species carrier_write_back rescales (hydrogen: H I, H2+, H3+; helium: He I with its 2³S; HeH+ never). With every heavy particle at the same sensible enthalpy the exchange moves only the electrons: $q_{st} = h_e \sum_k Z_k E_k$, $h_e = \gamma/(\gamma-1) kT$, exact for helium and corrected for the H2+/H3+ share of hydrogen. Routine ionization_stage_enthalpy_face_flux; the recoil set is formed once (recoiling_nucleus_enthalpies) for this and the molecular term (carrier_enthalpy_face_flux renamed molecular_carrier_enthalpy_face_flux); carrier_enthalpy_divergence_of_state returns the sum, so the stationary residual, the marching update and the advected column energy solve take it from one routine. Under Interdiffusion enthalpy flux (default on), only with Ionization transport: True; state token stage_enth. Validity at the code site: zero current, one temperature, no stage drift beyond E_k (the ion-neutral drift and its enthalpy are not modeled), no Dufour term.
- Tests: new src/tests/ionization_stage_enthalpy_flux/ 13 PASS (closed ends exactly 0, column sum 1.1e-16, manufactured q 7.7e-16, H2+/H3+ recoil 2.3e-15, zero for uniform fractions, iontrans off and key off); carrier_enthalpy_flux 13 PASS, ionization_stage_flux 45 PASS; an independent recomputation from the B2 state files agrees with the production Sidf difference to 1.8e-13 of its maximum. Default path (wasp_he23off) identical; hp_front (iontrans on) moves by up to 7.9e-6, identical with the key off, i.e. the new term.
- On the certified He/H 2.09 transport state (B2): the term is up to 1.7e-3 of the energy-row scale (1.05-1.2 R_p), above 1e-6 in 369 of 500 cells, about 12 times smaller than the phase-1 estimate on a local-balance state. Resolved with it ($R_{B2_stage_hold0.32}$): certified in 11 passes; $T -6.2e-5$, $v -2.3e-4$, $x(H+) +1.5e-4$, $He\ 2^3S +4.1e-4$ at most; log Mdot 7.80532 -> 7.80523; EW solution 0.5829 (was 0.5830), advected 0.5679.
- **Movement-bound rule** (bound_step_harmed, EXHALE_main.f90): a rise of the joint distance counts as harm of the step only if the update moved the cell that carries the joint distance by at least half of the largest movement (prog_cell, update_count_move); otherwise the rise is the wind's response there, logged as such, and the bound grows. In the refused run of section 68 the update had moved those cells by 0.002-0.17 of the largest movement. Without information the old rule applies. Verified: the He/H 2.09 iontrans switch without a trust hold now certifies in 28 passes (refused at 12 before; the same solution as B2 to ld ln nl 4.7e-7); the atomic phase-2 route 11 passes and the conduction rung $s = 0.32 \rightarrow 0.64$ 28 passes, both bitwise equal to the control build.
- Every iontrans state certified before this section (B2, B3, C, C2) solved the previous energy equation. The docs (md/input_schema.md K44, the manual's Interdiffusion enthalpy flux entry, docs/physics_overview/) are updated; TO_BE_DONE.md (j) is marked fixed for the stages.
- make test on the merged tree: 45 of 48 entries PASS. Besides the two steady suites that need EXHALE_STEADY_RUNLOGS, grid_and_gates fails stationary_evaluate_*: the stored certified state of backup/regression/wasp_full_newton, evaluated with the secondary ionization armed, now moves OIII by 6.0e-5 and is refused. With the control build of section 71 (before the recombination photoelectron partition) the same test passes (3.8e-10, certified, exit 0): the stored state is a solution of the previous equations. Its refresh

belongs with the deferred refreshes.

- Merged ifx binary for the next runs: LHS1140b/models/.P2/EXHALE_6af10ba4.x (manifest .P2/BINARY_MANIFEST_6af10ba4*.txt). The He/H 2.09 reference is being re-solved on it from B2 (..._HeH2.09_ref_merged01, no trust hold, Restart option change: stage_enth).

73. The He/H 2.09 reference on the merged binary (2026-10-03)

Re-solved from B2 on EXHALE_6af10ba4.x (sections 66-72 in it), no trust hold, Restart option change: stage_enth: certified in 21 outer passes (.P2/ew_fit/molecular_photochem_gj1132_kzzprofile_HeH2.09_ref_merged01, checked certified=T cert_reason=certified_in_wind). EW (phase3/ew_ref_merged01): solution spectrum 0.5793, advected product 0.5643 percent-angstrom (B2: 0.5830 / 0.5680); observed 1.1084 +- 0.0289. Next: the matching-level temperature sensitivity (plan section 8) and the Photochem/wind elemental-flux closure on this physics (plan phase 3 step 4).

74. Photochem/wind elemental-flux closure on the new physics (2026-10-03)

LHS1140b/models/.P2/ew_fit/phase3/closure/NOTES.md; driver unchanged (element_flux_closure.py passes the reference's keys and env; checked in the running process environment), binary 6af10ba4, Photochem 0.9.0 with climate on, omega 0.5, tolerance 5%.

- Iteration 0 (the reference's profile, trial H 2.566e7, He 5.335e7 g/s from the no-conduction closure): measured H 2.1209e7, He 4.2318e7 g/s (overlap window, spread 2e-7): eps 0.21 / 0.26. With conduction and carried stages the wind takes 17% less H and 21% less He.
- Converged at k = 2: eps 0.026 / 0.033 below 5%, window spread < 1e-7; three wind solves certified (4-6 passes each), log Mdot 7.800.
- The measured fluxes do not respond to the trial: over the iterations they moved by 1e-5 while the trial moved by 15-18%; eps fell only by the damping. The fixed point is the measured flux (H 2.12095e7, He 4.23172e7 g/s); the accepted trial is 2.6% / 3.3% above it (one undamped step closes it, for phase 6).
- At the 1 microbar match: He/H -1.7e-5, T unchanged, q_H2 +1.4e-5; the column above changes by at most 0.16%; the wind by < 1e-5 (He 2^3S -4.5e-4 near the H2 front). EW of k02: solution 0.5793, advected 0.5643, equal to the reference to the four digits recorded.
- Not done: the undamped step; the Kn, Mach and HeH+ Da profiles of the final state; the reproduction of the 2026-09-16 profile by the present adapter.

75. The EW depends on the matching-level temperature: the energy coupling to Photochem is required (2026-10-03)

LHS1140b/models/.P2/ew_fit/phase3/T0_sensitivity/NOTES.md. On the He/H 2.09 reference (section 73) the lower-atmosphere profile was raised by Delta T at and above the match and tapered below it with the column response phase 1 solved (all-band shape, 1/e 0.19 H, for 120 K; rotational-band shape, 1/e 0.41 H, for 240 K), levels inserted

every 0.02 H to 3 H, r re-integrated hydrostatically (the match rises by $2.6e-4$ and $1.1e-3 R_p$, so R_0 changes), composition and K_{zz} held at the reference values; headers marked as diagnostic modifications. The reference state was mapped onto each new grid (`map_state_to_grid.py --ic`) and re-solved with the reference controls; both certified (24 and 21 passes).

Delta T [K]	EW solution / advected [percent-angstrom]	log10 Mdot	base conductive flux [erg $\text{cm}^{-2} \text{s}^{-1}$]
0	0.5793 / 0.5643	7.8029	10.0
120	0.6998 / 0.6855	7.8720	9.53
240	0.8098 / 0.7964	7.9261	8.43

- dEW/dT_0 is about $1.0e-3$ percent-angstrom per K: a 6 K change of the matching-level temperature is the numerical target. The EW rises through Mdot (+17%, +33%) and a denser outer flow ($n(2^3S) +17/+32\%$ at $3 R_p$, +40/+80% at $20 R_p$), not through the base layer.
- The reference conducts $10.0 \text{ erg cm}^{-2} \text{s}^{-1}$ into its base (0.52 of the absorbed heating), not the 4.25 measured on the He/H 9 state (section 66); the column's actual warming under this flux was not solved and is likely larger than the imposed range.
- By plan section 8 the two-way energy coupling is required. Not separated: the effect of the raised R_0 from that of the raised T_0 ; the warmed column's composition was held fixed.

76. The wind writes the heat it conducts into the lower atmosphere (2026-10-03)

Phase 1 defect 2 (LHS1140b/models/.P2/ew_fit/phase1/NOTES_phase1.md section 5): the conductive flux through the base face carried 35-52 % of the wind's absorbed heat and was written nowhere; it was recovered from the budget's Sene.

- `viscous_conduction.f90: heat_conducted_into_lower_atmosphere(Tcell, f_sp)` forms the base face's own term of row 1 of `thermal_conduction_coeffs`, $q = -\kappa_{1/2}(T_1 - T_b)/(r_1 - r_0)$, T_b the bath at the ghost centre $r(0)$ (`conduction_base_level_T`), converted with $p_0 v_0 = n_0 \mu v_0^3$, sign: **positive into the lower atmosphere**. It is held apart from `conduction_base_heat_flux`, which follows the last residual evaluation (a perturbed state inside Newton).
- `certification.f90` calls it on the state it certifies (the evaluation made on the state about to be written, as for the elemental flux windows); gates nothing.
- `write_setup_report.f90`, EXHALE_resolved.out with Conduction on: `base_conduction_state` (unmeasured before the wind exists, then measured), `base_conductive_flux_cgs`, `base_conduction_T_bath_K`, `base_conduction_T_cell1_K`. One place, the record the closure driver already reads. Documented in `md/input_schema.md` and the user manual (EXHALE_resolved.out section).
- Binary LHS1140b/models/.P2/EXHALE_20021893.x (ifx, md5 2002189393ed2fa9012355fc8d2cd3fb, manifest .P2/BINARY_MANIFEST_2002189393ed.txt; it differs from 6af10ba4 in these three files only).

Checks.

- A stationary evaluate of a copy of the He/H 2.09 reference (ref_merged01) writes 10.01598 erg cm⁻² s⁻¹, T_{bath} 185.022 K, T₁ 339.796 K: the 10.0160 of routes S and K of section 75 (MEASURED; agreement to the 4 decimals that note gives). Closure k02: 10.01605.
- src/tests/boundary_state/run.sh contact_law (gfortran private build): five new assertions (the recorded flux is the base face term reversed times p₀ v₀ to 1e-14; the two temperatures; unmeasured without conduction) and the existing 23 all PASS. The test now sets p₀, which it did not (input_read sets p₀ = n₀ mu v₀² at every run).
- Not run: the regression matrix and the other suites. The change adds a measurement that writes module variables only and two output lines; nothing the solve reads changes.

77. Column energy balance under the conducted flux; the energy-coupled closure (2026-10-03)

Plan section 8: the two-way energy coupling. Notes: LHS1140b/models/.P2/ew_fit/phase3/energy_coupling/NOTES.md.

Column. src/utils/conducted_heat_column_balance.py solves, below p_{match} and down to where the response is below 1e-3 K and the bottom flux below 1e-6 of F (or the tropopause),

$$(1/r^2) d/dz (r^2 \kappa dT/dz) = C(T, p) - C(T_0(p), p),$$

$$\kappa dT/dz = F \text{ at } p_{\text{match}}, T = T_0 \text{ at the bottom},$$

in $s = -\ln p$ at fixed pressure (layer mass p/g fixed, thickness following T), 0.01 in $\ln p$, conservative finite volumes, Newton. κ is the mixture conductivity of viscous_conduction.f90; C the optically thin band emission of CH₄, NH₃, H₂O, C₂H₂, CO, CO₂ from the climate model's own k -distributions; only $C(T) - C(T_0)$ enters, so $F = 0$ returns the climate T bit for bit. Rotational bands in LTE (NH₃ rotation-inversion lines $A \leq 4.8$ s⁻¹ against a pressure-broadening rate of 1.4e4 s⁻¹ at 1 microbar, HITRAN). A vibrational band is a two-level system, net cooling ϵ_{LTE}

- a), $\epsilon_{\text{LTE}} = C_{10}/(C_{10} + A_{10})$, $a = \epsilon_{\text{LTE}}(T_0)/2$ (the upward intensity of the isothermal stratosphere), but only when its V-T rates for both He and H₂ are published; otherwise it stays in LTE and is flagged. Treatments: nlte, nlte_partial (incomplete rate sets used with the partners that are published, a lower bound on that band's emission), lte, rotational. The thin assumption is refused when more than 10 % of a bin's emission excess (bins carrying > 1 % of the total) would be absorbed by the column above (escape $\exp(-1.66 \tau_{\text{bar}})$).

Published inputs (published versions, read):

- CH₄ dyad + H₂: Halthore, Allen & De Cola 1994 (1994ApJ...424L..61H), Eq. (5): $C_{20} = 4.315e6 \text{ p[torr]} \exp(-40/T^{(1/3)})$; one measured temperature (~296 K), slope from pure CH₄ 150-500 K.
- CH₄ nu₄ A₁₀ = 2.12 s⁻¹: Yelle 1991 (1991ApJ...383..380Y) Table 2.

- Band intensities, 296 K, HITRAN line sums (hitran.org, 2026-10-03): NH3 rotation-inversion $1.718\text{e-}17$, NH3 nu2 $2.247\text{e-}17$ (Weaver & Mumma 1984, 1984ApJ...276..782W, Table 3: $2.2\text{e-}17$), CH4 dyad $5.195\text{e-}18$ (Yelle 1991: $5.42\text{e-}18$ for nu4) $\text{cm}^{-1}/(\text{molecule cm}^{-2})$.
- NOT obtainable (publisher refuses this host): CH4 + He (Yardley, Fertig & Moore 1970; Siddles, Wilson & Simpson 1994; Menard-Bourcin et al. 2005) and NH3 nu2 + He (Hovis & Moore 1978, 1980); no NH3 + H2 measurement found. On LHS 1140 b therefore $n_{\text{lte}} = \text{lte}$. (Superseded by section 78: the papers were obtained, and n_{lte} now carries them.)

k-table checks (MEASURED): band strength of the k-tables / HITRAN sum at 296 K: NH3 rotation 0.987, NH3 nu2 1.005, CH4 dyad 1.006. k-table LTE emission of a fundamental / two-level $h c \nu_0 A_{10} g_1 \exp(-h c \nu_0/kT)$: CH4 dyad 1.04, 1.03, 1.06 and NH3 nu2 1.11, 1.13, 1.20 at 185, 300, 500 K (the bin-mean product $\sigma_{\text{bar}} \times B_{\text{bar}}$ and the inversion structure of nu2).

Tests `src/tests/conducted_heat_column/run.sh`: 16 PASS ($F = 0$ identity bit for bit; manufactured column with constant κ and linear cooling against the analytic exponential, $7.8\text{e-}7$, order 2.000; budget: emission excess equals F to $3\text{e-}16$ in three treatments; collision limit of the two-level band equals LTE to $6\text{e-}9$ K; optically thick column refused).

Adapter. `photochem_to_lower_profile.py --conducted-heat-flux F` (default 0: profile and `base.inp` byte-identical, MEASURED against closure `k02` at its own arguments): levels inserted every 0.05 in $\ln p$ over 8 H below the match in the column handed to the chemistry; chemistry $\rightarrow$ column balance with its composition $\rightarrow$ chemistry re-solved on the new T from its previous solution (`reinitialize_to_new_climate_PT`, the escape boundary condition re-imposed, the grid re-pinned) $\rightarrow$ until T_{match} moves < 1 K $\rightarrow$ profile with the column solution's nodes inserted where the rise exceeds $1\text{e-}3$ K, T of the column solution, n and ρ from $p/(kT)$, r re-integrated hydrostatically. Header keys `conducted_heat_flux_cgs`, `column_energy_treatment`, `column_T_match_K`; they enter the `solution_id`. Record `conducted_heat_column.txt`. `lower_profile_schema.py` writes the keys only when present.

$F = 10 \text{ erg cm}^{-2} \text{ s}^{-1}$ on the `k02` column (MEASURED):

treatment	T_{match} [K]	$1/e$ depth [H]
<code>nlte (= LTE)</code>	337.5	0.070
<code>nlte_partial</code>	345.6	0.073
<code>rotational</code>	556.3	0.137

NH3 nu2 61 %, CH4 dyad 22 %, NH3 rotation 14 % of the flux (LTE). Emission-weighted τ above the match ≤ 0.039 ; up to 7.9 % of a band's excess absorbed above. The bounds 337-556 K differ by 36 times the 6 K target: the NH3 nu2 and CH4 + He rates decide the answer.

Closure. `element_flux_closure.py`, `config energy_coupling` (default `false`; `energy_coupling_omega` 0.5, `T_match_tol` 6 K): the wind's `base_conductive_flux_cgs` damped into the next column's `--conducted-heat-flux`; history columns `F_cond_trial`, `F_cond_meas`, `T_match`; convergence also needs $|T_{\text{match}}(k) - T_{\text{match}}(k-1)| < T_{\text{match_tol}}$; `--fcond0` or the seed run's record starts it. Also fixed: the `--print-config-template` output was not valid JSON (an escaped quote in a non-raw string). Running in

phase3/energy_coupling/ (lart4, 12 threads, nlte = LTE: the lower bound of T_match), seeded from closure k02.

78. The published V-T rates in the column energy balance (2026-10-03)

The five papers section 77 could not obtain were supplied and read in their published versions (ExoAtmosphere/references/; tables checked on the page images). conducted_heat_column_balance.py, section PUBLISHED RATES, every rate cited at its code site:

band, partner	rate [cm ³ s ⁻¹]	process	T range	source
CH4 dyad + He	1.7e-14 (295) ... 1.7e-15 (140), 7 points	coupled nu2/nu4 pair -> ground, relaxation rate	140-295 K	Siddles, Wilson & Simpson 1994, Chem. Phys. 188, 99, Table 5
CH4 dyad + H2	1.44e-13 (193), 3.68e-13 (296)	nu4 (= nu2) -> ground, elementary V-T	193-296 K	Menard-Bourcin et al. 2005, J. Phys. Chem. A 109, 3111, Table 4
NH3 nu2 + He	1.1e-13 (198), 2.8e-13 (293), 4.1e-13 (398)	nu2 -> ground V-T,R, relaxation rate	198-398 K	Hovis & Moore 1980, J. Chem. Phys. 72, 2397, Table I (293 K: Hovis & Moore 1978, J. Chem. Phys. 69, 4947, Table I)
NH3 nu2 + H2	not measured; ESTIMATE = He rate x k(H2)/k(4He) of the CD4 bending modes (27.5 at 295 K, 36 at 240 K, 49 at 190 K)	-	-	analog: Siddles et al. 1994 Tables 1-2

- A relaxation rate (k10 + k01) is turned into k10 by the population factor $1 + \sum (g_u/g_0) \exp(-E_u/kT)$ (≤ 3 per cent in the measured ranges).
- Inside the measured range: $\ln k$ interpolated linearly in $T^{(-1/3)}$ (every published value reproduced). Outside it: the Landau-Teller law continued from the nearest point, slope fitted to all points of the pair ($b = -54.6, -40.8, -37.5 K^{(1/3)}$ for CH4-He, CH4-H2, NH3-He; Halthore et al. 1994 used -40 for CH4). The column spans 185-560 K, so NH3-He is extrapolated 160 K above its data and CH4-H2 260 K.
- The analog argument for NH3-H2: the H2 enhancement is V-R transfer into the widely spaced H2 rotational levels, set by the quantum removed (CD4 bend 996 cm⁻¹ against NH3 nu2 950 cm⁻¹); the CH4 dyad gives 22-27. Uncertain by a factor of a few.
- A10 from the HITRAN lines with the rotational levels thermalized, T-dependent (NH3 nu2 13.2 s⁻¹, CH4 dyad 1.96 s⁻¹ at 185 K; published 15 and 2.12).
- Treatments redefined: nlte uses the measured and the estimated rates, nlte_partial the measured ones only.

- Cross-checks: Menard-Bourcin CH₄-He 1.69e-14 (296 K) against Siddles 1.7e-14; at 193 K 1.4 times Siddles. Yardley, Fertig & Moore 1970 Table I, $p \tau = 2.3$ (He) and 0.12 (H₂) microsecond atm at room temperature, i.e. 1.75e-14 and 3.4e-13.

Tests (src/tests/conducted_heat_column): 33 PASS. New: every tabulated published value reproduced to 1e-12; the NH₃-H₂ estimate is the He rate times the CD₄ ratio; the Landau-Teller continuation is continuous at the ends of the data; nlte treats the two bands that carry the flux; the collision limit equals LTE.

Adapter alone at F = 10.016 (k02 configuration, MEASURED):

treatment	T_match [K]	1/e depth [H]	NH ₃ nu ₂ / CH ₄ dyad / NH ₃ rotation
all LTE	337.50	0.070	61 / 22 / 14 %
nlte	353.24	0.077	62 / 16 / 17 %
nlte_partial (NH ₃ -H ₂ left out)	380.23	0.091	38 / 28 / 26 %

eps at the match: NH₃ nu₂ 0.78 (0.32 without H₂), CH₄ dyad 0.60. nlte - LTE = +15.7 K, above the 6 K target. The NH₃-H₂ estimate moves T_match by 27 K, 4.5 times the target: it is the largest remaining uncertainty.

Closures. The all-LTE closure (phase3/energy_coupling/) was stopped at k = 2: the rates entered the module while it ran, so its nlte would have changed meaning mid-record. Its k00-k01 stand as the all-LTE record (T_match 337.7, 335.8 K; log₁₀ Mdot 7.880, 7.870 against 7.800 without the coupling). The closure with the published rates runs in phase3/energy_coupling_nlte/ (lart4, 12 threads), seeded as before.

79. The EW of the energy-coupled state; the NH₃ nu₂ + H₂ rate as an option and its bracket (2026-10-03)

The coupled closure converged (phase3/energy_coupling_nlte/, section 78) at k = 2: eps_H 0.015, eps_He 0.043 (tol 0.05), T_match moved 1.15 K (tol 6 K). Its EW, by the reference operator (ew_of_state.sh, EW_DIAGNOSTIC=1, binary 6af10ba4, R = 68000, solution and adv), is energy_coupling_nlte/ew_k02/EW.txt.

--nh3-h2-rate-analog (photochem_to_lower_profile.py; module conducted_heat_column_balance.py, NH3_H2_RATE_ANALOGS, ColumnResponse(nh3_h2_rate_analog=...)). The unmeasured NH₃ nu₂ + H₂ de-excitation rate is the measured NH₃ nu₂ + He rate times k(H₂)/k(He) of an analog:

- cd4_bending (default): CD₄ bending modes, Siddles et al. 1994 Tables 1-2, 27.5 (295 K) to 49 (190 K);
- water_bending_overtone: H₂O 2nu₂ at 295 K, k(H₂) = 2.9e-12 cm³ s⁻¹ (Zittel & Masturzo 1991, J. Chem. Phys. 95, 8005, Table I, H₂(16)O) over k(He) = 3.2e-13 (Hovis & Moore 1980, Table III, H₂(18)O, 296 K) = 9.06, held constant in T (the H₂ rate exists at 295 K only). Zittel & Masturzo 1989 Table III has no He value. Both table values checked on the published text. Used by treatment nlte only. The profile header and conducted_heat_column.txt record the analog (nh3_h2_rate_analog) for the nlte treatment; the EXHALE

reader ignores the key. The solution_id does not include it (it includes column_T_match, which differs). Documented in README_HOWTO.md with --column-energy-treatment.

Tests (src/tests/conducted_heat_column): 41 PASS. New: the water estimate equals the He rate times $2.9 \times 10^{-12}/3.2 \times 10^{-13}$ at 185, 295 and 560 K; the default column solution equals the explicit cd4_bending one bit for bit; the water analog lies between CD4 and no H2 term in T_match; the analog does not enter nlte_partial; an unknown analog is refused. RED before the change (module reconstructed without the option): 29 PASS, then a TypeError at the first new assertion. The default nlte, nlte_partial and lte column solutions of the test column are bit-identical to the module before the change (T_match 352.358, 380.122, 336.988 K).

Bracket (phase3/energy_coupling_bracket/, NOTES.md there). Two single wind solves at the coupled state's fluxes, not new closures: the closure driver with --kmax 1, the k02 trial elemental fluxes, $F_{\text{cond}} =$ the 9.6264×10^2 measured, seeded from k02's output through map_state_to_grid.py --ic, the closure's binary, keys and environment; only the adapter option changed.

case	T_match [K]	F_cond trial / measured [erg cm ⁻² s ⁻¹]	log10 Mdot [g/s]	EW solution [%A]	EW adv [%A]	eps_H / eps_He
uncoupled reference	185.0	- / 10.016	7.8029	0.5793	0.5643	-
coupled, CD4 analog (converged)	349.88	9.7267 / 9.6264	7.8854	0.7255	0.7114	0.015 / 0.043
water 2nu2 analog (single solve)	358.01	9.6264 / 9.6227	7.8928	0.7401	0.7262	0.023 / 0.063
no NH3 + H2 term (single solve)	375.29	9.6264 / 9.5429	7.9041	0.7629	0.7491	0.035 / 0.092

(MEASURED; log10 Mdot at cell N-20 recomputed from the states, the history's 7.890 for k02 being the two-decimal post-processing print; the reference's 10.016 is the section 76 evaluate.)

- The energy coupling raises the EW by 0.146 %A (+25 per cent), 25 times the numerical target 0.0058, and Mdot by 21 per cent. The EW stays below the observed 1.108 ± 0.029 .
- The NH3 nu2 + H2 estimate: water analog +0.015, no H2 term +0.037 %A against CD4, 2.5 and 6.4 times the target. It remains the largest single uncertainty of the coupled EW.
- The neglected re-iteration: the conducted flux is closed (out/in -0.04 and -0.9 per cent, about 1 K of T_match); the He flux is not (eps_He 0.063 and 0.092). Its effect on the EW was not measured.
- Figures: [energy_coupling_bracket/fig_coupled_profiles{,_inner}.png](#), [fig_bracket_profiles{,_inner}.png](#), [fig_bracket_differences.png](#). The coupled wind is 18-37

per cent denser beyond 1.1 R_p , its temperature peak about 80 K higher, and $n(\text{He } 2^3\text{S})$ up to 1.5 times higher at 30 R_p .

80. The NH3 nu2 + H2 bracket re-closed: both cases converge (2026-10-04)

The two bracket cases of section 79 were continued from their single wind solves as full energy-coupled closures, identical to phase3/energy_coupling_nlte/ (binary 20021893, Photochem arguments, exhale_env, keys, Resid tol 4e-6, 12 OpenMP threads, omega 0.5, omega_E 0.5, tol 0.05, T_match_tol 6 K, kmax 12) except the column option: phase3/energy_coupling_water_analog/ (--nh3-h2-rate-analog water_bending_overtone) and phase3/energy_coupling_no_h2/ (--column-energy-treatment nlte_partial). Each was seeded from its bracket k00/output, with first trial elemental fluxes and F_{cond} equal to the driver's own damped update from that k00 (trial + 0.5 (measured - trial); F_{cond} 9.624561 and 9.584647 $\text{erg cm}^{-2} \text{s}^{-1}$). Both converged at $k = 1$ (two wind solves each, info 0, about 2-3 min each). EW by the operator of section 79 (ew_of_state.sh, EW_DIAGNOSTIC=1, binary 6af10ba4). Comparison: phase3/energy_coupling_bracket_converged/ (compare.py, compare.txt, NOTES.md).

case	k	T_match [K]	F_{cond} trial / measured [$\text{erg cm}^{-2} \text{s}^{-1}$]	log10 Mdot [g/s]	EW solution [%A]	EW adv [%A]	eps_H / eps_He
CD4 analog, converged	2	349.88	9.7267 / 9.6264	7.8854	0.7255	0.7114	0.015 / 0.043
water 2nu2 analog, converged	1	357.97	9.6235 / 9.6221	7.8928	0.7400	0.7261	0.006 / 0.016
no NH3 + H2 term, converged	1	374.62	9.5744 / 9.5692	7.9048	0.7643	0.7505	0.009 / 0.024
water 2nu2 analog, single solve (section 79)	0	358.01	9.6264 / 9.6227	7.8928	0.7401	0.7262	0.023 / 0.063
no NH3 + H2 term, single solve (section 79)	0	375.29	9.6264 / 9.5429	7.9041	0.7629	0.7491	0.035 / 0.092

(MEASURED; $\log_{10} \text{Mdot} = 4 \pi \rho v r^2$ at cell N-20 of the written state.)

- Re-closing changes the single-solve values by: water T_match -0.03 K, $\log_{10} \text{Mdot}$ -0.00004, EW -0.0001 (solution and adv); no H2 term -0.67 K, +0.0007, +0.0014. The re-iteration left open in section 79 moves the EW by at most 0.0014 %A, a quarter of the numerical target 0.0058.

- The bracket against the converged CD4 state: water +0.0145 / +0.0147, no H2 term +0.0388 / +0.0391 %A (2.5 and 6.7 times the target); T_match +8.1 and +24.7 K.
- The water k01 state: the solve certified it as written (mass row 2.17e-12 against 3.0e-12), but the evaluate of the reloaded state on 6af10ba4 refuses it on the mass row alone (3.316e-12 against 3.0e-12 at cell 253, r = 1.377 R_p; evaluate exit 2), so its EW products carry the label diagnostic. The EW is 1e-4 from the certified single solve of the same case. The cause of the reload difference was not investigated.
- Figures energy_coupling_bracket_converged/fig_converged_profiles{,_inner}.png: against CD4 the water and no-H2 winds are warmer below 1.1 R_p (up to +11, +29 K), cooler at 1.19 R_p (-11, -32 K), warmer beyond (+13, +33 K near 2.2 R_p; +3, +8 K at 30 R_p); n(He 2³S) is 0.96 and 0.89 times the CD4 value just outside the front at 1.15 R_p and 1.04 and 1.10 times at 30 R_p.

81. He 2³S transport: the metastable carried as a fourth helium state (2026-10-04)

Plan md/PLAN_20261003_lhs1140b_ew.md phase 3 step 3 (decision 4); design md/He23S_transport_design_20261003.md, approved with the key name He 2³S transport and Ionization transport: True required. Phase 1 (LHS1140b/models/.P2/ew_fit/phase1/NOTES_phase1.md section 1) had measured that the local balance of the level is exact inside 2 R_p and 1-4 % off at 3-30 R_p, where the 10830 line forms.

What the option does. He 2³S transport: True (default off) makes $x_3 = n(2^3S)/n(\text{He nuclei})$ carrier ic_HeTR (index 8) of the transport-chemistry operator. It rides on the helium nucleus flux with the stage flux of ionization_stage_transport.f90 equation (1), its source is the level balance tr_triplet_row (row 4 atomic, row 8 molecular with the H2 quench) at the trial densities, the singlet closes the three carried helium states, and every sweep is handed the transported value (x_hetr_fix replaces the level row). It is the steady advection balance of Oklopčić & Hirata (2018, Eq. 15) and Lampon et al. (2020, Eq. 11) with the stage eddy term.

Files.

- parameters.f90 (he23s_transport; the stale comment of ionization_transport, which said H+ alone and "requires Molecular carrier transport", corrected to the code: three stages, atomic gas allowed); input_read.f90 (key, refusals without Include He23S or Ionization transport); load_IC.f90 (token he23strans, the 26th, not a layout token, F when absent; coupling tag and notes); utilities.f90 (he23strans=T on the coupling line); write_setup_report.f90 (setup text; he23s_transport in EXHALE_resolved.out and the parse dump only when on).
- diffusive_photochemistry.f90: index, name He2³S, set and registration, the unknown, lookups, carrier_is_nucleus_fraction (the three stages and the level: the carriers written per element nucleus on the nucleus flux; carrier_is_ionization_stage keeps naming the three stages), the face state, the nucleus sum with nk = 3, the helium availability without the reservation of the level, the singlet closure with the trial level, the source row and its ledger, the FD guard and the projection on the three-dimensional simplex, the write-back of the level from its own row, the headroom of the level (the helium nucleus density: element_box_side refuses an excited level), and helium_metastable_excitation_energy_divergence (a measurement, item 3

below).

- `binary_element_diffusion.f90` (`n_car_max = n_bsp`), `ion_cell_state.f90` (`x_hetr_fixed`, `x_hetr_row`, `x_hetr_fix`), `ion_residual_core.f90` (the imposed level row), `ionization_equilibrium.f90` (flags, seed, release in the rate matrix, OpenMP shared list), `constrained_chemical_equilibrium.f90` (fixed species, seed, debug record), `attempted_step.f90` (hash), `certification.f90`, `ionization_stage_transport.f90` (header).
- Tests: new `src/tests/helium_metastable_transport/` (run.sh and driver);
 `src/tests/certification/certification_contexts.f90` (below).

Decisions taken in the implementation.

1. Entry name `carrier balance He2^3S` (no blank: the carrier name is one whitespace-delimited field of the row-term record).
2. Certification: `carrier balance He2^3S` at `cert_tol_carrier_wind`; `level balance He 2^3S` not applicable under the key; `ionization stage nucleus sum He` with `nk = 3`.
3. The excitation-energy flux of the level relative to He I is NOT a term of the energy row, contrary to design section 4 ("added if it exceeds $1e-3$ "): the 19.82 eV are chemical energy, booked where the level's reactions run (q13 cooling, q31g heating, Penning heat) at the transported population, so adding the flux would count them twice. It is printed by every certification as a reported-only line. MEASURED on the validation state: $\max |d_{q_{xl}} / (\text{energy row's largest term})| = 3.0e-7$ over the column (cell 198, $1.14 R_p$), $6.6e-9$ at $r \geq 1.2 R_p$; written at the code site.

Tests (gfortran private build; all MEASURED).

- `helium_metastable_transport`: 22 assertions, all PASS (RED before: the driver does not compile against the entry text, which has no `ic_HeTR`). The level is the solved last carrier on the nucleus flux; the available helium reserves `HeH+` alone ($1e-14$); a partition between the old reserved bound and the new simplex is left unchanged, one outside is returned to it; the write-back keeps H and He nuclei and mass to $1.1e-16$ and writes the level from its row; the ledger `sprod - sloss = src` holds for the level in molecular and atomic gas; at the closed-form root of the level balance the row reads $7.4e-17$ of its terms; $d \text{src}(\text{He II})/d n_3$ is the analytic $g_3 - g_1 + n_e (b_3 - b_1)$ to $1e-6$. Manufactured three-state helium column, 250/500/1000: 2^3S row $7.29e-7$, $9.68e-8$, $1.25e-8$ (observed order 2.935), extrapolated $7.5e-8$ at the anchoring spacing $1.8447e-3 R_p$ and $5.0e-7$ at the gating spacing $3.5215e-3 R_p$ (gate $1e-5$); nucleus-sum identity $4.9e-16$ against the `nk = 3` bound $1.5e-14$.
- The 22 existing suites whose drivers link the changed modules (`adv_static_limit`, `carrier_boundary_jacobian`, `carrier_constraint_attribution`, `carrier_enthalpy_flux`, `carrier_helium_inventory`, `carrier_outer_boundary`, `carrier_reference_scales`, `carrier_retry`, `carrier_returned_state_acceptance`, `certification`, `coupled_block_handover`, `coupled_block_jacobian`, `element_operator`, `grid_and_gates`, `interdiffusion_enthalpy_flux`, `ionization_imposed_fractions`, `ionization_stage_enthalpy_flux`, `ionization_stage_flux`, `krylov_and_dogleg`, `species_masses`, `stage_row_balance`, `steady_species_rows`): all pass except two items. (a) `certification_contexts` assumed an even report capacity (its last entry's tolerance and verdict were

hard-coded to the helium entry); the capacity grew by one carrier and became odd, so the expectation is now taken from the same parity as the name and measure (test defect; 145 PASS after). (b) `grid_and_gates/restart_intent_and_metadata.sh` fails three `stationary_evaluate_*` assertions (a rebuilt O III of $6.0\text{e-}5$ against $1\text{e-}8$, the stored certified fixture re-certified F, exit 2) with the tree's own EXHALE.x built before this change (md5 60fa14bf) and identically with the new build: pre-existing, the stored fixture is no longer certified under the current equations; not fixed (it needs the fixture re-made).

- Default off, one stored case: the EW evaluate (`ew_of_state.sh`, `EW_DIAGNOSTIC=1`) of `energy_coupling_nlte/k02` with the new binary writes `Hydro_ioniz`, `Ion_species` and their `_adv` pair with data rows bitwise identical to the reference evaluate made with 6af10ba4 (`energy_coupling_nlte/ew_k02`); the headers differ only by the run time and the new token `he23strans=F`; `EW 0.7255 / 0.7114 % A` as before. (The first build of this evaluate stopped in the all-carrier headroom loop on the level, `element_box_side`; that is the headroom fix above, made before any result was taken.)
- Restart contract with the binary: off -> on accepted with the token named and refused unnamed; on -> off likewise; the key without `Ionization transport` or without `Include He23S` refused at startup.

Binary. LHS1140b/models/.P2/EXHALE_4c0e8603.x (ifx 2026.1.0, md5 4c0e86033617eca84914079aa61a8edf; manifest .P2/BINARY_MANIFEST_4c0e86033617.txt, source copy .P2/build_he23s_ifx/): the source of 20021893 (the tree still held its changes, checked file by file against that manifest) plus this section.

Validation (LHS1140b/models/.P2/ew_fit/phase3/he23s_transport/). `solve/` restarts the converged `energy_coupling_nlte/k02` state with `He 2^3S transport: True` and `Restart option change: he23strans`, the closure's keys and environment (`Resid tol 4.0e-6`, 12 threads, Well balanced, stationary intent, Newton, Secondary ionization Immediate). It is CERTIFIED in 10 outer passes; the binding row of every pass was carrier balance `He2^3S` at cell 217 ($r = 1.20 R_p$), $5.8\text{e-}2$ on pass 1 and $8.59\text{e-}6$ on pass 10 (gate $1\text{e-}5$); below the gate the row reads $1.48\text{e-}4$ at cell 1 (reported). EWs with the same operator and binary 4c0e8603 for both states (the transported state's input carries the new key, which 6af10ba4 does not read):

	local balance (k02)	transported
EW solution [% A]	0.7255	0.7334
EW adv [% A]	0.7114	0.7114
log10 Mdot [g/s], cell N-20	7.885432	7.885431
$n(2^3S) / \text{local}, 3.02 R_p$	1	0.99951
10.03 R_p	1	1.03947
29.03 R_p (cell N)	1	1.16845
level entry	level balance He 2^3S $9.5\text{e-}21$ (READ, k02/run.log)	carrier balance He 2^3S $8.59\text{e-}6$

(MEASURED except where marked.) $n(2^3S)$ moves by less than $4\text{e-}5$ inside $2 R_p$, by -0.45% at most near $5.7 R_p$, and by $+18 \%$ at most near $25 R_p$; T moves by at most $3.1\text{e-}6$ and v by $1.1\text{e-}5$ relatively. The solution EW rises by 0.0079% A ($+1.1 \%$), 1.4 times the numerical target 0.0058 . The phase-1 estimate ($+0.013$ to $+0.019 \%$ A) was made on a different state (He/H 9 deep, EW 8.56% A), so the two are not directly comparable; the shape is the same

(negative at 3-7 R_p , positive beyond), the outer rise is larger here (+17 % at 29 R_p against +4 % there). The adv product is unchanged (0.7114): it rebuilds x_3 along streamlines from the hydrodynamic state, which moved by $1e-5$. Against it, the Lagrangian x_3 of `post_process_adv` on the transported state lies 1-4 % below the transported x_3 at 1.5-10 R_p and 1.8 % below at 29 R_p ; it reads 3.2-3.7 % below the local balance at 2-8 R_p , where the local balance is exact, which appears to come from the adv product's own H/He partition on its molecule-free background (module header of `post_process_adv.f90`) rather than from the level's transport; not investigated further. [Traced in section 82: the adv product advected every element on the bulk velocity; corrected there.] Elemental fluxes moved by $-7e-7$ (H) and $-1.8e-6$ (He) relatively and the base conducted heat by $+2e-7$, far inside the 5 % closure tolerance (READ, `closure.json`), so the Photochem closure was not re-run. Figure `he23s_profile_comparison.png` ($n(2^3S)$ of the two states and the adv product; ratios to the local balance).

Not verified. The option in an atomic gas end to end (the unit tests cover the atomic rows; the validation run is molecular); the marching route with the option (the advected register of the level and the conversion of its advected mass fraction into a fraction of the helium nuclei belong to the marching step, which no run of this section took); the inner ghost value of x_3 is the sweep's local root as for the stages (not re-examined).

Records. `md/input_schema.md` (K15f2, K44), `docs/EXHALE_user_manual.tex` (the new key; the Ionization transport entry rewritten to the code: three stages, atomic gas allowed, Coupled carrier solve refused, which the old text had the other way round), `README_HOWTO.md`, `docs/physics_overview/{ionization,molecules_metals,summary_table}.tex` (the stale "fifth carrier on the mass flux" paragraph rewritten), `md/He23S_transport_design_20261003.md` section 11.

82. The advected product steps each element on the velocity of its own nuclei (2026-10-04)

Finding. Section 81 left open why the `_adv` product of the He 2^3S -transported reference (LHS1140b/models/.P2/ew_fit/phase3/he23s_transport/) reads $n(2^3S)$ 3.2-3.7 % below the solution at 2-8 R_p , where the level is in local balance. The level is not the cause: $n(2^3S)/n(He^+)$ of the adv product agrees with the solution's to 0.1 % there (MEASURED), and the 2^3S deficit follows a He^+ deficit (He^+ 0.963-0.970 of the solution, H^+ 1.056-1.062). The stages are transport-dominated at those radii (the He^+ carrier row of the solution balances its net source against the stage-flux divergence, `solve/output/carrier_row_terms.txt`), so their values depend on the transport law, and the two products used different ones:

- the solution carries every stage on its element's nucleus flux (`ionization_stage_transport.f90`, equation (1)), i.e. on v_H and v_{He} ;
- the adv product stepped every row with $c1 = g h/v$, the mass-weighted velocity (`post_process_adv.f90` `composition_step`, the old line `adv_cell%c1 = step_width(jc, bdf2)*R0/(v(jc)*v0)`; the adv systems `System_implicit_adv_{H,HeH,HeH_TR}.f90` used that one $c1$ for every row). With $He_{diffusion}$ on and the strong separation of this state (He/H 2.09 at the base, 0.59 at 29 R_p) the nuclei do not move with v : $v_{He}/v = 0.82-0.97$ and $v_H/v = 2.1-1.08$ at 1.3-4 R_p (MEASURED from `element_flux_profile.txt`). The stage eddy term of (1), which the marching step also lacks, is small: $K |d \ln x/d r|/v_{el} = 0.035$ (He^+) at 1.3 R_p , 0.005 at 1.5 R_p , below 0.001 beyond 1.75 R_p ($K = 1e9 \text{ cm}^2 \text{ s}^{-1}$). Directions agree: the adv helium rode too fast

(less ionized), the adv hydrogen too slow (more ionized). **Verdict: a physics defect of the adv product whenever element diffusion is on**; the fractions it steps are fractions of one element's nuclei, which move with that element's velocity.

Fix. With element continuity $\text{div}(n_{\text{el}} v_{\text{el}}) = 0$ the steady stage balance $\text{div}(x n_{\text{el}} v_{\text{el}}) = S$ is $v_{\text{el}} dx/dr = S/n_{\text{el}}$.

- `post_process_adv.f90`: $v_{\text{He}} = v + J/(\rho Y)$ and $v_{\text{H}} = v - J/(\rho (1 - Y))$ per cell, J the diffusive helium mass flux of the state (`helium_diffusive_face_flux`, the flux the run's element row is the divergence of; mean of the two faces), Y the helium mass fraction (`element_mass_fractions`), ρ from `mixture_mass_sum`; both equal v without `He_diffusion`. `composition_step` sets $c1 = g h/v_{\text{H}}$ and the new $c1_{\text{he}} = g h/v_{\text{He}}$. The validity conditions use them: inflow of either element refuses the cell (condition (i)), and the Damkohler number of condition (ii) takes each element's rates with its own residence time. The omitted stage eddy term and its measured size are written at the code site.
- `ion_cell_state.f90` (`adv_rates%c1_he`, the `step` statement), `System_implicit_adv_HeH.f90` and `System_implicit_adv_HeH_TR.f90` (the helium rows on $c1_{\text{he}}$), `System_implicit_adv_H.f90` (comment), `charge_exchange.f90` (`he_h_cx_fvec_adv` takes $c1$ and $c1_{\text{he}}$: the H row on $c1$, the He rows on $c1_{\text{he}}$).

Tests. New suite `src/tests/adv_element_velocity/` (driver `element_velocity_rate_weights.f90`): on both adv layouts, with $(c1, c1_{\text{he}}) = (c, c)$ and $(c, 2c)$ the rate part of the hydrogen row does not move and every helium row doubles, charge-exchange pair included; with $(2c, c)$ the reverse. 9 assertions PASS on the private build (relative change 0 against $1e-12$); RED on the entry text (compile error: $c1_{\text{he}}$ not a member of `adv_rates`). `src/tests/charge_exchange_rows/advection_pair_reservoir.f90` sets $c1_{\text{he}}$. Suites whose drivers use the changed modules, private gfortran build: `adv_static_limit` 64, `charge_exchange_rows` 257, `charge_exchange_detailed_balance` 298, `attempted_step` 70, `carrier_constraint_attribution` 10, `carrier_helium_inventory` 15, `carrier_reference_scales` 16, `carrier_retry` 156, `constrained_network_layout` 8, `ionization_imposed_fractions` 48, `stage_row_balance` 10, `physics_probe` 2499 PASS; 0 FAIL. Without element diffusion the change is the identity: the evaluate of the stored backup/regression/`wasp_full_newton` state (`he_diffusion F`) with a gfortran control build of the entry text and the gfortran build of the change writes `Hydro_ioniz_adv.txt` and `Ion_species_adv.txt` with data rows bitwise identical (504 rows, 503 corrected cells).

Effect on the reference (`he23s_transport/ew_solve_advvel/`, binary `LHS1140b/models/.P2/EXHALE_8cc7b546.x`, ifx, md5 `8cc7b5462cc2390eefed1bb3658e307f`, manifest `BINARY_MANIFEST_8cc7b5462cc2.txt`; the source of `4c0e8603` plus this section). MEASURED, adv product against the solution:

$r [R_p]$	$n(^3\text{S})$ bulk v	$n(^3\text{S})$ element v	He+ bulk / element	H+ bulk / element
2.00	0.9680	1.0122	0.9668 / 1.0121	1.0480 / 0.9604
3.02	0.9635	1.0066	0.9629 / 1.0082	1.0615 / 0.9679
4.98	0.9662	1.0028	0.9666 / 1.0063	1.0604 / 0.9726
7.95	0.9686	1.0014	0.9698 / 1.0054	1.0576 / 0.9750
20.10	0.9728	1.0009	0.9743 / 1.0045	1.0524 / 0.9779

adv EW 0.7114 -> 0.7369 % A (+0.0255, 4.4 times the numerical target 0.0058); the solution EW 0.7334 does not depend on the adv product. The remaining offset starts at the first corrected cell (1.157 R_p), at the He ionization front, where He⁺ rises by a factor 17 in seven cells and the first adv step jumps by +15 % in He⁺ against the solution (it did so with the bulk velocity as well, +14 %); the frozen-in stages carry it outward. It is a difference between two discretizations of one equation across a front resolved by a few cells (upwind limited face fluxes against a backward-Euler start of the BDF2 march), not a missing term; it was not reduced further. Figure [he23s_transport/acceptance/fig_adv_vs_solution.png](#).

Superseded. Every adv EW of a run with He_diffusion: True made before this section was stepped on the bulk velocity: the adv columns of sections 79-81 (0.7114, 0.7261, 0.7505, 0.5643 and the bracket's) and every adv number of the LHS 1140 b catalog with element diffusion. They are not recomputed here.

83. The EW evaluate reloads the exact restart (2026-10-04)

Finding. Section 80 recorded that the water-analog state (phase3/energy_coupling_water_analog/k01), certified when written (mass row 2.17e-12 against 3.0e-12 at cell 255), was refused by the EW evaluate (ew_k01/: mass row 3.316e-12 at cell 253, exit 2, EW labels diagnostic). The evaluate did not load the exact restart: LHS1140b/models/.P1/fiducial_conduction/ew_of_state.sh staged only the dimensional pair, Hydro_ioniz.txt and Ion_species.txt as the _IC files, and never output/conserved_state.txt, which section 63 writes beside every 'eq' write and which init.f90 reads as output/conserved_state_IC.txt. The run therefore rebuilt (rho, rho v, E) from 16-digit rho, v, p with the energy re-formed from p through the caloric equation of state, and the mass row in the wind, which sits at the rounding level of the face flux difference there, moved past its 3.0e-12 floor at one cell. A defect of the script, not of the code. The generation=absent of EW.txt is not related: the state_id field is written only on a pass-state generation, and a final 'eq' write carries none (the script's header now says so).

Fix. ew_of_state.sh stages conserved_state.txt as conserved_state_IC.txt when the state has one, records its md5 (exact_state) and whether the evaluate read it (exact_state_read, from the reader's own message or its IGNORED notice) in EW.txt. element_flux_closure.py already staged it (section 63). LHS1140b/models/.P2/ew_fit/phase1/run_after.sh stages the pair only; it is the record of a finished measurement and is left as it ran.

Verification (MEASURED; same binaries as before).

state	binary	evaluate	certification on reload	EW solution / adv [% A]
water k01, text reload (ew_k01/)	6af10ba4	exit 2	REFUSED, mass 3.316e-12 / 3.0e-12, cell 253	0.74001762 / 0.72605621 (diagnostic)
water k01, exact (ew_k01_exact/)	6af10ba4	exit 0	CERTIFIED, mass 1.348e-11 / 1.6e-11 (rounding anchor), cell 207	0.74001762 / 0.72605621 (result / conditional)

state	binary	evaluate	certification on reload	EW solution / adv [% A]
reference, text reload (he23s_transport/ew_solve/)	4c0e8603	exit 0	CERTIFIED, mass 2.83e-12 / 3.2e-12, cell 250	0.73339944 / 0.71142279
reference, exact (he23s_transport/ew_solve_exact/)	4c0e8603	exit 0	CERTIFIED, mass 2.00e-12 / 3.0e-12, cell 252	0.73339944 / 0.71142279

With the exact state the physical cells of u are restored bitwise (the evaluate's own conserved_state.txt against the solve's: U rows identical except the two lower ghosts, which the boundary rebuilds). The species fractions are not bitwise: the stationary evaluate takes one composition sweep from the loaded composition (its log: chemical residual $1.6\text{e-}16$, particle count $4.5\text{e-}16$), which moves f_{sp} by at most $4.4\text{e-}13$ relative (text reload: $2.2\text{e-}12$), and the mass-row verdict cell moves with it (the solve wrote cell 255 at $2.17\text{e-}12$; the exact reload binds at cell 207). The mass row in the wind is a rounding-level quantity against a floor of the same order; that is the certification's design (the rounding anchor of each cell), not changed here.

84. Phase 3 of the LHS 1140 b EW plan accepted; TO_BE_DONE (AH) closed (2026-10-04)

Plan md/PLAN_20261003_lhs1140b_ew.md phase 3 steps 4-6 on the reference LHS1140b/models/.P2/ew_fit/phase3/he23s_transport/solve/ (section 81; binary 4c0e8603). Record and figures: he23s_transport/acceptance/ (NOTES.md, acceptance.py, acceptance_numbers.txt, five figures, all inspected).

Step 4, the elemental-flux closure and (AH). The closure measures on element_flux_profile.txt, whose face fluxes are the Riemann face mass flux of the certified state (write_element_flux_profile, called from certification.f90 with $Frho_{\text{cert}}$), reduced over the overlap window by the code (lower_profile_F_* of EXHALE_resolved.out). Re-measured from the profile itself on the reference: window 1.00058-1.01892 R_p , 82 faces (not empty); F_H $2.32443\text{e}7$ g/s, F_{He} $5.35726\text{e}7$ g/s, spreads $3.8\text{e-}9$ and $3.1\text{e-}9$ (the face-mean measure of Phase E gave 60-700 %); the fluxes are constant through the whole column to the 8 printed digits. Against the Photochem trial fluxes of the closure's last iterate ($2.28992\text{e}7$, $5.12443\text{e}7$): 1.48 % (H) and 4.35 % (He), inside the 5 % tolerance, so the closure's 5 % agreement holds on the Riemann measure. TO_BE_DONE (AH) closed with these numbers; the older atomic_photochem_*_kzzprofile rungs it named were not re-evaluated (superseded by the molecular closure).

Step 5, spectrum and EW. Solution 0.7334 % A (decision 2's acceptance basis), adv 0.7369 (section 82); observed 1.1084 ± 0.0289 , so the He/H 2.09 reference lies 0.375 % A (13 sigma, independent-sample error) below the observation with the full equation set. The line is twice as deep and three times narrower than observed (peak 2.52 % against 1.28 %, FWHM 0.24 A against 0.76 A on a 3-sample running mean of the observation), and the observed peak sits 0.15 A redward of the model's.

Step 6, acceptance.

- Certified wind under unchanged tolerances (solve/run.log, final state): mass $3.36\text{e-}12$ / $4.3\text{e-}12$ (rounding-anchored, cell 242), momentum $2.3\text{e-}11$, energy $1.5\text{e-}8$, carriers H2 $1.7\text{e-}7$, H+ $4.0\text{e-}9$, He+ $6.3\text{e-}8$, He++ $1.7\text{e-}7$, He²3S $8.59\text{e-}6$ (gate $1\text{e-}5$, $r \geq 1.2 R_p$), nucleus sums $2.2\text{e-}16$ / $4.3\text{e-}16$, elemental transport $2.7\text{e-}12$. Re-certified on the exact reload.
- Photochem at steady state: reached_steady_state T, iteration 2; the profile is the kO2 one byte for byte.
- Flux agreement over the stated overlap region: above.
- EW, line profile and full profiles against the observation: figures fig_line_profile.png, fig_profiles.png, fig_adv_vs_solution.png.
- Kn, Mach, HeH+ Damkohler (fig_validity.png): Kn_bulk = 0.1 at $5.69 R_p$ and 1 at $17.21 R_p$; sonic point $25.7 R_p$, Mach 1.08 at the boundary; HeH+ Da (code rates R16, R18, R19; advection and eddy times) $3.4\text{e}6$ at the n(²3S) peak ($1.40 R_p$), 94 at $5 R_p$, 3.4 at $10 R_p$, 0.12 at $29 R_p$; Da < 10 in 82 cells holding 2.0 % of the radial He ²3S column.

Not verified. The outer region (beyond Kn_bulk 0.1, where most of the line forms; exobase inside the domain) is phase 4; the ion-neutral drift and the NH3 nu2 + H2 bracket are not in the EW; the HeH+ Damkohler number leaves out molecular diffusion.

Carried into phase 4. State he23s_transport/solve/; binary for new calculations .P2/EXHALE_8cc7b546.x (4c0e8603 plus section 82, which changes only the adv product: the evaluate of the reference writes Hydro_ioniz.txt, Ion_species.txt and Heating_breakdown.txt with data rows bitwise identical on 4c0e8603 and 8cc7b546, MEASURED; a full solve on 8cc7b546 was not run).

85. Phase 4 of the LHS 1140 b EW plan: outer domain, outer boundary, transit-weighted shares, Roche geometry (2026-10-04)

Plan md/PLAN_20261003_lhs1140b_ew.md phase 4 on the phase-3 reference (section 84). Record, scripts and figures (all inspected): LHS1140b/models/.P2/ew_fit/phase4/NOTES.md. Binary .P2/EXHALE_8cc7b546.x (ifx), the reference's keys and environment, Photochem closure not re-run. EW in percent-angstrom (% Å); numerical target 0.0058. All MEASURED unless marked READ.

Roche geometry (code unit R0 = 0.16253525 R_J): Hill radius $38.01 R_0$, L1 37.61 , volume-equivalent Roche-lobe radius (Eggleton) 26.72 (nominal R_p : 39.18 , 38.77 , 27.54). The reference runs Domain mode: Spherical. EXHALE_transit.py ends the impact parameters at $R_*/R_p = 12.93$.

EW against the outer radius. Same keys; base block kept; for 15 and 20 the cell count that keeps the reference's stretch ratio (widths at shared radii within $7\text{e-}4$); beyond 30, Outer_shells [r_face, cells]: 37.6 15 (face at L1, interior bit for bit the reference's); seeds by map_state_to_grid.py --ic. All certified (21-23 passes); every final pass bound by carrier balance He²3S at cell 217.

outer radius	grid	EW sol / adv	log10 Mdot	sonic point	Kn = 0.1	Kn = 1
15	458, face 14.65	0.7565 / 0.7602	7.885528	none (Mach 0.24 at edge)	5.97	not reached

outer radius	grid	EW sol / adv	log10 Mdot	sonic point	Kn = 0.1	Kn = 1
20	476, face 19.52	0.7044 / 0.7078	7.885798	none (Mach 0.57)	5.62	not reached
30 (reference)	500, face 29.27	0.7334 / 0.7369	7.885456	25.69	5.69	17.21
37.6	515, face 37.60	0.7360 / 0.7396	7.885425	25.73	5.70	17.29

Elemental fluxes move by $\leq 1e-3$ relative and the conducted heat by $\leq 2.5e-4$ against the reference, all inside the 5 % closure tolerance (the neglected re-closure). With the face past the sonic point the extension to L1 changes the EW by +0.0026 (0.45 of the target); faces inside it (15, 20) put the flow on another subsonic branch beyond 5 R_p and change the EW by +3.2 % and -4.0 % (not monotonic).

Outer boundary condition. The code has one rule for each outer closure and no key or environment variable selecting another (READ): hydrostatic free-outflow ghost (the edge is supersonic here, Mach 1.08); zero diffusive flux for conduction ($1.2e-3$ to $3.0e-3$ of the advected enthalpy flux at the last interior faces), for the element operator (a one-cell layer, X_He -0.12 % at face 499) and for the carriers (eddy/advective $1.7e-8$ to $2.8e-7$). No admissible alternative exists to run; the last two cells carry $1.5e-4$ of the EW, so the element wall of TO_BE_DONE (m) / section 67 cannot move the EW of this reference by more than $1e-6$ relative (the proposed outflow option is not needed here and was not implemented). The cell-500 momentum row of section 67: the momentum row is largest at the last cell on every domain, $2.3e-11$ (30) and $5.5e-11$ (37.6) against $1e-8$; it does not affect this reference.

Transit-weighted shares ($n(2^3S)$ truncated beyond R in the transit synthesis of the solution state, EW differenced): beyond Kn = 0.1 (5.69 R_0) 39.8 % (radial column 5.3 %); beyond R_*/R_p 4.5 %; beyond Kn = 1 (17.21) 1.09 %; beyond the sonic point 0.094 %; beyond the volume-equivalent Roche radius 0.050 %.

Roche geometry of the transit. EXHALE_transit.py has no He 10830 triaxial branch; phase4/roche/roche_he10830.py restates the tool's triplet transit with the state on Roche equipotentials (roche_recon.ReconMap); spherical mode reproduces the tool to $2.7e-7$ in transmission (EW 0.733390 against 0.733399). Option (b) on the reference: 0.7247 (-0.0087, -1.2 %). The reference re-solved in the Roche potential (Domain mode: Roche, certified in 19 passes, sonic point 22.35): 0.7240 with the spherical transit and 0.7156 (-0.0178, -2.4 %) with the Roche reconstruction, the self-consistent pair. Option (c): Kn < 0.1 alone gives 0.4412, a bracket [0.441, 0.733] 50 times the target. Recommendation (for decision 3, not applied): keep the 30 R_p spherical domain for the scan and state the tidal geometry as a measured interval [Roche-consistent, spherical] at the two models bracketing the crossing; the full comparison is NOTES section 6.

Corrected on the way. EXHALE_transit.py, triaxial branch (triaxial_depth, off by default): the impact parameters ran to the outermost grid radius while the depth was normalized to the annulus up to $R_{ib} = \min(r_N, R_*/R_p)$, so for $r_N > R_*/R_p$ the mean transmission of the wider annulus was applied to the narrower one. The impact parameters now end at R_{ib} . Spherical products data-identical before and after on the reference (He 10830, H-alpha, Ly-alpha); the triaxial H-alpha line-center depth of the reference moves 0.001 $\rightarrow$ 0.004 %; unchanged by construction when $r_N \leq R_*/R_p$ (checked with $R_* = 1 R_{sun}$). No stored product used the branch.

Not verified. Re-closure of the new domains; refinement of the shells and of the 15/20 grids; the transit tool's central-transit, uniform-disk geometry against the real impact parameter; wind direction along the potential gradient in the reconstruction; any domain beyond L1.

86. Phase 5 of the LHS 1140 b EW plan, first step: the EW falls with He/H; the column energy balance limits the low side (2026-10-04)

Plan md/PLAN_20261003_lhs1140b_ew.md phase 5 step 2 on the two neighbors of the phase-3 reference (section 84). Record, script and figures (both inspected): LHS1140b/models/.P2/ew_fit/phase5/NOTES.md. Binary .P2/EXHALE_8cc7b546.x (ifx), lart4, 12 threads. No code changed. EW in percent-angstrom (% A), observed 1.1084 +- 0.0289. All MEASURED.

Route. Each He/H runs the energy-coupled elemental-flux closure of the reference (energy_coupling_nlte/closure.json with only --abundances He=<v>, the binary and the template changed; tol 0.05, T_match tol 6 K, omega 0.5, kmax 12) with He 2³S transport: True (resolved on in every k00), seeded by continuation in He/H from the certified reference state: the driver maps it onto the new reservoir (map_state_to_grid.py --ic --reservoir He/H); no molecular state is rescaled.

deep He/H	closure	T_match [K]	F_cond [erg cm ⁻² s ⁻¹]	log10 Mdot	EW sol / adv	red peak, FWHM	Kn_bulk 0.1 / 1 / sonic [R_p]
1.5	refused at k00 by the column energy balance (below)						
1.8	converged k = 3, certified	359.5	11.65	7.9359	0.7975 / 0.8015	2.77 %, 0.270 A	6.11 / 18.93 / 26.26
2.09	reference	349.9	9.63	7.8854	0.7334 / 0.7369	2.52 %, 0.273 A	5.69 / 17.21 / 25.69
3.0	stopped at k02 (below); not converged	340.5 (k02)	6.73	7.8190	0.6873 / 0.6900 (k02, diagnostic); 0.6977 / 0.7004 (k01)	2.28 %, 0.282 A	5.03 / 14.55 / 24.50

(Observation: red peak 1.28 %, FWHM 0.80 A on a 3-sample running mean; the widths are those of phase5/compare.py, which gives 0.273 A on the reference where section 84 quoted 0.24 A from another reading.)

Direction and slope. The EW falls as He/H rises: dEW/d(He/H) = -0.221 % A per unit He/H on 1.8-2.09 (both accepted) and -0.051 on 2.09-3.0 (diagnostic k02; -0.039 with k01). At lower He/H the wind is cooler (T_max 4795,

4960, 5352 K) and denser, and $n(\text{He } 2^3\text{S})$ is larger beyond about 3 R_p . The line deepens at a nearly fixed width (0.27-0.28 Å against 0.80 Å observed). The crossing of 1.108 by extrapolation alone: He/H 0.39 (linear), 0.87 (linear in $\ln \text{He/H}$), 0.98-1.20 (quadratics); all below 1.5.

Validity limit of the column treatment (He/H 1.5). The column energy balance under the conducted flux (section 77) refuses its own optically thin treatment at He/H 1.5 (ThinLayerRefusal, HeH1.5/k00/adapter.log): the column above absorbs 10.8 % of the emission excess of the $\text{CH}_4 \nu_4 + \nu_2$ bin 1275-1350 cm^{-1} and 10.3 % of the NH_3 rotational bin 100-160 cm^{-1} , against the 10 % limit. Band maxima of that fraction (CH_4 dyad / NH_3 rotational): He/H 3.0 5.4-5.8 / 5.5-5.6 %; 2.09 7.7 / 7.5 %; 1.8 8.5-8.9 / 8.4-8.5 %; 1.5 10.8 / 10.3 % (refused). The fraction grows as He/H falls (lower mean molecular weight, higher CH_4 and NH_3 mixing ratios). Every extrapolated crossing lies below the He/H at which the treatment holds, so bracketing 1.108 needs radiative transfer in the column bands (absorption by the overlying column) in place of the optically thin limit; the limit itself was not changed. SUPERSEDED (section 90): these fractions used the flux transmission $\exp(-1.66 \tau_{\text{bar}})$ where the escape of locally emitted photons is E_2 -type. With the correct escape (adapter alone, the same final arguments) the band maxima CH_4 dyad / NH_3 rotational are: 3.0 (k02) 16.1 / 13.1 %; 2.09 20.3 / 16.5 %; 1.8 (k03) 22.2 / 18.1 %; 1.5 (thin k01) 24.6 / 20.1 %. Every state of this section is past the 10 % limit.

He/H 3.0 refused at k02 on the mass row at its rounding floor. The wind solve certified its final state (cell 246, $r = 1.333 R_p$: 3.686e-12 against 3.7e-12, distance 0.987); the closure's stationary evaluate of that exact state measures 3.936e-12 (1.054) and exits 2, deterministically (12 and 8 threads give the same, scratch re-evaluation); the evaluate refreshes the composition by one sweep (up to 3.3e-6 relative), and its written state re-evaluates at 3.686e-12 and certifies. The certification verdict of the solve is therefore not reproduced by the evaluate of the exact restart at a cell whose tolerance is its rounding anchor: a reproducibility defect of the mass-row floor, reported, not changed (shared acceptance interface). The closure was not converged at k02 in any case (eps_He 0.070, T_{match} moved 10.5 K). The branch was stopped as the plan prescribes; states kept.

Not verified. A converged He/H 3.0 closure; any model below He/H 1.8; the $\text{NH}_3 \nu_2 + \text{H}_2$ bracket, the drift and the Roche interval at the new He/H; the 0.24 / 0.273 Å FWHM difference on the reference.

87. LHS 1140 b He 10830: the width-matching velocity kernel in the EW chain (2026-10-04)

User decision 2026-10-04: the line width is treated by convolution with an added Gaussian velocity kernel whose FWHM f is adjusted, for each model, so that the model line reaches the measured width, by the convention of docs/lhs1140b_exhale_vs_pwinds.tex Sect. sec:broadening; the crossing EW is the window EW of the model convolved with its own f . Record, script, numbers and figure (inspected): LHS1140b/models/.P2/ew_fit/phase5/nonthermal_kernel/NOTES.md. All MEASURED; EW in percent-angstrom, observed 1.1084 $\pm$ 0.0289.

What the kernel is. A Gaussian line-of-sight velocity distribution standing for the velocity field of the outflow that the 1-D model does not contain; not a computed physics term. Convention: the excess $(\max T - T)/\max T$ of the $R = 68,000$ instrument-convolved curve (column 3 of tpm_He10830.txt, T_{instr} ; corrected 2026-10-05: this entry first said "instrument- and rotation-convolved", but column 3 carries no planet rotation; for LHS 1140 b the rotation column gives an EW equal to 1e-6 %Å) convolved with a Gaussian of FWHM f in velocity (gaussian_filter1d, reflecting

boundary); depth, width and red/blue read by the three-Gaussian extractor used on the data; f solved so that the blended red-pair FWHM is the measured 0.841 Å; EW by the window operator (vacuum 10832.60-10834.20 Å, observation samples, trapezoid weights). The width audit's $\sigma = 9.33$ km/s is kept as a reference value only.

Code. LHS1140b/he10830_equivalent_width.py: broaden_excess, broadened_line_metrics, width_matching_fwhm_kms (moved unchanged from LHS1140b/make_memo_figures.py, which now imports them), and width_matched_product / CLI --width-matched (writes tpm_He10830_broadened.txt with the record in its header). LHS1140b/models/.P1/fiducial_conduction/ew_of_state.sh: option EW_NONTHERMAL_KERNEL=width_match (or a FWHM in km/s) adds a nonthermal_kernel line and, on each product line, kernel: fwhm_kms= σ _kms= EW_kernel= EW_nokernel= red= blue= red_blue= fwhm_Å=; EW= stays the unbroadened EW. Default unset: unchanged. EXHALE_transit.py and exhale_transit_lib.py unchanged.

Tests. Kernel off: EW.txt of HeH1.8/k03 identical to the stored one (up to the state-directory prefix), tpm_He10830.txt data bitwise equal; in the kernel runs all six unbroadened products equal the stored ones bitwise. The kernel conserves the absorption summed over the grid to $1.6\text{e-}16$ - $5.4\text{e-}16$ relative (the trapezoid moves by $-1.0\text{e-}4$ because the broadened blue wing reaches the grid end). The moved functions return bitwise the numbers of the replaced definitions. The window EW falls by 3.31-3.32 %: the red pair loses 4.0 % through the window edges, the blue component leaks 0.005-0.006 in.

Results (solution; adv in the NOTES).

He/H	f [km/s] (σ)	red [%]	blue [%]	red/blue	FWHM [Å]	EW no kernel / kernel
1.8	22.05 (9.36)	0.890	0.117	7.60	0.841	0.7975 / 0.7710
2.09	22.02 (9.35)	0.818	0.108	7.60	0.841	0.7334 / 0.7091
3.0 (k01, unconverged closure)	21.92 (9.31)	0.779	0.103	7.56	0.841	0.6977 / 0.6746
observed		1.254	0.198	6.35	0.841	1.1084

f is 21.9-22.1 km/s at every He/H (22.22 on the 2026-09 solutions of the memo). At that f the shape matches and the depth is 0.62-0.71 of the observed: the kernel lowers the window EW by 3.3 %, moving every model further below 1.108. Cross-checks (χ^2 on the 129 samples, v_{wind} 2.26 km/s): free f with the EW free 15.7-17.4 km/s (χ^2 147-241); with the window EW held at 1.1084 by an amplitude scale, 22.3-22.4 km/s (χ^2 18.7-18.8), close to the width-matching f .

Not verified. The uncertainty of f from the measured width; the blue depth near the reflecting grid end; the memo and its figures were not regenerated (make_memo_figures.py changed only by the import, not run).

88. The stationary evaluate certifies the stored pair (2026-10-04)

Finding. Section 86 recorded that the He/H 3.0 k02 state (LHS1140b/models/.P2/ew_fit/phase5/HeH3.0/k02) was certified by its wind solve (mass row at cell 246, $r = 1.333 R_p$: $3.686e-12$ against the rounding-anchored $3.7e-12$, distance 0.987) and refused by the closure's Restart intent: stationary evaluate of the same exact state ($3.936e-12$, distance 1.054, exit 2), while the state that evaluate wrote re-certified. The evaluate took one composition sweep from the stored composition BEFORE it certified, and certified, wrote and exited on the swept state. The verdict was therefore not a function of the stored state. MEASURED (binary EXHALE_99a76cb0.x, below) the sweep moves the physical composition very little: max ldf/f over species carrying at least $1e-12$ of a cell's mass $4.88e-12$ (cell 499, C I), max over cells of $\sum_k ldf_{kl}$ $1.51e-16$, particle count $5.2e-16$. The $3.3e-6$ relative change in Ion_species.txt quoted in section 86 sits in the two lower ghost rows (row 2, He 2^3S , $r = 0.99990 R_p$), which the boundary derives; the physical rows of the text files differ by less than $1e-8$. The mass row of the wind is a rounding-level quantity judged against a rounding-anchored tolerance, and a change of the caloric map at the $1e-16$ level moves it by 7 % at cell 246.

Rule adopted. The certificate a file carries is the one measured on the state written in it, and the evaluate writes the state it loaded.

- stationary_state_of_the_loaded_restart (EXHALE_main.f90) certifies the STORED pair first: the conserved state and the composition exactly as loaded, with the heating and cooling of that composition in the field it attenuates. They come from one sweep with the composition of the cells above the two lower ghosts held (ioniz_eq_sources_of_held_composition, ionization_equilibrium.f90): the rates are built from the entry composition ($xuv_field_block_cells = 0$, $xuv_self_field_passes = 1$, so every rate is a function of that composition and T), heat and cool are contracted with the entry densities, the interior composition is returned bit for bit (asserted; the run stops otherwise), and the two lower ghosts, which no file stores, are solved and derived by the boundary as on every route. That certification is the verdict: the certified= pair of every file written, the exit status (0, 2 a refused claim, 7 a rejected derived product) and stored pair verdict in the log.
- evaluate writes Hydro_ioniz.txt, Ion_species.txt, conserved_state.txt, the channel breakdowns and the _adv pair from the stored pair (the _adv pair carries the stored pair's certificate as derived_from provenance), then takes one composition sweep from the loaded state as it was before the held sweep, lower ghosts included (composition_sweep_movement_of_the_stored_pair), and reports its movement and its certification (record_verdict = .false. in certification_evaluate, which then sets no file pair, no exit status, no resolved-configuration record and writes no row-term or flux-profile file), with the two certifications printed entry beside entry (certification_reports_side_by_side). The swept composition is never written.
- Restart intent: stationary (the solve) makes the same stored-pair certification (the answer to the file's claim) and starts the solve from the swept state, entered at the loaded state as before.
- The solve routes already certify what they write: the stationary restart, the direct steady route (certify_the_state_to_be_written) and the final marching write assemble the residual from (u, f_sp, heat, cool) and write those same arrays. The heat and cool are those of the last sweep, taken at its entry temperature (the documented one-sweep lag); on k02 the stored-pair evaluate, whose sources are the stored composition's at the state's own temperature, reads the energy row at $2.06661e-8$ (cell 58) where the solve's final certification read

2.067e-8 (cell 58).

- Found while checking this: the periodic marching write (every 1000 steps) wrote the run's last certified pair, measured at an earlier monitor step on another state. It now writes `certified=F` `cert_reason=marching_snapshot_s<step>` and restores the run's pair after the write, as the pass-state snapshot already did. The final write is unaffected.
- `ew_of_state.sh` records the written state's pair as `written_state` (was `work_state`).

Binary. LHS1140b/models/.P2/EXHALE_99a76cb0.x, md5 99a76cb05e56ad3a6ee80e6c433814fd, ifx 2026.1.0, built from .P2/build_storedpair_ifx/; manifest .P2/BINARY_MANIFEST_99a76cb05e56.txt. To be used for the phase 5-6 acceptance of the EW plan. Against BINARY_MANIFEST_8cc7b5462cc2.txt exactly three sources differ (EXHALE_main.f90, certification.f90, ionization_equilibrium.f90), and the tree matched that manifest before the change.

Verification (MEASURED, lart4, 12 OpenMP threads unless stated).

state	certification	binding cell	measure / tol	distance	exit
HeH3.0 k02, solve's final (run.log, 8cc7b546)	solve	246	3.686e-12 / 3.7e-12	0.987	
HeH3.0 k02, old evaluate (8cc7b546)	after sweep	246	3.936e-12 / 3.7e-12	1.054	2
HeH3.0 k02, new evaluate, 12 and 8 threads	stored pair (verdict)	246	3.686e-12 / 3.7e-12	0.987	0
same run	after sweep (reported)	246	3.936e-12 / 3.7e-12	1.054	
reference phase3/he23s_transport/solve, solve's final	solve	242	3.360e-12 / 4.3e-12	0.790	
reference, new evaluate	stored pair (verdict)	242	3.360e-12 / 4.3e-12	0.790	0
same run	after sweep (reported)	252	2.001e-12 / 3.0e-12	0.667	

The stored-pair verdict reproduces the solve's own final certification at both states, and the after-sweep reading reproduces the old evaluate entry by entry (largest measures, cells, binding cells); of the volume averages only the energy row's moves, 8.886e-10 (old) against 8.865e-10, not traced further (the after-sweep sweep runs after another sweep has filled the frozen background and the boundary caches). Every other entry of k02 is within its tolerance on both. The k02 evaluate writes the stored state: `conserved_state.txt` U and F rows of the physical cells and upper ghosts bitwise

equal to the input, Ion_species.txt physical rows equal, Hydro_ioniz.txt p and T within $1.9\text{e-}15$ relative of the solve's text (re-derived from u), heat within $1.6\text{e-}14$ and cool within $7.4\text{e-}13$ (the sources of the stored composition at its own temperature against those of the solve's last sweep). The 12- and 8-thread Ion_species.txt are identical. An evaluate of the evaluate's own output writes the same data (Hydro_ioniz, Ion_species, conserved_state, both _adv files, both breakdowns) and the same verdict. Reference EW (ew_of_state.sh with the new binary): solution 0.7334, adv 0.7369; with EW_NONTHERMAL_KERNEL=width_match 0.7091 and 0.7125, as recorded with 8cc7b546 (phase5/nonthermal_kernel/ew_HeH2.09); the solution spectra are bitwise equal to that record, the adv spectra within $4.5\text{e-}16$ (unbroadened) and $6.4\text{e-}14$ (broadened) relative. Scratch: ~/ .claude/jobs/f24ea156/tmp/evalfix/.
Tests (MEASURED; the gfortran tree build, md5 817048a812c0594155159ee4ba14e480, against the control build of the entry text, 843c97f955b072158ccc8454f2cee6a8).

- src/tests/certification: 145 PASS, 0 FAIL before and after.
- src/tests/grid_and_gates: 420 PASS, 3 FAIL before; 424 PASS, 3 FAIL after. The three failures are the same rows of restart_intent_and_metadata.sh on its stale fixture (backup/regression/wasp_full_newton, solved under older physics, which the binary refuses): stationary_evaluate_exit_status (2), stationary_evaluate_certification_unchanged (writer T, re-evaluated F) and stationary_evaluate_rebuilt_within_sweep_budget, whose worst column moved from O III $6.01\text{e-}5$ (the swept composition) to heat $5.83\text{e-}5$ (the species now come back as stored; the heat of that fixture was written by an older binary). stationary_evaluate_products.sh was rewritten to the rule: four rows added (header_certificate_is_the_stored_pairs, exit_status_is_the_stored_pairs_verdict, products_are_made_on_the_stored_pair, the loaded state bit for bit, changed_option_swept_reading_reported), the dump points renamed (stored_before_products, stored_after_products), the claim rows read the new log lines.
- src/tests/boundary_state (the evaluate-route scripts boundary_of_the_installed_composition, interior_composition_held, ghost_composition_fixed_point, ghost_seed_and_ghost_record): 27 PASS, 0 FAIL before and after. An intermediate version that entered the reported sweep at the ghosts of the held sweep and rebuilt the boundary unconditionally failed face_state_gap_of_the_entry_composition_boundary and ghost_seed_of_the_installed_rows_is_inert; both pass with the version described above.
- src/tests/adv_static_limit: 64 PASS before and after (the derived column's worst residual $2.98\text{e-}11 \rightarrow 2.86\text{e-}11$, now derived from the stored pair); src/tests/coupled_block_jacobian (the solve route): 8 PASS before and after, every measured value identical.

Not verified. The solve route (Restart intent: stationary without evaluate) now takes the held sweep before its work-state sweep; the work state is entered at the loaded state, but caches the held sweep filled (the frozen background, the boundary record) can move it at the level of the volume-averaged energy row above; no solve was run with the new binary. The HeH3.0 k02 closure was not resumed (its branch stays stopped, section 86; its closure was unconverged in any case).

89. The column energy balance with absorption by the overlying column; the thin limit flags instead of refusing; phase 5 at He/H 1.5 and 1.2 (2026-10-04)

The column energy balance below the matching level (`src/utls/conducted_heat_column_balance.py`, section 86) treated the band emission as optically thin and refused a column in which the gas above absorbed more than 10 per cent of a bin's emission excess. That refusal stopped He/H 1.5. Records: LHS1140b/models/.P2/ew_fit/phase5/column_transfer/NOTES.md (measurements, figure inspected), phase5/HeH1.5_thick/, HeH1.5_thin/, HeH1.2_thick/ (NOTES.md each), phase5/compare_column_transfer.py and column_transfer_numbers.txt, fig_column_transfer_{line,profiles}.png (inspected). All MEASURED unless marked READ. EW in percent-angstrom.

Formulation (new option, default off). The adapter option is `--column-band-transfer cooling_to_space` (default thin). The form is the cooling-to-space approximation as generalized to a two-level band by Dickinson (1972, J. Atmos. Sci. 29, 1531), READ in the published version: section 3c Eq. 12 (cooling to space, $(1/2) n h \nu r T(z, \text{inf})$) and section 3g Eq. 33 (net rate $[1 + (1/2)(A/\lambda) T(z, \text{inf})]^{-1} [R_{\text{ext}} + A E(z) - (1/2) A T g e^{-(h \nu/kT)}]$, with $E(z)$ the exchange function of Eq. 30). Rodgers & Walshaw (1966, QJRM 92, 67) was not read: the publisher's page could not be retrieved, and it is cited only through Dickinson. Lopez-Puertas & Taylor (2001) and Goody & Yung (1989) were not read and are not used.

$E(z)$ is split in two parts:

- Toward the optically thick climate column below, the source function is held at its climate value. This part adds the downward half $1/2$ to the escape probability, and the field from below ($a = e_{\text{LTE}}(T_0)/2$, as before) to R_{ext} .
- Toward the gas above and inside the warmed layer, it is neglected (the cooling-to-space approximation; the gas above the match is the base of the wind, at least as warm as the match).

Escape of node i in bin b : $\beta = 1/2 + \sum_g w_g k_g u_g / \sum_g w_g k_g$, with $u_g = [E_3(\tau_{\text{t}}) - E_3(\tau_{\text{b}})] / (2(\tau_{\text{b}} - \tau_{\text{t}}))$ the layer integral of $(1/2) E_2(\tau)$. The optical depths are measured from the top of the profile with the k -distribution at the climate state.

- LTE bands: $\text{excess} = \sum_{\text{bins}} \beta (e_{\text{bin}}(T) - e_{\text{bin}}(T_0))$.
- Two-level bands: $\beta = \sum \beta e_{\text{bin}} / e_{\text{band}}$ and $\text{eps}_{\beta} = C_{10} / (C_{10} + \beta A_{10})$, so a band with $\beta A_{10} \ll C_{10}$ thermalizes; $\text{excess} = \text{eps}_{\beta}(T)(\beta e(T) - a) - \text{eps}_{\beta}(T_0)(\beta e(T_0) - a)$.

With no absorbing column this is the thin treatment bit for bit.

Validity (it replaces the 10 per cent refusal for this option). The exchange inside the domain that the approximation neglects is computed on each solution:

- exact slab exchange, E_3 of the optical-depth differences between faces, every g -point;
- layers thinner than $1e-6$ in optical depth are treated as sheets with E_2 fluxes (the slab form loses them to round-off);
- over the nodes warmed by more than $1e-6$ of the match rise.

Its first-order effect on T_{match} (one solve with the Jacobian of the solution) must not exceed `--column-exchange-shift-max` (default 1 K), else `ColumnTransferRefusal`. The value and the escape probability of each band are written to `conducted_heat_column.txt`; the choice enters the profile header (`column_band_transfer`) and, off the default, the `solution_id`.

Is the exchange term needed? `ColumnResponse.solve_with_exchange` solves with the full exchange of the whole 6 H domain. On the He/H 1.8 column it adds +0.493 K (nlte) and +0.605 K (lte) to T_{match} over cooling to space, whose first-order estimates are 0.395 and 0.480 K; on the test column it adds +0.466 and +0.569, against estimates of 0.373 and 0.452. The estimate is 0.79-0.80 of the full value in all four cases. The exchange deposits the downward emission over 1.5-400 microbar and warms that region by about 2 K. Not needed at these optical depths (0.5 K against the 6 K closure tolerance and the 2.4 K of the transfer itself); measured and bounded on every solution.

Thin limit: flag, not refusal (user decision 2026-10-04). Under thin, a column past the 10 per cent limit is kept. The 10 per cent stays the stated validity threshold at the code site (`THIN_ESCAPE_MIN`). The adapter log prints a WARNING. `conducted_heat_column.txt` and the profile header carry `thin_layer_limit_exceeded` T|F, `thin_layer_max_absorbed_fraction` and `thin_layer_max_absorbed_bin`. The profile header keys are written by `lower_profile_schema.write_profile`; EXHALE ignores header keys it has no consumer for, READ from `lower_atmosphere_profile.f90`. The closure driver does not repeat the flag. `ThinLayerRefusal` is removed.

Found and documented, not changed: the thin test's escape factor $\exp(-1.66 \tau_{\text{bar}})$ is the diffusivity form of the flux transmission $2 E_3$. The escape of locally emitted photons is $E_2(\tau)/2$, whose absorbed share is larger at small τ (14.6 against 6.4 per cent at $\tau = 0.04$), so the test understates the absorption. It is kept as the thin treatment's stated criterion (thin unchanged by instruction), noted at the code site. SUPERSEDED (section 90): the check now uses the cooling-to-space escape; every thin absorbed fraction quoted in this section is replaced there.

Tests (`src/tests/conducted_heat_column/run.sh`, 66 PASS, 0 FAIL):

- the thin limit of the new transfer is the thin solution bit for bit (lte, nlte);
- energy budget to 6e-11;
- the new transfer is warmer than thin, with escape between 1/2 and 1;
- NH3 x5 is flagged by the thin test and solved within the bound, while NH3 x300 is refused by the bound;
- a grey optically thick emitter (one bin at 2100 cm^{-1} , τ_2 at the match): escape probabilities against independent quadrature of E_2 to $1e-5$, τ against quadrature to $1e-5$, and T_{match} against an independent shooting solution (`solve_ivp`, bisection) of the continuous equations. The discrete solution converges at order 1.95, and its Richardson value agrees to $1.6e-3$ K ($2.8e-5$ of the rise);
- the diagnostic switch below;
- the existing refusal test now expects the flag.

The thin path is bitwise unchanged against the entry text (lte, nlte, nlte_partial, rotational on the test column; the thin adapter runs reproduce the stored T_{match} 349.876 and 359.546).

Thin against cooling to space, adapter alone (k-final chemistry arguments of each closure):

He/H	thin	cooling to space	difference
2.09	349.876	351.922	+2.05 K
1.8	359.546	361.949	+2.40 K
1.2	382.735 (flagged)	386.371	+3.64 K

All are below the 6 K closure tolerance, so the converged He/H 1.8 and 2.09 models need not be redone on this ground (decision for the main session).

The overlying gas. Both transfers take the column above p_match from the Photochem CH₄ and NH₃ above the match. In the coupled model that gas belongs to the wind, which carries C, N and O as atoms and is warmer. Diagnostic include_gas_above_match = False:

He/H	thin absorbed fraction, with / without	cooling to space - thin, with / without
1.8	0.0853 / 0.0099	+2.42 / +0.12 K
1.5	0.1015 / 0.0105	+3.08 / +0.13 K
1.2	0.1130 / 0.0112	+3.68 / +0.14 K

SUPERSEDED (section 90): with the correct escape the thin absorbed fractions with / without the gas above are 0.2219 / 0.0293 (1.8), 0.2522 / 0.0307 (1.5), 0.2790 / 0.0325 (1.2), so 87-88 per cent from that gas; the temperature columns are unchanged. The thin flag F at 9.60 per cent of the He/H 1.5 thin state below becomes T at 24.6 per cent, and the 1.2 thin value 0.1129 becomes 0.2788.

About 90 per cent of the absorbed fraction that triggered the thin limit, and 95-96 per cent of the cooling-to-space correction, come from molecules the wind model does not contain. Which overlying column is physical (the Photochem gas, or the wind's atomic gas with no band absorbers) is not settled here.

Pinning passes. At He/H 1.5 and 1.2 the Photochem model-top pinning on the warmed column converges by alternating passes and needed 7-11 of them; three first attempts stopped at the 6-pass cap. New adapter option --top-pressure-passes (default 6, unchanged); the phase-5 closures at 1.5 and 1.2 pass 12. The 1e-5 agreement is unchanged.

Phase 5 at He/H 1.5 and 1.2 (continued in He/H from HeH1.8/k03, settings of HeH1.8/closure.json except He/H, transfer and passes). The EW verdicts all come from binary 99a76cb0 (written_state certified=T), EW_NONTHERMAL_KERNEL=width_match:

He/H, transfer	closure (binary)	T_match	log10 Mdot (N-20)	EW sol / adv	EW width matched sol / adv (f km/s)	Kn_bulk 0.1 / 1 / sonic [R_p]
1.8, thin (section 86)	k = 3 (8cc7b546)	359.55	7.9359	0.7975 / 0.8015	0.7710 (section 87)	6.11 / 18.93 / 26.26

He/H, transfer	closure (binary)	T_match	log10 Mdot (N-20)	EW sol / adv	EW width matched sol / adv (f km/s)	Kn_bulk 0.1 / 1 / sonic [R_p]
1.5, thin	k = 1, eps 4.3e-4 (8cc7b546); flag F, 0.0960	373.54	7.9766	0.8318 / 0.8364	0.8042 / 0.8087 (22.09)	6.54 / 20.71 / 26.83
1.5, cooling to space	k = 1, eps 8.2e-3 (99a76cb0)	375.98	7.9770	0.8327 / 0.8373	0.8050 / 0.8096 (22.09)	6.55 / 20.72 / 26.83
1.2, cooling to space	k = 1, eps 0.049, dT 5.7 K (8cc7b546)	386.37	8.0218	0.8593 / 0.8648	0.8308 / 0.8361 (22.14)	7.08 / 22.95 / 27.45

- At He/H 1.5 the thin-treatment error is +2.44 K in T_match and 0.0009 (0.1 per cent) in the EW. The converged thin state is inside its limit (9.60 per cent); intermediate iterates passed it (10.15 per cent).
- He/H 1.2 was reached from 1.8 in one jump (no rung). Its closure converged just inside both tolerances.
- Stops, all kept in attempt*/ or k*.incomplete.*: the pinning cap (three times), and the evaluate of binary 8cc7b546 refusing solve-certified states on the mass row at the rounding floor (He/H 1.5 thick k01, 3.197e-12 against 3.0e-12; He/H 1.5 thin k00, 6.853e-12 against 6.7e-12; the defect of section 88). The He/H 1.5 thick closure was restarted on 99a76cb0 from its own k01 state.

Where the crossing is. The width-matched EW rises as He/H falls (0.7091, 0.7710, 0.8050, 0.8308 at 2.09, 1.8, 1.5, 1.2), but the step shrinks: +0.034 from 1.8 to 1.5, +0.026 from 1.5 to 1.2. A linear continuation of the last step reaches the observed 1.108 only at a negative He/H. No crossing exists at He/H $\geq$ 1.2 in this model family, and the trend gives no sign of one below it (an extrapolation, not a model). The line deepens at a fixed width; the hot peak cools (4795, 4568, 4272 K at 1.8, 1.5, 1.2) as the wind densifies.

Not verified. For a two-level band the exchange weighting by eps_beta at emitter and absorber is approximate; it is exact for LTE bands. The first order of the validity bound reads 0.8 of the full exchange. The gas-above-match question is open, as stated. The He/H 1.2 state lies near its closure tolerances.

90. Phase 5 closed: the EW has no crossing in He/H; the thin check uses the escape of emitted photons; the model-top pinning averages a 2-cycle (2026-10-04)

User decision (2026-10-04): run He/H 0.8 and 0.5 to measure the upper bound of the EW in He/H and close phase 5 on measured values. Records: LHS1140b/models/.P2/ew_fit/phase5/NOTES.md section 8, the NOTES.md of HeH1.0/, HeH0.8/, HeH0.65/, HeH0.5/, column_transfer/thin_check_escape/NOTES.md; compare_lowHeH.py, lowHeH_numbers.txt, fig_ew_vs_heh.png, fig_lowHeH_line.png (inspected). All numbers MEASURED. EW in percent-angstrom; the crossing quantity is the width-matched window EW (section 87).

1. Thin-treatment validity check (fixed). Physically wrong: `optical_depth_check` of `src/utils/conducted_heat_column_balance.py` took the escape of the band photons emitted in the column as the flux transmission $\exp(-1.66 \tau_{\text{bar}})$, the diffusivity form of $2 E_3(\tau)$, i.e. the transmission of a beam leaving the column. Photons emitted isotropically at depth τ escape upward with probability $E_2(\tau)$ (Dickinson 1972, Eq. 5 for the transmission to space), and integrated over a layer and the k-distribution g-points this is the escape the `cooling_to_space` transfer already uses (`escape_factors`). The check now takes the absorbed share of the upward emission excess as $\sum_i \text{ex}_i^2 (1 - \beta_i) / \sum_i \text{ex}_i$, β_i the cooling-to-space escape of node i 's control layer (one function for both). Only the flag and the recorded fraction change. The docstring paragraph on the overlying gas was re-measured. `DIFFUSIVITY` is removed (no other user).

- Tests (`src/tests/conducted_heat_column/run.sh`, 66 assertions): new `grey_thin_test_absorbed_share_vs_E2_quadrature`, the grey emitter (τ^2 at the match) against an independent 16-point Gauss-Legendre quadrature of E_2 over every control layer: entry text FAIL (0.96661 against 0.96476), corrected PASS (to $1e-6$). `thin_column_is_not_flagged` failed on the corrected check (the test column absorbs 20.9 per cent), so it now uses that column with a tenth of the CH₄ and NH₃ (3.8 per cent). Entry text 65 PASS / 1 FAIL, corrected 66 PASS / 0 FAIL.
- Thin solution bit for bit: on the stored column inputs of He/H 1.8 and 1.2, entry text against corrected module, treatments `nlte`, `lte`, `nlte_partial`, rotational: `T` and `q` array_equal in all eight; the adapter-alone thin runs reproduce the data rows of `conducted_heat_column.txt` of the entry-text runs (md5) at 2.09, 1.8 and 1.2 and the stored `T_match` at all five He/H.
- Recomputed absorbed shares (adapter alone, the last chemistry arguments of each closure, transfer thin), CH₄ $\nu_4 + \nu_2$ dyad / NH₃ rotational, old in brackets: 3.0 (k02) 16.1 / 13.1 % (5.76 / 5.60); 2.09 20.3 / 16.5 % (7.68 / 7.48); 1.8 (k03) 22.2 / 18.1 % (8.52 / 8.39); 1.5 (thin k01) 24.6 / 20.1 % (largest 9.60); 1.2 (k01) 27.9 / 22.8 % (largest 11.29). Every thin state of phases 3 and 5 is past the 10 % limit, by 1.6-2.8 times; what that costs is measured by the cooling-to-space transfer (+2.05, +2.40, +2.44, +3.64 K in `T_match` at 2.09, 1.8, 1.5, 1.2; 0.0009 in the width-matched EW at 1.5; section 89), below the 6 K closure tolerance. Without the gas above `p_match` the largest share is 2.9-3.7 % at every He/H measured (2.09-0.65), so 87-90 % of it comes from that gas. The numbers of sections 86 and 89 and of `md/NH3_CH4_ISSUES_20261004.md` are marked superseded where they stand.

2. Model-top pinning: averaged once the passes alternate (fixed). `steady_state_at_stated_model_top` (`src/utils/photochem_to_lower_profile.py`) iterates the map $z \rightarrow z'(z)$ of the Photochem domain top (pin, converge, read the altitude of the stated pressure). At He/H 0.5 on the warmed column the passes alternated without contracting ($+1.7e-2$, $-1.1e-2$, ..., $+6.2e-3$ at pass 12 against $1e-5$; `HeH0.5/attempt2_pinning_cycle/`), a 2-cycle of the undamped iteration (slope near -1). Once two successive moves have opposite signs the top is pinned at $(z + z')/2$: the same fixed point and the same $1e-5$ test. On the same arguments the entry text refuses after 12 passes and the corrected adapter converges at pass 9 ($1.8e-6$); passes that do not alternate are unchanged (the first pinning sequence of that run is identical line for line). Converged closures before this change are not affected (their stored states satisfy the same test); the He/H 1.0 and 0.8 closures and the 0.65 iterates after k00 ran with it. No suite drives this function;

the check is the He/H 0.5 run itself (HeH0.5/attempt4_from_0.65/k00/adapter.log, (top averaged) lines).

3. Climate starting value. At He/H 0.8 and 0.5 the clima root solve for the deep temperature does not bracket from the default 400 K. The root does not depend on the starting value (20 bar: 648.4 K at 0.8 from 600 and 650 K; 679.1 K at 0.5 from 650 and 700 K), so the low-He/H closures pass --climate-t-deep-guess 650; nothing else changed.

4. The scan (cooling to space, settings of HeH1.2_thick/closure.json, binary 99a76cb0 for every wind and EW verdict, written_state certified=T, evaluate exit 0):

deep He/H	closure k	T_match [K]	F_cond	log10 Mdot (N-20)	EW sol / adv	EW width matched sol / adv (f km/s)	Kn_bulk 0.1 / exobase / sonic [R_p]	thin flag (largest share)
1.2 (section 89)	1	386.37	17.16	8.0218	0.8593 / 0.8648	0.8308 / 0.8361 (22.14)	7.08 / 22.95 / 27.45	T (0.279)
1.0 (rung)	1	399.39	19.82	8.0531	0.8655 / 0.8714	0.8368 / 0.8425 (22.17)	7.50 / 24.74 / 27.89	T (0.307)
0.8	1	410.53	22.95	8.0930	0.8734 / 0.8798	0.8444 / 0.8507 (22.21)	8.04 / 27.08 / 28.40	T (0.343)
0.65 (rung)	2	412.59	25.87	8.1325	0.8822 / 0.8891	0.8529 / 0.8596 (22.24)	8.57 / 28.63 / 28.79	T (0.377)
0.5	not converged	398.51 (k00, diagnostic)	29.51	8.179	0.8773 / 0.8845	0.8482 / 0.8552 (22.28), diagnostic	9.24 / above the domain / none in the domain	

Elemental fluxes at the match (F_H / F_{He} , g/s, overlap-window spreads $1e-9$ to $2.4e-9$): 1.0 $4.636e7 / 6.665e7$; 0.8 $5.527e7 / 6.862e7$; 0.65 $6.479e7 / 7.088e7$. Neglected-exchange shift 0.53, 0.59, 0.64 K (bound 1 K); cooling to space - thin in T_match +4.40, +5.36, +6.16 K at 1.0, 0.8, 0.65 (+0.15 to +0.18 K without the gas above p_match).

- He/H 0.8: the jump from 1.2 stopped at k00 (wind info 1; the He 2^3S carrier balance at the first gated cell, $r = 1.204 R_p$, $4.8e-5$ against $1e-5$ after the 60 outer passes); converged from the 1.0 rung.
- He/H 0.5: NOT CONVERGED in four attempts, all kept. After the two fixes above, k01 from 1.2 and k00 from the converged 0.65 rung both stopped after the 60 outer passes on the He 2^3S carrier balance at the first gated cell ($r = 1.248 R_p$; $1.1e-4$ and $1.6e-4$ against $1e-5$). The same row needed 30-40 passes at He/H 1.2 and 1.5. Tolerances and the pass budget were not changed. The k00 diagnostic of the attempt from 1.2 (closure residual 0.27, T_match 398.5 K) has no sonic point in the domain (Mach 0.996 at $29.03 R_p$): the sonic point moves

outward as He/H falls (26.26, 27.45, 27.89, 28.40, 28.79 R_p at 1.8, 1.2, 1.0, 0.8, 0.65) and leaves the 30 R_p domain between 0.65 and 0.5.

5. The upper bound. Below He/H 1.2 the width-matched EW rises by 0.006-0.009 a step, 0.04-0.06 per unit He/H (2.09 to 1.2: 0.07-0.21 per unit). Fits on the four models with He/H ≤ 1.2 (compare_lowHeH.py): linear, rms 1.1e-3, 0.877 at He/H 0; quadratic, rms 2.6e-4, 0.905 at He/H 0 (EXTRAPOLATIONS beyond the last model, labeled so in fig_ew_vs_heh.png). Since the EW must vanish as He/H → 0, these are upper estimates of the maximum provided the slope does not steepen below 0.65 (an assumption). Three forms with EW(0) = 0 and a maximum fit with rms 0.008-0.010, five times the scatter of the models: the models show no turnover down to 0.65. The largest model EW, 0.8529 (adv 0.8596) at He/H 0.65, is 8.8 sigma below the observed 1.1084 ± 0.0289, the extrapolated 0.88-0.91 7-8 sigma below; 1.108 would need about 0.39 per unit He/H below 0.65, seven to ten times the measured slope. The width-matched lines peak near 0.98 per cent against the observed 1.25 and converge with each other at low He/H (fig_lowHeH_line.png). **No He/H of this model family reaches the observed EW; phase 5 closes without a crossing.**

Not verified. The closure of 0.65 stopped with eps_H 0.045 (tolerance 0.05) and that of 1.2 near both tolerances; at about 1e-3 per K (section 75) and the 6 K T_{match} tolerance the step-to-step EW differences below 1.2 (0.006-0.009) are of the order of the closure tolerance itself. The He 2³S carrier-row stall at the first gated cell is not diagnosed. The gas-above-match question (section 89) stays open and now carries 87-90 % of a thin absorbed share that is past its limit at every He/H.

91. LHS 1140 b: why the He 10830 EW rises as He/H falls, dEW decomposed by field and by balance term (2026-10-04)

User request: verify numerically the explanation that lower He/H raises the EW through a denser, cooler outer wind with larger n_e and $n(\text{He}^+)$, $n(2^3\text{S})$ tracking $n_e \times n(\text{He}^+)$. Records: LHS1140b/models/.P2/ew_fit/phase5/dEW_decomposition/NOTES.md, scripts, dEW_table.txt, figures (inspected) fig_dEW_factors.png, fig_n3_balance.png, fig_line_buildup.png, fig_mdot_drivers.png. All numbers MEASURED; width-matched window EW (section 87), f re-measured for every profile (22.02-22.24 km/s). No source file of the code changed; the stored states were only read.

Method. The He I 10830 branch of EXHALE_transit.py restated as a function (he10830_line.py): on the four stored states (He/H 2.09, 1.8, 1.2, 0.65) its instrument-convolved curve equals the stored tpm_He10830.txt to 4.4e-16 and the EW, width-matched EW and f equal the EW.txt records; on three substituted profiles EXHALE_transit.py itself agrees to 2e-14. $n(2^3\text{S})$ is written as C times the local root of the code's balance (tr_triplet_row plus the H2 sink), rate coefficients from the code's own elemental functions of Cool_coeff.f90 (a driver linked against a private object directory, removed afterwards); the 2³S photoionization rate P3 is measured from carrier_row_terms.txt and is the same in the four states to 0.3 per cent (6.41-6.54e-6 s⁻¹). $C = n_3/n_{3_loc}$ is 0.9955-1.0072 over 1.2-8 R_p and up to 1.18 above 8 R_p; the local form (C = 1) lowers the width-matched EW by 0.004-0.008. The level is fed by recombination only (the 1¹S collisional feed is below 1e-14 of the gain where n_3 matters). Shapley values over nine factor fields (n_H , local $y = n_{\text{He}}/n_H$, He fractions, H fractions, n_e , T, v, C with P3, R0), 512 syntheses for each

pair, order-independent and summing exactly to dEW (residual 3e-16); a radial zone Shapley; a first-order factor by zone split; and a second Shapley over the terms of the balance, $n_3 = C n_e n_{\text{HeII}} a_{\text{eff}} / (L_P + L_e + L_r)$.

Contributions (width-matched EW).

factor	2.09 -> 1.8	1.8 -> 1.2	1.2 -> 0.65	2.09 -> 0.65
n_H (density)	+0.0624	+0.1353	+0.1510	+0.3524
y = n_He/n_H local	-0.0566	-0.2223	-0.3428	-0.6261
He ionization fractions	-0.0257	-0.0489	-0.0652	-0.1406
H fractions	-0.0005	-0.0016	-0.0025	-0.0046
n_e	+0.0645	+0.1273	+0.1761	+0.3687
T	+0.0189	+0.0708	+0.1030	+0.1946
v	-0.0000	-0.0001	-0.0001	-0.0002
transport C, P3	-0.0019	-0.0040	-0.0043	-0.0112
R0	+0.0009	+0.0032	+0.0069	+0.0109
dEW	+0.0619	+0.0598	+0.0221	+0.1438

The change is a small difference of large terms: density, electrons and the lower temperature add +0.92, the lower helium abundance of the flow (He diffusion: top-of-domain y 0.59 at 2.09, 0.28 at 0.65) and the lower He+ share of the helium remove -0.77. By radius (zone Shapley, 2.09 -> 0.65) the rise comes from beyond 6 R_p (+0.177); inside 3.5 R_p the line weakens (-0.059; the n_3 peak falls from 52.5 to 22.5 cm^{-3}).

The $n_e \times n(\text{He}^+)$ explanation holds in part. Balance terms (2.09 -> 0.65): $n_e n_{\text{HeII}}$ +0.703, recombination coefficient $a_{\text{eff}}(T)$ +0.156, $\text{He}(2^3\text{S})+\text{H}$ ionization loss $n_{\text{HI}} Q_{31}$ -0.666, electron-collision loss -0.045, A31 + P3 + H2 and transport -0.011, thermal width -0.005, v -0.0002, R0 +0.011. Over 1.5-8 R_p , where most of the line forms, the $\text{He}(2^3\text{S})+\text{H}$ ionization takes 50-93 per cent of the 2^3S loss rate and A31 takes more than half only above 9.9-13.8 R_p . At fixed r, $n_e n_{\text{HeII}}$ rises x1.2-2.9 from 2.09 to 0.65 but n_{HI} x3.3-4.6, so $n_3/(n_e n_{\text{HeII}})$ at 0.65 is 0.41-0.51 of the 2.09 value over 2-8 R_p and reaches it only above ~20 R_p . The denser, more neutral hydrogen cancels 95 per cent of the gain of the $n_e \times n(\text{He}^+)$ product; the net comes from the cooler gas (larger recombination coefficient) and from the outer wind where A31 replaces the Penning loss. Velocity contributes nothing.

Density and Mdot. $Q_{\text{net}} - L_{\text{cond}}$ equals the energy the flow carries out to 0.04-0.62 per cent in all four states, at $1.95\text{-}1.98\text{e}12$ erg/g ($G M_p/R_0 = 1.91\text{-}1.92\text{e}12$). The heat deposited below 1.02 R_p (mostly the molecular-chemistry channel) is close to the conducted loss, which rises x2.7 but does not drive the wind. The heat deposited above 1.02 R_p rises x1.41 (2.09 -> 0.65) and Mdot per unit of it x1.26 (Mdot x1.77); the heating per unit mass in the wind is nearly unchanged, the heat is deposited higher (the radius halving it 1.35 -> 1.53 R_p), μ at T_{max} falls 2.62 -> 1.91 and the scale height there rises 0.174 -> 0.215 R_p ; the mass density beyond 2 R_p is 2.0-2.2 times larger.

Not verified. The cause of the Mdot rise: the absorbed stellar XUV power is not written by the code and no solve changing the mean molecular weight alone was run, so "lighter gas, larger scale height, heat deposited over more mass" is an interpretation consistent with the numbers. The rate coefficients were evaluated from the working tree (dd4a591 with uncommitted edits), not from the 99a76cb0 build that solved the states; their agreement is inferred from the smooth, state-independent measured P3. The radial factor split is first order (its zone sums differ from the Shapley

values by up to 0.036 on 2.09 -> 0.65).

92. Phase 6 of the LHS 1140 b EW plan: numerical and observational validation at He/H 0.65 (2026-10-04)

Decision taken with the user: phase 5 found no crossing in He/H (section 90), so phase 6 validates the numerics at the model with the largest EW, deep He/H 0.65 (phase5/HeH0.65/, closure $k = 2$), to show that the EW deficit is robust. Record: LHS1140b/models/.P2/ew_fit/phase6/NOTES.md (runs, attempts, figures). All numbers MEASURED; binary 99a76cb0 for every solve and evaluate; EW = width-matched window EW of the solution product (section 87; $f = 22.24$ km/s on every state, an added kernel, not a computed physics term). Reference 0.852918 (unbroadened 0.882189), observed 1.1084 ± 0.0289 , target 0.0058 for each test.

Numerical tests.

test	EW	change	below 0.0058?
grid 500 -> 1000 cells (all spacings halved, outer face kept), certified	0.845660	-0.00726	NO (1.25 x)
grid 1000 -> 2000 cells, diagnostic (not certified, below)	0.841623	-0.00404	yes (0.70 x)
Resid tol 4e-6 -> 1e-6, 1e-7, 3e-8, 1e-8, all certified	0.852918	0	yes
closure tol 0.05 -> 0.01, T_match tol 6 -> 2 K, converged	0.853294	+0.00038	yes
domain to L1 (Outer shells: 37.6 15), certified	0.854864	+0.00195	yes

- Grid. 1000 cells: Base grid [dr, cells]: $9.9999997473787516e-5$ 100, Grid cells: 1000, Outer radius [R_p]: 29.638956766027057 (the reference's outer face 29.273416 R_p to $1e-15$); cell widths 0.499-0.508 of the reference's. The seed needed one correction: the mapper interpolated the first refined cell from the reservoir ghost row (415 K against 576 K in the reference's cell 1), and the JFNK stalled on cell 1; with that cell set to the reference's cell 1 the solve converged, and certified after a continuation (the first run ended on the 60-pass cap with only the He 2^3S carrier row of the first gated cell above tolerance; continued with 150 passes, accepted at pass 116; tolerances unchanged). The refined state has $\log_{10} \dot{M}$ lower by 0.0034 (-0.8 %), ρ and $n(2^3S)$ lower by 0.8 and 0.8-1.5 % from 3 to 28 R_p: the grid error enters through the steep base layer that sets $\dot{M}$. The EW falls with refinement, as in the 1000/2000-cell study of 2026-09-24 (first order, -3 % there), so the 500-cell EW is an overestimate and the deficit larger in the continuum limit. 2000 cells (Base grid $4.999999873689376e-05$ 200, same face), seeded from the certified 1000-cell state: not certified after 300 outer passes (He 2^3S carrier row at the first gated cell $4.7e-5$ against $1e-5$, and the mass row of one cell at its rounding

floor), with the EW unchanged to $1e-6$ between pass 150 and 300: 0.841623 (diagnostic). Successive differences -0.00726, -0.00404, ratio 0.556, observed order 0.85; Richardson continuum (DERIVED) 0.8366, 0.016 below the 500-cell value (2.8 x the target). Mdot falls 0.8 % and then 0.4 % with each halving.

- Wind tolerance. Resid tol is the JFNK target, not a certification tolerance; the certification gates are compile-time (mass row $3e-12$ floor or ten times the cell's rounding, momentum $1e-8$, energy $1e-6$, closure and level $1e-6$, carriers and elemental transport $1e-5$ at $r \geq 1.2 R_p$). The JFNK reaches its rounding floor ($\|R\|$ 1.2-1.6e-8) at every target from $4e-6$ to $3e-8$, so those states are identical; $1e-8$ (below the floor, 21 of 32 solves end info 2) is accepted and moves the state by $\leq 6.5e-9$ (hydro), $1.6e-7$ ($n(2^3S)$). Tightest certifying target $1e-8$; EW change 0.
- Closure. Resuming the damped closure from k02 at tol 0.01 stopped twice at k03 on the wind (He 2^3S carrier row at cell 225: $2.1e-5$ and $1.15e-4$ against $1e-5$ after 60 and 150 outer passes; kept). A fresh closure started at the fluxes k03 measured converged at $k = 1$ (eps $4.7e-5 / 1.2e-4$, T_{match} 415.86 K, moved 0.053 K). Fixed point F_H 6.4803e7, F_{He} 7.0914e7 g/s, T_{match} +3.27 K over k02. The k02 eps_H 0.045 was the trial-measured gap; k02's measured flux was within $2e-4$ of the fixed point. Settled.
- Domain. L1 extension: sonic point 28.80 (28.79), log10 Mdot unchanged to $2e-5$, profile changes $\leq 1e-4$ inside $12.9 R_p$. The $30 R_p$ edge is $0.48 R_p$ beyond the sonic point and moving it does not change the branch.
- Roche (decision 3, option b'). Domain mode: Roche re-solve: certified by the solve (pass 28), but the evaluate of the reloaded state refuses it on the mass row of cell 1 ($3.514e-9$ against the cell's anchor $3.3e-9$; the solve read $3.101e-9$ there), so its EWs are diagnostic. Spherical synthesis 0.834533; Roche synthesis (phase4/roche/roche_he10830.py, sub 4, 8 azimuths) 0.820499. Geometric interval [Roche-consistent, spherical] = [0.8205, 0.8529] (-3.8 %; -2.4 % at He/H 2.09).

Observation. The width-matched model against the released Cherubim et al. (2026) spectrum (three-Gaussian extractor): red depth 0.985 against 1.254 %, blue 0.128 against 0.198 %, red/blue 7.67 against 6.35, FWHM 0.841 (matched) against 0.841 Å, shift -0.04 against +2.85 km/s. Residual χ^2 136 for 44 window samples (1962 for a zero model); the deficit sits in the red pair (mean residual 0.224 $\pm$ 0.025 % over 10832.9-10833.7 Å against 0.051 $\pm$ 0.031 % over the blue component). Frame: the model line has no shift; the data prefer +0.081 Å (+2.24 km/s) for the model's shape under independent errors (χ^2 16.9 against 61.2 at zero), the published fit with correlated noise 2.0 (+2.0/-2.2) km/s (READ, dataset README). The window is not symmetric about the red pair, so the shift moves the model EW by +0.0115 (at +0.081 Å) to +0.0137 (+0.104 Å), 2.0-2.4 times the target; the assumption holds at 1 sigma of the published shift and is carried as a systematic. Figure phase6/observation/HeH0.65_k02_solution.png (inspected).

Uncertainty budget at He/H 0.65 (changes of the EW): grid -0.00726 (1000 $\rightarrow$ 2000 -0.00404; to the continuum -0.0164, derived); Resid tol 0; closure +0.00038; domain +0.00195 (numerical, in quadrature 0.0075, or 0.0165 with the grid's continuum estimate); NH3 nu2 + H2 rate: water analog -0.00037 (T_{match} +7.6 K), no H2 term +0.00105 (+39.0 K), single solves at k02's chemistry arguments, 15-40 times smaller than at He/H 2.09 because the EW is nearly insensitive to T_{match} here; ion-neutral drift of He I (phase 1 method on this state, which carries He 2^3S : a fixed-background He 2^3S solve with the Stefan-Maxwell drift flux, lambda_3 from 63 stage-channel evaluates) +0.000037 (A) / +0.000034 (B), so no drift treatment is needed (plan section 8); geometry [Roche-consistent] -0.0324 (diagnostic); frame +0.0115 to +0.0137. Every positive item together gives at most 0.870, still 8.2 sigma below the observation: the

deficit is robust to every item measured. Validity: 31.4 % of the EW forms beyond $\text{Kn}_{\text{bulk}} = 0.1$ (8.57 R_p), 9.1 % beyond R_*/R_p , 0.01 % beyond the exobase (28.63 R_p , below the sonic point 28.79); max Kn 1.3.

Found, not fixed (need a decision). (1) In Roche mode the evaluate does not reproduce the solve's certification of the written state (cell-1 mass row, above); 99a76cb0 does in spherical mode. (2) The He 2³S carrier row at the first gated cell ($r = 1.2 R_p$) decides every hard ending: k03, the first 1000-cell run, the 2000-cell run, He/H 0.5 of phase 5; k02 itself passed it at 9.65e-6. (3) Refining the base block needs the seed correction above (phase6/grid/base_cell_seed.py); map_state_to_grid.py is unchanged.

Not verified. The 2000-cell state is not certified (its EW is diagnostic). The NH3-H2 cases were not re-closed. The drift estimate is a fixed-background linear estimate of a drift that is itself outside its Chapman-Enskog validity where $w/v \sim 1$.

93. Closure acceptance: the conducted heat flux has its own residual; Codex's resume and solve-exit refusals reviewed (2026-10-04)

Review md/LHS1140b_NH3_CH4_EW_REVIEW_20261004.md (Codex, 2026-10-04, interrupted by its usage limit). Codex also edited src/utls/element_flux_closure.py and added src/tests/element_flux_closure/ (6 tests): a window whose H or He flux or spread is missing or non-finite is unresolved (flux_window_resolved); a --resume refuses a last history row without a resolved measurement at the requested tolerance and checks it against the saved wind's own measurement; an EXHALE that exits non-zero is refused before post-processing even when its log says info=0. Read here, compiled, and the suite passes (6/6); kept.

Added here (the review's finding that the energy-coupled acceptance has no explicit conducted-flux test): $\text{conducted_flux_residual} = |F_{\text{meas}} - F_{\text{trial}}|/|F_{\text{meas}}|$ of the conducted base heat flux is held to the same tol as the elemental fluxes, both in the iteration and in the resume check, beside the T_match change. Measured on every stored energy-coupled closure (last history row): 0.0001-0.0375 on all converged ones (largest He/H 1.2, 0.0375), so no stored convergence verdict changes; phase5/HeH3.0 (not converged) has 0.218.

Also recorded from the review: the provenance of the phase-1 molecular emission estimate is the He/H 9.0103 state and six molecules (corrected in md/NH3_CH4_ISSUES_20261004.md); sed/scale_sed.py scales the whole SED, not the XUV alone (corrected in md/PLAN_20261004_lhs1140b_heh_high_and_xuv.md); cross-section-weighted photoionization rates at the b orbit, GJ 699 over GJ 1132: H I 38.7, He I 6.44, He 2³S 2.13 (md/oct4_review_evidence/stellar_photoionization.json); a uniform 1.30x of $n(\text{He } 2^3\text{S})$ on the He/H 0.65 reference gives width-matched EW 1.0957 (a scale of the required opacity, not a model).

94. The three phase-6 problems fixed: the mapper's base cells, the He 2³S carrier row, the certificate of a written state (2026-10-04)

User decision of 2026-10-04 (phase 6 of md/PLAN_20261003_lhs1140b_ew.md, section 92 and LHS1140b/models/.P2/ew_fit/phase6/NOTES.md section 4; review md/LHS1140b_NH3_CH4_EW_REVIEW_20261004.md section 6). Each problem was reproduced before it was fixed. Binary of every run below: LHS1140b/models/.P2/EXHALE_c37b3671.x (ifx 2026.1.0,

md5 c37b3671aa518ec7ccb75bf2718adaba, manifest .P2/BINARY_MANIFEST_c37b3671aa51.txt,
 source copy .P2/build_p6fixes_ifx/; against 99a76cb0 exactly three sources differ:
 EXHALE_main.f90, diffusive_photochemistry.f90, ion_residual_core.f90). Scratch runs:
 ~/.claude/jobs/f24ea156/tmp/p6fix/.lart4. All numbers MEASURED.

94.1 The mapper seeds the first refined cell from the reservoir row

Reproduced. map_state_to_grid.py --ic of phase5/HeH0.65/k02 onto the 1000-cell grid writes the data rows of phase6/grid/attempt1_interpolated_seed bit for bit: the first target center (1.0000491 R_p) lies between the source's lower ghost (0.99990 R_p, the reservoir, 415 K) and its first cell (1.0000967 R_p, 576 K) and was interpolated across the base face (536 K); that seed stalled the JFNK (attempt 1).

Rule adopted (src/utils/map_state_to_grid.py, map_rows, module header "THE FIRST PHYSICAL CELLS ARE MAPPED FROM PHYSICAL CELLS"): a target physical center that lies between the base face and the source's first physical center is inside the source's first cell and takes that cell's values (its average over the volume that holds the center); every other physical row is interpolated between the two source rows that bracket it, as before. The lower ghost rows are the reservoir, boundary data that load_IC.f90 does not read, never an interpolation partner of a physical cell. The upper ghosts are the free-outflow continuation of the flow and stay interpolation partners (the shifted-grid round trip of the suite needs that; holding the last cell there moved the density round trip from below $2e-5$ to $5.5e-4$). The # mapped: line counts the rows that took cell 1's values. An unrefined mapping holds no row and is unchanged.

Verification. The fixed tool reproduces the hand correction phase6/grid/base_cell_seed.py exactly: the k02 -> 1000-cell seed equals the seed of attempt 4 (Hydro bit for bit, species within $1.2e-16$ relative in row 3), the 1000 -> 2000-cell seed equals that of grid2000/solve bit for bit; the workaround is no longer needed (kept as a record). element_flux_closure.py maps onto the seed's own centers (a rewritten # grid R0 and --reservoir), which holds no row: suite row unrefined_stored_same_grid_reservoir_byte_identical... below, and element_flux_closure 6/6.

Test (src/tests/state_mapper, section 7 added): a manufactured base layer (cold reservoir ghosts, steep T) refined 2x: the first target cell is the source's cell 1 ($1e-14$), no base cell carries a value outside the first two source cells, the second cell is interpolated, the mapping line counts the held row; the stored k02 state onto the 1000-cell grid (when the tree carries it): first cell = source cell 1, not colder than it; three unrefined mappings (identity, base-layer identity, the stored state on its own centers with --reservoir He/H) byte-identical to the all-rows mapping. RED (entry text): 7 FAIL (4 manufactured, 3 stored, e.g. first cell 536.28 K against 575.80 K); GREEN: 90 PASS, 0 FAIL.

94.2 The He 2^3S carrier row at the first gated cell

Reproduced from the stored runs. On phase6/grid2000/continued the gated measure of carrier balance He 2^3S sits at cell 894 (1.230 R_p) in every one of 150 passes and wanders between $1.6e-5$ and $2.1e-3$ without contracting (the 1000-cell continuation and the closure k03 likewise at cells 448 and 225; the accepted states took a pass that happened to fall below $1e-5$).

What the row is there (row terms of the stored state; the level's gross halves recorded by this section, below): production 41.2515 and loss $41.2515 \text{ cm}^{-3} \text{ s}^{-1}$, net $8.797e-7$, transport $8.796e-7$ (the face flux changes sign in this cell:

8.44e-7 in, 3.5e-8 out), residual -8.36e-11. The level is in local balance to 2e-8 of its channels and the row is solved to 2e-12 of them; the measure divided that residual by $|J_{in}| + |J_{out}| + |net| = 1.76e-6$, i.e. by 4e-8 of the row's own terms, and so asked the coupled state for 4e-13 of the channel rates. The JFNK stops at $\|R\| \sim 6e-8$; the composition and temperature it leaves move the two halves at that order from pass to pass, and the measure moves across 1e-5 with them. Not a property of the equations, not a resolution question (the measure is 2.7e-6 at 1.217 R_p , 4.7e-5 where the flux crosses zero, 2e-7 at 1.37 R_p), not the gate's position, not the Jacobian (the row is linear in n_3 at a fixed background; the forward-difference step 1e-12 n_{He} is exact for it; helium_metastable_transport checks $d\text{src}(\text{He II})/dn_3$ to 1e-6): the scale of the measure.

Cause (code defect). carrier_residual (diffusive_photochemistry.f90) formed the physical scale row_terms_phys from `lsrcl`, the NET source, while the declaration of `carrier_accept_tol` states the measure as the row residual over "the transport flux divergence and the reaction production and loss", and the certification prints it as "the sum of the row's own terms". For a row that hands out only its net (helium stages, atomic proton) the two agree; for the level, whose balance is one closed expression of sign-definite channels, the net is the difference of the terms and not a term.

Fix.

1. ion_residual_core.f90: `tr_triplet_row_channels`, the production $n_e (n_{HeII} a_3 + n_{SI} q_{13})$ and loss $n_3 (P_3 + n_e (q_{31g} + q_{31a} + q_{31b} + b_3) + A_{31} + n_{HI} Q_{31})$ of `tr_triplet_row`'s own terms; `carrier_source` records them as `sprod/sloss` of the level (molecular row: + the $\text{He}(2^3S) + \text{H}_2$ quench), `sprod - sloss = src` to rounding.
2. carrier_residual: the physical scale takes `srpc + srcl` as its reaction terms. Where a row hands out only its net, `srpc + srcl = lsrcl` bit for bit, so the helium stage rows and the atomic proton row are unchanged; H_2 , the molecular proton row and the oxygen carriers now read their gross channels, as the stated definition says. The Newton's own scale (`dsc`) keeps `lsrcl`. The tolerance (1e-5 in the wind) is unchanged.
3. Found on the way and fixed: the carrier relaxation declared "the carrier block closed at this background" after ONE transport step with the level row at 9e-5 of its terms (every pass of `grid2000`). Its fixed-point test compared the largest change of any carrier with the largest fraction of ALL carriers ($x_{ref} \sim 1$, from the helium stages), against which a step that moved the level (5e-11 of helium) by 20 % of itself read 1e-11 < 1e-10. Now (`relax_photochemical_composition`) each carrier is judged against its own largest fraction (`carrier_displacement_on_own_scale`; the old `carrier_composition_displacement`, no longer called, removed), and the fixed point is also asked of the steady carrier residual of the present rows on the certification's measure, at or below `carrier_accept_tol` (1e-8): an implicit step's displacement bounds its residual only by n/dt . MEASURED with item 3 alone (scale unchanged), `grid2000`: the relaxation closes the block (9.7e-9 after 189 steps) but the measure after the next wind solve still wanders (3.0e-1 on the first pass after the closure, then 4.7e-5, 4.2e-5, 9.5e-5, 1.1e-5, 4.3e-5, 3.9e-5, 4.2e-5, accepted on an 8.7e-6 at pass 14), which is what isolated item 2 as the cause.

Verification (binary c37b3671, the phase-5 solve environment).

- phase6/grid2000 restarted from continued: certified at pass 4 (worst gated row 8.0e-8 (He^+), then He^{++})

- 6.3e-4, 6.8e-5, 2.9e-6, a geometric contraction); the stored-pair evaluate of the written state reproduces the solve (mass row binding cell 843, 3.559e-11 against 3.7e-11; exit 0). Width-matched EW 0.841623, equal to the earlier diagnostic value, now on a certified state; so 1000 -> 2000 cells = -0.004037 stands as measured.
- phase5/HeH0.5, the wind of attempt 4 (seed from the 0.65 rung, the k00 inputs): certified at pass 13 (He++ row 1.0 for six passes, then 1.2e-1, 2.3e-2, 4.8e-3, 9.6e-4, 2.0e-4, 3.9e-5, 8.3e-7); stored-pair evaluate certified, exit 0. Only the wind was re-solved; the He/H 0.5 closure was not re-run (its earlier k00 had no sonic point in the 30 R_p domain).
 - phase5/HeH0.65/k02 re-solved from its own seed: certified at pass 1, the state within 1.6e-12 (hydro) and 8.2e-13 (species) of the stored k02, width-matched EW 0.852918 (unbroadened 0.882189), identical to the reference to the printed digits. The equations are unchanged (the scale is the measure's, not the state's), so "unchanged within rounding" applies.
 - Gross channels recorded: on the stored 2000-cell state the level row of cell 894 reads production 41.2515, loss 41.2515, measure 1.45e-9 at most over the gated cells (was 4.75e-5); k02 evaluate: He2^3S 2.9e-4 (cell 111, below the gate) -> 1.7e-10; H2 4.0e-9 -> 3.7e-9, H+ 1.1e-7 -> 8.8e-9; He+ and He++ identical (net rows).
 - What the gate now asks of the level: its residual against its whole turnover, which in the wind is the transport plus two halves that cancel to 1e-8 at 1.2-1.4 R_p and to 1-4 % in the line-forming region (phase 1); the gate is therefore weaker against a transport defect of the level where the level is in local balance, which is where transport does not set it.

94.3 The certificate of a written state and of its reload

Reproduced. The stored-pair evaluate of phase6/roche/solve with 99a76cb0 reads the cell-1 mass row 3.514150e-9 against its anchor 3.3e-9 (exit 2); the solve's own final certification read 3.101e-9. A re-run of the Roche solve from its seed with a control binary (the 99a76cb0 source plus measurement records in `certify_the_state_to_be_written`) writes the stored files bit for bit and shows the cause.

Cause. Not the Roche geometry: the physical cells of the written and the reloaded state are identical (u, f_sp, n_tot, n_e), and the spherical k02 shows the same disagreement at its base cells (cell-2 mass row 6.219e-10 in the solve, 7.407e-10 on reload), only far below that cell's anchor. The two differ in the LOWER GHOST: the solve's final certification stood on the ghost the last sweep of the solve had returned (taken at the temperature of an earlier iterate), the reload on the ghost solved at the state's own temperature. Ghost T differs by 1.8e-12, the base face mass flux F_lower by 4.1e-16 (4e-10 of itself: the base continuity row reads the ghost with a gain of order 1/Mach, Mach ~ 1e-6 there), the cell-1 row by 13 % of its anchor. The Roche state sat 0.93 of its anchor and crossed; the spherical states did not. A certificate taken on quantities that depend on the iteration history is not a statement about the state written.

Fix (EXHALE_main.f90). `sources_and_ghosts_of_the_held_composition`: the temperature from u through the caloric map of f_sp, the heating and cooling and the lower ghost from one sweep that holds the composition (`ioniz_eq_sources_of_held_composition`, the interior returned bit for bit, asserted), i.e. the stored-pair evaluation of section 88 as one routine. It is called by that evaluation (unchanged in effect), by the joint test of every stationary outer pass before its certification, and by `certify_the_state_to_be_written`, so a pass is accepted, and a file is certified, on the numbers its reload measures. Measurement records added at the final certification

(EXHALE_EVAL_STATE_DUMP, EXHALE_BOUNDARY_TRACE, EXHALE_GHOST_RECORD, default off), the same records the evaluation writes.

Verification.

- phase6/roche re-solved from its seed: certified at pass 8; the stored-pair evaluate of the written state reproduces the final certification entry by entry (mass row max 4.663e-10 at cell 17, binding cell 218 2.654e-11 against 2.9e-11 in both; volume average 7.392e-11 against 7.469e-11), exit 0. Spherical-synthesis EW 0.834533 (as the diagnostic value); the state moved by at most 7.3e-6 (hydro) and 1.7e-7 (species) from the stored one. The Roche synthesis (roche_he10830.py) was not re-run on it.
- Spherical evaluates bitwise unchanged: phase3/he23s_transport/solve and phase5/HeH0.65/k02 evaluated with 99a76cb0 and with c37b3671 write Hydro_ioniz, Ion_species, conserved_state, both _adv files and both breakdowns with identical data rows; hydrodynamic entries identical; the carrier entries change as in 94.2. (A first version that repeated the derivation of step (1) inside the evaluation moved the two lower ghost rows of k02 by 1e-14; the routine takes the derivation as done there.)

94.4 Tests (gfortran tree build, md5 7a48413c1ec482c0305cb69c08a976a9)

state_mapper 90/0 (RED 7 FAIL above); helium_metastable_transport 24/0 (two rows added: the production and loss halves of the level equal the closed form at the local root, 1e-12; RED against the entry text: both FAIL, 0 and 5.7e-11 against 3.85e5); element_flux_closure 6/6; certification 145/0; ionization_stage_flux 45/0; stage_row_balance 10/0; ionization_stage_enthalpy_flux 13/0; carrier_boundary_jacobian 12/0; carrier_constraint_attribution 10/0; carrier_enthalpy_flux 12/0; carrier_helium_inventory 15/0; carrier_outer_boundary 16/0; carrier_reference_scales 16/0; carrier_retry 156/0; carrier_returned_state_acceptance 36/0; coupled_block_handover 33/0; coupled_block_jacobian 8/0; element_operator 65/0; interdiffusion_enthalpy_flux 18/0; ionization_imposed_fractions 48/0; krylov_and_dogleg 347/0; species_masses 9/0; steady_species_rows 197/0; adv_static_limit 64/0; boundary_state 43/0; grid_and_gates 424/3, the same three rows of restart_intent_and_metadata.sh on its stale fixture as in section 88 (worst column heat 5.83e-5, exit 2, certification F), unchanged.

Not verified. The Roche synthesis on the re-solved Roche state; the He/H 0.5 closure; the solve trajectories of molecular runs other than these four (the relaxation now runs to the closed block, 35-189 transport steps a pass on grid2000); the regression matrix (not run, by rule).

95. Route B: the GJ 699 proxy spectrum puts the He 10830 EW three times above the observation; the escape boundary condition's hydrogen carrier split made self-consistent (2026-10-04)

Plan md/PLAN_20261004_lhs1140b_heh_high_and_xuv.md, route B step 1 (user decision 2026-10-04: run GJ 699). Records: LHS1140b/models/.P2/ew_fit/route_xuv/NOTES.md, compare_route_xuv.py, route_xuv_numbers.txt, figures (inspected) fig_route_xuv_line.png, fig_route_xuv_profiles_HeH{0.65,2.09,5.0,10.0}.png, fig_route_xuv_triplet_balance.png, fig_route_xuv_ew_vs_heh.png. Binary 99a76cb0 for every closure; every final state also evaluated with c37b3671

(section 94): EW, width-matched EW and f agree to all printed digits. All numbers MEASURED; "EW wm" is the width-matched window EW in percent-angstrom (added Gaussian velocity kernel of FWHM f, not a computed physics term). Observed 1.1084 +- 0.0289.

The GJ 699 product is not LHS 1140's bolometric flux. Scaled to the LHS 1140 X-ray flux ($A_{699} = 0.056$) the GJ 699 product carries $1.767e6 \text{ erg cm}^{-2} \text{ s}^{-1}$ at the b orbit, 3.25 times the GJ 1132 product and 3.1 times $L_{\text{bol}}/(4 \pi a^2)$ of LHS 1140; the Photochem climate integrates the whole file. With it the climate is another planet (He/H 2.09: deep 790 K against 561 K, tropopause 248 K against 185 K, no water cold trap). Both were run: the pure GJ 699 file in the chemistry and the wind (as instructed), and a split sensitivity construction (make_split_sed.py: GJ 699 at $\lambda \leq 2600 \text{ \AA}$, GJ 1132 above; not a stellar model) in the chemistry with the pure GJ 699 file in the wind (the wind uses only $E \geq 4.77 \text{ eV}$). The split climate reproduces the GJ 1132 one. The GJ 1132 control at He/H 2.09 with the cooling-to-space transfer was run because no such state existed.

spectrum	deep He/H	closure	T_match [K]	log10 Mdot	EW wm sol / adv (f km/s)	Kn_bulk 0.1 / sonic [R_p]
GJ 1132	2.09	k = 1	350.71	7.8864	0.7109 / 0.7143 (22.02)	5.70 / 25.69
GJ 699	2.09	k00 wind, residuals 2.5e-5 / 1.0e-4 / 0.0054, T_match change 0.20 K (k01 chemistry); k01 wind not run (mapper)	329.74	8.7080	3.6977 / 3.6915 (21.52)	8.46 / 15.01
split	2.09	k = 5	264.59	8.6544	3.3876 / 3.3835 (21.52)	7.88 / 15.18
GJ 1132	0.65	k = 2 (section 90)	412.59	8.1325	0.8529 / 0.8596 (22.24)	8.57 / 28.79
GJ 699	0.65	REFUSED k00 (cooling- to-space exchange bound, 1.72 and 1.60 K > 1 K)				
split	0.65	k = 4	401.53	8.8152	3.7065 / 3.7114 (21.62)	11.54 / 14.54

spectrum	deep He/H	closure	T_match [K]	log10 Mdot	EW wm sol / adv (f km/s)	Kn_bulk 0.1 / sonic [R_p]
split	5.0	k = 3	271.78	8.6686	3.4270 / 3.4202 (21.41)	7.23 / 15.46
split	10.0	k = 3	280.31	8.6617	3.3276 / 3.3203 (21.34)	6.82 / 15.57

The sonic point lies inside the 30 R_p domain in every state. The GJ 699 lines are 3.0-3.5 times deeper than observed (width-matched red peak 3.8-4.3 per cent against 1.28), at the GJ 1132 line position (10833.27 Å, 0.15 Å blueward of the observed peak) and red/blue ratio (7.2-7.3). Thin flag T in every state but He/H 10 (largest absorbed share 0.08-0.76, the pure-GJ 699 0.763 from uncondensed water); without the gas above p_{match} 0.03-0.04.

No crossing in He/H on GJ 699. The width-matched EW is 3.71 (0.65), 3.39 (2.09), 3.43 (5), 3.33 (10) on the split spectrum and 3.70 at 2.09 on the pure file: 2.2-2.6 percent-angstrom above the observation everywhere, weakly dependent on He/H. The observation lies between the two proxy spectra (GJ 1132 0.71-0.85, GJ 699 3.33-3.71); the crossing is in the stellar input, not in He/H.

Penning loss. At He/H 2.09, GJ 699 against GJ 1132 at fixed r (1.5-15 R_p): $n(2^3S)$ $\times 2.3$ -15.6; recombination gain $n_e n(\text{He}^+)$ $\times 3.2$ -26.5; $n(\text{H I})$ $\times 0.61$ -1.14 ($x(\text{H}^+)$ rises 5-7 times but $\dot{M}$ $\times 6.6$), so the Penning loss rate falls by at most 40 per cent inside 3 R_p and not beyond 8 R_p ; the electron-collision loss rises $\times 2.8$ -11.9 and takes the Penning share of the loss from 0.5-0.7 to 0.2-0.3. P_3 rises $\times 2.12$ ($1.389\text{e-}5$ against $6.54\text{e-}6 \text{ s}^{-1}$) and stays below 5 per cent of the loss inside 8 R_p . The 39x H I photoionization does not raise $n(2^3S)$ by cutting the Penning loss; the larger He I photoionization ($\times 6.4$) and the hotter, more ionized wind raise its source.

Fixed: the hydrogen carrier split of the escape boundary condition (src/utls/photochem_to_lower_profile.py).

The elemental hydrogen flux is put on H2 when the atomic-H share of the hydrogen nuclei at the model top is below 1 per cent and split between H and H2 otherwise, but the share was measured on the composition the imposition sees, on a fresh start the initial one ($1\text{e-}40$): the split branch could not be reached, and a converged top above 1 per cent was refused (split He/H 2.09 k00: $1.009\text{e-}2$). Now the converged share is compared with the share the boundary condition carries (LAST_IMPOSITION); where they differ by more than 1 per cent the converged share is imposed (CARRIER_SHARE_STATED, needed because a re-solve on a new temperature re-initializes the composition from the climate grid) and the chemistry (under conduction, the column iteration) re-solved, up to 8 rounds; from the second round the imposed share is the mean of the imposed and the converged one (the map has slope about -0.75 at He/H 30: 0 $\rightarrow$ 0.098 $\rightarrow$ 0.025 $\rightarrow$ 0.079 $\rightarrow$ 0.039). RED: split 2.09 k00 refused; GREEN: one round, imposed 0.0101, converged 0.00914. A case needing no round runs the unchanged path: the GJ 1132 2.09 k00 arguments give a profile byte-identical to the closure's (cmp).

Found here, resolved in section 96. map_state_to_grid.py --reservoir refused a carry whose first physical cell departs from the requested C/H by more than $1\text{e-}12$ relative; the pure GJ 699 closure at 2.09 was stopped twice by it ($1.23\text{e-}12$, $1.03\text{e-}12$). The cause is not CO: the k00 state carries no OH, H2O or CO column (oxychem=F); its cell-1 C/H stands $1.03\text{e-}12$ from its own # reservoir line, and the factor new/old carries that departure onto the new value (section 96).

He/H 10 and 30 jumped directly from 2.09 stalled in the first wind passes and were stopped (kept); 10 converged from 5, and 30 from 10 was stopped in its first wind pass after 34 min and not pursued.

Not verified. The k01 wind of the pure GJ 699 2.09 closure; pure GJ 699 at any other He/H; He/H above 10; the region beyond $\text{Kn}_{\text{bulk}} = 0.1$ (6.8-11.5 R_p).

96. The mapper's base-row check takes the consumer's tolerance, 1e-6 (2026-10-04)

What refused the route-B carry (MEASURED). `map_state_to_grid.py --reservoir El/H v` multiplies the ionization-stage columns of El by $k = v/\text{old}$, old the El/H of the source's # reservoir line, and then refused the map when the base rows (He: the two inner ghosts and cell 1; a metal: cell 1) came out more than $1e-12$ relative from v. The GJ 699 He/H 2.09 k00 state (LHS1140b/models/.P2/ew_fit/route_xuv/GJ699_HeH2.09/k00/output) carries no OH, H₂O or CO column (# options ... oxychem=F), so the CO explanation of section 95 does not apply. Its cell-1 C/H is $2.77551410664327e-4$ against the line's $2.7755141066461295e-4$, a relative departure of $-1.0303e-12$ that the solve leaves (C/H varies by $2.5e-10$ over the physical cells); k carries it onto the new value unchanged, and the check refused at $1.03e-12$. The attempt-3 carry refused at $1.23e-12$ the same way. Both reproduced with the script as it stood, on copies.

Why 1e-6 (READ, src/modules/files_I0/load_IC.f90). The consumer of the mapped state accepts the loaded He/H within `heh_dev_tol = 1.0d-6` (line 76, the check at line 784), and renormalizes every metal column the handoff states by one factor (handoff El/H)/(El/H at the base cell) (the "elements the handoff states" loop, lines 969-1030), so a metal arrives at the run's El/H at the base cell whatever the mapper wrote. A $1e-12$ check was a million times stricter than the run that reads the file. The base-row check now uses one named constant, `BASE_ROW_RATIO_TOL = 1.0e-6`, for helium and for the metals, with a comment pointing to `heh_dev_tol` and to the base-cell renormalization; the refusal message states the tolerance and its source. The check still catches the gross failure it exists for: a species holding nuclei of two elements that the element's one factor does not move with it (HeH⁺; OH, H₂O and CO, which the tool does not multiply). The mapping itself is unchanged.

Tests (src/tests/state_mapper/, MEASURED). New: a metal state whose cell-1 C/H stands $1.03e-12$ from its reservoir line maps and its departure comes out unchanged ($-1.03040e-12$ against the source's $-1.03018e-12$; bound $1e-15$ on the difference); the same state at $2e-6$ is refused and the message names `heh_dev_tol`; HeH⁺ at 0.8 n_H (cell 1) with He/H 1.6 → 2.13 is refused (the existing 0.05 case is still refused); the route-B k00 → k01 carry with its closure arguments maps, every requested El/H at cell 1 within $4.4e-12$. RED with the previous script: 5 failures (the $1e-12$ case refused, the route-B carry refused, three message checks). GREEN: `run.sh 98 PASS`, 0 failures (the stored-state cases included); `element_flux_closure/run.sh 6 tests OK` (it does not call the mapper). Byte identity: the 34 output files of the suite's previously passing maps are identical between the two scripts (`cmp`), and 43 mapper invocations recorded in the `route_xuv/*/closure.log` and `phase5/*/closure.log` files, rerun with both scripts on their recorded sources into scratch directories, give 42 byte-identical pairs of outputs and the route-B carry, refused before, now written. The attempt-3 carry ($1.23e-12$) also maps now.

Installed at 2026-10-04 15:17:40 KST by an atomic replace (a temporary file in `src/utls/` moved over the script), because the `route_heh_high` closures invoke the script afresh at each iteration.

Not verified. The k01 wind of the route-B closure from the carried state (not run here); a state with oxygen-chemistry

carriers through --reservoir (no such state exists in the campaign; it is still refused when their share moves an E/H by more than $1e-6$).

97. Code audit of 2026-10-04: a non-finite restart center is refused; a zero conducted flux can close (2026-10-04)

Audit ../EXHALE_audit_20261004/CODE_AUDIT_20261004.md (independent copy of the 15:55 source; read in full).

B1, fixed. `verify_restart_radii` (`load_IC.f90`) compared `lr_file - r/lr` against a running maximum; a NaN center compares false, so the maximum stayed 0 and a file with a NaN radius passed the grid guard (reproduced by the audit on the real module). A non-finite relative difference is now set to `huge()` before the comparison, so it is refused as a grid mismatch. Test `src/tests/restart_radius_guard/` (the audit's driver, linked against the tree's objects): finite -> exit 0, different -> exit 1, NaN -> exit 1 (before the fix NaN returned 0). Valid states are untouched: the change acts only on non-finite values.

B2, fixed. `conducted_flux_residual` (`element_flux_closure.py`) returned inf whenever the measured conducted flux was zero, so an exact zero/zero fixed point could never close. It now returns 0 for two exact zeros, inf for a nonzero trial against a zero measurement, inf for a missing or non-finite value, and the relative residual otherwise. Tests added to `src/tests/element_flux_closure/` (10 pass). No stored closure has a zero flux, so no verdict changes.

P1, open (needs a decision). With Low-energy electron partition: `Furlanetto2010` on, any $n(H_2) > 0$ switches the atomic partition off in that cell (`low_energy_electron_degradation.f90`, the H_2 gate), so the deposition is discontinuous between zero and a trace H_2 density, and the tables are those of an atomic H/He mixture. The option is off in every LHS 1140 b run, so no stored result is affected. Recorded as `TO_BE_DONE` (AJ).

P2, P3 are the documented limitations already recorded: the dissociative-recombination recipients (section 72 range statement at the code site) and the NH_3/CH_4 overlap (`md/NH3_CH4_ISSUES_20261004.md`, `TO_BE_DONE` (AI)).

The tree's gfortran build was rebuilt; the running ifx closures use the already-built binaries and are unaffected (the `load_IC` change concerns only non-finite input).

98. Route B step 2: the observed He 10830 EW needs 1.33-1.39 times the GJ 1132 ionizing XUV at He/H 2.09 (2026-10-04)

Plan `md/PLAN_20261004_lhs1140b_heh_high_and_xuv.md`, route B step 2 (user decision 2026-10-04). Records: `LHS1140b/models/.P2/ew_fit/route_xuv_scaled/NOTES.md`, `compare_xuv_scaled.py`, `xuv_scaled_numbers.txt`, figures (inspected) `fig_xuv_scaled_line.png`, `fig_xuv_scaled_profiles.png`, `fig_xuv_scaled_ew_vs_s.png`. All solves and evaluates on `lart3`, binary `c37b3671`. All numbers MEASURED; "EW wm" is the width-matched window EW in percent-angstrom (added Gaussian velocity kernel of FWHM f , not a computed physics term). Observed 1.1084 ± 0.0289 .

The spectrum (a sensitivity construction, not a stellar model). `make_xuv_scaled_sed.py`: the GJ 1132 product with every row at $\lambda \leq 911.75$ Å (X-ray and hydrogen-ionizing EUV, the band that photoionizes the H I and He I ground states and in which the two proxies differ by 7x in F_{EUV}) multiplied by s , and every row above by one factor c holding the bolometric flux (trapezoid over the file) at the GJ 1132 value: $c - 1 = -1.3e-5$ at $s = 1.36$, $-1.07e-4$ at $s = 4$;

the written files (17 digits) reproduce the bolometric $5.439384522884e5 \text{ erg cm}^{-2} \text{ s}^{-1}$ to 0 relative. The 911.75-2600 Å band (Lyman alpha, FUV, NUV: the He 2^3S photoionizing band and the H2 Lyman-Werner band) keeps its shape, so the scan raises the He+ recombination source of 2^3S and not its photoionization loss (P_He23S at the orbit +0.6 per cent at $s = 1.36$, from the part of the 2^3S cross section below 911.75 Å; P_HI and P_HeI scale as s). Both the Photochem adapter and the wind read the scaled file: EXHALE_setup.out Log EUV / Log X equal the control's plus $\log_{10} s$ on every state, and the profile's stellar_flux record names the file with its sha256.

Settings identical to the GJ 1132 He/H 2.09 cooling_to_space control of section 95 (route_xuv/GJ1132_HeH2.09, $k = 1$, EW wm 0.7109) except the SED path and the binary; continuation in s from the nearest converged s ; no iteration budget raised, no tolerance changed, no climate starting value needed.

s	F_EUV(100-closure k 911)		T_match [K]	log10 Mdot	T_max [K]	EW wm sol / adv (f km/s)	wm red peak [%]	red/blue	Kn_bulk 0.1 / sonic [R_p]
1 (control)	10.05	1	350.71	7.8864	4961	0.7109 / 0.7143 (22.02)	0.821	7.60	5.70 / 25.69
1.3	13.07	1	350.29	8.0399	5212	1.0452 / 1.0489 (21.98)	1.207	7.52	6.06 / 24.06
1.36	13.67	1	350.39	8.0652	5250	1.1071 / 1.1107 (21.97)	1.278	7.51	6.12 / 23.77
1.4	14.07	1	350.31	8.0812	5274	1.1472 / 1.1509 (21.96)	1.325	7.50	6.16 / 23.58
1.5	15.08	1	349.95	8.1188	5329	1.2444 / 1.2481 (21.95)	1.437	7.49	6.25 / 23.12
2	20.10	3	345.81	8.2691	5531	1.6703 / 1.6735 (21.90)	1.928	7.42	6.64 / 21.22
4	40.20	3	339.95	8.6069	5894	2.7923 / 2.7931 (21.67)	3.222	7.29	7.85 / 17.14

Every closure converged (tol 0.05, T_match tol 6 K, conducted-flux residual ≤ 0.05); every wind certified as written; every evaluate exit 0 with written_state certified=T; sonic point inside the 30 R_p domain everywhere; thin flag T in every state (largest absorbed share 0.203-0.205, CH4 dyad, as the control; 0.028 without the gas above p_match), exchange shift 0.36 K. The $s = 1$ control re-evaluated on lart3 gives the section-95 EW, EW wm and f to all printed

digits.

The crossing. EW wm rises monotonically with s . Linear in $\ln s$ between the bracketing converged states: the observed 1.1084 at $s = 1.361$ (solution; 1.358 adv), the 1 sigma band 1.0795-1.1373 at $s = 1.333$ -1.390, i.e. $F_{\text{EUV}}(100\text{-}911) = 13.4\text{-}14.0 \text{ erg cm}^{-2} \text{ s}^{-1}$ at the b orbit (crossing 13.68), 0.195 of the GJ 699 value (70.18) and 1.36 times the GJ 1132 value. The converged $s = 1.36$ state is itself 0.0013 (solution) / 0.0023 (adv) from the observation. At that state the width-matched red peak is 1.278 per cent against the observed 1.275 (expected once EW and FWHM are both matched; not an independent test), the red/blue depth ratio 7.51 against the observed 6.35 (same extractor, section 92; 7.6 was quoted here in error, section 99), and the peak 0.15 Å blueward of the observed one, as at every s . The lower column hardly moves over the crossing range (T_{match} 349.95-350.39 K up to $s = 1.5$). Interpretation: the EW grows through He⁺ and n_e (the recombination source of 2³S) at every radius; $n(\text{H I})$ is nearly unchanged.

Not verified. The grid error at these states (about a quarter of the observed 1 sigma at He/H 0.65, section 90); the crossing at other He/H; the Roche-mode evaluate; the region beyond $\text{Kn}_{\text{bulk}} = 0.1$ (6.1 R_p at the crossing). The interval reflects the observed 1 sigma only.

99. Route B step 3: the observed He 10830 EW fixes $F_{\text{EUV}}(100\text{-}911)$ at 12.9-13.7 erg cm⁻² s⁻¹ over deep He/H 0.65-5 (2026-10-04)

Plan md/PLAN_20261004_lhs1140b_heh_high_and_xuv.md, route B (user decision 2026-10-04: repeat the section-98 crossing search at deep He/H 0.65, 1.2 and 5). Records: LHS1140b/models/.P2/ew_fit/route_xuv_scaled/NOTES.md section 6, compare_heh_match.py, heh_match_numbers.txt, figures (inspected) fig_heh_match_feuv.png, fig_heh_match_line.png, fig_heh_match_ew_vs_s.png. All solves and evaluates on lart3, binary c37b3671, the construction and settings of section 98 (cooling_to_space, energy coupling, tol 0.05, T_{match} tol 6 K); no iteration budget raised, no tolerance changed. All numbers MEASURED; "EW wm" is the width-matched window EW in percent-angstrom (added Gaussian velocity kernel, not a computed physics term). Observed 1.1084 ± 0.0289.

Seeds. The $s = 1$ converged GJ 1132 state at each He/H: phase5/HeH0.65 k02 (keeps the control's --climate-t-deep-guess 650), phase5/HeH1.2_thick k01, route_heh_high/HeH5.0 k01 (read only); later s from the nearest converged s at the same He/H. The two controls solved with older binaries re-evaluate with c37b3671 on lart3 to all printed digits. setup_case.py takes --control; cases HeH<x>/xuv_x<s>/. 14 closures, every one converged at $k = 1$ and certified, every evaluate exit 0 with written_state certified=T, sonic point inside the 30 R_p domain everywhere.

deep He/H	s for	F_EUV(100-911)	nearest converged state: s, EW wm sol / adv (f km/s)	log10 Mdot (N-20)	T_match / T_max [K]	red/blue (depth)	red peak [A vac]	sonic / Kn_bulk 0.1 / exobase [R_p]
0.65	1.300 (1.264-1.336)	13.07 (12.71-13.43)	1.3, 1.1084 / 1.1159 (22.20)	8.2408	414.9 / 3626	7.60	10833.276	26.20 / 8.78 / above the domain
1.2	1.280 (1.249-1.311)	12.86 (12.55-13.17)	1.28, 1.1088 / 1.1146 (22.10)	8.1379	390.4 / 4437	7.56	10833.265	25.32 / 7.31 / 24.99
2.09 (section 98)	1.361 (1.333-1.390)	13.68 (13.40-13.97)	1.36, 1.1071 / 1.1107 (21.97)	8.0652	350.4 / 5250	7.51	10833.272	23.77 / 6.12 / 19.90
5	1.336 (1.309-1.363)	13.43 (13.16-13.70)	1.34, 1.1129 / 1.1142 (21.79)	8.0000	326.8 / 6188	7.46	10833.266	22.67 / 5.09 / 15.30

Brackets (linear in $\ln s$): He/H 0.65 $s = 1.25$ -1.29 / 1.3-1.32 / 1.32-1.36 for obs - 1 sigma / obs / obs + 1 sigma (states at 1.15, 1.25, 1.29, 1.3, 1.32, 1.36); He/H 1.2 1.17-1.25 / 1.25-1.28 / 1.28-1.32 (1.17, 1.25, 1.28, 1.32); He/H 5 1.3-1.34 / 1.3-1.34 / 1.34-1.37 (1.3, 1.34, 1.37, 1.42). The adv crossings lie 0.001-0.009 lower in s .

The degeneracy line. The F_{EUV} that reproduces the observed EW is 12.9-13.7 $\text{erg cm}^{-2} \text{s}^{-1}$ at the orbit ($s = 1.28$ -1.36) over He/H 0.65-5, not monotonic (lowest at 1.2, as the $s = 1$ EW 0.853 / 0.831 / 0.711 / 0.715 is not monotonic). The spread of the four crossings, ± 3 per cent about 13.3, is of the order of each one's 1 sigma half-width (± 2.0 -2.8 per cent); the union of the 1 sigma intervals is 12.55-13.97. Interpretation: on these equations and this construction the measured EW fixes $F_{\text{EUV}}(100-911)$ at about 13.3 $\pm 0.7 \text{ erg cm}^{-2} \text{s}^{-1}$ and leaves deep He/H undetermined over 0.65-5; along the line $\log_{10} \text{Mdot}$ falls from 8.24 to 8.00 and T_{max} rises from 3630 to 6190 K.

Unfitted properties. The four matched lines are nearly identical. The red/blue depth ratio falls from 7.60 to 7.46 with He/H; the same three-Gaussian extractor on the released spectrum gives 6.35 (16-84 percentiles 5.25-7.89, Monte Carlo of the tabulated sigma with independent samples, nominal). The model red peak is at 10833.27 Å at every He/H, 0.14-0.15 Å blueward of the observed 10833.42 Å (extractor shift of the data +0.103 Å, 0.048-0.142). Neither distinguishes the He/H.

Correction (documents). Section 98 and the two route-B NOTES compared the model red/blue with an "observed 7.6"; the observed value from the same extractor is 6.35 (section 92; the published 6.7 ± 12.7 /-3.1, READ

from `he_line_metrics.py`). The quantity is the ratio of the fitted red depth to the blue depth, not of equivalent widths. Corrected in `route_xuv/NOTES.md`, `route_xuv_scaled/NOTES.md`, section 98 above and the column labels of `compare_xuv_scaled.py` / `xuv_scaled_numbers.txt`; no number of a model changes.

Not verified. Column checks (thin flag, gas above `p_match`) at the new states; the grid error (about a quarter of the observed 1 sigma at He/H 0.65, section 90); the Roche-mode evaluate; the region beyond $\text{Kn}_{\text{bulk}} = 0.1$ (5.1-8.8 R_p at the matched states). The intervals carry the observed 1 sigma only.

100. LHS 1140 b: heat conduction in the wind makes the mass-loss rate depend on composition (2026-10-04)

Question: $\log_{10} \dot{M}$ rises from 7.886 (deep He/H 2.09) to 8.133 (0.65) in the present model (section 91) and was flat across composition in the 2026-09 models; which physics does it? Records: `LHS1140b/models/.P2/ew_fit/mdot_attribution/NOTES.md`, `attribution.py`, `conduction_budget.py`, `conduction_ladder.sh`, `make_case.py`, `tables attribution.txt`, `conduction_budget.txt`, `escape_fluxes.txt`, `cooling_channels.txt`, `mach.txt`, figures (inspected) `fig_mdots_ew_vs_heh.png`, `fig_profiles.png`, `fig_dEW_factors.png`. Binary `c37b3671`, `lart4`, GJ 1132 SED. All numbers MEASURED; EW is the width-matched window EW (section 87; f 22.02-22.26 km/s, not a computed physics term). No source file changed.

Configurations (each at deep He/H 2.09 and 0.65; ionization transport, He 2^3S transport, molecular carriers, element diffusion, grid, domain, outer boundary, Resid tol $4e-6$ and closure tolerance 0.05 identical): A, the references (`route_xuv/GJ1132_HeH2.09/k01` and `phase6/closure_direct/k01`, solved with `99a76cb0`, re-evaluated with `c37b3671`, EW unchanged); B, the elemental-flux closure without `energy_coupling` (uncoupled column, conduction in the wind on: the closure before section 76); C, Conduction: False (the energy-row conduction source and the base conductive flux off; no flux to pass, so the column is the uncoupled one; element, carrier and stage enthalpy fluxes on, the code ties them to Interdiffusion enthalpy flux, not to conduction); D, conduction on, uncoupled column, Interdiffusion enthalpy flux: False (all three enthalpy fluxes off). Every closure converged (B $k = 0 / 2$, C $k = 0 / 1$, D $k = 1 / 0$) and every final state was certified. C at 2.09 needed a seed from a conductivity continuation on the fixed B column (EXHALE_CONDUCTION_SCALE 0.5, 0.2, 0.05, then off; the direct switch stalled at $\|R\| \sim 2$ and is kept).

config	He/H	$\log_{10} \dot{M}$	$\dot{M}$ dot ratio 0.65/2.09	EW wm sol / adv	T_{match} [K]	T_{max} [K]	$\mu(T_{\text{max}})$
A	2.09 / 0.65	7.8864 /	1.763	0.7109 /	350.7 /	4961 / 3501	2.621 /
		8.1326		0.7143; 0.8533 / 0.8600	415.9		1.906
B	2.09 / 0.65	7.8029 /	2.112	0.5664 /	185.0 /	4889 / 3503	2.587 /
		8.1276		0.5694; 0.8413 / 0.8479	184.6		1.900

config	He/H	log10 Mdot	Mdot ratio 0.65/2.09	EW wm sol / adv	T_match [K]	T_max [K]	mu(T_max)
C	2.09 / 0.65	8.0495 / 8.1659	1.307	1.3510 / 1.3540; 1.0764 / 1.0829	185.0 / 184.6	5353 / 3619	2.610 / 1.897
D	2.09 / 0.65	7.8276 / 8.1263	1.990	0.6860 / 0.6895; 0.8982 / 0.9054	185.0 / 184.6	4996 / 3527	2.626 / 1.917

dEW(0.65 - 2.09): A +0.142, B +0.275, **C -0.275**, D +0.212. The energy carried through cell N-20 equals $Q_{\text{net}} - L_{\text{cond}}$ to 0.983-1.005 in all eight states.

Attribution (log10 of the Mdot ratio): wind conduction C $\rightarrow$ B +0.208 dex (it lowers Mdot by 0.247 dex at 2.09 and 0.038 dex at 0.65); energy coupling B $\rightarrow$ A -0.079 dex (it raises Mdot by 0.084 dex at 2.09, 0.005 at 0.65: it reduces the composition dependence); enthalpy fluxes D $\rightarrow$ B +0.026 dex; remainder without conduction and coupling (C) 0.116 dex (x1.31), with $\mu(T_{\text{max}})$ 2.61 $\rightarrow$ 1.90, consistent with candidate (3) but not isolated. Heat conduction in the wind is what makes Mdot depend strongly on composition. Mdot follows the net heat of the gas above 1.02 R_p (heating - cooling + conduction source - enthalpy-flux divergence): B/C 0.587 at 2.09 (1.21e20 against 2.06e20 erg/s; Mdot 0.567) and 0.930 at 0.65 (Mdot 0.916). At 2.09 conduction keeps the cool layer above 1.02 R_p (1311 K at 1.02 R_p against 2304 K in C), where H3+ infrared cooling takes 2.75e19 erg/s (1.3e15 in C), the heating above 1.02 R_p falls by 0.38e20 and the conduction source removes 0.15e20; at 0.65 heating is unchanged and conduction removes 0.11e20 (interpretation of these budgets: the heavier gas keeps the hot layer close to the cold base).

Shapley (dEW, 9 factors, decompose_dEW.py unchanged): n_H +0.353 / +0.340 / +0.442 / +0.362 (A / B / C / D); y -0.628 / -0.470 / -1.142 / -0.560; He fractions -0.140 / -0.161 / -0.128 / -0.175; n_e +0.368 / +0.423 / +0.203 / +0.419; T +0.195 / +0.150 / +0.348 / +0.172; rest below 0.02. No factor changes sign; the net flips in C because the helium-abundance term doubles and the electron term halves: without conduction the 2.09 wind carries more helium (escaping He/H 0.73 against 0.50 in B; F_{He} 8.3e7 against 4.2e7 g/s) and the hydrogen flux rises less toward 0.65 (x2.36 against x3.04). Synthesis endpoints equal the EW records in all eight states.

Not verified. The C winds are subsonic inside the 30 R_p domain (isothermal Mach at most 0.66-0.68; A, B, D become supersonic at 18.5-21.5 R_p), so the C mass-loss rates depend on the outer boundary, which was held fixed; no larger domain was run. Candidate (3) not isolated by a fixed-opacity μ solve. The mechanism sentence is an interpretation.

101. Route A: deep He/H 2.09 to 1e4 on the GJ 1132 spectrum; the He 10830 EW passes a minimum and turns up but stays below the observation (2026-10-04)

Plan md/PLAN_20261004_lhs1140b_heh_high_and_xuv.md, route A (user decision 2026-10-04), extended to He/H 1e4 at the user's request of the same day. Records: LHS1140b/models/.P2/ew_fit/route_heh_high/NOTES.md, compare_route_heh_high.py, route_heh_high_numbers.txt, n3_column_split.txt, figures (in-

spected) `fig_heh_high_ew_vs_heh.png`, `fig_heh_high_line.png`, `fig_heh_high_profiles.png`, `fig_heh_high_triplet_shares.png`. Binary `c37b3671` (section 94) for every closure and EW, lart4, 12 threads. No source file changed. All numbers MEASURED; "EW wm" is the width-matched window EW (section 87; an added Gaussian velocity kernel, f 21.4-22.0 km/s, not a computed physics term). Observed 1.1084 ± 0.0289 .

Setup. Every case is the closure of the GJ 1132 control at He/H 2.09 (`route_xuv/GJ1132_HeH2.09/closure.json`: `cooling_to_space`, top-pressure passes 12, `tol` 0.05, `T_match` `tol` 6 K, energy coupling) with only `--abundances` `He=<x>`, the binary and the template changed; continuation in He/H (the mapper carries the reservoir; no molecular state rescaled). He/H 3.0 was started fresh from the control, not from the old-treatment phase5/HeH3.0 k01. No tolerance and no iteration budget was changed (the model-top pinning needed at most 7 of its 12 passes). Jumps of $\times 1.33$ - $\times 2$ solved; three jumps stalled in the first wind pass (JFNK $\|R\|$ 1.5-3 for 400-830 iterations) and are kept: 2.09 $\rightarrow$ 5, 400 $\rightarrow$ 1000, 1000 $\rightarrow$ 3000.

Scan (every row: closure converged, written state certified in the wind, EW evaluate certified, exit 0; sonic point inside 30 R_p):

deep He/H	closure k	T_match [K]	log10 Mdot	T_max [K]	EW sol / adv	EW wm sol / adv (f km/s)	Kn_bulk 0.1 / exobase / sonic [R_p]
2.09 (control, section 95)	1	350.71	7.8864	4961	0.7353 / 0.7388	0.7109 / 0.7143 (22.02)	5.70 / 17.23 / 25.69
3.0	1	332.83	7.8255	5358	0.6993 / 0.7020	0.6761 / 0.6787 (21.92)	5.06 / 14.67 / 24.54
4.0	1	326.05	7.8007	5657	0.7054 / 0.7076	0.6820 / 0.6841 (21.85)	4.72 / 13.28 / 23.82
5.0	1	321.73	7.8012	5877	0.7399 / 0.7418	0.7154 / 0.7172 (21.80)	4.57 / 12.64 / 23.49
7.0	1	328.84	7.8135	6159	0.8058 / 0.8072	0.7791 / 0.7805 (21.73)	4.45 / 12.03 / 23.23
10	1	334.06	7.8136	6384	0.8398 / 0.8407	0.8121 / 0.8129 (21.67)	4.31 / 11.38 / 22.98
14	1	335.13	7.8067	6541	0.8493 / 0.8498	0.8212 / 0.8217 (21.62)	4.18 / 10.81 / 22.76

deep He/H	closure k	T_match [K]	log10 Mdot	T_max [K]	EW sol / adv	EW wm sol / adv (f km/s)	Kn_bulk 0.1 / exobase / sonic [R_p]
20	1	343.02	7.7936	6660	0.8416 / 0.8418 (21.57)	0.8137 / 0.8139 (21.57)	4.06 / 10.26 / 22.53
30	1	357.62	7.7749	6754	0.8212 / 0.8212 (21.53)	0.7941 / 0.7941 (21.53)	3.93 / 9.71 / 22.28
40	2	371.07	7.7661	6803	0.8137 / 0.8135 (21.51)	0.7868 / 0.7866 (21.51)	3.87 / 9.44 / 22.16
60	2	394.30	7.7599	6852	0.8130 / 0.8126 (21.48)	0.7862 / 0.7858 (21.48)	3.82 / 9.21 / 22.07
80	2	412.25	7.7579	6875	0.8160 / 0.8156 (21.47)	0.7891 / 0.7887 (21.47)	3.80 / 9.11 / 22.04
100	2	428.94	7.7576	6888	0.8201 / 0.8195 (21.46)	0.7931 / 0.7925 (21.46)	3.79 / 9.06 / 22.03
200	1	484.29	7.7596	6909	0.8362 / 0.8354 (21.44)	0.8086 / 0.8079 (21.44)	3.78 / 8.99 / 22.03
400	1	546.76	7.7657	6915	0.8573 / 0.8562 (21.44)	0.8291 / 0.8281 (21.44)	3.79 / 9.01 / 22.07
700	1	602.09	7.7731	6914	0.8779 / 0.8768 (21.44)	0.8491 / 0.8480 (21.44)	3.81 / 9.08 / 22.13
1000	1	635.41	7.7781	6912	0.8913 / 0.8901 (21.44)	0.8621 / 0.8609 (21.44)	3.82 / 9.13 / 22.17
1400	1	667.00	7.7833	6910	0.9047 / 0.9034 (21.44)	0.8750 / 0.8738 (21.44)	3.84 / 9.19 / 22.21
2000	1	703.55	7.7897	6906	0.9205 / 0.9191 (21.44)	0.8904 / 0.8890 (21.44)	3.86 / 9.27 / 22.26

deep He/H	closure k	T_match [K]	log10 Mdot	T_max [K]	EW sol / adv	EW wm sol / adv (f km/s)	Kn_bulk 0.1 / exobase / sonic [R_p]
3000	1	748.33	7.7979	6901	0.9406 / 0.9391	0.9098 / 0.9084 (21.44)	3.88 / 9.36 / 22.32
4000	1	779.89	7.8038	6897	0.9550 / 0.9535	0.9238 / 0.9223 (21.45)	3.90 / 9.44 / 22.37
6000	1	816.20	7.8108	6892	0.9725 / 0.9709	0.9407 / 0.9392 (21.45)	3.93 / 9.52 / 22.42
7000	1	826.69	7.8131	6891	0.9781 / 0.9765	0.9461 / 0.9446 (21.45)	3.93 / 9.55 / 22.44
8000	1	837.95	7.8153	6889	0.9836 / 0.9820	0.9515 / 0.9499 (21.45)	3.94 / 9.58 / 22.45
1e4	1	855.53	7.8188	6886	0.9923 / 0.9907	0.9600 / 0.9584 (21.45)	3.95 / 9.62 / 22.48

The EW against He/H. The width-matched EW passes a minimum at He/H 3-4 (0.676), rises to a local maximum at 14 (0.821), falls to 0.786-0.789 at 40-80 and rises again above 100, about 0.10 per decade above 1000, to 0.9600 at 1e4: 0.148 (5.1 nominal sigma) below the observed EW. **No He/H between 2.09 and 1e4 reaches the observed EW on the GJ 1132 spectrum.** Two paths to the same He/H agree (10 from 5 and from 7: 0.8121 / 0.8106; 20 from 10 and from 14: 0.8137 / 0.8141). The lines keep one shape and position (10833.27 Å, 0.15 Å blueward of the observed peak); the width-matched peak grows to 1.107 per cent at 1e4 against the observed 1.28.

He 2³S loss as hydrogen disappears (the balance terms of phase5/dEW_decomposition/triplet_balance.py; P3 6.54e-6 s⁻¹ measured): Penning ionization by H I, 0.64 of the loss at 1.5-3 R_p at He/H 2.09, gives way to electron collisions between He/H 3 and 4 (0.03 at 100, 3e-4 at 1e4); electron collisions then take 0.97-1.00 inside 3 R_p and radiative decay A31 0.58-0.61 at 8 R_p; P3 stays below 0.05 and the H2 quench below 1e-4. The loss rate at 1.5 R_p falls only to 0.62 of its 2.09 value (the electron-collision floor) while n(He⁺) rises 1.8 times; n(2³S) there rises 2.7 times. Above He/H 100 the wind above 1.2 R_p is otherwise unchanged (T_{max} 6886-6915 K) but Mdot rises 15 per cent (log10 7.758 -> 7.819) with the warming base (T_{match} 429 -> 856 K: the conducted flux heats a column whose CH₄, NH₃ and H₂O scale with hydrogen), and the He⁺ front and the 2³S peak move down to the base (n₃ peak 263 cm⁻³ at 1.14 R_p at He/H 100, 2.6e3 at 1.03 R_p at 1e4). The dEW was not decomposed on the high side; the mechanism statements are interpretations of the balance terms.

Validity. Column: thin flag T up to He/H 5, F from 7 on (largest absorbed share 0.20 at 2.09, 0.004-0.006 at 1000-

1e4); exchange shift 0.36 → 0.02 K. Hydrogen-poor molecular network: the gated carrier rows (H₂, H⁺, He⁺, He⁺⁺, He 2³S) certify in the wind at every He/H (largest He⁺⁺, 9.6e-6 against 1e-5); match q_H2 0.193 → 5.0e-5 at 1e4. Climate and Photochem succeeded at every He/H (deep 20-bar temperature 561 K at 2.09, 252 K at 400; tropopause 1.0 → 9.3 bar). The share of the radial 2³S column inside the ungated base band $r < 1.2 R_p$ grows from 0.014 (2.09) to 0.231 (100), 0.439 (1000) and 0.598 (1e4); the worst ungated species row of those states is 0.4e-4 to 1.8e-4 (He⁺⁺ at 1.15 R_p), 4-18 times the wind's 1e-5. The rise above He/H 100 is therefore carried increasingly by cells the certificate does not gate (reported, not changed).

He/H 8000 refused once, by a hydrogen-specific step. Seeded from 4000, the k00 wind stopped in its first sweep: "(ioniz_eq) STOP: the base handoff partition of a lower ghost did not close against its own ionization balance" (30 passes, last move of the imposed $2 n(\text{H}_2)/n_{\text{H}} 5.10\text{e-}4$ against 1e-10, residual 3.45e-4, pin error 0; EXHALE exit 128). Budget and tolerance are compile-time parameters of `ionization_equilibrium.f90`, unchanged. The same pair closed in at most 4 passes (to 1e-13) at 4000, 6000, 7000, 8000 (seeded from 6000) and 1e4, so the refusal depends on the seed's radiation field at the base, not on He/H; kept in `HeH8000.0/attempt1_from_4000/`. The last good He/H is 1e4; nothing failed for good.

Not verified. The base band below 1.2 R_p (above); the region beyond $\text{Kn}_{\text{bulk}} = 0.1$ (3.8-5.7 R_p); a dEW decomposition of the minimum at 3-4 and the maximum at 14; why the 4000-seeded ghost at 8000 did not close.

102. The EUV that matches the observed EW, deep He/H 0.2 to 1e4; the exchange inside the warmed column solved as a transfer option (2026-10-05)

Directory `LHS1140b/models/.P2/ew_fit/route_xuv_scaled/` (NOTES section 7; `setup_lowheh_case.py`, `compare_euv_match_vs_heh.py`, `euv_match_vs_heh.txt`, `fig_euv_match_vs_heh.png`). User request 2026-10-04: lower He/H to 0.1 and find F_{EUV} matching the EW at each; and (user) the $s = 1$ route-A states at He/H 14, 100 and 1e4. Same construction and settings as sections 98-99 (GJ 1132 with $\lambda \leq 911.75 \text{ \AA}$ times s , bolometric held; binary `c37b3671`; `cooling_to_space`; energy coupling; tol 0.05; T_{match} tol 6 K; every solve on `lart3`). Seeds: the $s = 1.3$ state at the next He/H for the descent 0.65 → 0.45 → 0.3 → 0.2 → 0.1 (continuation in He/H at fixed s), the converged $s = 1$ states of `route_heh_high/` for 14, 100 and 1e4. The adapter climate start `--climate-t-deep-guess` 680-750 K at He/H 0.45-0.1 (the deep temperature rises as He/H falls; only a starting value). No tolerance changed.

Result (width-matched solution EW interpolated in $\ln s$ to 1.1084; 1 sigma interval from 1.0795 and 1.1373; every He/H has a converged state within 0.0058 of the observation except where noted):

deep He/H	s (1 sigma)	$F_{\text{EUV}}(100-911) [\text{erg cm}^{-2} \text{ s}^{-1}]$	nearest converged state: s , EW - obs
0.1	2.532 (2.43-2.64)	25.45 (24.42-26.50)	2.53, -0.0005
0.2	1.656 (1.60-1.71)	16.65 (16.07-17.23)	1.66, +0.0020
0.3	1.448 (1.40-1.49)	14.56 (14.09-15.02)	1.45, +0.0012
0.45	1.339 (1.30-1.38)	13.46 (13.07-13.86)	1.34, +0.0006
0.65	1.300 (1.26-1.34)	13.07 (12.71-13.43)	1.30, 0.0000
1.2	1.280 (1.25-1.31)	12.86 (12.55-13.18)	1.28, +0.0004

deep He/H	s (1 sigma)	F_EUV(100-911) [erg cm ⁻² s ⁻¹]	nearest converged state: s, EW - obs
2.09	1.361 (1.33-1.39)	13.68 (13.40-13.97)	1.36, -0.0013
5	1.336 (1.31-1.36)	13.43 (13.16-13.70)	1.34, +0.0045
14	1.271 (1.24-1.30)	12.77 (12.48-13.08)	1.27, -0.0008
100	1.331 (1.30-1.37)	13.38 (13.04-13.72)	1.33, -0.0008
1000	1.253 (1.22-1.29)	12.60 (12.29-12.92)	1.25, -0.0031
1e4	1.143 (1.11-1.17)	11.49 (11.20-11.79)	1.14, -0.0031

GJ 1132 has $F_{\text{EUV}} = 10.05$, GJ 699 70.18. The required F_{EUV} exceeds the GJ 1132 value at every He/H (by at least 14 per cent, at $1e4$), is nearly flat (12.8-13.7) over He/H 0.45-100, rises steeply below 0.45 (14.6 at 0.3, 16.7 at 0.2: the $s = 1$ EW falls below He/H 0.65, so the same EUV gives less line) and falls toward $1e4$ (11.5, where the $s = 1$ EW has turned up, section 101). Interpolated intervals carry the observed 1 sigma only. States at He/H 0.3 and 0.2 have the sonic point inside the domain (26.2-27.7 and 25.4-27.7 R_p), the exobase above it, $\text{Kn}_{\text{bulk}} = 0.1$ at 10.9-12.6 R_p .

He/H 0.1. The first attempt (cooling_to_space) stopped at k01: the exchange of the emission excess inside the warmed layer, neglected by that transfer, moves T_{match} by 1.003 K to first order against its bound of 1 K (0.89 K at He/H 0.2). The bound was not changed; the transfer was extended instead:

New option --column-band-transfer cooling_to_space_exchange (conducted_heat_column_balance.py, solve_with_included_exchange; default unchanged: thin). It is cooling_to_space with the exchange of the emission excess between all warmed nodes of the domain solved (fixed point on the exchange heating, the existing solve_with_exchange) instead of neglected, so it has no bound of that kind; the record carries included, the T_{match} shift from the exchange, heating/F and the first node. Validity as the module docstring states it (exact slab E_3 fractions for every g-point; for a two-level band weighted by eps_{beta} at both ends, exact for LTE bands). Tests (src/tests/conducted_heat_column, all pass): bit-identical to solve_with_exchange, shift = the history difference, shift ≥ 0 , budget closes to $8.5e-15$ (lte and nlte), an $\text{NH}_3 \times 5$ column's solved exchange is 1.38 times the first-order estimate that cooling_to_space measures (the audit measured 1.25 on the LHS 1140 b columns), the thin limit has no exchange. An $\text{NH}_3 \times 300$ column's fixed point settles slowly (30 iterations do not suffice: the increments fall like $n^{-1.5}$); the option's iteration cap is 100 and a non-settling solve raises, it does not return. The second attempt at He/H 0.1 with this transfer passed the column and then stopped in the model-top pinning (12 re-pinned solves left a monotone $-8.7e-5$ against the $1e-5$ test), so --top-pressure-passes was raised to 24 (the test unchanged), as for He/H 0.5 in section 90. Results at He/H 0.1 (cooling_to_space_exchange, top-pressure passes 24, $\text{HeH}0.1/\text{xuv}_x$): the EW (width matched, solution) is 0.7137 at $s = 1.3$, 0.9829 at 2.1, 1.0422 at 2.3, 1.0995 at 2.5 and 1.1079 at 2.53 (-0.0005 from the observation; closure converged at $k = 1-4$; sonic point 22.7 R_p , $\text{Kn}_{\text{bulk}} = 0.1$ at 16.4 R_p , exobase above the domain, red depth 1.279 per cent, red/blue 7.70). EW follows $s^{0.643}$ over 2.1-2.5. The match is $s = 2.532$ (1 sigma 2.43-2.64), $F_{\text{EUV}} = 25.4$, i.e. 2.5 times the GJ 1132 value. The $s = 2.7$ run was stopped by the operator during its first chemistry step to relieve the load of lart3 (its k00 log says so). $s = 2.6$ (closure converged at $k = 2$) gives 1.1273 (+0.0189), so the match $s = 2.532$ is now bracketed by two converged states (2.53 and 2.6); the upper 1-sigma end 2.636 lies just past the bracket.

He/H 1000. Run on lart4 from the $s = 1$ state of route_heh_high/: EW wm 0.8621 ($s = 1$), 1.1053 ($s = 1.25$, -0.0031), 1.1148 (1.26), 1.1590 (1.31); match $s = 1.253$, $F_{\text{EUV}} = 12.60$.

Spectrum figures. The overlays of figures 2 and 7 of docs/lhs1140b_exhale_vs_pwinds.tex show the model moved by the C26 fit $v_{\text{wind}} = +2.26$ km/s, as every earlier overlay of the memo does (make_memo_figures.py); the shift is for the overlay only and every EW, metric and χ^2 is computed from the unshifted spectra (checked: the numbers of phase6/observation/HeH0.65_k02_solution.json are unchanged). Figures 5 and 6 gain a right axis with F_{EUV} over the GJ 1132 value ($= s$).

Not verified: the column checks (thin flag, gas above p_{match}) at the states of this section; the grid error; the Roche-mode evaluate; the region beyond $\text{Kn} = 0.1$. The He/H 0.1 closure uses a different column transfer from the others (its T_{match} differs from the cooling_to_space value by about the exchange shift, 1 K).

103. The physics-and-algorithms document audited against the source (2026-10-05)

docs/EXHALE_physics_and_algorithms.tex and every docs/physics_overview/*.tex were checked statement by statement against src/ (and the src/utils/ scripts they describe) and brought to the state of sections 1-102. Corrections of what the document said the code does, each verified in the routine named:

- A Crank-Nicolson conduction stage that fails (floor, non-finite, pivot) stops the run (conduction_report, error stop 1); it is not retaken at a shorter step as the energy step is. The base conductive flux (eq:qbase), its bath at $r(0)$ (conduction_base_level_T, the profile's temperature where a profile is in use) and its EXHALE_resolved.out record are stated.
- The heating efficiency eta is formed but written to no output file.
- momentum_row_terms_of_cell is in RK_rhs.f90; sed_band_fluxes replaces a routine name that does not exist; the molecular rows no longer carry the retired R20; Penning Q31 is Movre & Meyer (1997) in the closed form of Garcia Munoz (2025); the sweep's XUV field is one block (xuv_field_block_cells = 0); the Ly-alpha default is jlya_mode = 0.
- The stationary residual's energy row carries the carrier and stage enthalpy divergence; the certification report holds 33 entries; the PTC start is $\text{dtau0} = 1$ on the marching handoff and the stationary restart, the CFL step only on the direct EXHALE_PTC=1 route.
- New text: carried ionization stages and He 2^3S transport (rows, fluxes, simplex, certification scales), the 10.2-30 eV partition of Furlanetto & Stoeve (2010) with its H2 gate and the (AJ) limitation, the elemental-flux closure and its acceptance rules, the escape boundary condition's hydrogen carrier split, the column energy balance under the conducted heat (two-way energy coupling), the restart mapper rules and verify_restart_radii, the element nucleus velocities of the _adv step, the stored-pair certificate, the width-matching velocity kernel (post-processing, Sec. 10.5), and the environment variables that select physics or continuation factors. Numbers quoted from earlier sections are labeled with their section; the $q_{31g}/(q_{31a}+q_{31b})$ ratios were re-measured (0.054 at $1e4$ K). Source comments found stale and left for their owners: the header of viscous_conduction.f90 section 5 and the conduction_report message ("B3a will reject and retry"), diffusive_photochemistry.f90 sections 2 and 6 (ion stages "not transported", "4x4 blocks"), ionization_stage_transport.f90 "RELATION TO THE

PROTON CARRIER", low_energy_electron_degradation.f90 (parameter name), species_table.f90 (f_sp columns 7..33), steady_newton.f90 set_transported_species_rows, the --column-band-transfer help text of photochem_to_lower_profile.py (thin is flagged, not refused), and EXHALE_transit.py lines 97-100 (bulk velocity). PDF rebuilt, 102 pages, 0 LaTeX errors, no undefined references.

104. The user documents audited against the source (2026-10-05)

docs/EXHALE_user_manual.tex, docs/transmission_spectrum.tex, README.md, README_HOWTO.md, md/input_schema.md, docs/lhs1140b_catalog_state.tex, docs/Update_EXHALE_appendix.tex and the workspace CLAUDE.md were checked against src/, the src/utills/ scripts and the LHS 1140 b EW tools for what sections 39-102 added or changed; every statement of a key, default, token, output field or option was read in the routine that implements it. Wrong before, corrected:

- The option-token count of the restart contract (24 in README.md, 23 in README_HOWTO.md, 20 in docs/composition_restart_and_base_handoff.tex): load_IC.f90 has 26 (opt_name); the vocabulary is now listed in the manual, the HOWTO and md/input_schema.md appendix D.2.
- The conductivity of Conduction was still the Watson et al. (1981) atomic-hydrogen value in README_HOWTO.md and md/input_schema.md K34b; it is the mixture value of section 41 (thermal_conductivity), the viscosity alone keeping Watson's (dynamic_viscosity).
- The _adv step was described on the bulk velocity (manual, transit section); it steps each element on its own nuclei velocity (section 82, post_process_adv.f90).
- EXHALE_PTC_DTAU0 also replaces the start 1.0 of the stationary restart (stationary_state_of_the_loaded_restart), not only that of EXHALE_PTC=1.
- README.md called carried ionization "H+ as a fifth carrier"; it is the three stages on their element's nucleus flux, in an atomic gas too.
- The README_HOWTO.md statement of the stationary evaluate (the measured state written back) and "the closure driver, which does not exist yet".
- The manual lacked examples 18 and 19, and docs/transmission_spectrum.tex had a wrong n2_populations signature and notebook path. Added: He 2³S transport, Interdiffusion enthalpy flux (with the carrier_enth / stage_enth tokens), H2 double ionization, H2 neutral dissociation in the schema; the EXHALE_resolved.out records (base_conduction_*, he23s_transport, low_energy_electron_partition); the cert_reason values (marching_snapshot_s<step>, pass_snapshot_p<k>, projected_seed, option_change_seed, legacy_factor_token, mapped_seed); EXHALE_PTC_SER with its two companions, the row-term, budget and evaluation dump keys; the column energy balance options of photochem_to_lower_profile.py, the profile header keys and which of them enter the solution_id; the energy coupling, acceptance and resume rules of element_flux_closure.py; the mapper's first-cell rule and BASE_ROW_RATIO_TOL; the width-matching kernel, the EW operator and the overlay-only v_wind shift (transmission document section "Comparison with an observed line profile", manual and README for every planet). The appendix was re-measured with codesize.py (core 236 files,

109,539 code lines, 31.2 times ATEs's 3,514); the catalog summary gained a section with the energy-coupled He/H and EUV scans (values read from the run files). Dated memos that stated a current state that is no longer true carry a one-line stale note: `kompot_comparison`, `koskinen2022_model_a_comparison`, `composition_restart_and_base_handoff`, `lower_atmosphere_coupling`.

Found in sources outside these files and left for their owners: section 87 above calls column 3 of `tpm_He10830.txt` the "instrument- and rotation-convolved" curve, and so does the step-1 comment of `LHS1140b/he10830_equivalent_width.py`; `EXHALE_transit.py` writes column 3 as `T_instr` (instrument only, no planet rotation) and column 4 as `T_rot+instr`, so the EW and the kernel act on the curve without rotation. Also stale: the `cond_on` comment of `parameters.f90` (Watson conductivity), the docstring of `element_flux_closure.py` (convergence stated without the conducted-flux residual the code holds to `tol`), the `--column-band-transfer` help text (thin is flagged, not refused), the H2 neutral dissociation comment of `input_read.f90` (33-41 eV; the ramp is 32-41.5 eV), and the "no element drift" header comment of `EXHALE_transit.py` and the `transit_state_files` docstring. Builds: the manual (125 pages), the transmission document (15), the catalog summary (20), the stage 3 log with the appendix and the four noted memos, all without LaTeX errors; the changed pages were rendered and inspected.

105. The lower-ghost chemistry of LHS 1140 b at He/H 2e4: the reservoir H2 partition solved with its own ionization, and the molecular rows normalized by the turnover of their element (2026-10-06)

Request of 2026-10-06: the He/H 2e4 rung of the LHS 1140 b He/H ascent at the fixed XUV scale $s = 1.14$ (GJ 1132 SED) stopped in its first closure iteration (`LHS1140b/models/.P2/ew_fit/route_xuv_scaled/HeH20000/xuv_x1p14/k00`, binary `EXHALE_c37b3671.x`) with `ioniz_eq`: persistent non-root chemical equilibrium: the two lower ghost cells (0 and -1, 900 K, $n_{\text{H}} 4.07e8$, $n_{\text{He}} 8.13e12 \text{ cm}^{-3}$) were non-roots from the first sweep, 2000 NON-ROOT lines in `run.log`, the stop after 1000 consecutive sweeps, while the interior JFNK reached $\|R\| 5e-8$. He/H 1e4 had none. Two defects, both in the code, each measured before it was fixed. Scratch runs: `~/claude/jobs/f24ea156/tmp/heh2e4_diag/`, `lart4`. All numbers MEASURED unless marked READ.

105.1 The reservoir H2 partition was pinned at the ionization of the previous state

The base handoff states $x_2 = 2q(1 + \text{He/H})/(1 + q)$ of the NON-IONIZED hydrogen (`base_h2_nuclei_fraction`), so the ghost owes $x_{\text{H2}} = x_2 (1 - x_{\text{ion}})$, x_{ion} the hydrogen in H^+ , H_2^+ , H_3^+ and HeH^+ . The sweep pinned row 4 at $x_2 (1 - x_{\text{ion}})$ with the x_{ion} of the state it was HANDED and closed the pair by passes with a secant step. A pinned $x(4)$ leaves $1 - x(4)$ of the hydrogen for H I and every ion together, and where the ionization balance asks for more ions than that, no composition satisfies the rows. At He/H 2e4 the H^+ of the ghost is made by $\text{He}^+ + \text{H}_2 \rightarrow \text{H}^+ + \text{H} + \text{He}$ at the rate the helium is photoionized, which doubles against the hydrogen from He/H 1e4 to 2e4. At the first sweep the pin was $x(4) = 0.9698$ (room 0.0302); the root of the same rows with the partition solved together with the ionization has $x_{\text{ion}} = 0.03136$ (test below). The only root `hybrd1` found with the pin (attempt 1, info 1, residual $2e-16$) had $x(\text{H}^+) + x(\text{H}_2) = 1.0011$, negative H I, and $\text{O}^+/\text{O} = 1.044$ (element violation $4.4e-2$); every bounded iterate of the three attempts

was a non-root (residual $1e-4$ to $5e-3$), and the retained non-root (x_{ion} $3.6e-4$) fed the next pin $x(4) = 0.991$, which left less room still.

Fix (`ion_cell_state.f90`, `System_HeH_mol.f90`, `System_HeH_mol_metals.f90`, `ionization_equilibrium.f90`, `constrained_chemical_equilibrium.f90`): the row IS the equation, $fvec(4) = x(4) - x_2 (1 - x_{ion}(x))$, with x_{ion} an unknown of the same solve (`x_h2_neutral_partition_fixed`, `h2_fraction_of_reservoir`, `ionized_hydrogen_nuclei_fraction`), in the fraction systems and in the constrained continuation, where H2 is now an unknown with that row instead of a held density. The neutral hydrogen left, $(1 - x_2)(1 - x_{ion})$, is never negative. `impose_transport_fractions` places a starting point and a returned state on the row; the closure passes, the secant and the infeasibility stop are removed (the closure check on the returned state stays). With this alone the ghosts reached roots, but the JFNK refused 275 trial states for ghosts without a root (worst residual $5e-5$, just above the $1e-6$ tolerance) and stopped with no descent at $\|R\| = 1.9$, which is the second defect.

105.2 The rows were scaled by a bound that places trace ions at the whole of their element

The molecular rows were divided by the sum of the magnitudes their terms reach with every species at the whole of its element (`set_mol_turnover_rates`). Where a trace ion of one element reacts with another element, the bound exceeds the turnover by the inverse of the ion's fraction. The H+ row carries $He^+ + H_2$ (R17) and $He(2^3S) + H$ (Q31) with He^+ and the metastable at all of the helium: at the ghost root the bound was $3.9e12 \text{ cm}^{-3} \text{ s}^{-1}$ against a gross H+ rate P + L of $1.3e3$, so the row reached the solve and the acceptance test at $3e-10$ of its weight. The Jacobian of the scaled system, its columns times $|x|$, had its singular value along H+ at $9.8e-11$; from the converged ghost at one JFNK perturbation the Newton step asked for 12 per cent of H+, its full length raised the residual from $1.3e-5$ to 1.0, and `hybrd1` returned info 4 at the starting point. Ten per cent of H+ off the root moved the scaled H+ row by $2.7e-11$: the tolerance $1e-6$ admitted an H+ imbalance of $3e3$ times its gross rate. He^+ , H_2^+ , H_3^+ and HeH^+ carry the same construction.

Fix: every row of the molecular system is divided by the CHEMICAL TURNOVER OF ITS ELEMENT, $G_E = \text{sum over the rows of element E of } (P_j + L_j)$ (hydrogen: H+, H2, H2+, H3+, HeH+; helium: He+, He++, He 2^3S ; each metal its two stage rows, oxygen with OH and H2O). The gross rates come from the row routines themselves (`mol_heh_rows`, `oxygen_carrier_rows`, `metal_rows`, `cx_add_to_fvec`, `he_h_cx_fvec`, optional argument `gross`); `element_turnover_reciprocal` and `scale_molecular_rows` (`System_HeH_mol.f90`) and `set_element_turnover_scales` (`System_HeH_mol_metals.f90`) apply them. The scale is taken at the starting point of each solve and HELD through it, and a candidate is judged with the scale taken at the candidate (`normalized_reaction_residual`, which no longer widens the molecular rows by a separate charge-exchange bound: the gross rates contain it). A row with no reaction keeps the old bound (1 for an identity row). The zeros of the rows are unchanged. Two variants were measured and rejected: (i) each row by its own P + L asks a species at $1e-22$ of its element (H_3^+ in the outer wind) to balance to $1e-6$ of itself, and 350 cells of the He/H $2e4$ run were left without a root (`hybrd1` info 1 with H2+, H3+, HeH+ rows at $1e-6$ to $5e-1$); (ii) the element turnover re-taken at every iterate bounds every row by one and leaves no gradient back from an iterate far outside the simplex: on the cold start of the `mol_metals` regression gate 214 cells were left without a root, attempts went out to fractions of $2e5$, and the ghost fixed point stopped the run. The constrained continuation keeps the bound on its own path; its candidate is judged by

the same measure as every other. The rate matrix of the decay-rate diagnostic now asks the systems for their rows as volume rates (`mol_rows_as_volume_rates`) instead of undoing a scale.

Also fixed: `cce_probe.x` stopped at once (`keq_H_H_to_H2` called before `h2_thermochemistry_init`); the driver now builds the table first. The two charge-exchange reservoir tests that undid the scale by setting `mol_inv_turnover` = 1 ask for volume rates instead. The cell-state hash of `attempted_step.f90` covers the two new fields. Documents: the normalization and the reservoir row are written into `docs/physics_overview/ionization.tex` and `molecules_metals.tex`; `docs/EXHALE_physics_and_algorithms.pdf` rebuilt.

105.3 Tests

New suite `src/tests/ghost_reservoir_chemistry/` (fixture: `ghost 0` of the failing run at its first sweep, from its `EXHALE_CCE_DUMP`): the reservoir row is $x(4) - x_2(1 - x_{\text{ion}})$ to the bit; from the state the earlier code retained, `hybrd1` reaches a root inside every element budget at $4.8\text{e-}10$ (tolerance $1\text{e-}6$); that root has $x_{\text{ion}} = 0.03136$ against the 0.03020 the pin left; ten per cent of H^+ off the root moves the H^+ row by $5.4\text{e-}2$ (bound $3.9\text{e}12$, gross $1.3\text{e}3$, $G_{\text{H}} 1.87\text{e}3 \text{ cm}^{-3} \text{ s}^{-1}$); the hydrogen rows share one divisor; the continuation solves H_2 as an unknown, its layout is square, and its H_2 row is the reservoir row. GREEN: 9 of 9. RED, the same assertions on the source of `EXHALE_c37b3671.x` with the pin at 0.9698 : `hybrd1` info 4, residual $2.8\text{e-}3$, outside the simplex, $x_{\text{ion}} 2.9\text{e-}3$, and the H^+ row $2.7\text{e-}11$ at ten per cent of H^+ : 4 FAIL. Suites run with the changed objects: `ghost_reservoir_chemistry` 9/9 (6/6 before the continuation rows were added), `charge_exchange_rows` 257/257 (22 FAIL before the two tests asked for volume rates), `ionization_imposed_fractions` 48/48, `constrained_network_layout` 8/8, `stage_row_balance` 10/10, `charge_exchange_detailed_balance` 298/298, `physics_probe` 2499/2499, `acceptance_classes` 22/22, `boundary_state` 43/43, `molecular_seed` 29/29, `attempted_step` 70/70, `carrier_returned_state_acceptance` 36/36, `carrier_retry` 156/156, `helium_metastable_transport` 24/24, `carrier_constraint_attribution` 10/10, `carrier_reference_scales` 16/16, `carrier_helium_inventory` 15/15, `carrier_boundary_jacobian` 12/12, and `grid_and_gates` 424 PASS, 3 FAIL: `stationary_evaluate_rebuilt_within_sweep_budget` ($5.832\text{e-}5$), `stationary_evaluate_certification_unchanged` and `stationary_evaluate_exit_status` (2) on the `wasp_full_newton` fixture, the same three with the same numbers from the source of `EXHALE_c37b3671.x`, so not of this change (an atomic fixture no longer certified by the current code; reported, not touched).

105.4 The actual cases

New frozen binary `LHS1140b/models/.P2/EXHALE_f6945afb.x` (ifx 2026.1.0, md5 `f6945afb87281affb90523b0fbb3823d`, manifest `.P2/BINARY_MANIFEST_f6945afb8728.txt`, source copy `.P2/build_ghostres_ifx/`). The He/H $2\text{e}4 \text{ k}00$ EXHALE stage rerun in scratch with the closure's environment (`closure.json exhale_env`, 12 threads): 0 NON-ROOT lines, 0 constrained-continuation rescues, 0 refused trials, 216 JFNK iterations, the stationary solve returned `info = 0`, CERTIFIED in the wind, 2.3 minutes. The old binary on the same input: 128 NON-ROOT lines in its first 90 s. The converged `ghost 0`: $x(\text{H}_2) = 0.9394$, $x(\text{H}^+) = 0.0534$, $n_{\text{e}} 2.51\text{e}7 \text{ cm}^{-3}$, $\text{He}^{++}/\text{He} 3.0\text{e-}14$ (the retained non-root had $\text{He}^{++}/\text{He} 2.6\text{e-}7 > \text{He}^+/\text{He} 4.2\text{e-}8$). $\log_{10} \text{Mdot}$ from the written profile, median over $r > 1.2 R_{\text{p}}$: 7.883 ($\text{He}/\text{H} 1\text{e}4 \text{ k}01$: 7.875).

He/H 1e4 (HeH10000/xuv_x1p14/k01 input and IC), EXHALE_c37b3671.x against EXHALE_f6945afb.x, as written (stationary evaluate) and as a stationary solve (both converge in one JFNK iteration, info = 0, no NON-ROOT): the ghost compositions of every evaluation (EXHALE_GHOST_RECORD) agree to 1e-14 relative in n_e and every fraction, Hydro_ioniz.txt to 1.9e-14 and Ion_species.txt to 8.2e-13. The accepted reaction residual of the ghosts reads 1.4e-17 - 3.8e-17 under the old normalization and 1.6e-15 - 7.2e-15 under the new one, the same roots measured in the new units.

Scoped impact on two regression cases (not the matrix, no golden touched), gfortran builds of the earlier and the new source, 12000 steps, single-threaded: lower_profile bit-identical in Hydro_ioniz, Ion_species and both _adv files; mol_metals (hot-Uranus gate with trace metals, molecular) within 7.0e-8 (Hydro_ioniz.txt), 1.1e-7 (Ion_species.txt) and 3.1e-6 (Hydro_ioniz_adv.txt) relative, log10 Mdot 10.58 in both, no non-root in either, and the 13 constrained-continuation rescues of the earlier normalization are 0. Marching speed of mol_metals, measured over the same four minutes at different steps: 115 (new) against 114 (earlier) steps per minute.

Open: the atomic systems' acceptance scale (normalized_reaction_residual, atomic branch) carries the same construction in its H+ row, the He <-> H charge exchange with He+ at all of the helium, (k1 + k2) n_H n_He; it is not changed here (the atomic systems are not solved at He/H 2e4 in this route).

106. A failed conduction stage is a refused and retaken step; the handoff-free base-level conflict warns; stale source comments corrected (2026-10-06)

User decisions of 2026-10-05 (items A and B) and the source comments the two documentation audits of sections 103 and 104 left for their owners (item C). Scratch runs: ~/.claude/jobs/f24ea156/tmp/condretry/, lart4. Numbers MEASURED unless marked READ.

106.1 A failed Crank-Nicolson stage refuses the attempted step

Before: conduction_report (viscous_conduction.f90) ended a failed stage (floor, non-finite input or solution, pivot breakdown) with error stop 1; the header (section 5) and the message said "B3a will reject and retry", which the marching loop never did. Now the stage prints its diagnostics, leaves its verdict in conduction_last_status and the status argument and returns no state; EXHALE_main passes status, and on a failure refuses the attempt with as_reject_conduction at as_op_conduction and leaves the trial, exactly as a failed energy update does (both run modes). The controller then restores the checkpoint and retakes the step at half dt (attempted_step_reduced_dt) within its budget of n_step_retry_max = 8 halvings; at dt/2^8 stop_on_exhausted_retry_budget stops the run with exit status 2 and the retry history. The refusal text names the stage's reason and cell. conduction_stop_on_failure is removed (only the unit driver set it). Why a shorter step is the remedy (header section 5): a mode of the conduction operator with decay rate lambda/dt is multiplied by (1 - lambda/2)/(1 + lambda/2), negative for lambda > 2, so a step longer than twice the diffusion time of a cell overshoots; halving dt halves lambda. The two remaining error stop statements of the module are not stage failures and stay: an EXHALE_CONDUCTION_SCALE outside [0, 1] (input) and h2_conductivity asked for at a non-positive or non-finite temperature (an argument guard also reached from the stationary residual; the stage's entry test covers every cell its stencil reads).

Test hook, default off: EXHALE_CONDUCTION_REFUSE_DT_ABOVE = t [s] makes the stage return the new status CONDUCTION_INJECTED (reason CONDUCTION_REASON_INJECTED) for every attempt whose longest cell interval exceeds t , and print that interval. New suite `src/tests/conduction_retry` (hot-Uranus gate `mol_carrier` with Conduction: True, 1 or 2 steps, 40 s), registered in `executable_identity`: (1) $t = 1e-300$: 9 refusals at the conduction step, every interval exactly half the one before (worst $\ln \text{ratio} - 1/2 = 0$), the exhaustion report naming the stage, exit 2; (2) $t = 0.3$ of the first interval (1.684 s): two refusals, the third attempt (1/4) accepted, exit 0; (3) $t = 1e300$: no refusal, `Hydro_ioniz.txt` and `Ion_species.txt` byte-identical to the run without the hook. 20 PASS with the tree build; with the control build (the tree before these edits) the rows of (1) and (2) FAIL (no refusal, exit 0). Unit level: `energy_update` `conduction_floor_tests` no longer sets the flag; with the new objects all 69 rows pass, with the control objects the floor case ends in ERROR STOP 1.

Byte identity where no stage fails: `mol_carrier + Conduction: True`, 300 steps, single thread, control build against the tree build: `Hydro_ioniz`, `Ion_species`, both `_adv` files, the heating and cooling breakdowns and the body of `conserved_state.txt` identical (its PAIR_CHECKSUM covers the provenance line). `attempted_step` with its whole-binary rows: 160 PASS, 4 FAIL with both builds, the same rows and values (`exhaustion_on_the_integration_error_*`, `temporal_order_of_the_transported_species_fractions` 1.9978/1.4453, `temporal_order_of_the_electron_fraction_and_T` 3.4109), so they predate this change; not investigated here.

Docs: `hydrodynamics.tex` (the stage paragraph), `numerics.tex` (the refused step statement, now naming the energy and conduction refusals in both modes), `src/tests/energy_update/README.md`, the EXHALE_main comment on the floor counters.

106.2 Base BC: pressure beside a density key without a handoff

`input_read.f90`: with no `base.inp` `p_base` and no profile, Base BC: pressure beside `Log10` lower boundary number density is still followed, now with a WARNING naming both levels (`n0` and pressure of each) and the one used; the overridden `n0` is kept in `base_density_key_overridden_n0` and `write_setup_report.f90` repeats the warning under "Base level". MEASURED on the LHS 1140 b atomic input of `.P1/fiducial_conduction/ew_fid_atomic_nocond`, which carries both keys: density key $n0 = 3.2063e13 \text{ cm}^{-3}$ ($1.0004e-6$ bar) against the pressure's $3.2049e13$ ($1.0000e-6$ bar); outputs of a 1-step run byte-identical to the control build, which printed no warning. Docs: `introduction.tex` (base-level paragraph), the manual's Base BC: entry, `md/input_schema.md` rows 2 and K32.

106.3 Stale source comments (each checked in the code first)

- `diffusive_photochemistry.f90` header: the transported set is the run's (`carrier_set_init`, up to 8 rows: H2, OH, H2O, CO; H+, He+, He++, He 2³S); the carried stages are charged and their ambipolar drift is neglected, not absent; blocks are 8x8 (`n_carrier_max`); a handoff's base H2 partition is imposed in the ghosts (section 5 said nothing was).
- `ionization_stage_transport.f90`: the carrier stage rows ARE formed from its equation (1) (`carrier_stage_face_flux`).

- `low_energy_electron_degradation.f90`: the switch is `use_low_energy_partition`.
- `species_table.f90`: metal columns 7..33.
- `util_ion_eq.f90` `HeITR_coeffs`: the $n \geq 3$ singlet share is 0.019 of the direct $q_{31a} + q_{31b}$ at $1e4$ K and 0.079 at $2e4$ K, evaluated from the coded fits (the comment said 0.023 and 0.09); the cascade share of q_{13} quoted there (0.10 and 0.33) was checked: 0.107 and 0.331.
- `ion_residual_core.f90` `heh_tr_rows`: the rows carry the electron-impact ionization of H I, He I (both levels) and He II.
- `steady_newton.f90` `set_transported_species_rows`: the rows are the elements, then every solved carrier (stages and He 2^3S included), carriers only in a molecular run with carrier transport.
- `photochem_to_lower_profile.py`: thin is flagged past 10 per cent, not refused.
- `parameters.f90` `cond_on`: the mixture conductivity.
- `element_flux_closure.py` docstring: convergence with energy coupling also holds the conducted-flux residual to `tol`.
- The H2 neutral dissociation window is 32-41.5 eV (`input_read.f90` comment and its log line, `parameters.f90`, `util_ion_eq.f90`, `h2_photo_channels.f90` header).
- `EXHALE_transit.py`, `exhale_transit_lib.py`: the `_adv` step marches each element on its own nuclei velocity (section 82); the J_{lya} estimate of Huang et al. (2017) is the order-of-magnitude estimate in the text after their Eq. (6), Section 2, page 3 (READ in the published PDF, [references/Huang_2017_ApJ_851_150.pdf](#); Eq. 6 is the $n(2l)/n(1s)$ ratio), and the recombination assumption is on page 3, not 11.
- `lya_rt.f90`, `excited_hydrogen.f90`: the $1/(1 + \tau)$ attenuation and the column-integrated F_{LyC} are EXHALE's closures, not the paper's. Also `README_HOWTO.md`: 54 suites (adds `conduction_retry` and `ghost_reservoir_chemistry`).

106.4 The atomic acceptance scale (measured, not changed)

`normalized_reaction_residual`, atomic branch, on the certified LHS 1140 b atomic state `ew_fid_atomic_nocond` (He/H 9.7, He 2^3S on), rows recorded with `EXHALE_STAGE_CHANNELS` at eight cells: the H+ row scale is $3.9e12$ (cell 2) to $2.1e2$ (cell 495, 26 R_p) times the gross H+ rate, and the term that sets it is the Penning bound $f_{Q31} n_{\text{He}} n_{\text{H}}$ (He 2^3S at all of the helium), 0.99 to 1.00 of the scale; the He $\leftrightarrow$ H charge exchange term that section 105 named is $1.6e-6$ to $2.4e-6$ of it. The He 2^3S row scale is $1.4e14$ to $3.9e3$ times its gross rate. With `ieq_res_tol = 1e-6` the test cannot refuse a non-root in these rows inside about 1.35 R_p . The roots on this state are tight (row residual $1e-16$ of the gross rate). Recorded as `md/TO_BE_DONE.md` (AM).

Build: bare make, tree EXHALE.x md5 3042c8726291e0b02d796c3325fa95cf. Documents rebuilt without LaTeX errors: the physics document (103 pages; the changed pages 4, 12 and 62 inspected) and the manual (125).

107. The helium carry of `map_state_to_grid.py` `--reservoir` holds the hydrogen-nucleus count of every row (2026-10-06)

The elemental-flux closure of the LHS 1140 b He/H ascent stopped at the He/H $7e4$ rung: the seed `route_xuv_scaled/HeH40000/xuv_x1p14/k01/output` (He/H $3.9974e4$) carried onto He/H 69936.32, C/H $2.758973e-4$, N/H $8.133062e-5$, O/H $6.023026e-4$ was refused, "C/H ... $2.758969E-04$... instead of $2.758973E-04$ (relative departure $1.52E-06$, above the tolerance $1E-06$)".

107.1 Cause

Not the file precision: the mapper writes every column at $\%24.16E$ and the check runs on its own float64 arrays before writing. The departure was a real inconsistency of the helium carry. `--reservoir` He/H multiplies every helium column by $k = \text{new/old}$, HeH+ included (`element_stage_columns`), and HeH+ holds one hydrogen nucleus, so every row gained $(k - 1) n(\text{HeH+})$ hydrogen nuclei and every El/H of the row moved by $-(k - 1) n(\text{HeH+})/n_H$: the metals that the same call carried by their own factor new/old, and He/H itself. The seed holds HeH+/H = $2.024e-6$ at cell 1 ($6.0e-7$ and $6.2e-7$ in the two lower ghosts; MEASURED from its `Ion_species.txt`), and the departure followed $(k - 1) \times 2.024e-6$ exactly (MEASURED with the entry mapper):

target He/H	k - 1	departure (C/H, cell 1)
4.5e4	0.126	passed
5e4	0.251	passed
6e4	0.501	$1.01e-6$
69936.32	0.750	$1.52e-6$
1e5	1.502	$3.04e-6$

Why the $2e4$ and $4e4$ rungs passed was not measured (their seeds' HeH+/H at cell 1 was not read).

107.2 Fix (`src/utils/map_state_to_grid.py`)

The hydrogen-nucleus count of every row is held. After the columns of an element are multiplied, the $(k - 1)$ Hs hydrogen nuclei they brought (Hs the hydrogen nuclei of the multiplied columns: HeH+ for helium, none for a metal) are taken out of the species holding only hydrogen (H I, H II, H2, H2+, H3+), all multiplied by one factor $g = 1 - (k - 1) \text{Hs}/P$, P their hydrogen nuclei, which keeps their ionization and molecular split; their mass (1 per hydrogen nucleus, `bsp_mass`) and particles plus electrons (H I 1, H II 2, H2 1, H2+ 2, H3+ 2) enter the density and the temperature written at fixed pressure. Every row then carries k times its source He/H and its source ratio of every other element exactly, in whatever order the elements are named. The `--uniform` row factor becomes $(\text{value } n_H - X)/S$ with n_H held. A HeH+ carry that the hydrogen-only species cannot supply ($g \leq 0$) is refused. The base-row test and its tolerance are unchanged (`BASE_ROW_RATIO_TOL` = $1e-6$, `heh_dev_tol` of `load_IC.f90`); it now catches only the oxygen carriers OH, H2O and CO, which hold nuclei of two elements and are not multiplied. Before this, a HeH+-bearing state was refused by design ("no single factor sets both counts"); it is now carried.

107.3 Tests (src/tests/state_mapper/)

The two assertions that a HeH+ state is refused are replaced by the carry with n_H held (HeH+/H 0.05 and 0.3: n_H equal to the control mapping to $4e-16$, HeH+ multiplied by k , base rows at the new He/H to $2e-16$, the diffused profile shape and the H II/H I ratio kept, $T = p/((n_{\text{tot}} + n_e) k_B)$ and ρ the species mass sum to $3e-15$), a refusal when the hydrogen-only species cannot supply the hydrogen (HeH+/H 0.3 carried by 6/1.6), the `--uniform` carry with HeH+, and a synthetic LHS 1140 b state (He/H 3.9974e4, HeH+/H 2.024e-6, C/H, O/H) carried to He/H 69936.32 and $1e5$ together with C/H and O/H. RED with the entry mapper: 5 failures (the two LHS carries refused at $1.52e-6$ and $3.04e-6$, the two HeH+ carries refused, n_H not held under `--uniform`, 0.29). GREEN after: 131 PASS, 0 failures. The suites that call the mapper without `--reservoir` (boundary_state, coupled_block_jacobian, grid_and_gates) were not run: the change lies inside the `--reservoir` branch only, and the suite's own no-option byte-identity assertions pass.

107.4 The real seeds (MEASURED)

The closure's command (the seed output of HeH40000/xuv_x1p14/k01 and of k00, onto the target grid header of HeH70000/xuv_x1p14/k00, C/H, N/H, O/H as in the closure log) for He/H 69936.32 and $1e5$ now writes the pair; recomputed from the written Ion_species_IC.txt the first-cell departures are, for k01: C/H $1.3e-14$, N/H $5e-15$, O/H $5e-15$, He/H $8.6e-13$ (base rows); for k00: C/H $1.9e-14$, N/H $2.7e-14$, O/H $1.7e-14$, He/H $3.2e-12$; the same at $1e5$. The frozen binary LHS1140b/models/.P2/EXHALE_f6945afb.x, run in scratch on the He/H $7e4$ pairs of both seeds with the rung's input_template.inp and the k00 lower_atmosphere_profile.dat, loads them: no restart-contract or He/H refusal, "base cells set to the input He/H = 6.9936E+04", C/H, O/H and N/H "column rescaled by 1.000000E+00". The $1e5$ pair was not loaded: no profile at He/H $1e5$ exists, and a copy of the $7e4$ profile with X_{He} multiplied is refused by input_read ("the base H2 fraction is not attainable at this He/H").

Noticed, not changed: input_read.f90:3159 and :3309 print He/H with F8.5, which shows ***** at He/H $\geq 1e3$ (the LHS 1140 b ascent) and q_H2(base) $\rightarrow 0.00001$ for $7.1e-6$.

Documents: docs/physics_overview/boundaries_initial.tex (the mapper paragraph) and docs/EXHALE_user_manual.tex (the mapper paragraph and the closure paragraph) state the held n_H ; rebuilt without LaTeX errors, the physics document (103 pages; page 74 inspected) and the manual (125; pages 10 and 83 inspected).

108. The EUV that matches the observed EW reaches He/H $1e5$ (2026-10-06)

The He/H ascent of section 102 above $1e4$ at fixed $s = 1.14$ (the s that matched at $1e4$), each rung seeded from the converged one: $2e4$ and $4e4$ converged at $k = 1$, $7e4$ and $1e5$ at $k = 2$ (max residual 0.0095, 0.0125, 0.0071, 0.0045; no non-root cell). It could be run only after two code defects were removed: the lower ghost chemistry at $2e4$ (section 105) and the mapper's helium carry at $7e4$ (section 107); the closures use the binary EXHALE_f6945afb.x. The width-matched solution EW at $s = 1.14$ is 1.1285 ($2e4$), 1.1607 ($4e4$), 1.2083 ($7e4$), 1.2425 ($1e5$) against 1.1084 observed: the $s = 1$ EW has turned up above $1e4$, so the matching s falls. At He/H $1e5$, $s = 1.06$ gives 1.1575, $s = 1.03$ gives 1.1247 (EW proportional to $s^{1.0}$ here) and the crossing at 1.015 was confirmed by a converged state: $s = 1.015$ gives 1.1082 (-0.0002; closure $k = 1$, residual 0.0108). $F_{\text{EUV}}(100-911) = 10.20$ (1.5 per cent above the

unscaled GJ 1132 value 10.05); the 1-sigma interval of s is 0.989-1.042 (lower end extrapolated). The $\text{He}/\text{H} = 0.1$ crossing was confirmed too: $s = 2.53$ gives 1.1079 (-0.0005) and $s = 2.6$ gives 1.1273, so the table has no extrapolated point left ($\text{He}/\text{H} = 1e3$ at $s = 1.25$: 1.1053). Figure 6 of docs/lhs1140b_exhale_vs_pwinds.tex now has thirteen He/H , 0.1 to $1e5$. The EW of states above $\text{He}/\text{H} = 1e4$ was evaluated with f6945afb (measure_case_f6945afb.sh, run_ew_f6945afb.sh, copies of the control scripts with the binary name changed). Record: NOTES.md section 7 of LHS1140b/models/.P2/ew_fit/route_xuv_scaled/.

Code-size summary: EXHALE vs. the original ATES (2026-06-07; updated 2026-10-05)

A line-by-line comparison of the Fortran source against the pristine upstream code (ATES/ATES-Code-main) quantifies the extent of the in-house extension. *Pure code lines (SLOC)* exclude blank lines and comments (the comment stripper is string-aware, so a `!` inside a quoted string is not treated as a comment); the line-by-line diff additionally normalizes whitespace and case, so reformatting and comment-only edits are *not* counted as changes. The measurement script is src/utils/codesize.py, which reprints the totals, the diff counts, every row of the two module tables and the byte-identical list of Table 55; the split of the core into the simulation build and the test drivers and the figures of the ten heaviest modified files are formed with its own counting functions. All numbers of this appendix are MEASURED on the working tree of 2026-10-05, after section 102 of the stage 3 log (md/Update_EXHALE_stage3.md). Up to 2026-09-22 (the previous edition, after section 16 of the stage 2 log) the extension comprised the acceptance and certification contracts, the attempted-step controller, the partitioned stationary solver, the oxygen and carbon chemistry, the molecular infrared bands, the ionization-stage transport and the charge-exchange rows, the base contact boundary and its trace, the molecular seed taken from an atomic state, the transition into the coupled carrier solve, the discrete conservation ledger, and the test suites those series brought with them. Stage 3 added the atomic-data audit and its repairs, the exact code-unit restart file (conserved_state_restart.f90), heat conduction with the conductivity of the mixture, the interdiffusion enthalpy flux, the enthalpy flux of the carried ionization stages and molecular carriers, the 10.2–30 eV photoelectron partition of Furlanetto & Stoevers (2010), the $\text{He } 2^3\text{S}$ transport, the advection-corrected systems stepped on the velocity of each element's nuclei, and 25 new test drivers.

What the script counts as *core*. Every .f90 file under src/ except those in src/modules/wind_ae/, which is the ported Wind-AE IC generator and is listed separately. That rule puts the standalone drivers under src/tests/ in the core figure, and they are now the larger part of it by file count, so the table splits the core row into the simulation build and those drivers. Of the 105 files outside src/tests/, 104 are in the SRC list of the Makefile and are compiled into EXHALE.x; the one exception is nonlinear_system_solver/constrained_equilibrium_probe.f90, which is built on its own. Each test suite likewise builds its own driver from src/tests/<suite>/run.sh.

	Fortran files	code lines (SLOC)
Original ATES (ATES-Code-main)	47	3 514
EXHALE core (excl. Wind-AE port)	236	109 539
of which the simulation build	105	70 592 ($\approx 20.1\times$)
of which the src/tests/ drivers	131	38 947
net change (core)	+189	+106 025 ($\approx 31.2\times$)
Wind-AE IC-generator port (wind_ae/)	+36	+4 756
EXHALE total	272	114 295 ($\approx 32.5\times$)

Line-by-line diff, core (comments + blanks excluded; whitespace/case normalized): **107 432** code lines added in EXHALE (new modules + the new side of modified lines) and **1 407** removed from the original (deleted + the old side of modified lines), for **108 839** total changed lines. At the file level, of the 47 original files **38 were modified**, **8 are unchanged** (all eight byte-for-byte; the list is Table 55) and **one was removed**: flux/speed_estimate_HLLC.f90, Toro's adaptive p_* estimate, was used by Num_Fluxes.f90 and never called, because the HLLC branch builds its own Davis/Einfeldt speeds inline. **190 new core modules** were added. Of those, 59 are part of the simulation build:

New module	Purpose	SLOC
time_step/steady_newton.f90	stationary solve of the hydrodynamic rows: residual $F(Y) = 0$ on the physical cells, Newton/Krylov with pseudo-transient continuation, line search, best-iterate ledger	11122
lower_atmosphere/diffusive_photochemistry.f90	vertical transport of the molecular carriers (H_2 , OH, H_2O , CO) and of the transported ionization stages (H^+ , He II, He III) and He 2^3S level, solved together with the chemistry that makes and destroys them, with the enthalpy fluxes they carry	4587
functions/binary_element_diffusion.f90	conservative binary H/He element transport (stage-resolved pair coefficients, donor-cell drift, metals slaved to H) and the interdiffusion enthalpy flux it carries	2610
lower_atmosphere/molecular_infrared_data.f90	band and line data of the molecular infrared coolant (generated by cooling_data/molecular_infrared_bands.py)	2182
time_step/certification.f90	one evaluator of the equations a run actually solves, and the contexts that read its verdict	1991
time_step/attempted_step.f90	the attempted-step controller: one checkpoint, one trial, one adoption boundary, and the rollback contract	1630
radiation/low_energy_electron_tables.f90	the electronic tables of Furlanetto & Stoeve (2010), energies 10.0–32.2 eV at their 14 ionized fractions (generated by src/Utils/furlanetto2010_tables_to_fortran.py)	1502
radiation/charge_exchange.f90	charge-exchange network (Huang et al. 2023, Table 4), as residual rows and as the stoichiometric sources of the transported stages	1498
nonlinear_system_solver/constrained_chemical_equilibrium.f90	chemical equilibrium of the molecular H/He network in logarithmic number densities, element conservation as explicit residual rows	1238
lower_atmosphere/h2_self_shielding_table.f90	H_2 Lyman–Werner dissociation cross section with overlapping lines, and its two branching ratios	952
states/base_boundary.f90	the lower boundary as a characteristic condition at the face, the number of conditions set by the Mach number there	723

New module	Purpose	SLOC
time_step/viscous_conduction.f90	Navier–Stokes viscous momentum force, its dissipation, and thermal conduction with the conductivity of the mixture (Crank–Nicolson)	671
init/molecular_seed_from_atomic_state.f90	the molecular state a solved atomic state of the same wind implies, written as an initialization file: an imposed H ₂ partition capped by each cell's own neutral hydrogen, closed on the caloric equation of state at fixed pressure or at fixed temperature	595
time_step/steady_residual.f90	finite-volume stationary residual $R = du/dt$ and its norms, shared by the diagnostic, the convergence monitor and the stationary solver	580
time_step/energy_semi_implicit.f90	implicit source step for the thermal energy of a cell, at fixed composition and heating	535
radiation/excited_hydrogen.f90	non-LTE H($n=2$) 2s/2p equilibrium (Christie, Arras & Li 2013) and the Balmer feedback	508
time_step/conservation_budget.f90	every term of the three finite-volume conservation rows of one assembly, with the geometry, the potential and the gravitational work the energy flux difference carries inside it	493
files_IO/conserved_state_restart.f90	the exact code-unit state $(\rho, \rho v, E, f_{sp})$ written beside the dimensional pair, and read back only when every recorded condition matches, so that a restart starts at the bits the run ended on	466
lower_atmosphere/element_inventory.f90	one stoichiometric map of the H, He, O and C nuclei of a cell, and the three constraints a composition state is judged by	448
nonlinear_system_solver/System_HeH_mol.f90	coupled H/He + molecular (H ₂ /H ₂ ⁺ /H ₃ ⁺ /HeH ⁺) system, turnover-scaled rows, with the gross production and loss of each row on request	448
lower_atmosphere/molecular_reaction_heat.f90	net chemical heat of the collisional reactions of the H ₂ /He network, with the dissociative recombinations that leave a product excited charged to that level and not to the gas	416
files_IO/lower_atmosphere_profile.f90	lower-atmosphere profile reader: log- p interpolation, elemental reservoirs, $K_{zz}(p)$	396
nonlinear_system_solver/ion_residual_core.f90	shared ionization-residual rows and analytic-Jacobian pieces of the System_* family	387
states/caloric_eos.f90	caloric equation of state: internal energy density, pressure and temperature, with the H ₂ level ladder	367
lower_atmosphere/oxygen_rates.f90	rate coefficients and thermodynamic data of the oxygen-hydrogen chemistry, with their validity ranges	316
functions/element_census.f90	one definition of the elemental and charge content of a composition state, and the invariant checks built on it	307
radiation/electron_energy_degradation.f90	degradation of a fast photoelectron in a partly ionized H/H ₂ /He gas: the heat fraction and the ionizations of each target	305
time_step/coupled_block_handover.f90	the state at which the wind-composition alternation stops on its movement bound, and the one recorded pair (u, f_{sp}) the coupled carrier solve is entered from	298
radiation/lya_rt.f90	Ly α transfer by the escape-probability closure, evaluated in line each step	261
functions/ionization_stage_transport.f90	transport of the ionization stages of one element on that element's own nucleus flux, the neutral stage closing the simplex, so that moving charge between stages moves no nucleus	235

New module	Purpose	SLOC
radiation/low_energy_electron_degradation.f90	partition of the 10.2–30 eV photoelectrons into heat, H I and He I ionizations and escaping line radiation (Furlanetto & Stoeve 2010), the option Low-energy electron partition	234
lower_atmosphere/molecular_infrared_cooling.f90	infrared exchange of the molecular layer: H ₂ O and CO vibration-rotation bands, H ₂ quadrupole and magnetic-dipole lines	231
lower_atmosphere/h3p_cooling.f90	optically thin H ₃ ⁺ infrared cooling (Miller et al. 2013)	228
functions/h2_photo_channels.f90	H ₂ photoabsorption resolved into mutually exclusive final-state channels, with one stoichiometric table	224
lower_atmosphere/mol_rates.f90	molecular reaction rates (Koskinen et al. 2022, Table 1)	206
files_IO/metals_input_read.f90	parse metals.inp: trace-metal abundances at runtime	191
radiation/opacity_models.f90	opacity-model dispatcher for the photoionization cross sections (A/C/P/T)	180
states/boundary_state_trace.f90	the distance between the cached base face state and the one the installed composition would give now, measured and never decided on	175
nonlinear_system_solver/System_HeH_metals.f90	coupled H/He+metals ionization (+ analytic Jacobian)	166
flux/species_face_flux.f90	advection of the species mass fractions on the same faces, with the same mass flux, as the density	163
functions/composition.f90	single point turning (ρ, f_{sp}) into every species density, n_e and n_{tot} , and converting pressure and temperature	162
time_step/hydrodynamic_rows.f90	the hydrodynamic rows of the stationary residual in double precision, instantiated from the kind-generic text	161
lower_atmosphere/h2_vibrational_relaxation.f90	the fate and the size of the vibrational energy left in an H ₂ molecule, for both of its drivers	155
nonlinear_system_solver/System_HeH_mol_metals.f90	molecular + metals merged ionization system	147
lower_atmosphere/lyman_werner.f90	H ₂ photodissociation in the Lyman and Werner bands (the Solomon process)	142
lower_atmosphere/co_photodissociation.f90	CO photodissociation on the 912–1201 Å band	127
nonlinear_system_solver/newton_solver.f90	damped Newton with an analytic Jacobian for the ionization systems, with the MINPACK hybrid1 retry	127
init/species_table.f90	metadata table of the metal stages and of the base, molecular and oxygen species	126
lower_atmosphere/water_photolysis.f90	H ₂ O and OH photodissociation in the four far-ultraviolet bands of the oxygen option	111
lower_atmosphere/co_self_shielding_table.f90	CO self-shielding function $\Theta[N(\text{CO}), N(\text{H}_2)]$ (Visser, van Dishoeck & Black 2009)	110
nonlinear_system_solver/ion_cell_state.f90	named-field cell state of the ionization residuals, in place of the position-packed params	110
states/stationary_operator.f90	the spatial operator a stationary state is a state of, and the face mass flux $r^2(\rho v)_f$ it puts through every face	107
radiation/hydrogen_n2_rates.f90	atomic data and rate coefficients of the H $n=2$ manifold and of the Ly α line, in one place	99
flux/low_mach_dissipation.f90	gated fourth-difference stress damping the two-cell velocity mode a contact-resolving flux leaves undamped	87
lower_atmosphere/lower_column.f90	analytic lower-atmosphere column from the 1-bar level to the base pressure (Koskinen et al. 2022)	87

New module	Purpose	SLOC
files_IO/opacity_input_read.f90	parse opacity.inp	85
init/blas_thread_policy.f90	the thread count of the LAPACK library the binary links, resolved at run time and set to one unless the run states it	73
nonlinear_system_solver/System_HeH_TR_metals.f90	merged He 2 ³ S+metals system	72
nonlinear_system_solver/constrained_equilibrium_probe.f90	standalone re-evaluation of one saved constrained-equilibrium state, without the hydrodynamics	15
simulation-build new-module total		42 168

The other 131 are the standalone test drivers under `src/tests/`, which did not exist in ATES at all. They state what a routine is allowed to return and are run by suite, not by the production binary:

Test driver	What it asserts	SLOC
tests/krylov_and_dogleg/krylov_and_dogleg_tests.f90	what the linear solve and the step geometry of the stationary solver are allowed to return	2755
tests/steady_species_rows/steady_species_rows_tests.f90	the unknown space of the stationary system, one row per transported balance	1826
tests/ionization_stage_flux/ionization_stage_flux_tests.f90	the stage fluxes sum to the element nucleus flux at every face, against the formula of the derivation and not against the module itself	1536
tests/carrier_retry/carrier_retry.f90	the carrier-local checkpoint and the rejection-and-retry controller, on a real molecular column	1463
tests/certification/certification_contexts.f90	the certification evaluator and its contexts, against the predicates they replace	1156
tests/diffusion_tests.f90	the binary H/He element-diffusion operator: analytic limits and conservation	1043
tests/grid_and_gates/hydrostatic_residual.f90	the momentum row returns zero on a hydrostatic state	1031
tests/charge_exchange_detailed_balance/charge_exchange_detailed_balance.f90	the forward and reverse rate coefficients of every charge-exchange pair obey microscopic reversibility with the partition functions of the four species	950
tests/element_operator/element_operator_tests.f90	the boundary the element transport operator is measured under	945
tests/stage_row_balance/stage_row_balance_tests.f90	the signed sum of the gross channels of an ion-stage row is that stage's source, in molecular and in atomic gas and with the He 2 ³ S level tracked and untracked	898
tests/carrier_retry/closure_returned_state.f90	what the thermochemical closure hands back on a molecular column: the composition, its chemical residual, the conditioning, and the state a refused trial puts back	890
tests/physics_probe/molecular_energy_recipients.f90	the enthalpy of a reaction is untouched by where its energy goes: heat, radiation and internal energy close on the one formation table, and the R12/R15 pair stays in detailed balance collider by collider	750
tests/grid_and_gates/base_boundary_continuity_probe.f90	the base face density does not move along a perturbation of the wind window once the branch is decided, and the two limits coincide at a vanishing window	592

Test driver	What it asserts	SLOC
tests/h2_level_ladder/h2_ladder_model.f90	a reduced statistical equilibrium of the H ₂ ground-state ladder from the published level, radiative and collisional data, which brackets the scalar thermalized fraction	561
tests/species_face_flux/species_face_flux_tests.f90	species advection on the face mass fluxes, and the element rows of the diffusion operator	549
tests/low_mach_stress_energy/low_mach_stress_energy_form.f90	the discrete kinetic-energy quadratic form of the gated fourth-difference stress, and the sign a coefficient carried outside both differences does not have	519
tests/energy_update/conduction_floor_tests.f90	what the Crank–Nicolson transport stage returns, refuses and conserves	482
tests/physics_probe/weno3_reconstruction_order.f90	convergence order of the face reconstruction against the exact face value	482
tests/helium_metastable_transport/helium_metastable_transport.f90	the He 2 ³ S population carried as a fourth helium state: the carried simplex, the write-back, the local limit, the source ledger and a manufactured three-state helium column	438
tests/charge_exchange_rows/reaction_source_orientation.f90	one electron transfer moves $-R$, $+R$, $-R$, $+R$ over four stages, creates no nucleus and makes no net charge, in each of the three equation bases	428
tests/grid_and_gates/base_reservoir_discrete_equilibrium.f90	whether the base boundary shares one discrete hydrostatic equilibrium with the well-balanced reconstruction, on static columns above the reservoir	420
tests/charge_exchange_rows/transported_stage_sources.f90	the charge-exchange source of a transported ionization stage is the source the local systems' rows carry, in each of the three bases the metal systems assemble	419
tests/spectrum_type/loaded_sed_table_semantics.f90	what a row of a loaded SED table means, and what the production consumers make of it	417
tests/attempted_step/attempted_step_tests.f90	the checkpoint and the step-size policy of the attempted-step controller	415
tests/a2_m1/a2_m1_rate_check.f90	the oxygen rate coefficients against their published sources	406
tests/spectrum_type/balmer_band_quadrature.f90	quadrature of the Balmer-continuum band integrals of excited_hydrogen	367
tests/coupled_block_jacobian/coupled_block_linear_system.f90	the directional action of the coupled block on one frozen iterate: its repeatability, and its perturbation ladder row group by row group	353
tests/physics_probe/metal_photoionization_fits.f90	every metal photoionization cross section against the published fit, and only where it is valid	343
tests/element_census_tests.f90	the elemental and charge invariants of a composition state	340
tests/interdiffusion_enthalpy_flux/interdiffusion_enthalpy_flux_tests.f90	the interdiffusion enthalpy flux $\sum_s h_s J_s$ (Cook 2009): conservative, zero where the element fluxes are, equal to the hand-evaluated face value, and absent with its key off	336
tests/energy_update/energy_update_tests.f90	the residual-controlled temperature solve of the implicit energy source step	334
tests/grid_and_gates/base_closure_candidates_probe.f90	two discriminants for the same characteristic base condition, read on one state under the production stationary operator	328
tests/physics_probe/recombination_photoelectron_partition.f90	the photoelectrons of the recombination radiation absorbed on the spot pass through the partition of the stellar photoelectrons	326

Test driver	What it asserts	SLOC
tests/boundary_state/contact_low_at_the_base.f90	the lower boundary as a contact: a column on the reservoir isentropic stays at rest, a pressure mismatch launches the acoustic response, and the entropy is upwinded on the matched face velocity	316
tests/carrier_helium_inventory/carrier_helium_inventory.f90	the helium a cell holds is the helium it had: the admissible stage partition leaves out the nucleus inside HeH^+ and the population in $\text{He } 2^3\text{S}$	310
tests/charge_exchange_rows/he_h_pair_reservoir.f90	the $\text{He} \leftrightarrow \text{H}$ charge-exchange pair reacts from the ground singlet, the same one in every system carrying helium and a metastable	304
tests/spectrum_type/loaded_sed_threshold_edges.f90	the photon grid built from a loaded SED, and the rates formed on it	304
tests/grid_and_gates/low_mach_face_mass_flux.f90	the face mass flux stays accurate relative to the wind as the Mach number of the lower layer falls to 10^{-9}	303
tests/ionization_imposed_fractions/imposed_ionization_rows.f90	an imposed ionization row is the identity $x - x_{\text{fix}}$ and is one number in all seven systems, and nothing depends on the carried values where nothing is imposed	289
tests/coupled_block_handover/coupled_block_handover_contract.f90	the transition into the coupled carrier solve is reached once and from the recorded entry state bit for bit, never from a live token the stalling pass cannot carry	285
tests/h2_level_ladder/h2_ladder_statistical_equilibrium.f90	the solved ladder against the first-event branching fraction: detailed balance and the thermal limit, and the four ways an injected formation energy leaves	275
tests/physics_probe/species_formation_energy_table.f90	the formation-energy table of the energy ledger, and the reaction energies that are its differences	275
tests/spectrum_type/wasp121b_sed_sensitivity.f90	the production band quantities of the three WASP-121 b spectrum candidates	273
tests/physics_probe/lya_band_transmission.f90	the stellar $\text{Ly}\alpha$ band reaching a cell through the atomic hydrogen column above it	272
tests/physics_probe/one_minus_exp_over_x_sites.f90	$(1 - e^{-x})/x$ of the whole code and the four quantities formed from it besides the absorbed fraction	263
tests/coupled_source_step/coupled_source_tests.f90	the coupled temperature-composition source step	260
tests/grid_and_gates/base_equilibrium_columns.f90	the static and the wind-carrying columns, and the one production evaluation, shared by the two measurements of the discrete equilibrium of the base	256
tests/physics_probe/inner_shell_photoabsorption_ledger.f90	the relaxation of an inner-shell metal vacancy (Kaastra & Mewe 1993), the energy ledger of the event, and the two-stage jump in the metal rows	246
tests/charge_exchange_rows/metal_helium_reservoir.f90	the metal + He reactions of group C react from the ground singlet, the same one in the three systems that assemble them	235
tests/physics_probe/lya_beam_cell_mean.f90	the stellar $\text{Ly}\alpha$ field of a cell is the beam mean over the cell's own optical depth	234
tests/physics_probe/lyman_werner_cell_mean.f90	the Lyman–Werner rate a cell carries is the mean of the local rate over that cell	234
tests/spectrum_type/balmer_continuum_field.f90	the two scalar Balmer-continuum rates of $\text{H}(n=2)$	227
tests/carrier_returned_state_acceptance/carrier_returned_state_acceptance.f90	what makes a returned carrier state a solution, and what does not	225

Test driver	What it asserts	SLOC
tests/physics_probe/positivity_limiter_scaling.f90	the positivity scaling of the reconstructed face states (Zhang & Shu 2010)	223
tests/physics_probe/ground_capture_and_cascade_data.f90	the atomic data of the recombination and $n=2$ physics against the published tables they come from	222
tests/low_energy_electron_degradation/low_energy_electron_partition.f90	the 10.2–30 eV photoelectron partition against the authors own interpolator of the same tables, and the energy budget the tables close	216
tests/grid_and_gates/photon_grid_threshold_edges.f90	every active absorber's threshold against the bin partition of the photon grid	213
tests/physics_probe/photoionization_field_substitution.f90	the order in which the field and the composition that attenuates it are brought into agreement	209
tests/adv_static_limit/enthalpy_flux_of_mass_divergence.f90	the third term of the advected energy equation	202
tests/adv_static_limit/stationarity_by_the_mass_row.f90	which test decides that the state the post-process was handed is stationary in a cell	202
tests/charge_exchange_rows/advection_pair_reservoir.f90	the advection-corrected system hands the He $\leftrightarrow$ H pair the same ground singlet the equilibrium system does	199
tests/carrier_enthalpy_flux/carrier_enthalpy_flux.f90	the sensible enthalpy flux of the transported molecular carriers: zero at the column ends, conservative, the carrier operators own face flux times the H ₂ enthalpy, and absent with its key off	198
tests/carrier_outer_boundary/carrier_outer_boundary.f90	the carrier column is closed to diffusion at both ends, one outflow ghost rule serves every evaluation, and the advective band entries are the operator's own derivative	196
tests/physics_probe/co_destruction.f90	the CO destruction rates, the shielding table, the deposited energies and the cell mean	195
tests/carrier_reference_scales/carrier_reference_scales.f90	the element a transported carrier is made of, and whether its chemistry rows can be differentiated	191
tests/ionization_stage_enthalpy_flux/ionization_stage_enthalpy_flux.f90	the sensible enthalpy the carried ionization stages move relative to their element, against a manufactured column	186
tests/spectrum_type/photon_grid_n2_floor.f90	the photon grid built when the excited-hydrogen coupling is armed	184
tests/carrier_constraint_attribution/carrier_constraint_attribution.f90	which residuals belong to the solver and which to a constraint	179
tests/physics_probe/recombination_radiation_on_the_spot.f90	the escape weights of the ground captures and the photon and energy accounting of the recombination radiation absorbed where it is emitted	176
tests/grid_and_gates/base_branch_discriminant.f90	what the base contact reads its direction off: the wind window wherever it carries one flux, the matched face velocity elsewhere, never the first cell's own product	174
tests/grid_and_gates/free_outflow_boundary.f90	the outer free-outflow boundary on a supersonic wind, and its ghost state	174
tests/spectrum_type/planck_photon_field.f90	the incident flux put on the photon grid for the photospheric blackbody type	173
tests/physics_probe/lya_escape_probability.f90	the escape probability of the two-level closure against the published slab solution	169
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Test driver	What it asserts	SLOC
tests/carrier_boundary_jacobian/carrier_boundary_jacobian.f90	the boundary derivatives of the carrier advection, a copied ghost carrying one and a stated one carrying none	165
tests/a2_m2/a2_m2_kinetics_check.f90	the kinetics quantities the oxygen option adds on top of the rate audit	163
tests/e1_h2/e1_h2_channel_check.f90	the mutually exclusive H ₂ photoabsorption channels	161
tests/physics_probe/roe_low_mach_velocity_jump.f90	the low-Mach scaling of the normal velocity jump on the ROE branch, against the analytic three-wave flux (Rieper 2011)	152
tests/physics_probe/photoionization_field_self_consistency.f90	the field one cell is solved in, against the composition that cell returns	148
tests/physics_probe/co_helium_ion_sink.f90	He ⁺ + CO as a species sink, not only as a heat source	147
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tests/grid_and_gates/photon_grid_quadrature.f90	the photoionization rates of one unattenuated cell against the band quadrature	146
tests/physics_probe/riemann_wave_speeds.f90	signal speeds at a face, and the state one conservative update with that flux leaves behind	143
tests/physics_probe/helium_level_cooling_ledger.f90	which helium level each cooling channel of eval_cool is charged to	140
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tests/physics_probe/heating_channel_closure.f90	one assembly of the volumetric heating rate, and the closure of its channels	138
tests/adv_static_limit/upstream_caloric_energy.f90	the upstream energy of the advected energy equation	137
tests/acceptance_classes/acceptance_class_contract.f90	one meaning of the ionization acceptance classes for every consumer	132
tests/physics_probe/metal_ionization_energy_ledger.f90	the energy metal collisional ionization and recombination take from the electron gas, and the ledger of electron gas, ionization energy and radiation they close	128
tests/physics_probe/absorbed_fraction_switch.f90	$(1 - e^{-d})/d$, the fraction of the photons entering a cell that the cell absorbs	126
tests/physics_probe/opacity_model_p_ledger.f90	the radiative ledger of a cell whose cross sections are broadened by opacity model P	124
tests/physics_probe/helium_584_resonance_line.f90	the fate of the He I 584 Å recombination photons in a cell with He I (Wood, Mathis & Ercolano 2004)	123
tests/grid_and_gates/outer_shells_base_identity.f90	the outer shells leave the constructed grid unchanged bit for bit, close on the stated face and widen in geometric progression	122
tests/physics_probe/cooling_assembly_identity.f90	the cooling the post-process temperature root balances is the cooling eval_cool reports, channel for channel	118
tests/physics_probe/photoevent_energy_ledger.f90	the energy an isolated photoabsorption event has to account for	117
tests/physics_probe/xuv_cell_mean_attenuation.f90	the beam attenuation the photoionization rate of a cell is evaluated with	115
tests/adv_element_velocity/element_velocity_rate_weights.f90	each element's rows of the advection-corrected systems are stepped on the velocity of that element's nuclei	112
tests/physics_probe/helium_bray_collision_strengths.f90	the electron-impact collision strengths of the He I metastable network against Bray et al. (2000), Tables 2 and 3	105

Test driver	What it asserts	SLOC
tests/grid_and_gates/grid_window_indices.f90	the two index statements that decide which cells a quantity is read from	101
tests/grid_and_gates/base_cell_width_provenance.f90	the base cell width is written as a decimal that reads back to the same bits, and the default and the pinned line state the widths their runs were made on	98
tests/physics_probe/h2_rovibrational_identity.f90	the identities tying the H ₂ partition function of the chemistry to the internal energy of the caloric EOS	96
tests/adv_static_limit/stationary_flux_refusal.f90	the sensitivity of the advected energy correction to a non-stationary mass flux	91
tests/physics_probe/ionization_threshold_turn_on.f90	one definition of each ionization threshold, and every cross section turning on exactly there	90
tests/grid_and_gates/base_grid_pinned_width.f90	which grid each of the four base cell widths builds, compared on the bits of every cell center	86
tests/physics_probe/voronov_table_rows.f90	every Voronov (1997) electron-impact ionization fit of the code against the published row	84
tests/grid_and_gates/grid_width_identity.f90	the cell-width and total-width identities of the radial grid	83
tests/physics_probe/h2_channel_detector_ratio.f90	the measured proton-to-H ₂ ⁺ ratio of Chung et al. (1993) against the four channels	83
tests/physics_probe/helium_metastable_detailed_balance.f90	the He I 1 ¹ S ↔ 2 ³ S electron collisions in detailed balance, finite where the excitation underflows	83
tests/physics_probe/fine_structure_escape_probability.f90	the escape probability of the ground-term fine-structure lines, and the depth it is fed	82
tests/physics_probe/hydrogen_helium_excitation_data.f90	the collisional-excitation cooling of H I, He I and He II against the coronal sums it is fitted to, and the shared H(<i>n</i> =2) rate set	81
tests/physics_probe/atomic_mass_and_radius_constants.f90	the helium mass and the astronomical constants of the input readers, one definition each	79
tests/physics_probe/recombination_energy_loss.f90	the kinetic energy the recombination cooling removes in each capture onto H II, He II and He III, $kT\alpha(3/2 + d\ln\alpha/d\ln T)$	79
tests/physics_probe/assertion_report.f90	one printed verdict line per assertion, and the failure count the driver exits on	74
tests/adv_static_limit/thermal_damkohler_limit.f90	the validity condition of the advection-corrected energy solve, as a function of the flow speed	72
tests/a2_m2/a2_m2_g3_run_check.f90	with every photolysis rate at zero, the solved O/OH/H ₂ O partition is the chemical equilibrium	71
tests/constrained_network_layout/transported_proton_constraint_layout.f90	which species are unknowns of the continuation, and whether the system built is square	71
tests/test_columns.f90	synthetic columns whose species carry the mass density they are measured against	67
tests/physics_probe/h2_infrared_line_populations.f90	the LTE populations weighting the H ₂ quadrupole lines are those of the caloric EOS ladder	65
tests/charge_exchange_rows/turnover_cell_rates.f90	the charge-exchange turnover bound is the loaded cell's, carries no history, and is each thread's own	63
tests/physics_probe/wind_ae_bridge_composition.f90	the composition an EXHALE input.inp hands the Wind-AE solver	54

Test driver	What it asserts	SLOC
tests/spectrum_type/fuv_band_ quadrature.f90	the band integral of a loaded spectrum against closed-form answers, with the band edges inside segments, on nodes and at the table ends	53
tests/physics_probe/voronov_carbon_ ionization.f90	the electron-impact ionization of C^0 and C^+ against the published Voronov (1997) rows	52
tests/physics_probe/water_photolysis_ lyman_alpha_yields.f90	the H_2O branching on the band that contains $Ly\alpha$, against the published yields	49
tests/physics_probe/h3p_recombination_ branching.f90	the two product channels of H_3^+ dissociative recombination	44
tests/energy_update/cooling_models.f90	analytic cooling laws for the tests of the implicit energy update	43
tests/species_masses/carrier_background_ masses.f90	the mass the carrier transport gives each of its background collision partners	36
tests/charge_exchange_rows/stale_cell_ rates_refused.f90	a residual assembled at a temperature the rate coefficients were not filled at is refused	31
tests/charge_exchange_rows/stale_cell_ turnover_refused.f90	a turnover bound taken at a temperature the rate coefficients were not filled at is refused	27
tests/certification/certification_ capacity.f90	one equation past a full certification report is refused by name, before any field is written	26
tests/restart_radius_guard/restart_radius_ guard_probe.f90	a restart whose recorded cell centers are not finite or differ from the runs grid is refused	26
tests/charge_exchange_rows/stale_cell_ stage_sources_refused.f90	a stage source evaluated at a temperature the rate coefficients were not filled at is refused	24
test-driver total		38 947
new-module total		81 115

So the 190 new core modules contribute 81 115 of the 107 432 added code lines; the remaining 26 317 added (and 1 338 of the 1 407 removed, the other 69 being the deleted HLLC estimator) are edits spread across the 38 modified files. The ten heaviest, by added code lines, are the main driver (EXHALE_main.f90, 5 110 added, 246 $\rightarrow$ 5 225 SLOC: the marching loop, the attempted-step adoption boundary, the run-mode contracts and the written-state assertions), the cooling module (Cool_coeff.f90, 3 837 added, 228 $\rightarrow$ 3 882 SLOC: CHIANTI metal-line fits, Fe tables, fine-structure statistical equilibrium, Penning and molecular rate coefficients), the ionization equilibrium and its diagnostics (ionization_equilibrium.f90 3 243 and util_ion_eq.f90 2 139: the metal, molecular and oxygen networks, the acceptance classes, the photoionization field and the heating channels), input parsing (input_read.f90, 1 986), the state reader and writer (load_IC.f90 1 774, write_output.f90 1 562, write_setup_report.f90 989), the advection-corrected post-processor (post_process_adv.f90, 1 533) and the cross sections (cross_sec.f90, 596). Those ten carry 22 769 of the 26 317. The rest of the edits are in the reconstruction, the fluxes, the boundary conditions, the grid, the time step, the energy equation and the System_* ionization systems, which wire the metals, the molecules, the opacity dispatcher, the Roche potential, the transonic/auto/Wind-AE IC, the excited-H and $Ly\alpha$ physics, the element diffusion and the Brent/Newton solver upgrades into the original solver.

In total the EXHALE core is now $\approx 31.2\times$ the original code base (3 514 $\rightarrow$ 109 539 SLOC), or $\approx 20.1\times$ counting only the simulation build (70 592 SLOC), touching almost all of the original files (38 of 47 modified, one deleted, plus 190 new); with the ported Wind-AE IC generator the full Fortran tree is 272 files / 114 295 SLOC ($\approx 32.5\times$). Beyond

Fortran, the tree also carries the Python tooling (GUI, the EXHALE_transit.py transmission post-processor, plotting/-analysis, the lower-atmosphere adapters and the elemental-flux closure driver) and 351 309 lines of documentation (460 md/tex files, notebook checkpoints excluded: the memos, the manual, the dated audit directories and the update logs) under docs/ and md/ (the Markdown memos moved to md/ on 2026-09-25).

Fortran source inherited unchanged from ATES

Of the 47 upstream Fortran files (ATES/ATES-Code-main), **8 are byte-for-byte identical** in EXHALE (re-measured 2026-10-05, and the same eight as on 2026-09-11 and 2026-09-22: nothing entered or left the set in between): the numerics EXHALE reuses without touching, which are now the linear-algebra part of the included MINPACK nonlinear solver plus the gravity pre-evaluation. Table 55 lists them with their role. No file is identical after whitespace and case normalization but not byte-identical, so the normalized-unchanged set and the byte-identical set are the same eight. Five files left the byte-identical set since the 2026-08-30 edition of this table, each for a documented reason rather than drift.

- `nonlinear_system_solver/hybrd.f90`, `hybrd1.f90` and `fdjac1.f90` carry a new `unit_step_floor` argument threaded from `hybrd` to `fdjac1`, which selects the forward-difference step of the Jacobian: `.false.` is MINPACK's own $h = \varepsilon |x_j|$ with the exact-zero fallback, `.true.` floors it at unit magnitude, $h = \varepsilon \max(1, |x_j|)$, which is the rule for unknowns already scaled so that unity is their natural size (the logarithmic unknowns of the constrained chemical equilibrium). The params dummy also became assumed-size, so one transport array serves systems of different length. `hybrd1` passes `.false.` and is therefore unchanged in behavior.
- `states/Source.f90` gained the well-balanced option of 2026-09-10 (item N37 of the stage-2 solver program): under `Well balanced: True` the gravitational source and the geometric pressure term are not evaluated here at all, because they cancel the equilibrium part of the momentum flux difference identically and `RK_rhs` assembles the row from what is left (Käppeli & Mishra 2014, J. Comput. Phys. 259, 199, eq. 2.26). The file also reads its pressure through `caloric_eos` instead of a constant γ .
- `radiation/J_inc.f90` carries the single-spectrum-type decision: `Spectrum type:` now states the spectrum of the whole photon grid, the XUV and the part below 13.6 eV alike, and no band is filled from a type the input did not select. The file also became the one definition of the eV-to-Hz conversion and of the Balmer-continuum head $e_{\text{th,HI}}/4$.

The two wave-speed estimators are no longer in the “effectively untouched” category either. `flux/speed_estimate_HLLC.f90` was deleted as dead code, as recorded above. `flux/speed_estimate_R0E.f90` was rewritten (99 code lines added, 43 removed) into a star-state estimate that returns a status: the primitive-variable, two-rarefaction and two-shock branches of Toro (2009) chapter 9 are selected by the pressure ratio, the vacuum condition of his eq. 4.76 is tested first, and a face with no admissible star state is reported rather than given a square root of a negative ratio.

Table 55: The 8 Fortran files byte-identical between the original ATES (ATES-Code-main) and EXHALE (measured 2026-10-05), grouped by role.

File	Purpose
<i>MINPACK linear algebra and constants</i> (Argonne; dir nonlinear_system_solver/)	
qrfac.f90	QR factorization via Householder transformations
qform.f90	Accumulate the orthogonal factor Q from the QR factorization
r1updt.f90	Rank-one update of the QR factorization
r1mpyq.f90	Apply the Givens-rotation sequence of the rank-one update
dogleg.f90	Dogleg (trust-region) step combining Gauss–Newton and steepest descent
enorm.f90	Euclidean norm of a vector (overflow/underflow safe)
dpmpar.f90	Machine floating-point precision / range constants
<i>Grid setup</i>	
init/set_gravity_grid.f90	Pre-evaluate gravity at cell centers and interfaces